

AN INTRODUCTION TO PROBABILITY AND STATISTICS

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Preface

Probability theory and statistics provide fundamental tools to measure chance, set expectations, estimate parameters, forecast future events, design experiments, collect, analyze, and incorporate empirical evidence into decision making.

In this book, we wish to provide an introductory level knowledge for probability theory and statistics to an individual who wants to understand the meaning of these subjects and has mature mathematical language skills, that is to say who knows limits, derivatives, integrals, vectors and matrices, at the level of an undergraduate college student.

Thus, this book can be used for a self study, or advanced undergraduate course, as well as a beginning level graduate course. It is strongly suggested that for a sound understanding of statistics, this book's information should be followed by multivariate statistical analysis and design of experiment subjects.

Also, it is expected that the reader is capable of learning and using various software packages, so that computers could be used as convenient support systems.

Examples and exercises are provided to enforce learning and enrich curiosity, many more practical problems are expected to be identified and solved by the reader.

We used $\text{\LaTeX}$ 2 _{ϵ} to type this book and occasionally used Excel, Maple, and R softwares to generate figures and tables.

Chapter 1

Meaning of Statistics

“Where all the children are above average...” - Garrison Keillor.

1.1 Introduction

Once upon a time, governments collected facts and kept records about their populations, such as number of births and deaths in a day, age and gender of the people, marriages, business transactions, etc., in order to utilize these facts to make proper decisions, so that they collect taxes from their population, or, could raise armies to go to war with their adversaries.

The knowledge of collecting, organizing, processing and analyzing facts, in order to acquire useful information to make proper decisions, is known as *statistics*.

It is obvious that all humans need to know statistics to carry out their daily activities by making reliable decisions based on facts.

In this day and age, the advent of technology makes it possible to collect immense amount of data continuously and process them to make almost instantaneous decisions. You may see many examples of these data collection technologies in our daily lives, such as Google Search trends or crowd-sourced traffic incident reports.

1.2 Why We Care about Statistics

This field of statistics can be separated into three major areas. The first one is called the *descriptive* statistics, the second one is called the *inferential* statistics, and the third one is called the *experiment design*.

Descriptive statistics refers to the body of knowledge for organizing and

representing facts, where observed raw facts are called data, plural form of the Latin word datum. In the Roman empire it meant daily record of events (data sometimes used as a singular word in English), to represent them in a more efficient and effective manner.

Inferential statistics refers to the body of knowledge that leads one to acquire information about a population of interest by utilizing data, in order to provide a basis for that information. That is, we carry induction on the sample (data) information to acquire some useful knowledge about the unknown population characteristics, also known as parameters (inferential statistics is also known as inductive statistics). Since this process is a generalization (induction), it is subject to the risk of making the wrong decision, based on the available evidence. Hence, we want to measure the chance of making mistakes, where the measure of chance is the probability measure.

Experiment design refers to the methods and techniques of building data, in order to generate the necessary and sufficient information, by optimally utilizing the resources. That is to say, minimizing the time and the cost of collecting necessary observations.

It is often the case that we say anything is possible, yet if we want to know how possible an event is, then we employ the probability measure to tell us the degree of the possibility, similar to a temperature measure, which gives us the degree of hotness. For example, you may say today is a hot day, but there is a difference between 99 degree Fahrenheit and 80 degree Fahrenheit days, even if they are both hot days.

The probability measure we employ is the Kolmogorov's set function, denoted as P , that assigns a numerical value between zero to one to the chance of an event, as the degree of possibility [6]. There are also alternatives to Kolmogorov's construct, for example see [16].

The term *population* has a generic meaning. It refers to a set of entities that we care to know about, or set of values of a measurable entity of interest. The term *sample* denotes an observable (measurable) subset of the population of interest. The simplest reason that we use a sample to learn about a population, instead of investigating the population itself, is the cost of learning. Each observation that reveals some information about the characteristics of a population inherently costs something. Sometimes it is not possible to study the whole population to learn about it, for example in order to assess the quality of a specific year's wine, you conduct a taste test of some randomly selected bottles, rather than testing every bottle of wine.

We construct and use models to represent reality, its structures and relationships. Thus, our models are abstractions that support our understanding of some entity. A model is not a perfect expression of our universe, and we wish to build clear, simple models to represent reality as closely as possible.

Historically, model builders have been using Occam's razor, which dictates that model building must be done with a very sharp instrument that cuts out all the unnecessary complexities and vaguenesses, in order to provide the simplest useful model.

Evidence based scientific models heavily use causality (cause and effect relationship), yet causal relationships are only embedded in the field of study of statistics, which seeks out probabilistic associations, based on the evidence. Thus, the person who utilizes statistics is the one with the responsibility to identify causality in the statistical model, otherwise one can easily make false statements, based on available evidence.

In this day and age, the technology is contributing very much to increase the efficiency of processing large data sets, such that computers are becoming much more economical to use and also effective to handle immense amount of data processing.

Computer software is instrumental to do data processing and statistical analysis. Among many others, Excel, SPSS, S-Plus, R and SAS are commonly available for statistical analysis. Some of these software packages are also used for data mining by practitioners. Data mining refers to automatically extracting patterns of interest from data-bases. Excel is a widely available spreadsheet software, even though it is not a dedicated statistical software such as SAS.

Practitioners are encouraged to learn and use statistical softwares, such as R, which is conveniently available as a freeware from the following website: <http://www.r-project.org>, and learn the similarities, as well as the differences, among them.

1.3 Descriptive Statistics

It is a common desire to understand the environment, based on observations. Statistics provides instruments to carry out this process of understanding, in order to facilitate probabilistic decision making.

The collection of entities that we are trying to understand is called a population. Sometimes we are able to observe the whole population, sometimes it is impossible to do so. If we cannot observe the whole population, then we utilize a sample, coming from this population, as a representative of the unknown population characteristics. Thus, we observe the sample, acquire information about the sample characteristics, and project this sample information to the unknown population characteristics that we are trying to understand, with some degree of acceptable risk of making the wrong judgement. The desire to observe, collect, and process data benefits from effective

and efficient procedures. The descriptive statistics provides means to process, organize and represent the data, so that the processed data can provide information in a simple to understand form.

We want to describe the data set by utilizing statistical measures (numerical values), in order to extract the necessary information for our decision making.

1.3.1 Frequency Distribution

The processing of the raw data set begins with organizing data into clusters or groups of similar observations, known as classes or categories, such that this sorting and arranging the values of the data set leads to a clearer picture of the observations. One common way of placing data in classes (categories) is to order the numerical values in a data set from its smallest value (minimum) to its largest value (maximum) and then break this range of values in the data set into mutually exclusive (disjoint), yet collectively exhaustive intervals of the values. That is, each and every one of these classes will capture some proportion of the ordered data values with no common (shared) observation, and collectively these classes will capture all possible values in the data set. If the data set consist of nonnumerical observations, then we can assign cardinal numbers (identifiers) to different characteristics. For example the sex of a group of individuals can be categorized by male and female and we can assign the number 1 to male and the number 2 to female category, or we may as well assign the number 2 to male and the number 1 to female category, with no order among these numbers (or, any two distinct numbers can be assigned to identify these two distinguishable characteristics). But, even with cardinal numbers, it is convenient to preserve the natural order among numbers.

Here, we introduce class boundaries by assigning a lower limit and an upper limit for the values in each class, and call the distance between the upper limit and the lower limit of a class as the class range, or class interval. Usually the midpoint in a class interval is called as the class mark.

After we allocate all the values in a data set in these classes, we count the number of values in each one of these classes, and we call it the class frequency (the number of observations in each class).

The frequency distribution is a relationship between the values in each and every one of these classes with their class frequencies, it is a function such that the class boundaries are its domain and the class frequencies are its range. A mid-point value of a class is known as the class mark. We can identify a frequency distribution function by its graphical representation, also known as a histogram. A histogram is a bar-graph of frequencies, that identifies a frequency distribution as the hight of a bar representing the frequency of a

class on the vertical, Y , axis and the length of a bar representing the class interval on the horizontal, X , axis, not necessarily but usually with constant (or equal) class intervals for all classes.

Example 1.1

In a classroom of 10 students, 6 of them were males and 4 of them were females. The frequency distribution of the sex in this particular classroom is:

Let X be the sex of an individual, such that $X = 1$ denotes a male, and $X = 2$ denotes a female. The following Table 1.1 illustrates the frequency distribution of the sexes.

Value (X)	Frequency(X)
1	6
2	4

Table 1.1: Frequency Distribution

A bar-graph that represents this relationship is called the histogram for this data set, where the horizontal axis refers to the class values and the vertical axis refers to their frequencies, as in Figure 1.1.

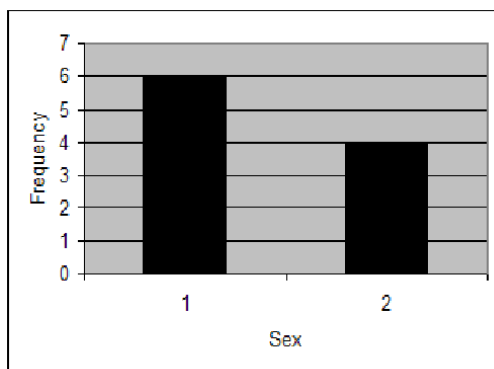


Figure 1.1: Histogram

Another graph representing a frequency distribution is called the frequency polygon, similar to a histogram, where line segments connect the points with coordinates (class mark, class frequency) for the frequency distribution, as in Figure 1.2.

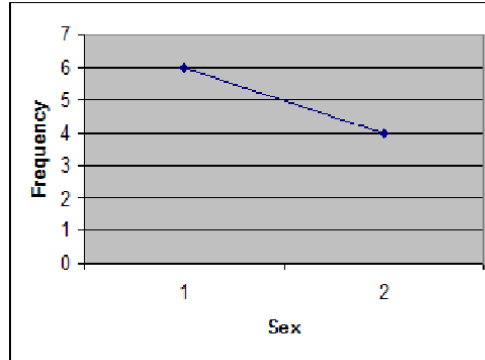


Figure 1.2: Frequency Polygon

The desire to make this relation to provide sufficient information for large data sets leads to the usage of relative frequency distribution, which converts class frequencies into relative class frequencies that shows percentage of the number of observations in each class, calculated as described below.

Let rf be the relative frequency for a class, cf be the class frequency, and n be the total number of observations, thus $rf = \frac{cf}{n}$.

For Example 1.1, we have

$$rf(1) = \frac{cf(1)}{10} \text{ and } rf(2) = \frac{cf(2)}{10}.$$

Table 1.2 gives the relative frequency distribution of the sexes.

Value (X)	Relative Frequency(X)
1	6/10
2	4/10

Table 1.2: Relative Frequency Distribution

As this relation implies, relative frequency distribution provides information about percentages (proportions) of observations, falling in each category.

This relationship is graphically represented by using a pie-chart, that is a circle divided into slices where a slice represents a class and the size of its area represents the relative frequency of that class, as in Figure 1.3.

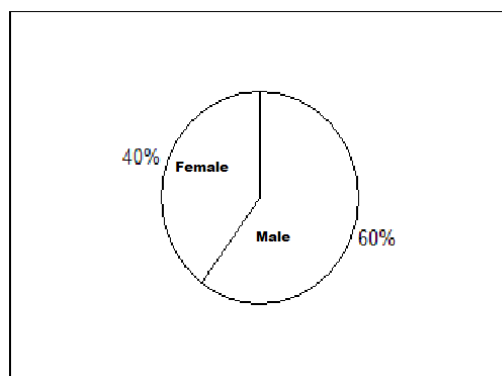


Figure 1.3: Pie Chart

A graph of a cumulative frequency distribution (or cumulative relative frequency distribution) can also be constructed, which is known as an ogive.

Usually horizontal axis refers to the classification values and the vertical axis refers to the cumulative frequency (or proportion) of the observations, as in Figure 1.4.

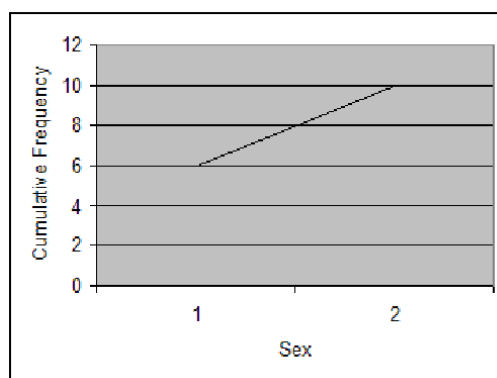


Figure 1.4: Ogive

There are data sets that have observations of ordered components, where these data are called multivariate data. A special case is a bivariate data set, where each observation is an ordered pair of variables, (X, Y) , such as (Age, Sex) pairs. The frequency distribution of countable bivariate data is tabulated in a contingency table.

Example 1.2

Let us assume that there are two variables for each observation of our data set, Age and Sex, such that Age is given in years, categorized in ten year intervals, and sex is categorized as Male and Female.

Age:	1 to 10	11 to 20	Right Margin
Female	20	15	35
Male	15	10	25
Bottom Margin	35	25	60

Table 1.3: Contingency Table

Then Table 1.3 would be the frequency distribution of this bivariate data set. The right margin and the bottom margin of this contingency table sum up over one factor, such as Age, or Sex. For example, you can find the frequency of all Females, as well as the frequency of all Males, on the right margin of this contingency table that eliminates the Age category. Similarly, the bottom margin would have the frequency of one to ten year olds, as well as the frequency of eleven to twenty year olds, without a Sex categorization. Therefore, the marginal information on any contingency table eliminates the effect of one category.

The intersection of margins yields the total number of observations. Hence, we could construct a relative frequency distribution of a bivariate data set, by dividing the frequency of each cell of the contingency table with the total number of observations, as in Table 1.4.

Age:	1 to 10	11 to 20	Right Margin
Female	20/60	15/60	35/60
Male	15/60	10/60	25/60
Bottom Margin	35/60	25/60	1

Table 1.4: Relative Frequency Table

1.4 Location Measures

The location measures, also known as the measures of central tendency, describe specific points as locations in a data set, where these measures identify the point-wise characteristics of the data.

1.4.1 Average

The most common location measure is the average, or the mean, of a data set. Unless declared otherwise, mean stands for the arithmetic average of the values in the data set. Let the values are denoted by $X = \{X_1, X_2, \dots, X_n\}$ where X identifies the character we are trying to learn and X_i identifies the yet unknown individual value of the i th observation in the data set, for $i = 1, 2, 3, \dots, n$. That is, there are finite number of n distinct observations in the data set and we assume that they all have finite values. Thus the mean of X (arithmetic mean), denoted by $\bar{X}$, is as follows:

$$\bar{X} = \left(\frac{1}{n}\right) \sum_{i=1}^n X_i = \left(\frac{1}{n}\right)(X_1 + X_2 + \dots + X_n). \quad (1.1)$$

We use the “bar” notation, $\bar{X}$, for the average, or mean.

When we have actual measurements for a data set, these observed values are known as “realizations”, and we use lower case letters to identify the measurements, such as $x = \{x_1, x_2, \dots, x_n\}$, where x_i identifies the measured value of the i th observation in the realization set, for $i = 1, 2, 3, \dots, n$.

Example 1.3

We randomly select an investment portfolio among many others and observe that it has ten different sources of investments to diversify with the following number of shares, identified by the set

$$\begin{aligned} x &= \{x_1, x_2, x_3, x_4, x_5, x_6, x_7, x_8, x_9, x_{10}\}, \\ x &= \{17, 20, 18, 17, 19, 21, 20, 17, 17, 18\}. \end{aligned}$$

Thus the mean (or average) number of shares of this portfolio is

$$\bar{x} = \left(\frac{1}{10}\right)(17 + 20 + 18 + 17 + 19 + 21 + 20 + 17 + 17 + 18) = 18.4.$$

Hence we say that the average number of shares of this group of investments is 18.4 shares per investment, yet as we saw from this data set none of the investments has 18.4 shares, since they are all integral (integer) numbers.

This implies that the mean may take a value that may not exists in the data set, and it is a value in between the minimum (smallest) and the maximum (largest) values in the data set. We can think of it as an equalizer point, or, an equilibrium point of the data set where all the elements are balanced about it, or the total collective value of the data set is redistributed equally, which is equal to the mean value, among all the members of the data set.

As if each and every one in this group of observations has 18.4 number

of shares (every investment has the same number of shares) their collective total number of shares would be 184. In other words, if you locate each individual investment's shares on a number line, these observations will balance themselves by placing the equilibrium point (fulcrum) at the point 18.4.

1.4.2 Weighted Average

If all n observations have different importances, then we use the weighted average, $\bar{X}_W$, by assigning weights denoted by w_i to each and every one of the observations. Thus giving

$$\bar{X}_W = \frac{\sum_{i=1}^n w_i X_i}{\sum_{i=1}^n w_i}. \quad (1.2)$$

Example 1.4

Let us assume that the investments observed in the previous example are working and providing return according to the corresponding number of shares. The observed return on investments, in thousand dollars a year unit, for this portfolio are as follows:

$w = \{7, 8, 7, 8, 10, 10, 9, 9, 10, 9\} = \{w_1, w_2, w_3, w_4, w_5, w_6, w_7, w_8, w_9, w_{10}\}$. Hence the weighted average return on investment in a year, for this portfolio is as follows:

$$\bar{x}_w = \frac{7*17+8*20+7*18+8*17+10*19+10*21+9*20+9*17+10*17+9*18}{7+8+7+8+10+10+9+9+10+9} = \frac{1606}{87} = 18.46.$$

Thus we say that this portfolio brings on the average \$18,460,000 per year.

1.4.3 Geometric Mean

If we have observations with positive values, $X_i > 0$ for $i = 1, 2, \dots, n$, then we define an average for their product values as follows:

$$X_{GM} = (X_1 X_2 X_3 \dots X_n)^{1/n}. \quad (1.3)$$

X_{GM} is called the geometric mean, with a relationship to the arithmetic mean of the data set, such that $X_{GM} \leq \bar{X}$. That is, the geometric mean underestimates the arithmetic mean for a given set of positive values.

Example 1.5

A commonly utilized version of the geometric mean in economics is the percent change in value, for a given time period.

Let V_1 be the value of an entity at the beginning of observation period, and V_n be the value at the end of the observation period. Then the average percentage change in the value per time period would be,

$$X_{GM} = \left(\frac{V_n}{V_1}\right)^{1/(n-1)} - 1.$$

That is, if the value of a car, when it was purchased, was $V_1 = \$27,000$, and after three years its value is $V_3 = \$3,000$. Then the average decrease in its value is,

$$\sqrt[3]{\frac{3,000}{27,000}} - 1 = \frac{1}{3} - 1 = -\frac{2}{3}.$$

Hence we say that the average depreciation of the value for this car is about 66 % per year.

1.4.4 Harmonic Mean

Let $X = \{X_1, X_2, \dots, X_n\}$ be a set of positive numbers, then the following formula provides the harmonic mean, X_H .

$$\frac{1}{X_H} = \left(\frac{1}{X_1} + \frac{1}{X_2} + \dots + \frac{1}{X_n}\right). \quad (1.4)$$

For the same data set of positive numbers, geometric mean overestimates the harmonic mean, and the arithmetic mean overestimates the geometric mean, i.e. we have an order of means, $\bar{X} \geq X_{GM} \geq X_H$.

1.4.5 Mode

In a data set, we call the most frequently observed value as the mode (fashion) of the data set. The mode can be unique (a single value with the highest frequency), or there may be two distinct values with the same highest frequency (in this case, the data set is called bimodal), or there may be many distinct values with the same highest frequency (data set is called multimodal), or there may not be a mode (no distinct value that has the highest frequency), also known as a “uniform distribution”.

Example 1.6

- a) Let $x = \{1, 2, 3\}$, x has no mode.
- b) Let $x = \{1, 1, 2\}$, x has a mode=1, since value 1 has the highest frequency (two).
- c) Let $x = \{1, 1, 2, 2, 3\}$, x is bimodal with modes equal to 1 and 2, since value 1 and value 2 have the highest frequency (two).
- d) Let $x = \{1, 1, 2, 2, 3, 3, 4, 5, 6, 7, 8, 9, 10\}$, x is multimodal with modes equal to 1 and 2 and 3, since value 1, value 2, and value 3 have the highest frequency (two).

1.4.6 Median

The middle position value of an ordered values for a data set, X , is called as its median (middle, or, center), and it may be denoted as $\tilde{X}$. The median is also used to measure the depth of a data set.

If the data set has an even number of values, then there will be no middle position value, that is its value is missing. Yet since the data set is ordered, median must be greater than or equal to the value on its left and it must be less than or equal to the value on its right. Thus, we may use the arithmetic average of the two middle position values of the ordered set as the missing median value.

Example 1.7

a) Let $x = \{2, 1, 5, 4, 9\}$, ordered data set $= \{1, 2, 4, 5, 9\}$, since the middle position value in this ordered data set is 4, $\tilde{x} = 4$.

b) Let $x = \{2, 1, 5, 10, 4, 9\}$, ordered data set $= \{1, 2, 4, 5, 9, 10\}$, since the two middle position values in this ordered data set are 4 and 5, $\tilde{x} = 4.5$.

The mean value of a data set is very sensitive to the extreme values, also known as outliers, where very few, or, even one observation may have an extreme value. Even a single extreme value induces bias to the value of the mean, and pulls the mean toward itself.

On the other hand, the median value of a data set is robust with respect to the outliers, since in an ordered data set the extreme values are located either at the beginning, or, at the end of the set. This is one of the reasons we utilize the median value, rather than the mean value, if the data set has extreme values in it.

We are interested in the symmetry (shape) of the frequency distribution, with respect to the mean. We call a data set, or its frequency distribution, or its relative frequency distribution, symmetric if its mean is equal to its median. If a data set is asymmetric (not symmetric), then we call it skewed. Two types of data skewness can happen.

(1) If the median is less than the mean, then the data set is called positively skewed. In this case the frequency distribution (or, the relative frequency distribution) of the data set is said to have a tail to the right, i.e. it has an outlier with a greater value than the rest of the observation values, thus inducing a “positive” bias to the mean.

(2) If the mean is less than the median, then the data set is called negatively skewed, and its frequency distribution (or, its relative frequency distribution) is said to have a tail to the left, i.e. it has an outlier with a smaller value than the rest of the observation values, thus inducing a “negative” bias to the mean.

1.4.7 Quantiles

We can identify special break point values of an ordered data set, by the proportion of data points that lies before them, such as quartiles which are the break point values, if we break (separate) the ordered data set into four equal parts of the data points. For example, first quartile is the value of the break point that has one quarter of the ordered data set observations before itself.

Similarly, percentiles which are the break point values, if the ordered data set is separated into hundred equal parts of data points. For example, 99th percentile is the value of the break point that has 99% of the ordered data set observations before itself. In general, these break point values are known as quantiles, thus median is the second quartile, or the 50th percentile, or 50th quantile of an ordered data set.

1.5 Spread Measures

The location measures are necessary points to identify and to simplify the characteristics of a data set, yet they are not sufficient.

The variations among the observation values in a data set impose a degree of risk for making decisions, based on the information coming from location measures. That is if we know that all observation values are very close to one another, then the value of the mean is very rich in information and quite reliable as a representative of the whole data set. On the other hand, if the values in the data set spread in a very high range of possible values, then the value of the mean is very poor in information and not a very reliable representative of the values in the data set. Hence, we have to measure the spread, or, dispersion, of the values in a data set, as well.

1.5.1 Range

The range is the easiest measure to construct in order to learn about the degree of the spread in a data set. It is defined as the difference between the maximum and the minimum values in a data set, X .

$$\text{Range}(X) = \max(X) - \min(X). \quad (1.5)$$

Example 1.8

For $x = \{1, 2, 3, 4, 5, 6, 10\}$,

$$\text{Range}(x) = 10 - 1 = 9,$$

where $\max(x) = 10$ and $\min(x) = 1$.

1.5.2 Interquartile Range

An i th quartile is the value of a point that captures $25\% \times i$ of the ordered data values and denoted by Q_i , for $i = 1, 2, 3$.

First quartile, Q_1 , refers to the value of the point that has 25% of the smaller observation values before it. Second quartile (median), Q_2 , identifies the observation value that has 50% of the smaller observation values before it. Third quartile, Q_3 , is the value of the observation point that has 75% of the smaller observations values before it.

The interquartile range is defined to be the difference between the third and the first quartiles, $Q_3 - Q_1$.

We derive a measure of the spread of data from the interquartile range as follows.

Quartile deviation is a measure of the spread of data defined by the half of the interquartile range, that is quartile deviation $= \frac{Q_3 - Q_1}{2}$.

Instead of dividing the range of ordered data into four equal parts, as in the case of quartiles, we can divide the ordered data range into hundred equal parts, and each one of the dividers is called the k th percentile.

The relative measures for the spread of the values in a data set is defined with respect to the mean of the data set. If we try to find the average spread in the values of the data set with respect to its mean, then we try to define a measure of the spread as follows.

Let the average spread with respect to the mean be

$$AVSP = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}) = \frac{1}{n} \sum_{i=1}^n X_i - \frac{1}{n} \sum_{i=1}^n \bar{X},$$

where X_i is the i th observation value in the data set X with n observations and $\bar{X}$ is the mean of the data set. Since $\bar{X}$ is a constant with respect to i , and also $\frac{1}{n} \sum_{i=1}^n X_i = \bar{X}$, clearly

$$AVSP = \bar{X} - \left(\frac{1}{n}\right)(n)(\bar{X}) = 0,$$

for any data set. Thus, AVSP is not a useful measure (provides no information) for the spread of values with respect to the mean.

Consequently, we use the following relative spread measures.

1.5.3 Mean Absolute Deviation

The average of the absolute deviations from the mean, for each observation value, is called the mean absolute deviation, denoted by MAD, and is defined as

$$MAD = \frac{1}{n} \sum_{i=1}^n |X_i - \bar{X}|, \quad (1.6)$$

where $|z|$ denotes the absolute value of a real number z .

Example 1.9

$x = \{1, 2, 3\}$, $\bar{x} = 2$ and

$$MAD = \frac{1}{3} [|1 - 2| + |2 - 2| + |3 - 2|] = \frac{1+0+1}{3} = \frac{2}{3}.$$

1.5.4 Variance

Variance may be the most commonly used measure for our purpose, it is defined for a sample and for a population in similar, yet different ways.

Population Variance: is denoted by the Greek letter σ^2 , and is defined as,

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (X_i - \mu)^2, \quad (1.7)$$

where N is the population size (number of elements in a population) and μ is the population mean. X_i is the i th value in the population.

Sample Variance: is denoted by S^2 , and is defined as,

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2, \quad (1.8)$$

where n is the sample size and $\bar{X}$ is the sample mean. X_i is the i th observation value in the sample.

Example 1.10

a) If the population $X = \{1, 2, 3\}$, then its variance

$$\begin{aligned} \sigma^2 &= \frac{1}{3} \sum_{i=1}^3 (X_i - 2)^2 = \frac{1}{3} [(1 - 2)^2 + (2 - 2)^2 + (3 - 2)^2] = \frac{1+0+1}{3} \\ &= 2/3 = 0.66666666666667. \end{aligned}$$

b) If the sample $x = \{1, 2, 3\}$, then its variance

$$s^2 = \frac{1}{3-1} \sum_{i=1}^3 (x_i - 2)^2 = \frac{1}{2} [(1 - 2)^2 + (2 - 2)^2 + (3 - 2)^2]$$

$$= \frac{1+0+1}{2} = 2/2 = 1.$$

Thus the same data set yields different variances for a population and its sample.

1.5.5 Standard Deviation

The unit of observation values, say number of people, in the definition of variance becomes squared, which may not have any meaning at all. Thus, in order to normalize the spread measure units back to their original meaning, we define a derived measure of spread as the positive square root of the variance, which is known as standard deviation. This would pull-back the squared units of observations into their original form.

Standard deviation is derived from the variance, hence we have two formulations for the standard deviation.

Population Standard Deviation: It is the square root of Population Variance, denoted by σ , and $\sigma = \sqrt{\sigma^2}$.

Example 1.11

In the previous example we calculated the population variance,

$$\sigma^2 = 2/3 = 0.666666666666667,$$

thus the standard deviation for this population would be

$$\sigma = \sqrt{2/3} = \sqrt{0.666666666666667} = 0.8164965809277.$$

Sample Standard Deviation: It is the positive square root of the Sample Variance, denoted by S , and $S = \sqrt{S^2}$.

Example 1.12

In the previous variance example, we calculated the sample variance as $s^2 = 1$, thus the standard deviation for this sample would be $s = \sqrt{1} = 1$.

1.5.6 Coefficient of Skewness

We can measure the degree of symmetry with respect to the mean, by utilizing the coefficient of skewness, defined as follows:

$$SK = \frac{3(Mean - Median)}{StandardDeviation}. \quad (1.9)$$

A positive value for SK indicates positive skewness, i.e. a tail to the right, and a negative value for SK indicates negative skewness, i.e. a tail to the left, with a degree of asymmetry. If SK is zero, then the data set is symmetric.

Example 1.13

The waiting time for a mail order computer on the internet is on the average four days with a median of two days, based on the historical sale data. The standard deviation was calculated to be three days.

What is the shape of the waiting time distribution?

The coefficient of skewness is,

$$SK = \frac{3(4-2)}{3} = 2.$$

Here, we have a positive skewness coefficient, thus the waiting time distribution is not symmetric, and it has a tail to the right, i.e. few customers are waiting much longer than the expected waiting time.

1.5.7 Coefficient of Variation

If we want to compare the spread of two different type of data sets, then different units make it similar to comparing apples and cars. Yet, we have a unit-less relative relationship between the measure of spread among these different type of observations, that is called the coefficient of variation, and it is defined as,

$$CV = \frac{\text{Standard Deviation}}{\text{Mean}}, \quad (1.10)$$

that indicates the degree of spread in data, with respect to its mean, and it is independent of unit. Thus we can compare the coefficient of variation for different types of observations, so that we can say which data set is more tightly distributed about its mean, relative to the other data set.

There is a convenient way of getting pictorial information about a quantitative data set, and it is known as the box plot, a.k.a. the five points summary. In this picture summary, we order the minimum value, first quartile, second quartile, third quartile, and the maximum value of a data set.

Example 1.14

Let the following random sample be the observations about the number of children a mother has, X .

$$x = \{2, 2, 2, 4, 2, 4, 1, 4, 3, 4, 2, 1, 2, 1, 2, 4, 1, 3, 5, 1, 3, 3, 4, 2, 12, 10, 6, 7, 4\}.$$

Outliers are also located outside of the box plot, as in Figure 1.5.

The central line segments that connect the minimum and the maximum values to the quartiles are called whiskers.

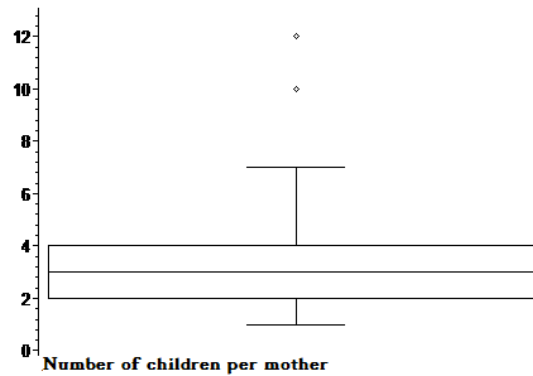


Figure 1.5: Box Plot

1.6 Covariance and Correlation

There may be more than one population that we are interested in learning their characteristics. If there are two populations, say X and Y , about which we collect observations, then we may wish to know how these two populations behave collectively. For that, we introduce some descriptive statistical measures. Let the sample coming from population X be a set, $\{X_1, \dots, X_n\}$ of ordered observations, ordered by the observation instant, “first observation,” ... “last observation.” Here, the last observation is the n th one. Similarly, let the sample coming from population Y be the set $\{Y_1, \dots, Y_n\}$, with the same number of observations as the data set X (or, we may say that these two finite data sets have the same cardinality).

We can now introduce a measure that will tell us the strength and the direction of statistical association between sample sets X and Y .

1.6.1 Covariance

If we construct the average of the simultaneous variations from their respective averages of the two sample sets, then we would have an idea about how strongly these two populations statistically associate, on the average. This measure is known as the sample covariance.

$$S_{XY} = Cov(X, Y) = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y}), \quad (1.11)$$

where n is the sample size, and $\bar{X}, \bar{Y}$ are the sample means of X and Y .

If we look at $(X - \bar{X})$ and $(Y - \bar{Y})$ as two vectors, generated as two data vectors shifted by their respective averages, where $X = (X_1, X_2, \dots, X_n)$ and $Y = (Y_1, Y_2, \dots, Y_n)$ are n -dimensional data vectors, then Covariance(X, Y) is the inner product of two vectors, $U = (X - \bar{X})$ and $V = (Y - \bar{Y})$, which is also known as the cosine law:

$$Covariance(X, Y) = \langle U, V \rangle = \|U\| \|V\| \cos \alpha,$$

where $\|\cdot\|$ is the euclidean norm (magnitude) of a vector, and α is the angle between these two vectors, U and V .

1.6.2 Correlation

A simpler measure to indicate the direction of statistical association between X and Y data vectors is constructed, based on the covariance measure, and it is known as correlation coefficient, denoted by r_{XY} . Such that

$$r_{XY} = \frac{S_{XY}}{S_X S_Y}. \quad (1.12)$$

Here, S_X and S_Y are the standard deviations of data vectors X and Y , respectively.

According to the inner product definition of covariance, the correlation coefficient corresponds to the cosine of the angle between $(X - \bar{X})$ and $(Y - \bar{Y})$ vectors, where the norms of these data vectors are sample standard deviations, S_X and S_Y , respectively.

Therefore, if these two vectors are aligned (colinear) and pointing the same direction, then the angle between them would be zero degree, and its cosine would be positive one. If the correlation coefficient turns out to be positive one, this indicates that these two data sets behave similarly, i.e. if one is an increasing set, the other one is also an increasing set, or, if one is a decreasing set, then the other one is also a decreasing set.

Similarly, if these two vectors are on the same line, yet pointing opposite directions, then the angle between them would be one hundred and eighty degrees, and its cosine would be negative one. Thus, if the correlation coefficient turns out to be negative one, that indicates that if one data set is increasing, then the other data set is decreasing.

But, if these two vectors are perpendicular, the angle between them would be either ninety degrees, or, two hundred and seventy degrees, and the cosine would be zero in both cases. If the correlation coefficient turns out to be zero, then these two data sets do not provide information about each other's behaviour.

As the cosine of an angle, the correlation coefficient, r_{XY} , can take values between negative one and positive one. Hence, it indicates the direction of statistical association between two data sets. This statistical association is a sufficient and necessary information for us to know, how two data sets behave simultaneously, on the average.

1.7 Index Numbers

The pursuit of disseminating information in a simple, yet reliable manner led us to utilize index numbers. They represent the relative behavior of populations as a single number, such as gross domestic product, consumers' price index, Dow Jones industrial average, Standard and Poor's 500 stock average, etc. These index numbers published regularly by the government and by businesses. Usually they are a measure of change in an entity at a desired time period with respect to a base period. This entity may very well be a collection of relevant yet different items.

Example 1.15

Consumer price index includes about 400 consumer items, and Dow Jones industrial index includes commonly traded stock prices of about 50 influential companies.

1.7.1 Simple Index

The simple index measures the change in a single item, over a time period. Let the quantity of the item at the base period be Q_b and the quantity of the item at the desired time period be Q_t , then the simple index is calculated to be.

$$I = \left(\frac{Q_t}{Q_b} \right) \times 100. \quad (1.13)$$

Example 1.16

The price of an introduction to statistics book was \$45.00 in 2000. The new edition of the same book is \$90.00 in 2010.

The relative change in the price of this book is given by the simple price index,

$$I = \left(\frac{90}{45}\right)100 = 200\%.$$

We say that the price is increased by $200\% - 100\% = 100\%$ in the period of 2000 to 2010 (in ten years).

If you want to move the base from one year to another year, then you compute the new index, with respect to the new base, and find the ratio of the old index to the new index, $\frac{I_{old}}{I_{new}} \times 100$, which implies what percentage of the change had happened, from the old base to the new base. The base can be a period of time itself, not necessarily a single year.

Example 1.17

The base is years 2000 and 2001, and the price of a computer game for these two years, $P_{2000}=\$20.00$, $P_{2001}=\$22.00$.

Thus the base price (average) would be,

$$P_b = \frac{P_{2000}+P_{2001}}{2} = \frac{20+22}{2} = \$21.00.$$

Then we can construct a simple index according to this base, say,

YEAR: 1999, 2000, 2001, 2002, 2003,

PRICE (\$): 18, 20, 22, 23, 38,

INDEX (%): $\left(\frac{18}{21}\right)100 = 85.7$, $\left(\frac{20}{21}\right)100 = 95.2$, $\left(\frac{22}{21}\right)100 = 104.8$,

$\left(\frac{23}{21}\right)100 = 109.5$, $\left(\frac{38}{21}\right)100 = 181$.

A summary of the data and index values are given in Table 1.5:

YEAR	PRICE	INDEX
1999	18	87.5
2000	20	95.2
2001	22	104.8
2002	23	109.5
2003	38	181

Table 1.5: Simple Index

Thus we may say that the price is increased by $181\% - 100\% = 81\%$ from the base period to the year 2003 (target period).

Even though an index is used for tracking the behavior of an item, it would be much more useful if it can provide a tracking device for a collection of

entities, preferably, even unrelated items. For this reason, we have aggregate index.

1.7.2 Simple Aggregate Index

We can also measure change in a collective items of similar kind, known as the simple average of value relatives, or, simple aggregate index as an average of the simple index of each and every one of the items in a collective, as follows:

$$I = \frac{\sum_{i=1}^n I_i}{n}, \quad (1.14)$$

where I_i is the simple index for the i th item in a collective of n related items, at the same base and at the same target period.

Example 1.18

Let us find a collective simple index for the following five items, as the base period of 2001 and the target period of 2002,

ITEM NUMBER: #1, #2, #3, #4, #5,

BASE VALUE (2001): 1, 2, 3, 10, 25,

TARGET VALUE (2002): 2, 1, 6, 22, 5,

INDEX (%): 200, 50, 200, 220, 20.

Thus $I = \frac{200+50+200+220+20}{5} = 690/5 = 138$.

Table 1.6 provides a summary of parameter values.

ITEM	BASE	TARGET	INDEX
#1	1	2	200
#2	2	1	50
#3	3	6	200
#4	10	22	220
#5	25	5	20

Table 1.6: Simple Aggregate Index

We can say that the average collective index increased by $138\% - 100\% = 38\%$, from the base to the target period, although some of the items in this collective has an increase, and some has a decrease in their value.

1.7.3 Aggregate Index

Sometimes it is necessary to account for the different importance between the items in a collective. Thus, we may assign different weights to different items and construct a weighted aggregate index.

There are two commonly used weighted aggregate indexes, known as Laspeyres price index,

$$I_L = \frac{\sum_{i=1}^n p_i^t q_i^b}{\sum_{i=1}^n p_i^b q_i^b} \times 100, \quad (1.15)$$

and Paasche price index,

$$I_P = \frac{\sum_{i=1}^n p_i^t q_i^t}{\sum_{i=1}^n p_i^b q_i^t} \times 100, \quad (1.16)$$

where p_i indicates the price and q_i indicates the quantity of the i th item, superscripts t indicates target and b indicates base time period.

Both of these include consumption quantities and unit prices for their construct. Laspeyres index focuses on the base time period consumption, for the price change in the target time period, yet Paasche index uses target time period consumption, to find the price change from the base period to the target period.

There are many more versions of index constructions, for different purposes, but all indices are designed to provide a simple way of identifying change in the value of a collection of various items.

1.8 Exercises

1. Describe the term statistics.
2. What are the two major parts of statistics, explain them in your own terms.
3. Start an Excel spreadsheet and enter your daily observations, such as exercise time, place, duration, intensity level, age, sex, body weight, sleep time; for four weeks period of time.
4. Download a copy of R software and repeat the above exercise.
5. Construct the frequency distribution and the relative frequency distribution of age in your classroom. Hint, you may break age into mutually exclusive classes of 0 to 10, 11 to 20, 21 to 30, 31 to 40, 41 to 50, 51

to 60, 61 to 70, 71 to 80, 81 to 90, 91 to 100, 101 to 110, 111 to 120, 121 to 130, 131 to 140, 141 to 150 years old. Also, pictorially represent them by using a histogram and a pie-chart.

6. Repeat the above gender (sex) study for your classroom. Construct the frequency and relative frequency distributions. Then, represent them by a histogram and a pie-chart.
7. Construct the frequency and relative frequency distributions for the midterm exam grades of your classroom. Represent them by a histogram and by a pie-chart.
8. Collect data about age, gender, weight and height of your statistics class. Find the mean, median, mode, variance and standard deviation for these variables (you can identify sex of an individual by setting female=1, male=2, if you like) consider that your class is a sample for student population in your school, for this year. Then plot histograms for these variables.
9. Repeat Problem 1, by considering your class is the population of interest.
10. Collect a similar data about another one of your classes, and repeat Problem 1. Calculate the coefficient of skewness for these data sets, then find the shape of the data for these two different data sets. Also calculate the coefficient of variations for these two data sets, which data set has larger spread about its mean, why?
11. Refer to your collected data; find the covariance and correlation of weight and height, age and weight, age and height, gender and weight, gender and height. Analyze your findings.
12. Repeat Problems 1 and 3, by using the R software.
13. Find the relative price of your textbook, ten years ago, and compare with today's price, by constructing a simple index.
14. Find the relative price of five items that you select, ten years ago, and compare to their current prices, by constructing a simple aggregate index.
15. Find the relative price and quantities of five items that you select, three years ago and today, then construct Lapeyres index and Paasche index.

Chapter 2

Probability Theory

2.1 Introduction

The measure of chance is the probability measure and it tells us the degree of our expectation of an event. Here, we will introduce Kolmogorov's [6] construct as the probability measure.

If an experiment happens only once, or infinitely many times, then the soon to be introduced frequentist construct would not be appropriate for the probability measure.

The random (or, chance) experiment is an observable and perhaps a repeatable phenomenon, such as a sunrise over the blue horizon, or tossing a coin in a gambling game.

The collection of all possible outcomes of an experiment (for the sake of simplicity we dropped the random prefix) is known as the sample space (of the underlying experiment) and denoted by S (also, Ω may denote the sample space. It is the last letter of the Greek alphabet, and may refer to the everything, or, universe). For example if you toss a coin, assume that you can either observe a head or a tail outcome, but not both, and nothing else, i.e. the coin would not disappear into thin air. In this case the sample space of the coin toss experiment would be $S = \{head, tail\}$.

In the case of compound experiments, that is a combination of many experiments, the sample space of the compound experiment is the cartesian product of the individual experiment's sample spaces.

For example, if you repeat the coin toss experiment many times, say n times, then the sample space for the n -repetition of the same coin toss experiment would be $S \times S \dots \times S$, or, S^n . That is, if you repeat the coin

toss experiment two times, then the sample space would be

$$\begin{aligned} & \{head, tail\} \times \{head, tail\} \\ &= \{(head, head), (head, tail), (tail, head), (tail, tail)\}. \end{aligned}$$

Here, the order of outcomes is significant, i.e. the $(head, tail)$ pair is different than the $(tail, head)$ pair, since the first component is a possible outcome of the first experiment and the second component is a possible outcome of the second experiment.

A subset of the sample space is called an event and we wish to measure the possibility of an event, as its probability measure. If one can repeat an experiment many times, under the same conditions, then it is convenient to compute the relative frequency of all events, i.e. *relative frequency*($event_i$) equals $\#\{event_i\}/\#\{\text{repetitions}\}$, for all i , where $\#\{\}$ denotes the count number, or cardinality, of a set.

As long as the experiment stays the same and we keep repeating it infinitely many times (ad infinitum), then we may say that the probability measure of the $event_i$ will take place is approximately the relative frequency of the $event_i$, as long as convergency conditions are satisfied for infinite sequences.

Again, assume that we are tossing a coin which is not going to change its characteristics (i.e. time and space invariant experiment) and observing the head and the tail events. After many repetitions of the coin toss experiment, we can construct a frequency distribution and then the relative frequency distribution of these two events. Therefore, if we repeat the experiment long enough, we may say that the probability measure of the head event is approximately equal to the relative frequency of the head event, and similarly the probability measure for the tail event is approximately equal to the relative frequency of the tail event.

It is clear that these relative frequencies add up to 100% for mutually exclusive (no common elements) and collectively exhaustive (collection generates the universe) events. This construct of the probability measure is known as the frequentist approach. Moreover, it implies that the probability measure for an event can only take numerical values between 0% to 100%.

It is obvious that a probability measure indicates a degree of expectation of a random event, which is not a perfect predictor for the future, since there exists no construct to perfectly predict the future.

We are interested in assigning a probability measure to events. Since events are subsets of the sample space, we need an algebra to operate on the sample space.

To this end, let $\mathcal{B}$ be a nonempty set of subsets of S (sample space). Then $\mathcal{B}$ is known as a σ -field (sigma-field) if:

1. $\emptyset \in \mathcal{B}$.
2. If $A \in \mathcal{B}$, then $A^c \in \mathcal{B}$.
3. If $A_i \in \mathcal{B}$, for all i , then $\cup_i A_i \in \mathcal{B}$.

Therefore, a σ -field is closed under complement and union. Due to DeMorgan's law,

$$\cup_{i=1}^n A_i = (\cap_{i=1}^n A_i^c)^c,$$

thus, it is also closed under intersection.

In general, σ refers to countability, or, additivity, according to the context. Obviously, σ -field contains S and $\emptyset$. For example, the power set of S (the set of all subsets of S) is a σ -field.

If I is the set of all open intervals in $\mathcal{R}$, then

$\mathcal{B}^0 = \cap\{\mathcal{C} | \mathcal{I} \subset \mathcal{C}, \mathcal{C} \text{ is a } \sigma\text{-field}\}$, is the Borel σ -field on the real line. It also contains closed and semi-closed intervals on the real line.

If $\mathcal{F}$ is a σ -field, then define a measure $\mu : \mathcal{F} \rightarrow \mathcal{R}_+ \cup \{0\}$, such that

$\mu(\emptyset) = 0$ and $\mu(\cup_{i=1}^n A_i) = \sum_{i=1}^n \mu(A_i)$, for mutually disjoint sets $A_i \in \mathcal{F}$.

Borel measure assigns the distance $b - a$ to the real interval $[a, b]$, as the measure, or the length, of this interval.

Lebesgue measure, as a generalized Borel measure, assigns the product of the length of intervals as the volume of a set, similar to the volume of a linear space that tightly contains the set (or, similar to the volume of a shoe box that contains a pair of shoes). But, not all subsets of $\mathcal{R}^n$ are Lebesgue measureable (related to the Banach-Tarski paradox). A more generalized version of a Borel measure is known as a Haar measure, a generalized Lebesgue measure which is defined on locally compact sets.

A measure is σ -finite if $\{A_i\} \in \mathcal{F}$ such that $S = \cup_{i=1}^n A_i$ and $\mu(A_i) < \infty$. For example $\mathcal{R}$ under the Lebesgue measure is σ -finite, but $\mathcal{R}$ is not finite.

Let S be a sample space and $\mathcal{B}$ be a σ -field of S . Also, let P be a real set function on $\mathcal{B}$, such that P translates elements of $\mathcal{B}$ (sets) onto $\mathcal{R}_{[0,1]}$, nonnegative real numbers in $[0, 1]$, then we can define the probability measure as this set function, P , if

1. $P(A) \geq 0$, for all $A \in \mathcal{B}$.
2. $P(S) = 1$.

3. If $\{A_i\}$ is a sequence of sets in $\mathcal{B}$ and $A_k \cap A_l = \emptyset$ for all $k \neq l$, then $P(\cup_i A_i) = \sum_i P(A_i)$.

We will simply refer to the probability set function as the probability measure, or, probability. The relationship between all possible events and their probabilities is known as the probability distribution (of a random experiment). Below we list some of the consequences of the above axioms.

Since, $A \cup A^c = S$ and $A \cap A^c = \emptyset$, we have

$$P(A \cup A^c) = 1 \text{ and } P(A \cup A^c) = P(A) + P(A^c).$$

Therefore, $P(A) + P(A^c) = 1$, or, $P(A^c) = 1 - P(A)$. Hence, $P(\emptyset) = 0$.

Probability preserves order, i.e. if we have ordered two events, such that $A_1 \subset A_2$, then $P(A_1) \leq P(A_2)$. This can be inferred since,

$$A_2 = A_1 \cup (A_2 \cap A_1^c) \text{ and } A_1 \cap (A_2 \cap A_1^c) = \emptyset,$$

we have

$$P(A_2) = P(A_1) + P(A_2 \cap A_1^c).$$

On the other hand, $P(A_2 \cap A_1^c) \geq 0$, thus giving $P(A_2) \geq P(A_1)$.

If we look at the collection of any two events, $A \cup B$, we can partition it as a union of disjoint sets, as follows:

$$A \cup B = A \cup (A^c \cap B) \text{ and } B = (A \cap B) \cup (A^c \cap B).$$

Therefore,

$$P(A \cup B) = P(A) + P(A^c \cap B) \text{ and } P(B) = P(A \cap B) + P(A^c \cap B).$$

Hence, $P(B) - P(A \cap B) = P(A^c \cap B)$ and $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

In general, we have $P(A_1 \cup \dots \cup A_n) = \sum_{i=1}^n (-1)^{i+1} \sigma_i$, where σ_i is the sum of probabilities for all possible intersections of i events. For example $\sigma_1 = P(A_1) + \dots + P(A_n)$, $\sigma_2 = P(A_1 \cap A_2) + P(A_1 \cap A_3) + \dots + P(A_{n-1} \cap A_n)$, $\sigma_3 = P(A_1 \cap A_2 \cap A_3) + \dots + P(A_{n-2} \cap A_{n-1} \cap A_n)$, etc. This formulation is known as the inclusion-exclusion formula, and it generates some useful inequalities. In general, we have $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_n \geq 0$.

Boole's inequality: $\sigma_1 \geq P(\cup_{i=1}^n A_i)$, where n could be ∞ . For $n = 2$, we have a special inequality, known as the Bonferroni's inequality, $\sigma_2 \geq \sigma_1 - 1$. Since, $1 \geq P(A_1 \cup A_2)$ and $P(A_1 \cup A_2) = P(A_1) + P(A_2) - P(A_1 \cap A_2)$, we have the Bonferroni's inequality.

If the sample space can be partitioned into k mutually exclusive (i.e. no common elements) and collectively exhaustive (i.e. collection generates the sample space) finite number of events such that the random experiment

assigns equal chance to each one of these events (fair experiment), then we call these events equally likely events. Under this assumption, since all the events have the same probability, then we can find the probability of an event as the reciprocal of the cardinality of the sample space.

For example, if you toss a fair coin once, then the probability of the head event would be $1/2$, as well as the probability of the tail event, since there are only two events in the sample space, and they have equal chances.

The probability distribution of the fair coin toss experiment would be given as in the following Table 2.1:

Event	Probability
HEAD	$1/2$
TAIL	$1/2$

Table 2.1: Fair Coin Toss Probability Distribution

The probability measure, as constructed here, has continuity property, that is, if $\{A_i\}$ is a nondecreasing sequence of events (sets), i.e. $A_i \subseteq A_{i+1}$, for all i , then

$$\lim_{i \rightarrow \infty} P(A_i) = P(\lim_{i \rightarrow \infty} A_i) = P(\cup_{i=1}^{\infty} A_i). \quad (2.1)$$

Similarly, for the nonincreasing sequence of events, $\{A_i\}$, i.e. $A_i \supseteq A_{i+1}$, for all i , we have

$$\lim_{i \rightarrow \infty} P(A_i) = P(\lim_{i \rightarrow \infty} A_i) = P(\cap_{i=1}^{\infty} A_i). \quad (2.2)$$

2.2 Odds Ratio

The odds of success refers to the number of successes versus number of losses, such that the “number of successes” divided by the “number of losses” denotes the odds ratio, OR .

We can define odds ratio for any event A , by using the probability measure of the event,

$$OR(A) = \frac{P(A)}{1 - P(A)}. \quad (2.3)$$

By rearranging the above formula, we write

$$OR(A)[1 - P(A)] = P(A),$$

and

$$OR(A) = [OR(A)][P(A)] + P(A) = OR(A) = [OR(A) + 1]P(A).$$

Hence, the inverse relation between the odds ratio and the probability measure is,

$$P(A) = \frac{OR(A)}{1 + OR(A)}. \quad (2.4)$$

For example, let us cast a fair die, and pick the event A as the event of getting a “six”, i.e. a face of the die with six dots.

$$\text{Thus, } P(A) = \frac{1}{6}, \text{ and } OR(A) = \frac{\frac{1}{6}}{1 - \frac{1}{6}} = \frac{1}{5}.$$

Here, we may say that the “odds are one-to-five” for a “six”.

Similarly if the odds are “one-to-five” for an event A , its probability would be $P(A) = \frac{\frac{1}{5}}{1 + \frac{1}{5}} = \frac{1}{6}$.

The odds ratio is an alternative tool to represent chance measurements.

2.3 Conditional Probability

We may be interested in the probability of an event that depends upon previously happened events. Thus, we restrict the sample space to those events that took place (subspace of the sample space). For example, if we have three eggs in a basket and only one of them is a rotten egg, yet we do not know which one, then if we randomly pick one egg, assuming that they are all look alike (equally likely selection), then we may say that the probability of picking the rotten one is $1/3$. After we picked one egg and cracked it, we would know for sure whether it is a rotten one or not.

If we pick another egg from the basket for the second time, then the probability of picking the rotten egg depends upon the status of the previously picked and cracked egg (we have additional information now, based on what we have observed, that refines the σ -field). That is, if the previously picked egg was the rotten egg, then the second egg will be a good one, on the other hand, if the first egg was a good one, then the probability of picking the rotten egg is $1/2$, for the second choice. This is a repetition of an experiment without replacement, and past versions of the experiment provide refining information for the probability of future events, generated by the experiment.

We denote the probability of an event, A , given that we have information about another event, B , as $P(A|B)$, and we call it conditional probability of

A , given that B had happened (a restriction on the sample space).

Obviously, $P(A|A) = 1$, and $P(A|A^c) = 0$.

Conditional probability measure satisfies the probability axioms, that is $P(A|B) \geq 0$, $P(B|B) = 1$, and $P(A_1 \cup A_2 \cup \dots | B) = \sum_{i=1,2,\dots} P(A_i|B)$, for conditionally exclusive events A_i and A_j , with respect to the event B , i.e. $\{A_i|B\} \cap \{A_j|B\} = \emptyset$, for all $i \neq j$.

Observe that, $P(A \cap B|B) = P(A|B)$, that is the conditioning event B is the restricted sample space for the probability of the event A . This intuitively implies that

$$P(A|B) = \frac{P(A \cap B)}{P(B)}. \quad (2.5)$$

Here, B behaves as a restricted sample space, whenever $P(B) > 0$.

On the other hand, if $P(B) = 0$, then also $P(A|B) = 0$.

By using this construct, one can see that the probability of A and B can happen simultaneously depends upon B , as $P(A \cap B) = P(A|B)P(B)$.

Hence, we can say that A and B are dependent events. We may refer to this relationship as the product rule of probability.

In general we have,

$$\begin{aligned} P(A_1 \cap A_2 \cap \dots \cap A_n) &= P((A_1 \cap A_2 \dots \cap A_{n-1}) \cap A_n) \\ &= P(A_n | A_1 \cap A_2 \dots \cap A_{n-1}) P(A_1 \cap \dots \cap A_{n-1}) \text{ and} \\ P(A_1 \cap \dots \cap A_{n-1}) &= P(A_{n-1} | A_1 \cap \dots \cap A_{n-2}) P(A_1 \cap \dots \cap A_{n-2}), \text{ etc.} \end{aligned}$$

Hence we have,

$$P(A_1 \cap A_2 \cap \dots \cap A_n) = P(A_n | A_1 \cap \dots \cap A_{n-1}) \dots P(A_2 | A_1) P(A_1). \quad (2.6)$$

Since $B = (B \cap A) \cup (B \cap A^c)$ is the union of two mutually exclusive events, we have,

$$P(B) = P(B \cap A) + P(B \cap A^c).$$

But, $P(B \cap A) = P(B|A)P(A)$ and $P(B \cap A^c) = P(B|A^c)P(A^c)$,

on the other hand, $P(A|B) = \frac{P(A \cap B)}{P(B)}$, thus,

$$\begin{aligned} P(A|B) &= \frac{P(A \cap B)}{P(B \cap A) + P(B \cap A^c)} = \frac{P(B \cap A)}{P(B|A)P(A) + P(B|A^c)P(A^c)} \quad (2.7) \\ &= \frac{P(B|A)P(A)}{P(B|A)P(A) + P(B|A^c)P(A^c)}. \end{aligned}$$

This formulation is known as Bayes' rule [1], and it gives us an instrument to refine the probability assessment of the event A , $P(A)$ (prior probability), with additional information provided by the event B , such as $P(A|B)$ (posterior probability).

If we partition a sample space, S , into finitely many disjoint events, $\{A_i\}$, for $i = 1, \dots, n$, then we have $B = B \cap S = B \cap \{\bigcup_{i=1}^n A_i\} = \bigcup_{i=1}^n \{B \cap A_i\}$, thus we have,

$$P(A_j|B) = \frac{P(B|A_j)P(A_j)}{\sum_{i=1}^n P(B|A_i)P(A_i)}, \quad (2.8)$$

which is known as the general form of the Bayes' rule.

There are experiments that produce stochastically (or, statistically) mutually independent events. For example, assume that you are simultaneously tossing two identical coins; the outcomes of one coin toss experiment do not depend upon the outcomes of the other coin toss experiment, that is these two identical coin toss experiments are independent experiments.

Hence, their outcomes are also independent, i.e., the probability of getting an event from one coin toss experiment is independent of what happens to the other coin toss experiment.

If we have two independent events A and B , coming from two independent experiments, then the conditional probability of A , given that B had happened, $P(A|B)$, would be the same as the individual probability of A , $P(A)$, since the information about B is irrelevant to the probability of A . In this case, $P(A \cap B) = P(A|B)P(B) = P(A)P(B)$, for independent events A and B . In general we have,

$$P(A_1 \cap \dots \cap A_n) = P(A_1)P(A_2) \dots P(A_n), \quad (2.9)$$

when A_i s are independent events, for $i = 1, 2, \dots, n$.

For example, assume that we are simultaneously tossing 100 identical fair coins. If we inquire about the probability of getting all (100) heads in this compound experiment, we may assume that each coin toss experiment is independent of the others, and since coins are fair, $P(HEAD) = \frac{1}{2}$, for one experiment. Hence, the probability of getting all heads for 100 identical, yet independent, coin toss experiments is

$$P(HEAD \cap \dots \cap HEAD) = P(HEAD) \times \dots \times P(HEAD) = \left(\frac{1}{2}\right)^{100}.$$

2.4 Random Variables

A random variable (denoted by r.v.) is a set function that maps the sample space into numbers (that could be any kind of number, including complex

numbers, but in general real numbers are sufficient), this induces a translation of the original sample space into a renamed sample space, where outcomes are identified by numbers. For the time being, we will restrict ourselves to the real line, i.e. the location of real numbers and integers, to identify events.

Thus, the domain of a random variable, X , is a sample space, S , and its range (images of the original outcomes) is the real line, $X : S \rightarrow \mathcal{R}$, or, $X(\omega) = x$, for $\omega \in S$ and $x \in \mathcal{R}$.

We use capital letters to indicate random variables, and lower case letters to indicate their numerical values (may be also referred as realization of a random variable, after it is observed).

If the sample space is countable, then the random variable is called a discrete random variable, if the sample space is not countable, but measurable (for example, Lebesgue measurable [11]), then the random variable is called a continuous random variable.

The original σ -field is translated into a new σ -field of intervals, induced by the random variable. Some intervals have countable number of points for discrete random variables, and some others have uncountably many points for continuous random variables, but the length of an interval is measurable.

Hence, intervals on the real line are sufficient for the induced sample space, or the σ -algebra, to assign probabilities to the events.

Example 2.1

If one tosses a fair coin, then the probability distribution would be given as in Table 2.1.

We may introduce a random variable, X , that will assign numerical values (real numbers) to the events of the sample space, as follows, let

$X(HEAD) = 1$, and $X(TAIL) = 0$, in fact you can pick any two distinct real numbers to identify these two events. Therefore, the probability distribution for the new sample space would be given as in Table 2.2,

X	P(X)
1	1/2
0	1/2

Table 2.2: Fair Coin Toss Probability Distribution Revisited

On the other hand, if one tosses a coin (not a particularly fair coin), and uses the same random variable X to identify the events, when the probability of heads is p and the probability of tails is q , where $p \geq 0$, $q \geq 0$ and $p+q = 1$, since they are the probabilities of two mutually exclusive and collectively exhaustive events, then the probability distribution would be as in Table 2.3

X	P(X)
1	p
0	q

Table 2.3: Coin Toss Probability Distribution

Example 2.2

Assume that you have lost your door key on the real line and searching for it. If you randomly pick a search interval on the real line, then the length of this interval is directly related to the probability of finding your lost key in it, i.e. if your search interval is the whole real line, then you would definitely find your key, even if it takes infinitely long time to search for it.

Here, we can introduce a random variable, Y , identifying the location of the key on the real line, and a search interval (a, b) on the real line, and ask ourselves, what is the probability of finding my key in this interval, $P(\text{key} \in (a, b)) = P(Y \in (a, b))$?

The answer to this question depends on the size of this search interval (a, b) . Here, Y is a continuous random variable. Obviously if $a = b$, then you will never expect to find the key, unless you are extremely lucky to look at a single point on the real line and find it there, i.e. “hit the bull’s eye”.

2.4.1 Cumulative Distribution Function

The cumulative distribution function (cdf) of a random variable, X , is the probability measure that the random variable will take values in a semi-open interval $(-\infty, x]$, a Borel set, and is denoted as $F_X(x) = P(X(\omega) \leq x)$, for all $\omega \in S$, or, briefly,

$$F_X(x) = P(X \leq x). \quad (2.10)$$

Hence, F_X is a right continuous function, that is

$$\lim_{x_n \rightarrow x} F_X(x_n) = F_X(x).$$

But, F_X is not necessarily left continuous, that is $\lim_{x \leftarrow x_n} F_X(x_n) \neq F_X(x)$.

The cumulative distribution function is also known as the distribution function. Observe that

$$F_X(-\infty) = P(X \leq -\infty) = 0, \quad (2.11)$$

and

$$F_X(\infty) = P(X \leq \infty) = 1, \quad (2.12)$$

and $F_X(x) \leq F_X(y)$, when $x \leq y$, i.e. F_X is a nondecreasing function, since $(-\infty, y] = (-\infty, x] + (x, y]$ and these two intervals are mutually exclusive, thus $P(X \in (-\infty, y]) = P(X \in (-\infty, x]) + P(X \in (x, y])$ and $P(X \in (-\infty, y]) \geq 0$, $P(X \in (-\infty, x]) \geq 0$, $P(X \in (x, y]) \geq 0$. Therefore, $F_X(y) \geq F_X(x)$.

The cumulative distribution function generates the same information provided by probability distribution of an experiment, i.e. for $x \in \mathcal{R}$, we will obtain all possible events and their probabilities by a cdf, such that $P(X \in (a, b]) = F_X(b) - F_X(a)$, for $a < b$.

Observe that $\{-\infty < X \leq b\} = \{-\infty < X \leq a\} \cup \{a < X \leq b\}$, where the right hand side of this equation is a union of two disjoint events, and it is valid for both discrete and continuous random variables. Also, since the cdf is a right continuous function, we have

$$P(X = x) = F_X(x) - F_X(x^-).$$

In general, we write $X \sim F_X$, which indicates that a random variable, X , is subject to a probability law, represented by its cdf, F_X .

Conditional Cumulative Distribution Functions

If an event, B , with $P(B) > 0$, is related to a random variable, X , then we incorporate this information into the cdf such that the conditional cdf of X is written as,

$$F_X(x|B) = \frac{P(\{X \leq x\} \cap B)}{P(B)}. \quad (2.13)$$

There are also other sources of information, beside the probability distribution and the c.d.f, that gives us the necessary and sufficient information about a random experiment.

2.5 Discrete Random Variables

A discrete random variable, X , is a function that assigns integer numbers to the events of a countable sample space, S , such that $X(\omega) : S \rightarrow \mathcal{Z}$, for all $\omega \in S$.

2.5.1 Probability Mass Function

The probability mass function (pmf) is defined for a discrete random variable, X , and denoted by f_X .

Thus, f_X gives a relationship between all possible values of X , together with their probabilities, and this information is sufficient to describe the characteristics of a countable sample space, such that $f_X(x) = P(X(\omega) = x)$, for all $\omega \in S$ and $x \in \mathcal{Z}$. Moreover, the pmf is related to the cdf by $F_X(x) = \sum_{\{x_i \leq x\}} f_X(x_i)$. Furthermore, the jump amount of the cdf at a point x_i is equal to the pmf value at this point.

For a generic discrete random variable that assigns integer values $\{x_i\}$ to all possible events, for $i = 1, \dots, n$, we say that the pmf, f_X , is given in Table 2.4.

X	f_X
x_1	$P(X = x_1)$
x_2	$P(X = x_2)$
$\vdots$	$\vdots$
x_n	$P(X = x_n)$

Table 2.4: Probability Mass Function

Observe that $\sum_{i=1}^n P(X = x_i) = 1$, since this table is a listing for the whole sample space.

Therefore, pmf is a nonnegative function, $f_X(x) \geq 0$, for all x , and $f_X(x_i) = P(X = x_i)$, such that $\sum_{i=1}^n f_X(x_i) = 1$.

2.5.2 Expected Value of Discrete Random Variables

The weighted average of all possible values of a discrete random variable, X , where the weights are the probabilities associated with these values, is called the expected value of X , and denoted by $E[X]$, also known as the mean of X , and denoted by μ_X , or, μ if it is a clear designation of the random variable. Thus, we compute the mean of X ,

$$E[X] = \frac{\sum_{i=1}^n P(X = x_i)x_i}{\sum_{i=1}^n P(X = x_i)}. \quad (2.14)$$

Here, the denominator is equal to one, thus we have,

$$E[X] = \sum_{i=1}^n P(X = x_i)x_i. \quad (2.15)$$

$E[X]$ is usually denoted by μ_X .

The expectation is a linear operator, that is $E[c] = c$, $E[cX] = cE[X]$, for a constant c . Also, for a sum of random variables, $Y = \sum_{i=1}^n X_i$, we have,

$$E[Y] = E\left[\sum_{i=1}^n X_i\right] = \sum_{i=1}^n E[X_i]. \quad (2.16)$$

Expected value of a random variable may not exist. For finite sample spaces (finite number of points and points have finite values), $E[X]$ exists (finite). For others, $E[X]$ exists only if $E[|X|] < \infty$.

For a nonnegative discrete random variable, X , if its mean exists, it is equal to

$$E[X] = \sum_{i=1}^{\infty} P(X \geq i). \quad (2.17)$$

This follows because,

$$\begin{aligned} \sum_{i=1}^{\infty} P(X \geq i) &= \sum_{i=1}^{\infty} \sum_{j=i}^{\infty} P(X = j) = \sum_{j=1}^{\infty} \sum_{i=1}^j P(X = j) \\ &= \sum_{j=1}^{\infty} jP(X = j) = E[X]. \end{aligned}$$

2.5.3 Variance of Discrete Random Variables

The weighted average variations from the mean of a discrete random variable X is known as the variance of X , denoted by $Var(X)$, or, σ_X^2 , and is computed by, $E[(X - E[X])^2]$. That is,

$$Var(X) = \sum_{i=1}^n P(X = x_i)(x_i - E[X])^2. \quad (2.18)$$

The positive square root of the variance of X is called the standard deviation of X , and denoted by σ_X . Variance is a quadratic operator.

For a constant c , we have $Var(c) = E[(c - E[c])^2] = E[(c - c)^2] = 0$, and

$$\begin{aligned} Var(cX) &= E[(cX - E[cX])^2] = E[(cX - cE[X])^2] \\ &= E[c^2(X - E[X])^2] = c^2 Var(X). \end{aligned}$$

The variance of a sum of independent random variables, $Y = \sum_i^n X_i$, is obtained as

$$\begin{aligned} Var(Y) &= E[(\sum_{i=1}^n X_i - \sum_{i=1}^n \mu_i)^2] = E[(\sum_{i=1}^n (X_i - \mu_i))^2] \\ &= \sum_{i=1}^n Var(X_i) + \sum_{i=1}^n \sum_{j \neq i}^n E[(X_i - \mu_i)(X_j - \mu_j)] = \sum_{i=1}^n Var(X_i), \end{aligned}$$

since X_i 's are independent and $E[X_i - \mu_i] = 0$.

Sometimes, it is convenient to use

$$\begin{aligned} Var(X) &= E[(X - E[X])^2] = E[X^2 - 2XE[X] + (E[X])^2] \\ &= E[X^2] - 2E[X]E[X] + (E[X])^2 = E[X^2] - (E[X])^2. \end{aligned}$$

Because the variance is nonnegative, this also implies that $E[X^2] \geq (E[X])^2$.

2.5.4 Moments of Discrete Random Variables

The k^{th} moment of a discrete random variable, about point a , for a nonnegative integer k is defined as,

$$\mu_a^k = E[(X - a)^k]. \quad (2.19)$$

That is,

$$\mu_a^k = \sum_i^n P(X = x_i)(x_i - a)^k. \quad (2.20)$$

The moment generating function (mgf) of a random variable X is defined as,

$$M_X(t) = E[e^{tX}], \quad (2.21)$$

for some finite t , provides the necessary information to find moments of a random variable, X , about zero.

Note that $M_X^{(k)}(0) = \mu_0^k$, where $M_X^{(k)}(0)$ is the k th derivative of $M_X(t)$ with respect to t , evaluated at $t = 0$.

For example, $M_X^{(1)}(0) = \frac{dM_X(t)}{dt}|_{t=0} = \frac{dE[e^{tX}]}{dt}|_{t=0} = E[Xe^{tX}]|_{t=0} = E[X]$, and

$$M_X^{(2)}(0) = \frac{d^2M_X(t)}{dt^2}|_{t=0} = \frac{d^2E[e^{tX}]}{dt^2}|_{t=0} = E[X^2e^{tX}]|_{t=0} = E[X^2].$$

The moment generating function may not always exist. On the other hand, the characteristic function, $\varphi_X(t)$, of a random variable X , defined as,

$$\varphi_X(t) = E[e^{itX}], \quad (2.22)$$

always exists and it provides the necessary and sufficient information about the random variable. To see this, note that since $\varphi_X(t) \leq |\varphi_X(t)|$, we have $\varphi_X(t) \leq |E[e^{itX}]| \leq E[|e^{itX}|]$. Moreover, by the Euler formula, we have

$$E[|e^{itX}|] = E[|\cos(tX) + i\sin(tX)|] = E[\cos^2(tX) + \sin^2(tX)] = E[1] = 1.$$

Hence, $\varphi_X(t)$ exists. Note that the characteristic function is the Fourier transform of the pmf of X . Similar to the moment generating functions, the moments of a random variable about zero can be obtained by taking the k -th derivative of the characteristic function,

$$i^k \mu_0^k = \varphi_X^{(k)}(0) = \frac{d^k \varphi(t)}{dt^k}|_{t=0}, \text{ where } i = \sqrt{-1}.$$

Let us study some of the most commonly used discrete random variables.

2.5.5 Bernoulli Random Variable

A Bernoulli trial is a random experiment that has only two possible outcomes, generically called success and failure (this does not necessarily imply real success, or, failure). Moreover, it is assumed that the probability of success, p , and the probability of failure, $q = 1 - p$, stay the same (stationary), no matter where and when you conduct this trial. Thus, the probability mass function of the Bernoulli trial would be given in Table 2.5,

EVENT	Probability
success	p
failure	q

Table 2.5: Probability Distribution of a Bernoulli Trial

The sample space of the Bernoulli trial S is $\{success, failure\}$ and it is countable, with a cardinality of two. Hence, if we use a discrete random variable, X , to identify the elements of S , say $X(success) = 1$ and

x	P(X=x)
1	p
0	q

Table 2.6: pmf of a Bernoulli Trial

$X(\text{failure}) = 0$, we can construct a pmf of the Bernoulli trial, as given in Table 2.6, where

$$f_X(x) = p^x q^{1-x}, \text{ for } x=0,1, \quad (2.23)$$

and $f_X(x) = 0$, for all other values of x .

The indicator function, I_A of a set A , is defined as $I_A(x) = 1$, if $x \in A$ and $I_A(x) = 0$, if $x \notin A$. Clearly, we may induce a probability measure on the indicator function of a random outcome X , by using a Bernoulli r.v., such that success may identify $X \in A$, and failure may identify $X \notin A$. We refer to this function as the indicator random variable.

Expected value of a Bernoulli r.v. X is given as

$\mu_X = E[X] = 1p + 0q = p$, that is if you repeat the experiment infinitely often, then on the average you expect to see success with $p \times 100$ percent of the time. This implies that the expected value indicates a long term average behavior of the probability measure.

Variance of a Bernoulli r.v. X is given as

$$\sigma_X^2 = E[(x - \mu_X)^2] = p(1 - p)^2 + q(0 - p)^2 = p(1 - p)(1 - p + p) = pq,$$

since $q = 1 - p$. Hence, the standard deviation of a Bernoulli r.v. is

$$\sigma_X = \sqrt{p(1 - p)} = \sqrt{pq}.$$

2.5.6 Binomial Random Variable

A binomial r.v., X , counts the number of successes in some repetitions, n , of a Bernoulli trial, with a given probability of a success, p , for each and every one of these trials. Thus, we have n identical, yet independent Bernoulli experiments, since the probability of a success is the same for all trials (identical), yet probabilities will not be influenced by the observations of previous trials (independence).

Hence, we have a random vector (components are random variables), $(X_1, X_2, \dots, X_n)$, where each component is a Bernoulli r.v. which may turn

out to be a success with probability p , or a failure with probability q , and

$$X = \sum_{i=1}^n X_i.$$

Let k be the number of successes in n trials. For a strange twist of fate, we may get k number of successes for the first k trials, and the rest of the trials would be failures. The probability of this event is

$$\underbrace{p \cdot p \cdot \dots \cdot p}_{k \text{ times}} \cdot \underbrace{q \cdot q \cdot \dots \cdot q}_{n-k \text{ times}} = p^k q^{n-k},$$

since all trials are independent and identical. On the other hand, there are $\binom{n}{k}$ different ways of choosing k successes in n trials, and these are mutually exclusive events, say A_i be the event of choosing k successes in n trials for the type i .

$$\text{Hence, } P(X = k) = P\left(\bigcup_{i=1}^{\binom{n}{k}} A_i\right) = \sum_{i=1}^{\binom{n}{k}} P(A_i),$$

with each $P(A_i) = p^k q^{n-k}$.

Therefore, $P(X = k) = \binom{n}{k} p^k q^{n-k}$, where $p + q = 1$, giving the pmf of the binomial r.v. as follows:

$$f_X(x) = \binom{n}{x} p^x q^{n-x}, \text{ for } x = 0, 1, \dots, n. \quad (2.24)$$

We denote a random variable, X , subject to the binomial probability law, by writing $X \sim B(n, p)$, which reads as; X is subject to binomial distribution with n repetitions, and each time we have a trial, the probability of success is p . Here, n and p are called the parameters of a binomial distribution.

Expected value of a binomial r.v. X :

As we have discussed before we can write $X = \sum_{i=1}^n X_i$. Thus, by linearity of the expectation operator,

$$E[X] = E[X_1 + X_2 + \dots + X_n] = E[X_1] + E[X_2] + \dots + E[X_n].$$

But, since each experiment is the Bernoulli trial,

$$E[X_i] = p, \text{ for } i = 1, 2, \dots, n.$$

Thus the expected number of successes in n trials, $E[X] = \mu_X = np$.

Variance of a binomial r.v. X :

Following the above reasoning, for independent trials we have

$$\text{Var}(X) = \text{Var}(X_1 + X_2 + \dots + X_n) = \text{Var}(X_1) + \text{Var}(X_2) + \dots + \text{Var}(X_n).$$

Since trials are identical and $Var(X_i) = pq$, for every i , the variance is equal to

$$\sigma_X^2 = Var(X) = npq = np(1 - p), \text{ and the standard deviation,}$$

$$\sigma_X = \sqrt{np(1 - p)} = \sqrt{npq}.$$

Example 2.3

Let a college student take a Japanese literature course, in Japanese, and without knowing the language at all. Assume that exams are multiple choice tests and in one test there are ten questions with four alternative answers, yet only one of them is correct. Thus, we may assume that since the student is completely ignorant about the subject, each one of the all four alternatives has the same chance to be selected by the student as a correct answer, i.e. $p = \frac{1}{4}$ and $q = \frac{3}{4}$.

The probability that the student will answer all ten questions correctly can be found by using the binomial model, where n is ten and p is 0.25.

$$\text{Hence, } P(X = 10) = \binom{10}{10}(0.25)^{10}(1 - 0.25)^{10-10} = (0.25)^{10}.$$

Similarly, the probability that the student will give all wrong answers,

$$P(X = 0) = \binom{10}{0}(0.25)^0(0.75)^{10-0} = (0.75)^{10}.$$

The expected number of correct answers would be $E[X] = np$, i.e., 10×0.25 , or, 2.5 correct answers expected for each test, if the student takes these tests forever and without learning anything, thus the expected value has a meaning in the long-term, as an average value per experiment.

The variance of the number of correct answers would be $Var(X) = npq$, i.e., $10 \times 0.25 \times 0.75$, or, 1.875 number of correct answers squared per test, if the student takes these tests forever. Since the unit of the variance is the number of correct answers per test squared, we want to pull this unit back to its original form, by taking the positive square root of the variance, a.k.a. “standard deviation,” which is approximately 1.37. This means that, on the long run average, we expect to see a difference of 1.37 number of correct answers, from the expected number of correct answers (2.5) for a test.

2.5.7 Geometric Random Variable

A geometric r.v., X , counts the number of failures before the first success, in a sequence of Bernoulli trials, where p denotes the probability of success and q denotes the probability of failure, such that $p + q = 1$.

The pmf of a geometric X is,

$$f_X(x) = P(X = x) = pq^x, \text{ for } x = 0, 1, 2, \dots \quad (2.25)$$

Geometric random variables have the property of being memoryless, that is the random experiment restarts each time the experiment is conducted, and thus does not remember its former states. A detailed investigation of this phenomenon is provided in Chapter 5.

Expected value of a geometric r.v. X :

$$E[X] = \sum_{x=0}^{\infty} x p q^x = p q \sum_{x=0}^{\infty} x q^{x-1},$$

but, $\sum_{x=0}^{\infty} x q^{x-1} = \frac{d}{dq} \sum_{x=0}^{\infty} q^x$, and $\sum_{x=0}^{\infty} q^x = \frac{1}{1-q}$, since $0 < q < 1$.

Hence, $\frac{d}{dq}(\frac{1}{1-q}) = \frac{1}{(1-q)^2}$. Therefore,

$$E[X] = p q \frac{1}{(1-q)^2} = p q \frac{1}{p^2} = \frac{q}{p} = \mu_X.$$

Observe that, we have $E[X] = \sum_{x=0}^{\infty} P(X \geq x)$.

Variance of a geometric r.v. X :

$$\text{Var}(X) = E[X^2] - (E[X])^2 = E[X(X-1)] + E[X] - (E[X])^2.$$

$$E[X(X-1)] = \sum_{x=0}^{\infty} x(x-1) p q^x = p q^2 \sum_{x=0}^{\infty} x(x-1) q^{x-2},$$

but, $\sum_{x=0}^{\infty} x(x-1) q^{x-2} = \frac{d^2}{dq^2} \sum_{x=0}^{\infty} q^x = \frac{d^2}{dq^2}(\frac{1}{1-q}) = \frac{2}{(1-q)^3}$.

Thus, $E[X(X-1)] = p q^2 \frac{2}{(1-q)^3} = \frac{2 p q^2}{p^3} = 2(\frac{q}{p})^2$, and

$$\text{Var}(X) = 2(\frac{q}{p})^2 + \frac{q}{p} - (\frac{q}{p})^2 = \frac{q}{p}(1 + \frac{q}{p}) = \frac{q}{p^2} = \sigma_X^2.$$

Example 2.4

What is the probability of having one failure before first success, in the previous multiple choice test? It is computed as

$$P(X = 1) = p q^1 = 0.25 \times 0.75, \text{ or, } 0.1875.$$

The expected number of failures before first success is 3.

Note that in some contexts, it may be convenient to define a shifted geometric r.v., Y , such that $Y = X + 1$. In this case, Y denotes the time of first success

$$f_Y(y) = p q^{y-1}, \text{ for } y = 1, 2, \dots, E[Y] = \mu_X = \frac{1}{p}, \text{Var}(Y) = \sigma_X^2 = \frac{q}{p^2}.$$

Example 2.5

What is the probability that first success occurs at the first trial? The answer is immediate as, $P(Y = 1) = p q^{1-1}$, or, 0.25.

The time of first success is 4, and of course there will be on the average 3 failures before first success.

Thus, these two versions correspond to the same situation, one in terms of the number of failures before first success and the other in terms of the

time of first success. Moreover, the geometric random variables associate with the number of successes until first failure and the time of first failure can be similarly defined.

2.5.8 Negative Binomial Random Variable

A negative binomial r.v., X , counts the number of trials, until the r th success happens, in a sequence of i.i.d. Bernoulli trials. Since each experiment is an independent and identically distributed Bernoulli trial, we just need to consider the probability of allocating $r - 1$ successes to the first $x - 1$ trials. But, this is the binomial distribution, given by $\binom{x-1}{r-1} p^{r-1} q^{x-r}$. And the last trial must be a success with probability p , hence giving the pmf of a negative binomial random variable as,

$$f_X(x) = \binom{x-1}{r-1} p^r q^{x-r}, \text{ for } x = r, r+1, \dots \quad (2.26)$$

Because all trials are independent and identically distributed, the negative binomial random variable can be formulated as a sum of r i.i.d. shifted geometric random variables, i.e.,

$$X = \sum_{i=1}^r Y_i,$$

where Y_i 's are i.i.d. shifted geometric random variables. Thus, the expected value and the variance of a negative binomial r.v. can be immediately derived as given below.

Expected value of a negative binomial r.v. X :

$$E[X] = \mu_X = \frac{r}{p}.$$

Variance of a negative binomial r.v. X :

$$\text{Var}(X) = \sigma_X^2 = \frac{rq}{p^2}.$$

Example 2.6

In the previous multiple choice test example, what is the probability that the student gives wrong answers three times, before the second correct answer?

$$P(X = 3) = \binom{3-1}{2-1} (0.25)^2 (0.75)^{3-2}, \text{ which is about } 0.09375.$$

The expected value of number of wrong answers, before the second correct answer, is 8.

2.5.9 Multinomial Random Variable

If we partition the sample space into m mutually exclusive events, $A_1, \dots, A_m$, such that $P(A_j) = p_j$, for $j = 1, 2, \dots, m$ and n independent repetitions of the same (identical) random experiment take place, then when k_j counts the number of times the event A_j occurs, the multinomial random vector $X = (X_1, X_2, \dots, X_m)$ counts the number of occurrences of all these partitions of the sample space, such that,

$$P(X_1 = k_1, X_2 = k_2, \dots, X_m = k_m) = \frac{n!}{k_1! k_2! \dots k_m!} p_1^{k_1} p_2^{k_2} \dots p_m^{k_m}, \quad (2.27)$$

with $\sum_{i=1}^m k_i = n$, and $\sum_{i=1}^m p_i = 1$.

2.5.10 Poisson Random Variable

The Poisson r.v., X , counts the number of events in a unit interval (temporal, spatial, or could be a multidimensional interval).

For example, the number of daily (per diem) arrivals in an airport, or, number of air bubbles in a cubic inch of glass, etc.

The pmf of a Poisson r.v. X is,

$$f_X(x) = \frac{\lambda^x}{x!} e^{-\lambda}, \text{ for } x = 0, 1, 2, \dots, \quad (2.28)$$

of which $\lambda > 0$, and it is called the arrival rate.

We denote a random variable, X , subject to a Poisson probability law, by writing $X \sim P(\lambda)$, that reads, X is subject to a Poisson distribution, with an arrival rate of λ . Here, λ is the parameter of the Poisson distribution.

Expected value of a Poisson r.v. X :

$$E[X] = \mu_X = \sum_{x=0}^{\infty} x \frac{\lambda^x}{x!} e^{-\lambda} = \lambda e^{-\lambda} \sum_{x=1}^{\infty} \frac{\lambda^{x-1}}{(x-1)!} = \lambda e^{-\lambda} e^{\lambda} = \lambda.$$

Variance of a Poisson r.v. X :

$$\text{Var}(X) = E[X(X-1)] + E[X] - (E[X])^2.$$

$$\begin{aligned} E[X(X-1)] &= \sum_{x=2}^{\infty} x(x-1) e^{-\lambda} \frac{\lambda^x}{x!} \\ &= e^{-\lambda} \lambda^2 \sum_{x=2}^{\infty} \frac{\lambda^{x-2}}{(x-2)!} = \lambda^2 e^{-\lambda} e^{\lambda} = \lambda^2. \end{aligned}$$

Hence,

$$\text{Var}(X) = \sum_{x=0}^{\infty} (x - \lambda)^2 \frac{\lambda^x}{x!} e^{-\lambda} = \lambda^2 + \lambda - (\lambda)^2 = \lambda = \sigma_X^2.$$

Therefore, the expected value and the variance are equal for Poisson random variables.

Example 2.7

Assume that in an airport three airplanes land in every ten minutes, on the average. What is the probability of three airplanes landing in the next ten minutes?

$$P(X = 3) = \frac{3^3}{3!}e^{-3}, \text{ or about } 0.22404.$$

On the other hand, the expected number of airplanes landing for the next ten minutes would be three.

Example 2.8

Assume that, on the average, there are twenty chocolate chips in a cup of ice cream. If you eat two cups of ice cream, then what is the probability that you will eat forty chocolate chips?

$$P(X = 40) = \frac{20^{40}}{40!}e^{-20}, \text{ or, about } 2.8 \times 10^{-5},$$

which is relatively a very small chance. On the other hand, the expected number of chocolate chips that you will eat is 40.

2.6 Continuous Random Variables

A continuous random variable, X , is a function that assigns real numbers to the events of a measurable sample space, S , such that $X(\omega) : S \rightarrow \mathcal{R}$, for all $\omega \in S$, and it has a continuous cdf F_X .

2.6.1 Probability Density Function

The probability density function, pdf, generates the necessary and sufficient information about the probability distribution of a continuous r.v., X , and it is denoted by $f_X(x)$, such that $P(X(\omega) \in A) = \int_{\{x \in A | \forall \omega \in S\}} f_X(x)dx$.

The cumulative distribution function is then written as,

$$F_X(x) = P(X \leq x) = \int_{-\infty}^x f_X(s)ds.$$

Moreover, if $F_X(x)$ is a smooth differentiable function, then $f_X(x) = \frac{dF_X}{dx}$.

Otherwise, we write $P(X(\omega) \in A) = \int_{\{x \in A | \forall \omega \in S\}} dF_X(x)$.

where $dF_X(x)$ may have jump discontinuities, representing the probability measure which is associated with those jump events.

The pdf is a nonnegative function, $f_X(x) \geq 0$, for all x , such that

$$\int_{-\infty}^{\infty} f_X(x)dx = 1.$$

Conditional pdf

The conditional pdf of a random variable X that depends on the event A , is given in terms of the conditional distribution function as $f_X(x|A) = \frac{dF_X(x|A)}{dx}$.

2.6.2 Expected Value of Continuous Random Variables

The expected value of a continuous r.v., X , is denoted by $E[X]$, or, μ_X , and computed by

$$E[X] = \int_{-\infty}^{\infty} x f_X(x) dx. \quad (2.29)$$

That is, it is the weighted average of all possible values of x .

If $E[|X|] < \infty$, then $E[X] < \infty$, i.e. expected value may not exist for every random variable.

Since the probability measure is associated with a random variable, we can define the expected value of a function of a random variable, $g(X)$, such that

$$E[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx,$$

when X is a nonnegative continuous random variable and if its expected value exists, then $E[X] = \int_0^{\infty} (1 - F_X(x)) dx$.

This immediately follows from

$$\begin{aligned} \int_0^{\infty} (1 - F_X(x)) dx &= \int_0^{\infty} P(X > x) dx = \int_0^{\infty} \left(\int_x^{\infty} f(s) ds \right) dx \\ &= \int_0^{\infty} \left(\int_0^x ds \right) f(x) dx = \int_0^{\infty} x f(x) dx = E[X]. \end{aligned}$$

Hence, we can say that for any continuous random variable, X , with a finite mean, we have $E[X] = \int_0^{\infty} P(X > x) dx - \int_0^{\infty} P(X < -x) dx$.

Similarly, this statement is also true for any discrete random variable with a finite mean, when the integral operator is replaced by the summation operator.

2.6.3 Variance of a Continuous Random Variable

The variance, $Var(X)$, or σ_X^2 , of a continuous r.v. X is defined as,

$$E[(X - \mu_X)^2] = \int_{-\infty}^{\infty} (x - \mu_X)^2 f_X(x) dx. \quad (2.30)$$

2.6.4 Moments of a Continuous Random Variable

The k th moment of a continuous r.v., about a point b , is defined as $\mu_b^k = E[(X - b)^k]$, for a nonnegative integer k . That is,

$$\mu_b^k = \int_{-\infty}^{\infty} (x - b)^k f_X(x) dx. \quad (2.31)$$

Moments about point zero can be obtained by using the moment generating function of a continuous random variable, if it exists, or by using the characteristic function which exists always, as indicated before,

$$|\varphi(t)| = \left| \int_{-\infty}^{\infty} e^{itx} f(x) dx \right| \leq \int_{-\infty}^{\infty} |e^{itx} f(x)| dx,$$

and $f_X(x) \geq 0$, for all x , we have,

$$|\varphi(t)| \leq \int_{-\infty}^{\infty} |e^{itx}| f(x) dx.$$

On the other hand, by the Euler formula, we have $|e^{itx}| = 1$.

Therefore, $|\varphi(t)| \leq \int_{-\infty}^{\infty} f(x) dx = 1$, for all real t . Hence, $\varphi(t) < \infty$, for all real t .

The first moment about zero, μ_0^1 , of a random variable X , is known as the mean, μ , of the random variable. The second moment, μ_μ^2 , of a random variable, X , about its mean μ , is known as the variance, σ^2 , of the random variable, and its positive square root is called the standard deviation, σ , of the random variable.

Moments would yield all the information about the random variable, thus they are a source of sufficient information.

The ratio, μ_μ^3/σ^3 , is known as the coefficient of skewness of a random variable, and it is a measure of symmetry, about the mean, for a probability distribution, if it exists.

On the other hand, the ratio, μ_μ^4/σ^4 , is known as the coefficient of kurtosis of a random variable, and it is a measure of peakedness about the mean, for a probability distribution. That is, higher the kurtosis, more jagged the probability distribution, if it exists.

For a normal, or mesokurtic distribution, this ratio would be three, for a jagged, or leptokurtic distribution, this ratio would be larger than three, and for a flat, or platykurtic distribution, this ratio would be less than three.

Let us look at some of the commonly used continuous random variables.

2.6.5 Uniform Random Variable

A uniform r.v., X , assigns equal chance to the events of a measurable sample space, say $S = [a, b]$, with the following pdf,

$$f_X(x) = \frac{1}{b-a}, \text{ for } x \in [a, b], \quad (2.32)$$

and $f_X(x) = 0$, for $x \notin [a, b]$.

We denote a random variable, subject to a uniform probability law, by writing $X \sim U(a, b)$; that reads, X is subject to a uniform distribution, from a to b . Here, a and b are the parameters of the uniform distribution.

Expected value of a uniform r.v. X :

$$E[X] = \mu_X = \int_{-\infty}^{\infty} x \frac{1}{b-a} dx = \int_a^b x \frac{1}{b-a} dx = \frac{a+b}{2}.$$

Variance of a uniform r.v. X :

$$\sigma_X^2 = \text{Var}(X) = \int_{-\infty}^{\infty} (x - \mu_X)^2 \frac{1}{b-a} dx = \frac{(b-a)^2}{12}.$$

The uniform r.v. may be used to represent complete ignorance, i.e., if we have no clue about the random experiments, then we may assume that anything is possible with the same probability.

Example 2.9

If we will receive a telephone call within the next hour, then what is the probability that we will wait 45 minutes for that call?

$$P(X \leq 45) = \int_0^{45} \frac{1}{60-0} dx = \frac{1}{60} \int_0^{45} dx = \frac{45}{60} = \frac{3}{4},$$

or, there is a 75% chance that we will wait 45 minutes.

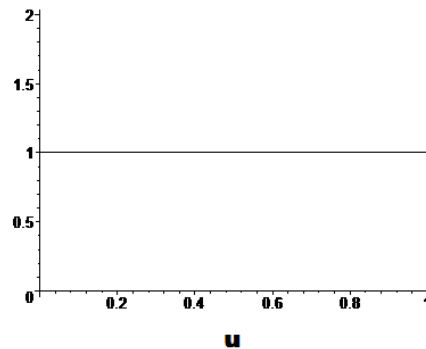


Figure 2.1: Standard Uniform pdf

A uniform random variable X can be translated into the standard uniform random variable, $U = \frac{X-a}{b-a}$, with a pdf graph is given as in Figure 2.1, where $f_U(u) = 1$, for $u \in [0, 1]$.

There is a discrete version of the uniform probability law, that is called the discrete uniform r.v., such that when there are n distinct points in an interval of $[a, b]$, we may assign equal likelihood to select a point, X , in that interval, by the following discrete uniform pdf,

$$f_X(x) = \frac{1}{n}, \text{ for } x = 1, 2, \dots, n,$$

with $E[X] = \frac{n+1}{2}$ and $Var(X) = \frac{n^2-1}{12}$.

2.6.6 Exponential Random Variable

An exponential random variable, X , is known as a memoryless r.v. and is used as a model for random lifetime, with the following pdf,

$$f_X(x) = \lambda e^{-\lambda x}, \text{ for } \lambda > 0 \text{ and } x \geq 0, \quad (2.33)$$

with parameter λ .

Expected value of an exponential r.v. X :

$$E[X] = \int_0^\infty x \lambda e^{-\lambda x} dx = \frac{1}{\lambda}.$$

The above equality clearly implies the immediate interpretation of parameter λ as the hazard rate, i.e., the number of failures per unit time, which is given by $\lambda = 1/E[X]$.

Variance of an exponential r.v. X :

$$E[X^2] = \int_0^\infty x^2 \lambda e^{-\lambda x} dx = \frac{2}{\lambda^2}.$$

$$Var(X) = E[X^2] - (E[X])^2 = \frac{1}{\lambda^2}.$$

2.6.7 Gaussian Random Variable

A Gaussian, a.k.a. normal, random variable X has the following pdf,

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \text{ for } \sigma > 0, \quad (2.34)$$

that is symmetric about the μ and it has a peak at the μ . Its inflection points are $\mu - \sigma$ and $\mu + \sigma$.

The expected value of a normal r.v. X :

$$E[X] = \int_{-\infty}^\infty x f_X(x) dx = \mu,$$

The variance of a normal r.v. X :

$$\text{Var}(X) = \int_{-\infty}^{\infty} x^2 f_X(x) dx = \sigma^2.$$

We denote a random variable, subject to a normal probability law, by writing $X \sim \mathcal{N}(\mu, \sigma^2)$; that reads, X is subject to a normal probability distribution, with mean μ and variance σ^2 . These parameters are necessary and sufficient information to describe a normal probability distribution.

2.6.8 Standard Normal Random Variable

A special case of the normal r.v. is the standard normal r.v., Z , with the following pdf, $f_Z(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}$, as in Figure 2.2,

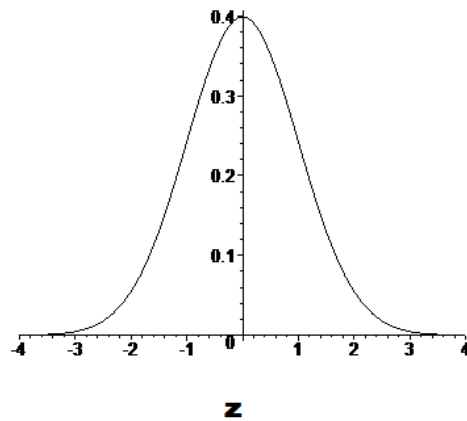


Figure 2.2: Standard Normal pdf

where, $E[Z] = 0$ and $\text{Var}(Z) = 1$. Its probability law is denoted by $Z \sim \mathcal{N}(0, 1)$.

In some context, the standard normal random variable may be called as the z -variate, $Z \sim \Phi(z)$. In fact, we can translate any normal random variable $X \sim \mathcal{N}(\mu, \sigma^2)$ into the standard normal random variable, by using the following translation, $Z = \frac{X-\mu}{\sigma}$, which is a probability measure preserving translation, i.e.,

$$P(a \leq X \leq b) = P\left(\frac{a-\mu}{\sigma} \leq Z \leq \frac{b-\mu}{\sigma}\right), \text{ for } b \geq a.$$

The computation of $P(z_1 \leq Z \leq z_2) = F_Z(z_2) - F_Z(z_1)$ uses the cdf of Z , $F_Z(z) = P(Z \leq z)$, which is conveniently denoted as $\Phi(z)$ and in tabular forms. Any statistical computer software package will provide the evaluation of $\Phi(z)$, and its inverse function $\Phi^{-1}(z)$.

Normal random variables are unimodal and symmetric with respect to their means, with a single mode located at the mean, thus $\mu = \text{median} = \text{mode}$, for normal random variables. Normal random variables are said to have a bell-shaped pdf (it looks like a cross-section of a bell), thus their pdfs may be called as the “bell-curve”.

2.6.9 Gamma Random Variable

A gamma random variable, X , is a flexible model for many random experiments, whenever there is a need to combine a number of i.i.d. exponential random variables. The pdf of a gamma random variable, X , is given as,

$$f_X(x) = \frac{\lambda(\lambda x)^{\alpha-1} e^{-\lambda x}}{\Gamma(\alpha)}, \text{ for } \alpha > 0, \lambda > 0 \text{ and } x \in \mathcal{R}^+. \quad (2.35)$$

where $\Gamma(s) = \int_0^\infty x^{s-1} e^{-x} dx$, for $s > 0$, with the following properties,

$\Gamma(s+1) = s\Gamma(s)$ is a recursive function, and

$\Gamma(n+1) = n!$, or $\Gamma(n) = (n-1)!$ for a positive integer n .

Since $\Gamma(1) = \int_0^\infty e^{-x} dx = 1$, this implies that $1 = \Gamma(1) = (1-1)!$,

or, $0! = 1$.

In order to evaluate the gamma function, $\Gamma(s)$, at specific values of s , say a , it is advisable to lift the function up to the complex plane first, by substituting $x \in \mathcal{R}^+$ with $z \in \mathcal{C}$, where $z = x + iy$, then using Cauchy's residue theorem to evaluate the integral, for $s = a$, and then pull the result back by taking the real part of the complex number value of the integral.

As mentioned before a gamma random variable can be interpreted as the sum of α i.i.d. exponential random variables with parameter λ , i.e.,

$$X = \sum_{i=1}^{\alpha} X_i,$$

with $X_i \sim \text{Exp}(\lambda)$. Thus, the mean and variance of the gamma r.v. are derived immediately.

Expected value of a gamma r.v. X :

$$E[X] = \int_0^\infty \frac{(\lambda x)^\alpha e^{-\lambda x}}{\Gamma(\alpha)} dx = \int_0^\infty \frac{(\lambda x)^\alpha e^{-\lambda x}}{\Gamma(\alpha)} \frac{\lambda}{\lambda} dx = \frac{\alpha \Gamma(\alpha)}{\lambda \Gamma(\alpha)} = \frac{\alpha}{\lambda}.$$

Variance of a gamma r.v. X :

$$E[X^2] = \int_0^\infty x \frac{(\lambda x)^\alpha e^{-\lambda x}}{\Gamma(\alpha)} dx = \frac{\alpha(\alpha+1)}{\lambda^2}.$$

Thus, $Var(X) = E[X^2] - (E[X])^2 = \frac{\alpha(\alpha+1)}{\lambda^2} - (\frac{\alpha}{\lambda})^2 = \frac{\alpha}{\lambda^2}$.

2.6.10 Chi-Square Random Variable

A special case of the gamma distribution is the chi-square, χ_r^2 , distribution where $\alpha = \frac{r}{2}$ and $\lambda = 1/2$, for an integer r , which is called the degrees of freedom of the chi-square distribution.

In fact, we can define the chi-square random variable with r degrees of freedom as the sum of independent, identically distributed, squared standard normal random variables, $\chi_r^2 = \sum_{i=1}^r Z_i^2$.

The pdf of the chi-square random variable, $X \sim \chi_r^2$, is

$$f_X(x) = \frac{1}{\Gamma(\frac{r}{2})2^{r/2}} x^{\frac{r}{2}-1} e^{-\frac{x}{2}}, \text{ for } 0 < x < \infty. \quad (2.36)$$

Expected value of a chi-square r.v. X :

$$E[X] = E[\sum_{i=1}^r Z_i^2] = \sum_{i=1}^r E[Z_i^2].$$

But, $E[Z_i^2] = Var(Z_i) = 1$ (show this as an exercise).

Thus, we have,

$$E[X] = \sum_{i=1}^r 1 = r.$$

Variance of a chi-square r.v. X :

$$\begin{aligned} Var(X) &= E[X^2] - (E[X])^2 = E[X(X-1)] + E[X] - (E[X])^2 \\ &= r^2 + 2r - (r)^2 = 2r. \end{aligned}$$

2.6.11 Beta Random Variable

A beta random variable, X , has the following pdf,

$$f_X(x) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1}, \text{ for } \alpha > 0, \beta > 0 \text{ and } 0 \leq x \leq 1. \quad (2.37)$$

Expected value of a beta r.v. X :

$$E[X] = \frac{\alpha}{\alpha+\beta}.$$

Variance of a beta r.v. X :

$$Var(X) = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}.$$

2.6.12 Laplace Random Variable

The Laplace random variable, X , is symmetric about a point θ and its pdf is

$$f_X(x) = \frac{\alpha}{2} e^{-\alpha|x-\theta|}, \text{ for } \alpha > 0 \text{ and } x \in \mathcal{R}. \quad (2.38)$$

The Laplace probability distribution is known as the double exponential.
Expected value of a Laplace r.v. X :

$$E[X] = \theta.$$

Variance of a Laplace r.v. X :

$$\text{Var}(X) = \frac{2}{\alpha^2}.$$

2.6.13 Cauchy Random Variable

The Cauchy random variable, X , has the following pdf, with no finite moment generating function, i.e. it has no variance,

$$f_X(x) = \frac{\alpha/\pi}{x^2 + \alpha^2}, \text{ for } \alpha > 0 \text{ and } x \in \mathcal{R}. \quad (2.39)$$

But, it is usually considered as a model for the ratio of standard normal random variables, and it has a mode and a median equal to 0.

2.6.14 Weibull Random Variable

The Weibull random variable, X , has the following pdf

$$f(x) = \alpha\beta x^{\beta-1} \exp(-\alpha x^\beta), \text{ for } x > 0 \text{ and } f(x) = 0, \text{ for } x \leq 0. \quad (2.40)$$

Here, the scale parameter, $\alpha > 0$, and the shape parameter, $\beta \geq 1$, identify the Weibull distribution, as a model representing *lifetime* of an organism. In contrast to the exponential r.v. that is memoryless, the probability that an organism will die during the next time interval depends on the age of the organism.

Expected value of a Weibull r.v. X :

$$E[X] = \alpha^{-\frac{1}{\beta}} [\Gamma(1 + \frac{1}{\beta})].$$

Variance of Weibull r.v. X :

$$\text{Var}(X) = \alpha^{-\frac{2}{\beta}} [\Gamma(1 + \frac{2}{\beta}) - \Gamma^2(1 + \frac{1}{\beta})].$$

2.6.15 Student's t-Distribution

The Student's t-distribution has a pdf with $\nu > 2$ degrees of freedom,

$$f(x) = \frac{1}{\sqrt{\nu\pi}} \frac{\Gamma(\frac{\nu+1}{2})}{\Gamma(\frac{\nu}{2})} \left[1 + \frac{x^2}{\nu}\right]^{-\frac{\nu+1}{2}}, \text{ for } -\infty < x < \infty, \quad (2.41)$$

that was investigated by William Sealy Gossett, who used a *pen-name*, “Student,” to publish his work [14]. Its pdf is symmetric about its mean, 0, and the variance is $\frac{\nu}{\nu-2}$, yet there is no moment generating function for the Student's t-variable.

Asymptotically, Student's t-distribution converges to the Standard Normal distribution as ν increases, but practically this convergency is expected for about $\nu > 25$.

2.6.16 F Distribution

The F distribution represents a ratio of two Chi-square random variables, identifying two independent variances, say U and V , with numerator and denominator degrees of freedoms, m and n respectively. This variance ratio originally studied by R. A. Fisher, and later its present form was constructed by G. W. Snedecor, who named it after the original investigator.

The pdf of F distribution,

$$f(x) = \frac{\Gamma(\frac{m+n}{2})}{\Gamma(\frac{m}{2})\Gamma(\frac{n}{2})} \left[\frac{m}{n}\right]^{\frac{m}{2}} \frac{x^{(\frac{m}{2}-1)}}{\left[1 + \frac{m}{n}x\right]^{\frac{m+n}{2}}}, \text{ for } x \geq 0, \quad (2.42)$$

and $f(x) = 0$, for $x < 0$.

We denote that a random variable X is subject to an F distribution with numerator degrees of freedom m and denominator degrees of freedom n , as $X \sim F_{m,n}$.

Expected Value of a r.v. $X \sim F_{m,n}$:

$$E[X] = \frac{n}{n-2}, \text{ for } n > 2.$$

Variance of a r.v. $X \sim F_{m,n}$:

$$Var(X) = \frac{2n^2(m+n-2)}{m(n-4)(n-2)^2}, \text{ for } n > 4.$$

Also, there is an inverse relationship between two F r.v.s, such as $\frac{U}{V}$ and $\frac{V}{U}$, with obviously interchanged degrees of freedoms, such that

$$F_{1-p;m,n} = \frac{1}{F_{p;n,m}},$$

where $F_{1-p;m,n}$ is the F r.v. with m numerator degrees of freedom and n denominator degrees of freedom that generates a right-tail area equals to $1 - p$, and $F_{p;n,m}$ is the F r.v. with n numerator degrees of freedom and m denominator degrees of freedom that generates a right-tail area equals to p .

The F-distribution and the t-distribution are called sampling distributions, since they arise in sampling from normal populations, and there is a relationship between them, such that

$$F_{1-p;1,n} = t_{1-\frac{p}{2};n}^2,$$

where, $t_{1-\frac{p}{2};n}$ is the Student's t random variable with n degrees of freedom that generates a right-tail area equals to $1 - \frac{p}{2}$.

Asymptotically, F distribution converges to χ^2 distribution such that,

$$F_{p;m,\infty} = \frac{\chi_{p;m}^2}{m}.$$

Here, $\chi_{p;m}^2$ is a χ^2 random variable with m degrees of freedom that generates a right-tail area which is equal to p .

2.7 Some Properties of Expectation

We have seen that the expectation is a linear operator. Now we can observe some more characteristics of expectation, related to a random experiment.

Jensen's Inequality 2.7.1. *For a random variable X , with $E[X]$ is finite:*

If g is a convex function, then $g(E[X]) \leq E[g(X)]$.

Similarly, if g is a concave function, then $g(E[X]) \geq E[g(X)]$.

Proof. If $g(x)$ is a real convex function over $[a, b]$, where $x_i \in [a, b]$ and $\lambda_i > 0$, for $i = 1, 2, \dots, n$, such that $\sum_{i=1}^n \lambda_i = 1$, then $g(\sum_{i=1}^n \lambda_i x_i) \leq \sum_{i=1}^n \lambda_i g(x_i)$.

Geometrically, this means that if $y = g(x)$, where y is a real convex function of x :

We have $\bar{x} = \sum_{i=1}^n \lambda_i x_i$ and $\bar{y} = \sum_{i=1}^n \lambda_i g(x_i)$.

Hence, $g(\bar{x}) \leq \bar{y}$.

When $g(x)$ is a concave function $-g(x)$ is a convex function, therefore $g(\bar{x}) \geq \bar{y}$.

Without loss of generality, let X be a discrete random variable, such that $E[X] = \sum_{i=1}^n P(X = x_i)x_i$ is finite.

Moreover, if $g(x)$ is a smooth (differentiable) function, then by using the Taylor series expansion about $E[X] = \mu$, we have

$$g(x) = \sum_{k=0,1,2,\dots} \frac{\partial^k g(x)}{\partial x^k} \Big|_{x=\mu} (x - \mu)^k.$$

A linear approximation of $g(x)$ is given as

$$g(x) \approx g(\mu) + \frac{\partial g(x)}{\partial x} \Big|_{x=\mu} (x - \mu),$$

where $g(\mu) = g(E[X])$.

Thus, $E[g(X)] = g(E[X])$, since $E[(X - \mu)] = 0$.

For the quadratic approximation, we have

$$g(x) \approx g(\mu) + \frac{\partial g(x)}{\partial x} \Big|_{x=\mu} (x - \mu) + \frac{\partial^2 g(x)}{\partial x^2} \Big|_{x=\mu} (x - \mu)^2.$$

Thus, $E[g(X)] = g(E[X]) + \frac{\partial^2 g(X)}{\partial X^2} \Big|_{X=\mu} E[(x - \mu)^2]$,

where $\frac{\partial^2 g(X)}{\partial X^2}$ is positive for a convex, or, negative for a concave g .

Therefore, $E[g(X)] \leq g(E[X])$, for a concave function g , and

$$E[g(X)] \geq g(E[X]), \text{ for a convex function } g. \quad \square$$

Hint:

$$E[g(X)] = g(E[X]), \text{ for a concave and convex, i.e. linear function } g.$$

Markov's Theorem 2.7.2. *Let $g(X)$ be a nonnegative function of a random variable X .*

$$\text{If } E[g(X)] \text{ exists, then we have } P(g(X) \geq c) \leq \frac{E[g(X)]}{c},$$

for a positive constant c .

Proof. For the sake of simplicity, assume X is a continuous r.v.

$$\text{Since } E[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx < \infty,$$

we can rewrite the integral as a sum of two finite integrals, such that,

$$\int_{-\infty}^{\infty} g(x) f_X(x) dx = \int_{\{x|g(x) \geq c\}} g(x) f_X(x) dx + \int_{\{x|g(x) < c\}} g(x) f_X(x) dx.$$

Since g and f_X are nonnegative functions over their level sets, then each one of these two right hand side integrals are nonnegative. Thus,

$$E[g(X)] \geq \int_{\{x|g(x) \geq c\}} g(x) f_X(x) dx \geq \int_{\{x|g(x) \geq c\}} c f_X(x) dx,$$

since $g(x)$ dominates c , for all x in this integral.

Therefore, $E[g(X)] \geq c \int_{\{x|g(x) \geq c\}} f_X(x) dx$, implying that,

$$E[g(X)] \geq cP(g(X) \geq c), \text{ or, } E[g(X)]/c \geq P(g(X) \geq c) \quad \square$$

Here, we have seen a relationship between the expectation and the probability measure of a nonnegative random variable.

Next theorem shows a relation between the mean and the standard deviation of a random variable.

Chebychev's Theorem 2.7.3. *Let there be a random variable X with a finite variance, σ^2 . Also, show that this is a sufficient condition for X having a finite mean, μ .*

Then $P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$, for a positive constant k .

Proof. Let $g(X) = (X - \mu)^2$ and $c = k^2\sigma^2$ in the above theorem 2.7.2.

Hence, $P((X - \mu)^2 \geq k^2\sigma^2) \leq \frac{E[(X - \mu)^2]}{k^2\sigma^2}$.

Since $\sigma^2 = E[(X - \mu)^2]$, we have, $P((X - \mu)^2 \geq k^2\sigma^2) \leq \frac{\sigma^2}{k^2\sigma^2}$.

On the other hand, observe that $|X - \mu| = +\sqrt{(X - \mu)^2}$.

Thus, $P((X - \mu)^2 \geq (k\sigma)^2) = P(|X - \mu| \geq k\sigma)$, by taking the positive square root of both terms of the event, in order to preserve the probability measure.

Therefore, $P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$ □

In Chebychev's theorem, we have seen that if you look at k -steps away from the mean value of a random variable, where each step size is as big as the standard deviation of the random variable, the probability of finding values for the random variable decreases as fast as $\frac{1}{k^2}$.

Chebychev's theorem can be seen as a particular case of a general theorem about the finite absolute moments of a random variable, around a point a , $E[|(X - a)^k|] < \infty$, such that

$$P(|X - a| \geq \epsilon) \leq \frac{E[|(X - a)^k|]}{\epsilon^k}, \text{ for any finite } \epsilon > 0.$$

2.7.1 Minimizers of Certain Expectations

We may be interested in a special point that minimizes the expected value of $|X - a|$, or a quadratic form $(X - b)^2$, as follows.

Note that

$$F_X(\text{Median}(X)) = \frac{1}{2}. \tag{2.43}$$

That is to say, $P(X \leq \text{Median}(X)) = P(X > \text{Median}(X)) = \frac{1}{2}$.

Median Theorem 2.7.4. *If $E[|X - a|] < \infty$, then the minimum of $E[|X - a|]$ is obtained when $a = \text{Median}(X)$.*

Proof. Without loss of generality, assume that X is a continuous r.v., then

$$\begin{aligned} \min_a \{E[|X - a|]\} &= \min_a \left\{ \int_{-\infty}^{\infty} |x - a| dF_X(x) \right\} \\ &= \min_a \left\{ \int_a^{\infty} (x - a) dF_X(x) - \int_{-\infty}^a (x - a) dF_X(x) \right\}. \end{aligned}$$

This implies that,

$$\frac{d}{da} \left[\int_a^{\infty} (x - a) dF_X(x) - \int_{-\infty}^a (x - a) dF_X(x) \right] = 0,$$

and also by the Leibniz integral rule,

$$\begin{aligned} \Rightarrow \int_a^{\infty} \frac{d}{da} (x - a) dF_X(x) - \int_{-\infty}^a \frac{d}{da} (x - a) dF_X(x) &= 0 \\ \Rightarrow - \int_a^{\infty} dF_X(x) + \int_{-\infty}^a dF_X(x) &= 0, \end{aligned}$$

or,

$$\int_{-\infty}^a dF_X(x) = \int_a^{\infty} dF_X(x).$$

Hence, $P(X \leq a) = P(X > a)$.

On the other hand, $P(X \leq a) + P(X > a) = 1$.

Therefore, we have $P(X \leq a) = P(X > a) = \frac{1}{2}$.

Consequently, we have $a = \text{Median}(X)$. $\square$

Average Theorem 2.7.5. *If $E[(X - b)^2] < \infty$, then the minimizer of $E[(X - b)^2]$ is $b = E[X]$.*

Proof. $E[(X - b)^2]$ is a convex function of X and b , thus, its minimum is found by the first order optimality conditions, i.e.,

$$\frac{d}{db} E[(X - b)^2] = 0,$$

that implies $E[2(X - b)(-1)] = -2E[X - b] = 0$.

Finally, this in turn gives the result as, $E[X - b] = 0$, i.e. $E[X] - b = 0$, or $b = E[X]$. $\square$

2.8 Some Approximations

The computational difficulties of the binomial probability distribution open doors to approximately practice the binomial probability law, as n gets larger, by a Poisson probability law, when np is moderate and stable, or, by a

standard normal probability law, when $np(1-p)$ is large and stable, i.e. $np(1-p)$ will not go to infinity as fast as n goes to infinity.

2.8.1 Poisson Approximation to Binomial

A Poisson random variable may be used as an approximation, when n gets larger, to a binomial random variable, X , with parameters n and p , such that $\lambda = np$ and np stays stable, by the Stirling's approximation:

$$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x} \approx \frac{\lambda^x}{x!} e^{-\lambda}, \text{ for } x = 0, 1, 2, \dots, n. \quad (2.44)$$

2.8.2 Normal Approximation to Binomial

A standard normal random variable may also be used as an approximation to a binomial random variable, X , when n gets larger, yet the binomial is a discrete r.v. and the standard normal is a continuous r.v.

We have to reinterpret a discrete event, in order to embed point events into the realm of continuous events, such that

$$P(X = x) = P(x - \epsilon \leq X \leq x + \epsilon), \text{ where } 0 < \epsilon < 1.$$

Since X is the number of successes in n independent and identically distributed (i.i.d.) Bernoulli trials, we can introduce DeMoivre-Laplace theorem, as follows.

DeMoivre-Laplace Theorem 2.8.1. *If X_n is the number of successes in n independent Bernoulli trials, then*

$$\lim_{n \rightarrow \infty} P(a \leq \frac{X_n - np}{\sqrt{np(1-p)}} \leq b) = \Phi(b) - \Phi(a), \text{ for any } a < b.$$

Theorem 2.8.1 opens a road to the central limit theorem, which is the hinge of using normal distribution for almost all statistical approximations.

Therefore,

$$\begin{aligned} P(X = x) &= \binom{n}{x} p^x (1-p)^{n-x} \\ &\approx \Phi\left(\frac{x + \epsilon - np}{\sqrt{np(1-p)}}\right) - \Phi\left(\frac{x - \epsilon - np}{\sqrt{np(1-p)}}\right), \text{ for } x = 0, 1, 2, \dots, n. \end{aligned} \quad (2.45)$$

Here, ϵ is a small positive number which is less than one, i.e., $\epsilon \in (0, 1)$

2.9 Exercises

1. Describe the terms sample space, event, probability, conditional probability, dependent and independent events.
2. Show the product rule of probability, by induction.
3. Let there be a set of two distinct elements (binary set). Construct the power set of it. What is the cardinality of the power set?
4. Prove the Boole's inequality.
5. A drawer contains three identical mechanical pencils, yet one of them is defective (not writing). If you pick a pencil from the drawer, then what is the probability that you picked the defective one? If you repeat the picking the pencil experiment for the second, and for the third time, then what are the probabilities of picking the defective pencil, respectively? Solve this problem once for repeating the experiment without replacement, and another time for repeating the experiment with replacement (putting the pencil back into the drawer and restarting the experiment), include all your assumptions.
6. Assume that 1% of a population has a disease that may be detected by a blood test, such that the test yields a positive result 95% of the time, when an ill person takes it. On the other hand, the test yields a false positive result with 2% rate, when a healthy person takes it. If a person takes the test and the test result turns out to be positive, then what is the probability that the person has the disease?
7. If $I_A(X)$ is the indicator function of the event $\{X \in A\}$, then show that $E[I_A(X)] = P(X \in A)$.
8. If a student takes a multiple choice test of 20 questions with 5 choices each, on a completely unknown subject to the student, then what is the probability that the student will answer all these questions correct?
9. If p is small and unknown we may wish to estimate the frequency of success by inverse sampling, that is we continue sampling until a certain number of successes are observed. What is the number of the sample (x) that maximizes the pmf of a negative binomial with respect to p , when $r > 1$?
10. John sends messages to Jane over a noisy telephone connection. When Jane detects an error, she asks John to resend the message. Let the

probability of a message transmission error is 5%, then what is the probability that a message would be transmitted more than three times?

11. If, on the average, there are ten undesirable telephone calls a day, then what is the probability that you will receive five undesirable telephone calls tomorrow?
12. If the lifetime is a continuous r.v. X , then find the conditional pdf of X , given that the event $A = \{X > t\}$, i.e. it is still alive at time t . Hint, first of all find the conditional cdf of X , when A is given.
13. Let us randomly pick a point in a unit circle, centered at origin. Also, let X be the random variable that measures the distance of the point from the origin. What is the probability that the randomly selected point would be in the ring with radii 0.5 and 0.75?
14. Show that the exponential random variable is memoryless, by showing $P(X > x + t \mid X > t) = P(X > x)$, for $t > 0$, and explain.
15. If the lifetime of a laptop computer battery is a normal random variable with an average life of three years, and a standard deviation of two years, then what is the probability that your laptop computer's battery will last somewhere between four to six years?
16. Show that $\int_{\mu}^{\infty} \frac{1}{\Gamma(k)} x^{k-1} e^{-x} dx = \sum_{x=0}^{k-1} \frac{\mu^x e^{-\mu}}{x!}$, for $k = 1, 2, \dots$
17. Show that Cauchy probability distribution does not have a mean, nor a variance.
18. Show Markov's theorem for discrete random variables.
19. Show Chebychev's theorem for discrete random variables.
20. Show DeMoivre-Laplace theorem.
21. If you call 1,000 customers a day, on the average 800 calls would be successes. If you are going to make 100 calls in this afternoon, then what is the probability that you would have more than seventy-five successes?

Chapter 3

Functions of Random Variables

In this chapter, we will study the functions of random variables and their probabilistic characteristics. In real life we encounter many such functions. For example, the revenue of a retailer depends on the demand for his goods, that might be uncertain; the total time that a clerk in the grocery store is busy serving customers continuously depends on the number customers requesting his services; the time to system failure in a serially designed multi-component system is the minimum time until first component failure.

We refer to the n -tuple of random variables, $\{X_1, \dots, X_n\}$, as a random vector, $\mathbf{X} = (X_1, X_2, \dots, X_n)$, such that each individual component of the random vector is a random variable.

Similarly, we consider the collection of random vectors as a random matrix, where each component (element) of that matrix is a random variable.

3.1 Univariate Functions

Let X be a random variable and $h(X)$ be a real function of X , such that $Y = h(X)$ is also a random variable. That is, X identifies the events of a sample space S as real numbers, and Y translates those real numbers to another set of real numbers, as an image of X through function h .

Hence, the probability distribution of random variable Y depends on the probability distribution of random variable X and function h , such that $P(Y \in B) = P(h(X) \in B) = P(X \in A)$, where event B is the image of event A , under function h . Moreover, if there exists an inverse function of h , then we can find the pre-image of B , such that $h^{-1}(B) = A$.

The expected value of function of a random variable is written by the

definition of the expectation as

$$E[Y] = E[h(X)] = \begin{cases} \int_{x=-\infty}^{\infty} h(x)f(x)dx, & \text{for continuous r.v. } X, \\ \sum_{x=-\infty}^{\infty} h(x)PX = x, & \text{for discrete r.v. } X, \end{cases}$$

that again will be simply formulated as

$$E[Y] = E[h(X)] = \int_{x=-\infty}^{\infty} h(x)dF(x).$$

Example 3.1

Let $Y = h(X) = aX + b$, for $a \neq 0$, be a linear function of r.v. X with given $F_X(x)$.

The cdf of r.v. Y , $F_Y(y) = P(Y \leq y)$, is obtained as

$$F_Y(y) = P(aX + b \leq y) = \begin{cases} P(X \leq \frac{y-b}{a}) = F_X(\frac{y-b}{a}), & \text{if } a > 0, \\ P(X \geq \frac{y-b}{a}) = 1 - F_X(\frac{y-b}{a}), & \text{if } a < 0. \end{cases}$$

Therefore, for $u = \frac{y-b}{a}$,

$$F_Y(y) = \begin{cases} F_X(u), & \text{if } a > 0, \\ 1 - F_X(u), & \text{if } a < 0. \end{cases}$$

Because $f_Y(y) = \frac{dF_Y(y)}{dy}$, using the chain rule, we express the density function of r.v. Y in terms of the density function of r.v. X ,

$$f_Y(y) = \begin{cases} \frac{1}{a}f_X(\frac{y-b}{a}), & \text{for } a > 0, \\ \frac{1}{-a}f_X(\frac{y-b}{a}), & \text{for } a < 0. \end{cases}$$

Briefly we write $f_Y(y) = \frac{1}{|a|}f_X(\frac{y-b}{a})$.

The expected value and the variance of r.v. Y are derived as

$$E[Y] = E[aX + b] = aE[X] + b,$$

and

$$Var(Y) = Var(h(X)) = a^2Var(X),$$

since

$$\begin{aligned} \text{Var}(Y) &= E[(Y - E[Y])^2] = E[(aX + b - aE[X] - b)^2] \\ &= E[a^2(X - E[X])^2] = a^2 \text{Var}(X). \end{aligned}$$

Special case: Let X be a normal r.v., such that $X \sim \mathcal{N}(\mu, \sigma^2)$, and $Y = aX + b$.

Thus, $f_Y(y) = \frac{1}{\sqrt{2\pi\sigma^2|a|}} e^{-\frac{(y-b-a\mu)^2}{2(a\sigma)^2}}$, i.e., $Y \sim \mathcal{N}(b + a\mu, (a\sigma)^2)$.

Fact: A linear function of a normal random variable is itself a normal random variable with the corresponding mean and variance.

Example 3.2

Let $Y = h(X) = X^2$, i.e., a quadratic function of a r.v. X with $F_X(x)$ given.

Then, $F_Y(y) = P(Y \leq y) = P(X^2 \leq y) = P(-\sqrt{y} \leq X \leq \sqrt{y})$, for $y \geq 0$.

Hence, $F_Y(y) = F_X(\sqrt{y}) - F_X(-\sqrt{y})$, for $y \geq 0$, implying that

$$f_Y(y) = \frac{f_X(\sqrt{y})}{2\sqrt{y}} + \frac{f_X(-\sqrt{y})}{2\sqrt{y}}, \text{ for } y > 0.$$

The mean and variance of r.v. Y are

$$E[Y] = E[X^2] = \text{Var}(X) + (E[X])^2,$$

and

$$\text{Var}(Y) = E[(X^2 - E[X^2])^2] = E[X^4 - 2X^2E[X^2] + E^2[X^2]],$$

that results in

$$\text{Var}(Y) = E[X^4] - E^2[X^2].$$

Special Case: Let Z be a standard normal random variable, and $Y = Z^2$,

then, $f_Y = \frac{e^{-\frac{y}{2}}}{\sqrt{2\pi y}}$.

Here, Y is known as a chi-square random variable, χ^2 , with one degree of freedom (there is only one source of information, Z , in order to describe the behavior of Y).

In general, if $y = h(x)$ has n solutions, $\{x_1, x_2, \dots, x_n\}$, then $f_Y(y) = \sum_{i=1}^n (f_X(x) / |\frac{dy}{dx}|) |_{x=x_i} = \sum_{i=1}^n (f_X(x) |\frac{dy}{dx}|^{-1}) |_{x=x_i}$.

Example 3.3

Again, let $y = x^2$, which has two solutions:

$$x_1 = \sqrt{y}, \text{ and } x_2 = -\sqrt{y}, \text{ for } y \geq 0.$$

Since $|\frac{dy}{dx}| = |2x|$, we have

$$f_Y(y) = f_X(\sqrt{y})|2(\sqrt{y})|^{-1} + f_X(-\sqrt{y})|2(-\sqrt{y})|^{-1} = \frac{f_X(\sqrt{y})}{2\sqrt{y}} + \frac{f_X(-\sqrt{y})}{2\sqrt{y}}.$$

3.2 Bivariate Functions

Let X_1 and X_2 be two random variables with a joint probability measure $P(X_1 \in A \cap X_2 \in B)$, and a joint cdf $F_{X_1 X_2}(x_1, x_2) = P(X_1 \leq x_1, X_2 \leq x_2)$.

If $f_{X_1 X_2}(x_1, x_2)$ is a differentiable function, then the joint pdf

$$f_{X_1 X_2} = \frac{\partial^2}{\partial x_1 \partial x_2} F_{X_1 X_2}(x_1, x_2), \quad (3.1)$$

where $f_{X_1 X_2}(x_1, x_2) \geq 0$, for all x_1 and x_2 , and $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X_1 X_2}(x_1, x_2) dx_1 dx_2 = 1$.

Moreover, the marginal pdf of the random variable X_j is obtained from the joint density function by summing over all possible outcomes of X_i r.v., as

$$\int_{-\infty}^{\infty} f_{X_1 X_2}(x_1, x_2) dx_i = f_{X_j}(x_j), \text{ for } i = 1, 2 \text{ and } j = 1, 2 \text{ and } i \neq j.$$

If X_1 and X_2 are independent random variables, the joint pdf would be the product of two marginal pdfs, $f_{X_1 X_2}(x_1, x_2) = f_{X_1}(x_1)f_{X_2}(x_2)$.

Let $Y_1 = h_1(X_1, X_2)$ and $Y_2 = h_2(X_1, X_2)$ be two random variables, such that h_1 and h_2 are differentiable real functions, for all (x_1, x_2) .

Assume that $y_1 = h_1(x_1, x_2)$ and $y_2 = h_2(x_1, x_2)$ have solutions,

$$x_1 = g_1(y_1, y_2) \text{ and } x_2 = g_2(y_1, y_2).$$

Thus, the joint pdf is given as

$$f_{Y_1, Y_2}(y_1, y_2) = f_{X_1, X_2}(x_1, x_2)|J(x_1, x_2)|^{-1}, \text{ for } J(x_1, x_2) \neq 0 \quad (3.2)$$

with $J(x_1, x_2)$ as the Jacobian determinant of h_1 and h_2 , such that

$$J(x_1, x_2) = \begin{vmatrix} \left(\frac{\partial h_1}{\partial x_1}\right) & \left(\frac{\partial h_1}{\partial x_2}\right) \\ \left(\frac{\partial h_2}{\partial x_1}\right) & \left(\frac{\partial h_2}{\partial x_2}\right) \end{vmatrix} \neq 0,$$

evaluated at $x_1 = g_1(y_1, y_2)$ and $x_2 = g_2(y_1, y_2)$.

Example 3.4

Let $Y_1 = X_1 + X_2$ and $Y_2 = X_1 - X_2$, with a joint pdf $f_{X_1 X_2}(x_1, x_2) = \frac{1}{\pi}$, on the unit disk, i.e. $x_1^2 + x_2^2 \leq 1$.

Since $h_1(x_1, x_2) = x_1 + x_2 = y_1$ and $h_2(x_1, x_2) = x_1 - x_2 = y_2$,

let $x_1 = g_1(y_1, y_2)$ and $x_2 = g_2(y_1, y_2)$.

Hence we have

$$J(x_1, x_2) = \begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix} = -2,$$

and $x_1 = \frac{1}{2}(y_1 + y_2)$, $x_2 = \frac{1}{2}(y_1 - y_2)$ are the unique solutions of g_1 and g_2 , for x_1 and x_2 .

Therefore, we have

$$f_{Y_1 Y_2}(y_1, y_2) = f_{X_1 X_2}(\frac{1}{2}(y_1 + y_2), \frac{1}{2}(y_1 - y_2)) | -2 |^{-1} = \frac{1}{2\pi},$$

for $0 \leq y_1 + y_2 \leq 2$, and $0 \leq y_1 - y_2 \leq 2$.

3.3 Multivariate Functions

Let $X_1, X_2, \dots, X_n$ be n random variables with a joint probability measure $P(X_1 \in A_1 \cap \dots \cap X_n \in A_n)$, and a joint cumulative distribution function

$$F_{X_1 X_2 \dots X_n}(x_1, x_2, \dots, x_n) = P(X_1 \leq x_1, X_2 \leq x_2, \dots, X_n \leq x_n).$$

If $F_{X_1 X_2 \dots X_n}(x_1, x_2, \dots, x_n)$ is a differentiable function, we have the joint pdf,

$$f_{X_1 X_2 \dots X_n}(x_1, x_2, \dots, x_n) = \frac{\partial^n}{\partial x_1 \partial x_2 \dots \partial x_n} F_{X_1 X_2 \dots X_n}(x_1, x_2, \dots, x_n). \quad (3.3)$$

The marginal pdf of a random variable X_j is,

$$f_{X_j}(x_j) = \underbrace{\int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty}}_{n-1 \text{ integrals}} f_{X_1 X_2 \dots X_n}(x_1, x_2, \dots, x_n) dx_1 \dots dx_{j-1} dx_{j+1} \dots dx_n,$$

for $j = 1, \dots, n$.

If X_i 's are (mutually) independent random variables, for $i = 1, 2, \dots, n$, then the joint pdf would be the product of their marginal pdfs,

$$f_{X_1 \dots X_n}(x_1, \dots, x_n) = \prod_{i=1}^n f_{X_i}(x_i). \quad (3.4)$$

Furthermore, if $X_1, X_2 \dots X_n$ are (mutually) independent random variables, then

$$E[h_1(X_1)h_2(X_2)\dots h_n(X_n)] = E[h_1(X_1)]E[h_2(X_2)]\dots E[h_n(X_n)]. \quad (3.5)$$

for h_i s being functions of X_i s. This follows immediately from Eq. 3.4 for continuous r.v.'s X_i 's, as

$$\begin{aligned} E[h_1(X_1)\dots h_n(X_n)] &= \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} h_1(x_1)\dots h_n(x_n)f_{X_1\dots X_n}(x_1,\dots,x_n)dx_1\dots dx_n \\ &= \int_{-\infty}^{\infty} h_1(x_1)f_{X_1}(x_1)dx_1 \int_{-\infty}^{\infty} h_2(x_2)f_{X_2}(x_2)dx_2 \dots \int_{-\infty}^{\infty} h_n(x_n)f_{X_n}(x_n)dx_n. \end{aligned}$$

Similarly, we observe that the above statement is also true for discrete random variables, as integrals are replaced by the appropriate summations.

Let $Y_i = h_i(X_1, X_2, \dots, X_n)$, for $i = 1, 2, \dots, n$ be a set of n random variables.

When the set of equations $\{y_i = h_i(x_1, x_2, \dots, x_n) | i = 1, \dots, n\}$, have a nonempty solution set, $\{x_i = g_i(y_1, y_2, \dots, y_n) | i = 1, \dots, n\}$, we have the joint pdf as follows:

$$f_{Y_1 Y_2 \dots Y_n}(y_1, y_2, \dots, y_n) = f_{X_1 X_2 \dots X_n}(x_1, x_2, \dots, x_n) |J(x_1, x_2, \dots, x_n)|^{-1}, \quad (3.6)$$

which is evaluated at the solution set, $\{x_i = g_i(y_1, y_2, \dots, y_n) | i = 1, \dots, n\}$, when the Jacobian is not zero,

$$J(x_1, \dots, x_n) = \begin{vmatrix} \frac{\partial h_1}{\partial x_1} & \dots & \frac{\partial h_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial h_n}{\partial x_1} & \dots & \frac{\partial h_n}{\partial x_n} \end{vmatrix} \neq 0.$$

3.4 Some Well-Known Functional Forms

One of the basic functional forms is the sum of random variables. Such functions arise when there is a need to combine the effects of uncertain entities. For example, we have mentioned before that if we perform n i.i.d. Bernoulli experiments and investigate the behavior of the total number of successes in these n trials, we obtain the Binomial random variable with its pmf.

Let $X_1, X_2, \dots, X_n$ be independent and identically distributed Bernoulli experiments with probability of success equal to p , and let $X_1^n = \sum_{i=1}^n X_i$.

In order to use the induction argument we will first obtain the pmf of X_1^2 as

$$\begin{aligned} P\{X_1^2 = 0\} &= P\{X_1 = 0, X_2 = 0\} = (1-p)^2, \\ P\{X_1^2 = 1\} &= P\{X_1 = 1 \text{ or } X_2 = 1\} = 2(1-p)p, \\ P\{X_1^2 = 2\} &= P\{X_1 = 1, X_2 = 1\} = p^2, \end{aligned}$$

thus implying that $X_1^2 \sim B(2, p)$. Now assume that $X_1^n \sim B(n, p)$, and show that $X_1^{n+1} = X_1^n + X_{n+1}$ is distributed as $B(n+1, p)$.

$$\begin{aligned} P\{X_1^{n+1} = k\} &= P\{X_1^n = k, X_{n+1} = 0\} + P\{X_1^n = k-1, X_{n+1} = 1\} \\ &= \binom{n}{k} p^k (1-p)^{n-k} (1-p) + \binom{n}{k-1} p^{k-1} (1-p)^{n-k+1} p \\ &= \frac{n!}{k!(n-k+1)!} p^k (1-p)^{n-k+1} ((n-k+1) + k) \\ &= \binom{n+1}{k} p^k (1-p)^{n-k+1}. \end{aligned}$$

Hence, giving the result.

In general, we obtain the distribution of the sum of random variables, e.g., $Z = X + Y$, for X and Y nonnegative discrete random variables, via the use of a convolution sum, i.e.,

$$P\{Z = k\} = \sum_{i=0}^k P\{X = i, Y = k-i\}.$$

Example 3.5

Let X and Y denote i.i.d. random variables with shifted geometric distribution with the same parameter, the probability of success, p . Meaning that, X and Y both represent the time of first success when i.i.d. Bernoulli experiments are performed, hence their pmfs are the same and equal to,

$$P\{X = k\} = P\{Y = k\} = (1-p)^{k-1}p, \text{ for } k = 1, 2, \dots$$

Then, $Z = X + Y$, represents the time of second success and it has the following pmf given by the above equality,

$$\begin{aligned} P\{Z = k\} &= \sum_{i=0}^k P\{X = i, Y = k-i\} = \sum_{i=1}^{k-1} P\{X = i\}P\{Y = k-i\} \\ &= \sum_{i=1}^{k-1} (1-p)^{i-1}p(1-p)^{k-i-1}p = (k-1)(1-p)^{k-2}p^2. \end{aligned}$$

The second equality is due to the independence of X and Y , while the last equality follows from combining the superscripts and noting that inside the summation nothing is left other than 1's, thus giving its sum as $k - 1$. As we have mentioned before, the last formula is nothing but the pmf of a negative binomial r.v. with parameters 2 and p .

Example 3.6

The pmf of $Z = X + Y$, for independent r.v.'s, $X \sim P(\lambda)$ and $Y \sim P(\mu)$, is derived as

$$\begin{aligned} P\{Z = k\} &= \sum_{i=0}^k P\{X = i\}P\{Y = k - i\} = \sum_{i=0}^k (e^{-\lambda} \frac{\lambda^i}{i!}) (e^{-\mu} \frac{\mu^{k-i}}{(k-i)!}) \\ &= \frac{e^{-\lambda-\mu}}{k!} \sum_{i=0}^k \frac{k!}{i!(k-i)!} \lambda^i \mu^{k-i} = e^{-\lambda-\mu} \frac{(\lambda + \mu)^k}{k!}, \end{aligned}$$

which is just the pmf of a Poisson r.v. with parameter $(\lambda + \mu)$. Note that the last equality follows directly from the Binomial theorem, i.e.,

$$\sum_{i=0}^k \frac{k!}{i!(k-i)!} a^i b^{k-i} = \sum_{i=0}^k \binom{k}{i} a^i b^{k-i} = (a + b)^k.$$

For nonnegative continuous random variables, we can obtain the density function of a sum, $Z = X + Y$, of two random variables, X and Y , as the following convolution integral in terms of the joint density function, $f_{X,Y}(x, y)$,

$$f_Z(z) = \int_{x=0}^z f_{X,Y}(x, z-x) dx,$$

which will be simplified to the following equation for the independent r.v.'s,

$$f_Z(z) = \int_{x=0}^z f_X(x) f_Y(z-x) dx.$$

Example 3.7

The pdf of the sum of two i.i.d. exponential random variables with parameter λ is given as

$$\begin{aligned} f_Z(z) &= \int_{x=0}^z f_{X,Y}(x, z-x) dx = \int_{x=0}^z (\lambda e^{-\lambda x})(\lambda e^{-\lambda(z-x)}) dx \\ &= z\lambda^2 e^{-\lambda z}, \end{aligned}$$

which is just the density function of $Gamma(2, \lambda)$ random variable.

Next, we present the distributions of the minimum and the maximum of random variables.

Let $Z = \min\{X, Y\}$ for some random variables X and Y , then,

$$P\{Z > z\} = P\{X > z, Y > z\},$$

and if X and Y are independent, then $P\{Z > z\} = P\{X > z\}P\{Y > z\}$.

Example 3.8

Consider a system consisting of two independent serially connected components. Each component may fail randomly according to a time to failure distribution. Let component 1's time to failure, X , is distributed as an exponential r.v. with parameter λ , and component 2's time to failure, Y , distributed as exponential r.v. with μ . Then, the tail distribution of the system time to failure, Z , is given by

$$P\{Z > z\} = P\{X > z\}P\{Y > z\} = e^{-\lambda z}e^{-\mu z} = e^{-(\lambda+\mu)},$$

which is again denotes an exponential random variable, but with parameter $(\lambda + \mu)$.

If we consider a system consisting of two components connected in parallel, then the system time to failure distribution is given by the maximum of the individual time to failures. Thus, $Z = \max\{X, Y\}$, with the following distribution function,

$$P\{Z < z\} = P\{X < z, Y < z\}.$$

3.5 Covariance and Correlation

The pairwise behavior of random variables are approached by covariance and correlation (originally co-relation) analysis. We have defined the covariance as the inner product of two vectors $X - \bar{X}$ and $Y - \bar{Y}$, before.

Now, we can say that the expected value is a weighted inner product (weights are the probabilities) of random variables, X and Y , such that $E[(X - E[X])(Y - E[Y])]$, then the covariance of X and Y would be this inner product, i.e., $Cov(X, Y) = E[(X - E[X])(Y - E[Y])]$

$$\begin{aligned} &= E[XY - XE[Y] - E[X]Y + E[X]E[Y]] \\ &= E[XY] - E[X]E[Y] - E[X]E[Y] + E[X]E[Y] = E[XY] - E[X]E[Y]. \end{aligned}$$

Covariance is also denoted by σ_{XY} .

As before, we can look at the cosine of the angle between $(X - E[X])$ and $(Y - E[Y])$, as the indicator of relative orientations of these two random variables, and as before, we call it the correlation of X and Y , i.e.

$$Cor(X, Y) = \frac{Cov(X, Y)}{\|X\| \|Y\|},$$

where the norm of random variables are defined as,

$$\|X\| = \sqrt{Var(X)} = \sigma_X, \text{ and } \|Y\| = \sqrt{Var(Y)} = \sigma_Y.$$

Usually ρ , or ρ_{XY} , denotes the linear correlation of X and Y , such that $-1 \leq \rho_{XY} \leq 1$, thus this provides a simple formulation for correlation as

$$\rho_{XY} = \frac{\sigma_{XY}}{\sigma_X \sigma_Y}. \quad (3.7)$$

Obviously, collinear random variables have $\rho_{XY} = 1$, or $\rho_{XY} = -1$, indicating that covariances and correlations can be negative.

On the other hand, independent random variables have,

$$Cov(X, Y) = E[XY] - E[X]E[Y] = E[X]E[Y] - E[X]E[Y] = 0.$$

Hence, if random variables are independent, then $\rho_{XY} = 0$. That is they are also uncorrelated.

3.6 Mean and Variance of a Finite Sum

When we have a sum of n random variables, $\sum_{i=1}^n X_i$, we can find the expected value of this sum as the sum of the individual random variables, i.e.,

$E[\sum_{i=1}^n X_i] = \sum_{i=1}^n E[X_i]$. Similarly, if we look at the variance of the sum of these random variables, we have the following observations,

$$Var(\sum_{i=1}^n X_i) = \sum_{i=1}^n Var(X_i) + 2 \sum \sum_{\{i < j\}} Cov(X_i, X_j).$$

Note that covariance is symmetric, $Cov(X, Y) = Cov(Y, X)$.

Moreover, when the random variables are independent, we have

$$Var(\sum_{i=1}^n X_i) = \sum_{i=1}^n Var(X_i).$$

3.7 Bivariate Normal

Two random variables, X_1 and X_2 , with means μ_1 and μ_2 , and variances σ_1 and σ_2 , respectively, and correlation ρ , are called bivariate normal random

variables, with the joint pdf

$$f_{X_1 X_2}(x_1, x_2) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} e^{-\frac{1}{2(1-\rho^2)}\left(\frac{(x_1-\mu_1)^2}{\sigma_1^2} - 2\rho\frac{(x_1-\mu_1)(x_2-\mu_2)}{\sigma_1\sigma_2} + \frac{(x_2-\mu_2)^2}{\sigma_2^2}\right)}. \quad (3.8)$$

The expected value of a bivariate normal distribution is $\boldsymbol{\mu} = E[X_1, X_2]$.

The pdf of a bivariate normal is an ellipsoid, centered at $\boldsymbol{\mu}$.

That is, if you cut the pdf with planes parallel to the $X_1 X_2$ plane, the cross-sections (level sets) would be ellipses and their eccentricity depends upon the standard deviation of the random variables, X_1 and X_2 . Also, the orientation of their axes depends upon the correlation of X_1 and X_2 .

Example 3.9

When X_1 and X_2 have $\mu_1 = \mu_2 = 0$, $\sigma_1^2 = \sigma_2^2 = 1$, and $\rho = 0$, we have the following bivariate normal pdf, as in Figure 3.1,

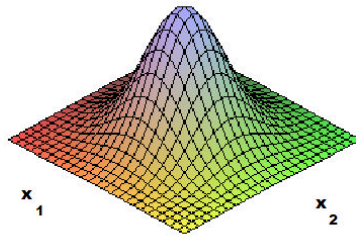


Figure 3.1: Bivariate Normal pdf

In the above example, since $\sigma_1 = \sigma_2$, and $\rho = 0$, these level sets (ellipses) become circles, centered at $\boldsymbol{\mu} = (0, 0)$.

3.8 Multivariate Normal

An n -dimensional column random vector $\mathbf{X} = (X_1, X_2, \dots, X_n)^T$ has a multivariate (or, n -variate) normal distribution with the pdf,

$$f_{\mathbf{X}}(\mathbf{x}) = \frac{1}{\sqrt{(2\pi)^n} \sqrt{|\Sigma|}} e^{\left(-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^T \Sigma^{-1}(\mathbf{x}-\boldsymbol{\mu})\right)}, \quad (3.9)$$

where $\mathbf{x} = (x_1, \dots, x_n)^T$ and $\boldsymbol{\mu} = (E[X_1], E[X_2], \dots, E[X_n])^T$, are n -dimensional column vectors.

Moreover,

$$\Sigma = [\text{Cov}(X_i, X_j)] = \begin{pmatrix} \text{Cov}(X_1, X_1) & \text{Cov}(X_1, X_2) & \dots & \text{Cov}(X_1, X_n) \\ \text{Cov}(X_2, X_1) & \text{Cov}(X_2, X_2) & \dots & \text{Cov}(X_2, X_n) \\ \vdots & \vdots & \ddots & \vdots \\ \text{Cov}(X_n, X_1) & \text{Cov}(X_n, X_2) & \dots & \text{Cov}(X_n, X_n) \end{pmatrix},$$

is the positive definite, symmetric variance-covariance matrix, a.k.a. covariance matrix, of the random vector $\mathbf{X}$.

The pdf of a multivariate normal distribution is a hyper-ellipsoid (an ellipsoid in dimensions higher than three), centered at the equilibrium point of this hyper-ellipsoid, $\boldsymbol{\mu}$.

3.9 Moment Generating Function

A moment generating function for a random vector, $\mathbf{X} = (X_1, \dots, X_n)$, is denoted as $M_{\mathbf{X}}(\mathbf{t}) = M_{\mathbf{X}}(t_1, t_2, \dots, t_n)$, and defined as,

$$M_{\mathbf{X}}(t_1, t_2, \dots, t_n) = E[e^{\sum_{i=1}^n t_i X_i}]. \quad (3.10)$$

A marginal moment generating function is the individual component for the random variable X_j , when $t_i = 0$, for $i \neq j$, such that

$$M_{X_j}(t_j) = M_{\mathbf{X}}(0, 0, \dots, t_j, 0, \dots, 0).$$

If the random variables are independent, then

$$M_{\mathbf{X}}(t_1, \dots, t_n) = \prod_{i=1}^n M_{X_i}(t_i). \quad (3.11)$$

A moment generating function for $\mathbf{X}$ may not exist.

We can similarly define the characteristic function for a random vector $\mathbf{X}$ as

$$\varphi_{\mathbf{X}}(\mathbf{t}) = \varphi_{\mathbf{X}}(t_1, t_2, \dots, t_n) = E[e^{i \sum_{j=1}^n t_j X_j}], \quad (3.12)$$

with $i = \sqrt{-1}$ denoting the imaginary number.

A marginal characteristic function of an individual random variable X_j is $\varphi_{X_j}(t_j) = \varphi_{\mathbf{X}}(0, 0, \dots, t_j, 0, \dots, 0)$.

If the random variables are independent, we have

$$\varphi_{\mathbf{X}}(t_1, \dots, t_n) = \prod_{j=1}^n \varphi_{X_j}(t_j). \quad (3.13)$$

A characteristic function for $\mathbf{X}$ always exists, over the field of real numbers, since $|\varphi_{\mathbf{X}}(\mathbf{t})| \leq \varphi_{\mathbf{X}}(|\mathbf{t}|) = 1$.

3.10 Conditioning

Conditioning is a very powerful tool to compute probability distributions and other moments. We will see its usage throughout the book.

We have introduced the conditional probability in Section 2.3. If in Eq.2.5, events A and B correspond to $\{X = i\}$ and $\{X < x\}$ for some i and x such that $i < x$, respectively, for a discrete r.v. X , the Bayes' formula becomes

$$P\{X = i | X < x\} = \frac{P\{X = i, X < x\}}{P\{X < x\}}, \quad \text{for } i < x.$$

When events A and B correspond to $\{X = i\}$ and $\{Y = k\}$ for some i and k and for discrete r.v.'s X and Y , respectively, then Bayes' formula becomes

$$P\{X = i | Y = k\} = \frac{P\{X = i, Y = k\}}{P\{Y = k\}}, \quad \text{for } i < x,$$

in terms of the joint, $P\{X = i, Y = k\}$, and the marginal $P\{Y = k\}$ pmf's. Similarly, the conditional cumulative distribution function is given as

$$F_{X|Y}(x|y) = \frac{F_{XY}(x, y)}{F_Y(y)},$$

thus giving

$$f_{X|Y}(x|y) = \frac{f_{XY}(x, y)}{f_Y(y)}.$$

with f as a pdf for continuous random variables, or a pmf for discrete random variables.

The conditional expectation of r.v. X given Y is then written in terms of the conditional density function as,

$$E[X|Y] = \begin{cases} \int_{-\infty}^{\infty} x f_{X|Y}(x|Y) dx, & \text{for a continuous random variable } X, \\ \sum_{\{\forall x\}} x f_{X|Y}(x|Y), & \text{for a discrete random variable } X, \end{cases}$$

when $f_Y(y) \neq 0$, for all y .

Here, $E[X|Y]$ is a function of random variable Y , evaluated at any point y in the support space of Y , i.e. evaluated at $Y = y$, for any $y \in S$.

Thus, $E[X|Y]$ is itself a random variable, yet once a point $Y = y$ is fixed, it becomes a constant, weighted average, for X .

Remember that the marginal pdf is $f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x, y) dx$, and the marginal pmf is $f_Y(y) = \sum_{\{\forall x\}} f_{XY}(x, y)$.

Conditioning preserves the properties of expectation, i.e.,

$E[h(X)|Y] = \int_{-\infty}^{\infty} h(x) f_{X|Y}(x|Y) dx$, for a continuous random variable X , and

$E[h(X)|Y] = \sum_{\{\forall x\}} h(x) f_{X|Y}(x|Y)$, for a discrete random variable X .

Also, $E[\sum_{i=1}^n X_i|Y] = \sum_{i=1}^n E[X_i|Y]$.

Observe that,

$$E[E[X|Y]] = \int_{-\infty}^{\infty} E[X|Y = y] f_Y(y) dy, \quad (3.14)$$

for a continuous random variable Y , and

$$E[E[X|Y]] = \sum_{\{\forall y\}} E[X|Y = y] f_Y(y), \quad (3.15)$$

for a discrete random variable Y , where $f_Y(y) = P(Y = y)$.

Thus,

$$E[E[X|Y]] = E[X]. \quad (3.16)$$

Example 3.10

Assume that you are in an unknown territory and there are two trails that you can follow throughout. One trail meanders two hours and returns back where you are now. The other trail will take you to known places in four hours. Since you are completely ignorant about this environment, you

can choose either path equally likely. How long do you expect to be there, without resting a moment, until you go to the known places?

Let X be the time that you will be there, until you will go to known places, but you shall not rest until you have reached the known places. Thus,

$$E[X] = E[E[X|T]] = E[X|T = 1]P(T = 1) + E[X|T = 2]P(T = 2),$$

where T denotes the trail that you will choose, such that $T = 1$ represents the restart event and $T = 2$ represents the going home event.

Since, $E[X|T = 1] = 2 + E[X]$ and $E[X|T = 2] = 4$, we have

$$E[X] = (2 + E[X])(\frac{1}{2}) + (4)(\frac{1}{2}) = 1 + 2 + \frac{1}{2}E[X].$$

That is, $E[X] = 6$ hours. A well deserved rest stop follows.

3.11 Minimum Mean Squared Error

When $h(X)$ is a function of an observable random variable X , such that $h(X)$ is predicting the values of another random variable Y , we would like to make this prediction as reliable as possible, i.e. we want $h(X)$ gets as close as possible to Y .

A commonly used distance metric is the mean squared differences between the predicted and the predictor, $E[(Y - h(X))^2]$.

The minimum mean squared difference estimator is the conditional expectation of Y , based on X ; that is among all possible functions, $h(X) = E[Y|X]$ would be the minimizer of $E[(Y - h(X))^2]$. One could reason that the estimation error is minimized in L_2 norm if the error vector is orthogonal to X , i.e.,

$$E[(Y - h(X))X] = 0,$$

implying that $E[YX] = E[h(X)X]$. This is a valid statement if and only if $h(X) = E[Y|X]$, since then,

$$E[XE[Y|X]] = E[E[XY|X]] = E[YX].$$

Extensive work on this subject was done by many, especially by Gauss in the nineteenth century, and for more information we suggest [3].

3.12 Convergence

A sequence of random variables, $X_1, X_2, \dots, X_n$, may converge to a random variable, X , in many different fashions, but we will look at three major convergence concepts.

3.12.1 Convergence in Probability

The random sequence converges in probability, if

$$\lim_{n \rightarrow \infty} P(|X_n - X| < \epsilon) = 1, \quad (3.17)$$

for any small positive ϵ .

Similarly, if

$$\lim_{n \rightarrow \infty} P(|X_n - X| \geq \epsilon) = 0, \quad (3.18)$$

for any small positive ϵ .

In this case we say that the sequence converges to X in probability,

denoted by $X_n \xrightarrow{P} X$.

Here, X could even be a degenerate random variable, i.e. a constant.

Also, if $X_n \xrightarrow{P} X$ and h is a real continuous function, then $h(X_n) \xrightarrow{P} h(X)$.

3.12.2 Convergence with Probability One

The random sequence converges with probability one (w.p.1), if

$$P\left(\lim_{n \rightarrow \infty} |X_n - X| < \epsilon\right) = 1, \quad (3.19)$$

for any small positive ϵ .

This is a stronger condition than convergence in probability, since this refers to the fact that the actual sequence of random variables converges to a limit.

3.12.3 Convergence in Distribution

Let F_{X_n} and F_X be the cdfs of X_n and X , respectively. If

$$\lim_{n \rightarrow \infty} F_{X_n}(x) = F_X(x), \text{ for all } x, \quad (3.20)$$

except at the point of discontinuities of $F_X(x)$, then we say that X_n converges to X in distribution, and denoted by $X_n \xrightarrow{D} X$.

In addition, if h is a real continuous function, then $h(X_n) \xrightarrow{D} h(X)$.

In fact, when the sequence of random variables has a moment generating function, $M_{X_n}(t)$, for all n , and the random variable X has a moment generating function, $M_X(t)$, and moreover

if $\lim_{n \rightarrow \infty} M_{X_n}(t) = M_X(t)$, then $X_n \xrightarrow{D} X$.

Similarly, for the characteristic functions of the sequence of random variables X_n and the fixed random variable X , $\varphi_{X_n}(t)$ and $\varphi_X(t)$, respectively, we have, if

$\lim_{n \rightarrow \infty} \varphi_{X_n}(t) = \varphi_X(t)$, then $X_n \xrightarrow{D} X$.

Convergence in probability refers to a sequence of random variables getting closer to a random variable, but convergence in distribution is the limiting distribution of a sequence of random variables.

Hence, $X_n \xrightarrow{P} X$ implies that $X_n \xrightarrow{D} X$, but the converse argument is not valid, i.e. $X_n \xrightarrow{D} X$ does not imply that $X_n \xrightarrow{P} X$.

We can simply denote the convergence in distribution, by using the following notation in which the limit random variable is replaced by its own probability distribution on the right hand side, i.e., $X_n \xrightarrow{D} F_X(x)$. However, this abuse of notation is the conventional way of writing convergence in distribution.

3.13 Law of Large Numbers

The sequences of random variables have some interesting behaviors. One of them is known as the law of large numbers. There are two versions of this law.

Weak Law of Large Numbers 3.13.1. *If $\{X_1, X_2, \dots, X_n\}$ is a sequence of independent and identically distributed (i.i.d.) random variables with a common finite mean, μ , and a common finite variance, σ^2 , the following holds*

$$\lim_{n \rightarrow \infty} P\left(\left|\frac{X_1 + \dots + X_n}{n} - \mu\right| \geq \varepsilon\right) = 0, \text{ for a positive small number } \varepsilon. \quad (3.21)$$

Proof. Since $\text{Var}\left(\frac{X_1 + \dots + X_n}{n}\right) = \frac{1}{n^2} \sum_{i=1}^n \sigma^2 = \frac{1}{n^2} (n\sigma^2) = \frac{\sigma^2}{n}$,

by theorems 2.7.2 and 2.7.3, we have

$$\lim_{n \rightarrow \infty} P\left(\left|\frac{X_1 + \dots + X_n}{n} - \mu\right| \geq \varepsilon\right) \leq \frac{\sigma^2}{n\varepsilon^2},$$

and $\frac{\sigma^2}{\varepsilon^2}$ is a finite positive constant. □

The other version of the law of large numbers is given below.

Strong Law of Large Numbers 3.13.2. *If $\{X_1, X_2, \dots, X_n\}$ is a sequence of i.i.d. random variables with a common finite mean, μ , then*

$$\lim_{n \rightarrow \infty} P\left(\frac{X_1 + \dots + X_n}{n} = \mu\right) = 1. \quad (3.22)$$

A modified form of the strong law of large numbers, known as the law of iterated logarithm, maybe useful in some cases.

Law of Iterated Logarithm 3.13.3. *Let $\{X_1, X_2, \dots, X_n\}$ be a sequence of bounded, i.i.d. random variables with a common finite mean, μ , and a common finite standard deviation, σ .*

Also, let $\xi_n = \frac{\sum_{i=1}^n (X_i - \mu)}{\sigma}$, then

$$P\left(\limsup_{n \rightarrow \infty} \frac{|\xi_n|}{\sqrt{2n \log \log n}} \leq 1\right) = 1. \quad (3.23)$$

If the random experiment is a Bernoulli trial, with $P(X_i = 1) = p$, and the success rate is the relative frequency of success, $\zeta = \frac{\sum_{i=1}^n X_i}{n}$, then $P\left(\limsup_{n \rightarrow \infty} \frac{|\zeta - p|}{\sqrt{\frac{2p(1-p) \log \log n}{n}}} \leq 1\right) = 1$, where “sup” identifies the least upper bound.

3.14 Central Limit Theorem

The central limit theorem of the probability theory opens avenues for productively approaching many statistical problems.

Central Limit Theorem 3.14.1. *If $\{X_1, X_2, \dots, X_n\}$ is a sequence of i.i.d. random variables with a common finite mean, μ , and a finite variance, σ^2 , then*

$$\lim_{n \rightarrow \infty} P\left(\frac{X_1 + \dots + X_n - n\mu}{\sqrt{n\sigma^2}} \leq z\right) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-\frac{x^2}{2}} dx. \quad (3.24)$$

This form of the central limit theorem implies that $\frac{X_1 + \dots + X_n - n\mu}{\sqrt{n\sigma^2}}$ converges to the standard normal random variable, no matter what the original governing probability law was for X_i s. Thus, this is a convergence in distribution.

On the other hand, if X_i 's are not identical, but all of them have finite means, μ_i , and finite variances, σ_i^2 , then we have another version of the central limit theorem, such that

$$\lim_{n \rightarrow \infty} P\left(\frac{\sum_{i=1}^n (X_i - \mu_i)}{\sqrt{\sum_{i=1}^n \sigma_i^2}} \leq z\right) = \Phi(z). \quad (3.25)$$

This is relaxing the original assumption about knowing specific probability laws, and making the central limit theorem a robust instrument for building generalized models.

3.15 Exercises

1. Let $Y_1 = X_1 + X_2$ and $Y_2 = X_1 - X_2$, where X_1 and X_2 are bivariate normal random variables with μ_1 and μ_2 means, σ_1^2 and σ_2^2 variances, respectively, and ρ correlation. Find the joint pdf $f_{Y_1 Y_2}(y_1, y_2)$.
2. Show that if X and Y are independent random variables, then $Cov(X, Y) = 0$, but the reverse statement is false. (Hint: an evidence of a counterexample is sufficient for disproving.)
3. Let $Y = AX + b$, where X is an n dimensional (column) random vector, with multivariate normal joint pdf, A is an $m \times n$ real matrix, and b is an m dimensional (column) real vector. Find the joint pdf for the m dimensional Y random (column) vector. Hint, a moment generating function is unique for any random variable.
4. Let $X = \sum_{i=1}^N X_i$ be a random sum of i.i.d. random variables, $\{X_i\}$, where N is an independent positive integer random variable. What is the moment generating function of X ? Also, what are the mean and variance of X ? Hint, use the moment generating function for the first and the second moments of X .
5. Prove that $E[E[X|Y]] = E[X]$ and $Var(E[X|Y]) \leq Var(X)$.
6. Let $X_1, X_2, \dots, X_n$ be i.i.d. random variables with finite mean μ and finite variance σ^2 , such that $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$.
 Prove that $E[\bar{X}] = \mu$, $Var(\bar{X}) = \frac{1}{n} \sigma^2$,
 and $E[\sum_{i=1}^n (X_i - \bar{X})^2] = (n-1)\sigma^2$.
7. What is the minimum mean square estimator $h(X)$, for $E[(Y - h(X))^2]$, when $h(X) = aX^2 + bX + c$. That is, what are the values of a, b, c ?

8. Let X_1 and X_2 be continuous random variables with a joint pdf of $f(x_1, x_2) = 2x_1x_2$, for $x_1 \in (0, 1)$ and $x_2 \in (0, 1)$, and zero elsewhere.

What is the joint pdf of $Y_1 = X_1^2$ and $Y_2 = X_1X_2$?

9. Show that $Y = \frac{(X-\mu)^2}{\sigma^2}$ is subject to a χ^2 distribution with one degree of freedom, when we have $X \sim \mathcal{N}(\mu, \sigma^2)$.
10. If $X_1, \dots, X_n$ are mutually independent χ^2 random variables with $\nu_1, \dots, \nu_n$ degrees of freedoms respectively, then show that the random variable $Y = \sum_{i=1}^n X_i$ is subject to a χ^2 distribution with $\nu = \sum_{i=1}^n \nu_i$ degrees of freedom.

Chapter 4

Sampling and Its Distributions

4.1 Sampling

Many of us may have already encountered or will soon experience situations that we need to learn and infer about a whole group of individuals, that we will call as the “population”. Yet it is usually impossible to study the whole population. For example, assume that you are a wine producer. Before setting a price, you wish to know the quality of your product, by taste testing them. If you drink all the wine, there will be nothing left to sell. Instead, you would settle on opening some bottles of your wine and tasting them, for the taste test (or, for the statistical quality control). This subset of the population that provides information about the whole population, to some degree, is known as the sample.

The moral of the story is that; one collects evidence, weighs it and makes a decision, based on the weight of the evidence. This decision making is a generalization of the sample information for the unknown population characteristic, i.e., the parameter.

It is clear that this induction, based on the measurable sample characteristic of the same kind, statistic, is open to generalization errors, or, risk of making the wrong decision based on the weight of the evidence. We wish to minimize that risk.

Estimating the parameter depends upon how good a sample represents the population. We assume that while we are looking at the sample data, i.e. collection of raw facts, the population is not going to change its characteristics. Hence, you must intelligently design a sampling method (experiment) to choose from the population, so that your sample data shall not mislead

you very much, while providing information about the unknown population characteristic.

Some experiment designers choose their sample from the easiest observable part of the population, since there is always a cost for the sampling involved, such as their friends and family, thus inducing a bias to their evidence. No matter what the intentions are, any bias in the sample would result in disaster. For example, quotas or any other seemingly reasonable ways of introducing similarity to the population, introduce bias into the sample.

An approach to minimize the effect of bias is to conduct random sampling, and the easiest design of experiment is to choose randomly without replacement.

Choosing samples randomly with replacement is another form of the random sampling. This will assign equal chance to every individual in the population, to be included in the sample, and that provides the largest spectrum of information about the unknown parameter.

Beside the bias, the decision risk is involved with the observation errors and random errors. The sample size plays a role in reducing the risk, to some degree, yet the sample size may not have any effect on the bias, this is the experiment designer's responsibility.

4.2 Random Sampling

A population is identified by a probability distribution, with a cdf $F_X(x|\theta)$, where the cdf is unknown, or, if the cdf has a known form, the parameter θ is unknown.

For the time being, we assume that the parameter is a number, such as p , or a real vector such as (μ, σ^2) .

In statistics, we introduce the parameter explicitly in the cdf, since it is unknown and we want to estimate it. On the other hand, in probability theory, we assume a probability distribution with a known parameter, well hidden under the background sample space.

In order to estimate the unknown parameter, we collect a random sample from the population, $\{X_1, X_2, \dots, X_n\}$, where each observation X_i is an independent random variable which inherits the population's probability distribution, that is our sample is an n -tuple of i.i.d. random variables, or a random vector, $X = (X_1, X_2, \dots, X_n)$.

After we measure every component, we have a realization, $x = (x_1, \dots, x_n)$, of the sample. If we repeat the sampling with the same number of observations, then we get another realization. Thus, the sample X has many realizations $\{x^{(i)} | i = 1, 2, \dots\}$ of the same size.

The joint cdf of the sample is

$$F_{X_1 X_2 \dots X_n}(x_1, x_2, \dots, x_n | \theta) = \prod_{i=1}^n F_{X_i}(x_i | \theta). \quad (4.1)$$

Since the outcome of each selection is independent of others, in a sample. Moreover, since each selection in the sample inherits the common collective cdf F , we have

$$F_{X_1 \dots X_n}(x_1, x_2, \dots, x_n | \theta) = \prod_{i=1}^n F(x_i | \theta). \quad (4.2)$$

Without loss of generality, we can simplify the original problem, by using the pdf, $f(x|\theta)$, for the population, where θ is the unknown parameter. In order to estimate the unknown parameter, we collect a random sample of size n from the population and we look for a best (in the sense of minimizing the risk of making errors, or acquiring the maximum information from it) function of the sample to estimate the parameter.

Any function of the sample that does not explicitly depend upon the unknown parameter is called a *statistic*. Since a statistic is a function of random variables, a statistic is also a random variable and it has a probability distribution, known as the sampling distribution. Hence, we can define the r -th moment of the sample, about a point a , as follows:

$$m_a^r = \frac{1}{n} \sum_{i=1}^n (X_i - a)^r, \text{ for } r = 1, 2, 3, \dots \quad (4.3)$$

For example, $m_0^1 = \frac{1}{n} \sum_{i=1}^n X_i$ is the sample mean, $\bar{X}$, and also we have $m_{\bar{X}}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$ as the sample variance, S^2 .

Before a sample realization is obtained, these are statistics (random variables as functions of random variables), and after a sample realization, we have $\bar{x}$ and s^2 , as real numbers, with no more randomness left in those moments.

We want to use a statistic $h(X)$ as an estimator, to estimate the unknown parameter θ , and an estimator is known to be unbiased if $E[h(X)] = \theta$.

Otherwise, if $E[h(X)] = \theta + b$, then we say that the estimator is biased by the amount of b .

For example, the sample mean is a statistic,

$$h(X) = \bar{X}, \quad (4.4)$$

and it is an unbiased estimator of the unknown population mean, μ , since $E[\bar{X}] = E[\frac{1}{n} \sum_{i=1}^n X_i] = \frac{1}{n} \sum_{i=1}^n E[X_i]$ and X_i s are i.i.d. random variables with the same common mean μ , i.e. $E[X_i] = \mu$.

Therefore,

$$E[\bar{X}] = \frac{1}{n} \sum_{i=1}^n \mu = \mu. \quad (4.5)$$

The variance of the sample mean is obtained as,

$$Var(\bar{X}) = Var(\frac{1}{n} \sum_{i=1}^n X_i) = \frac{1}{n^2} \sum_{i=1}^n Var(X_i),$$

since X_i s are i.i.d. random variables with an inherited common finite variance of the population, σ^2 , i.e. $Var(X_i) = \sigma^2$.

Hence,

$$Var(\bar{X}) = \frac{1}{n^2} \sum_{i=1}^n \sigma^2 = \frac{\sigma^2}{n}. \quad (4.6)$$

Therefore, the sample mean is an unbiased estimator of the unknown population mean, with a variance decreasing by the increasing sample size.

Similarly, the sample variance is a statistic (random variable),

$$h(X) = S^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2, \quad (4.7)$$

and it has an expectation,

$$\begin{aligned} E[S^2] &= E[\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2] = E[\frac{1}{n} \sum_{i=1}^n (X_i^2 - 2X_i\bar{X} + \bar{X}^2)] \\ &= E[\frac{1}{n} (\sum_{i=1}^n X_i^2 - 2\bar{X} \sum_{i=1}^n X_i + n\bar{X}^2)] \\ &= E[\frac{1}{n} (\sum_{i=1}^n X_i^2 - 2n\bar{X}^2 + n\bar{X}^2)] \\ &= E[\frac{1}{n} (\sum_{i=1}^n X_i^2 - n\bar{X}^2)] \\ &= \frac{1}{n} (\sum_{i=1}^n E[X_i^2] - nE[\bar{X}^2]). \end{aligned}$$

Since $E[X^2] = \sigma^2 + \mu^2$, we have

$$E[S^2] = \frac{1}{n} (n\sigma^2 + n\mu^2 - n\frac{\sigma^2}{n} - n\mu^2) = \frac{(n-1)}{n} \sigma^2. \quad (4.8)$$

Which is not the same as σ^2 , thus S^2 is a biased estimator of σ^2 .

However, $\lim_{n \rightarrow \infty} E[S^2] = \sigma^2$, i.e. $E[S^2]$ approaches to σ^2 , asymptotically.

In order to eliminate this bias, we can multiply S^2 by $\frac{n}{n-1}$,

and we redefine the sample variance as,

$$S_n^2 = \left(\frac{n}{n-1}\right) \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2,$$

which is an unbiased estimator of σ^2 .

Thus the redefined sample variance is an unbiased estimator of the population variance, since $E[S_n^2] = \sigma^2$.

4.3 Asymptotic Behavior of a Sample

The random sample $\{X_1, X_2, \dots, X_n\}$ refers to a set of measurable i.i.d. random variables with cardinality n , usually a finite and fixed integer number that depends upon your resources, also known as the sample size. If we could increase the sample size at will, then we would have very nice asymptotic properties for random samples.

The sample mean, $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$, of a set of i.i.d. random variables with a finite common mean μ and a finite common variance σ^2 , converges to μ in probability, $\bar{X}_n \xrightarrow{P} \mu$, by Theorem 3.13.1.

Let a population have a cdf, $F_X(x|\theta)$, and $\{X_1, \dots, X_n\}$ be a random sample, coming from this population. Also, let H_n be a statistic of this random sample.

If $H_n \xrightarrow{P} \theta$, then H_n is a consistent estimator of θ .

The sample mean, $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$,

and the sample variance, $S_n^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2$,

are two consistent estimators, i.e. $\bar{X}_n \xrightarrow{P} \mu$ and $S_n^2 \xrightarrow{P} \sigma^2$.

The probability distribution of a sample is known as the sampling distribution, which is the limiting distribution of the sequence $\{X_1, X_2, \dots, X_n\}$.

Theorem 3.14.1 indicates that $\frac{\sum_{i=1}^n X_i - n\mu}{\sqrt{n}\sigma} \xrightarrow{D} Z$.

Therefore, we say that the asymptotic distribution of $\frac{\sum_{i=1}^n X_i - n\mu}{\sqrt{n}\sigma}$ is the standard normal.

Hence, for large n , we can make probabilistic statements about events involving with the statistic $\bar{X}_n$.

Since $\frac{(\bar{X}_n - \mu)}{\frac{\sigma}{\sqrt{n}}} \xrightarrow{D} Z$, we have a limiting sampling distribution for the $\bar{X}_n$, that is $\lim_{n \rightarrow \infty} F_{\bar{X}_n} = \mathcal{N}(\mu, \frac{\sigma^2}{n})$. Or, $\bar{X}_n \xrightarrow{D} \mathcal{N}(\mu, \frac{\sigma^2}{n})$, by the conventional way of writing convergence in distribution.

When we have a random sample coming from a normal distribution, $\mathcal{N}(\mu, \sigma^2)$, we have the probability distributions for the sample mean

$$\bar{X}_n \xrightarrow{D} \mathcal{N}(\mu, \frac{\sigma^2}{n}), \quad (4.9)$$

and the sample variance

$$S_n^2 \xrightarrow{D} \frac{\sigma^2}{n} \chi^2(n-1), \quad (4.10)$$

which is a χ^2 distribution with $n-1$ degrees of freedom, where $\bar{X}_n$ and S_n^2 are independent random variables.

Moreover,

$$\frac{\bar{X}_n - \mu}{(\frac{\sigma}{\sqrt{n}})} \xrightarrow{D} \mathcal{N}(0, 1), \quad (4.11)$$

and

$$\frac{\bar{X}_n - \mu}{(\frac{S_n}{\sqrt{n}})} \xrightarrow{D} T_{df}, \quad (4.12)$$

which is a Student's t-distribution with $df = n-1$, degrees of freedom.

4.4 Student's t-Distribution

A family of sampling distributions, $\{T_{df}\}$, coming from a normal distribution, are used as an estimator of the population standard deviation. They are similar to the standard normal distribution, yet they depend upon the sample size $n > 3$, through a degrees of freedom, $df = n-1$.

Also, $T_{df} \xrightarrow{D} \mathcal{N}(0, 1)$.

That is, as the sample size gets larger, Student's t-distribution converges to standard normal distribution. Otherwise, for small sample sizes, it is symmetric about zero, bell-shaped, similar to the standard normal distribution.

The Student's t-distribution has heavier tails than the standard normal distribution, to indicate that the small samples leave more room for errors. William Sealey Gossett, who used the pen name Student when he published

this sampling distribution [14], as a ratio of the sample mean (a random variable) to the sample standard deviation (another random variable), such that $T_{df} = \frac{Z}{\sqrt{V/df}}$.

Here, $Z \sim \mathcal{N}(0, 1)$ and $V \sim \chi_{df}^2$ are independent random variables, such that for a t-distributed random variable, x , with df degrees of freedom we have the following pdf,

$$f_{T_{df}}(x) = \frac{1}{\sqrt{\pi df} \Gamma(\frac{df}{2})} \left(1 + \frac{x^2}{df}\right)^{-\frac{df+1}{2}} \Gamma\left(\frac{df+1}{2}\right), \quad (4.13)$$

with

Expected value: $E[T_{df}] = 0$, and

Variance: $Var(T_{df}) = \frac{df}{df-2}$.

4.5 Asymptotic Multivariate Sample

Let $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n$ be a sequence of m -dimensional random vectors, and $\mathbf{X}$ be an m -dimensional random vector.

If $\lim_{n \rightarrow \infty} P(\|\mathbf{X}_n - \mathbf{X}\| \geq \epsilon) = 0$, for any positive small ϵ , then $\mathbf{X}_n \xrightarrow{P} \mathbf{X}$,

which denotes component-wise convergence in probability, i.e. $X_{n,j} \xrightarrow{P} X_j$, for all $j = 1, 2, \dots, m$.

Example 4.1

Let $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n$ be a sequence of m -dimensional i.i.d. random vectors, such that

$\mathbf{X}_n \sim \mathcal{N}(\boldsymbol{\mu}_n, \boldsymbol{\Sigma}_n)$, where $\mathcal{N}$ is an m -variate normal distribution.

If $\mathbf{X}_n \xrightarrow{P} \mathbf{X}$, where $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, thus $\mathbf{X}_n \xrightarrow{D} \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$,

then $\lim_{n \rightarrow \infty} \boldsymbol{\mu}_n = \boldsymbol{\mu}$ and $\lim_{n \rightarrow \infty} \boldsymbol{\Sigma}_n = \boldsymbol{\Sigma}$.

4.6 Central Limit Theorem Revisited

Let $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n$ be a sequence of m -dimensional i.i.d. random vectors, with collective mean vector, $\boldsymbol{\mu}$, and collective variance-covariance matrix, $\boldsymbol{\Sigma}$, then

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n (\mathbf{X}_i - \boldsymbol{\mu}) \xrightarrow{D} \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}).$$

Hence,

$$\bar{\mathbf{X}}_n \xrightarrow{D} \mathcal{N}(\boldsymbol{\mu}, \frac{1}{n}\boldsymbol{\Sigma}).$$

4.7 Linear Transformation of a Random Sequence

Let $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n$ be a sequence of m -dimensional i.i.d. random vectors, such that $\mathbf{X}_n \xrightarrow{D} \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$.

If $\mathbf{A}$ is a $p \times m$ constant matrix, and $\mathbf{b}$ is a p -dimensional constant vector, then

$$\mathbf{A}\mathbf{X}_n + \mathbf{b} \xrightarrow{D} \mathcal{N}(\mathbf{A}\boldsymbol{\mu} + \mathbf{b}, \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^T).$$

4.8 Continuous Transformation of a Random Sequence

Let $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n$ be a sequence of m -dimensional i.i.d. random vectors, such that

$$\mathbf{X}_n \xrightarrow{D} \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma}).$$

If $\mathbf{h}(\mathbf{X}) = (h_1(\mathbf{X}), \dots, h_k(\mathbf{X}))$ is a k -dimensional multivariate function of $\mathbf{X}$, with $\mathbf{H} = [\frac{\partial h_i}{\partial \mu_j}]$ is a continuous $k \times m$ matrix of partial derivatives at $\boldsymbol{\mu}$, for $i = 1, 2, \dots, k$ and $j = 1, 2, \dots, m$;

then

$$\mathbf{h}(\mathbf{X}_n) \xrightarrow{D} \mathcal{N}(\mathbf{h}(\boldsymbol{\mu}), \mathbf{H}\boldsymbol{\Sigma}\mathbf{H}^T).$$

4.9 Exercises

1. If $X_1, X_2, \dots, X_n$ is a finite sample (that is n is finite and also X_i is finite for all i), coming from a common normal population, such that $X_i \sim \mathcal{N}(\mu, \sigma^2)$, for $i = 1, 2, \dots, n$, then find the probability distribution for the sample average $\bar{X}_n$.
2. If $X_1, X_2, \dots, X_n$ is a finite sample, coming from a common Bernoulli population with the probability of success of p , then find the expected value and the variance for the sample average $\bar{X}_n$.

3. Melanie is running for the class presidency and she believes that she can win the elections if she can collect at least 55% of the votes. She believes that 50% of voters will vote for her. If 100 voters would cast their votes, then what is the probability that Melanie will win?
4. If the standard deviation of the sample mean for 25 observations is 2.6, how many more observations must be collected to reduce the standard deviation to 1.56?
5. If $X \sim \mathcal{N}(25, 9)$, then what would be the sample size in order to have $P(24 \leq \bar{X}_n \leq 26) = 0.99$?
6. If $X_1, \dots, X_n$ are i.i.d. exponential random variables with parameter θ , then show that $Y = \sum_{i=1}^n X_i$ is a gamma random variable with parameters $\alpha = n$ and $\beta = \theta$.
7. A certain brand of cereal claims that the average saturated fat content is 0.2 grams. A random sample of this cereal yield saturated fat content of 0.3, 0.3, 0.2, 0.5, 0.2, 0.4, 0.6, 0.5, 0.1 grams. Assume a normal distribution for the saturated fat content, would you agree with this claim?
8. If S_1^2 and S_2^2 are the variances of independent samples of sizes $n_1 = 25$ and $n_2 = 49$, taken from independent normal populations with variances $\sigma_1^2 = 9$ and $\sigma_2^2 = 25$, what is $P(\frac{S_1^2}{S_2^2} \leq 1)$?
9. Assume a soda filling machine is used to fill bottles of soda. The filling specification is that each bottle should contain 10 oz. soda with 0.5 oz. tolerance. It is expected that at least 99% of the bottles should meet this specification. What proportion of these bottles are not acceptable when the filling process is $\mathcal{N}(10, 1)$?

Chapter 5

Parameter Estimation

5.1 Introduction

In this chapter, we assume that the cdf of the population has an unknown parameter, θ , as a real number, or as a real vector.

Thus, we want to estimate this unknown parameter, as well as we could, by an estimator, $\hat{\theta}$.

There may be other unknown parameters of the distribution, but if we are not interested in them, we may consider them as nuisance parameters.

There are some common characteristics to all parameter estimators, but two of them are quite important.

Bias:

An estimator $\hat{\theta}$ of a parameter θ is said to be an unbiased estimator if

$$E[\hat{\theta}] = \theta.$$

Consistence:

An estimator $\hat{\theta}_n$ of a parameter θ is said to be a consistent (asymptotically) estimator if

$$\hat{\theta}_n \xrightarrow{P} \theta.$$

5.2 Sufficient Statistic

We are using a sample to generate information about the unknown parameter θ , which is a point (maybe multi-dimensional) in a space. Hence, our sample

set of i.i.d. random variables (or, random vectors) $\{X_1, X_2, \dots, X_n\}$ should yield a point estimator $\hat{\theta}$.

Let $U = h(X_1, \dots, X_n)$ be a statistic and $V = g(X_1, \dots, X_n)$ be another statistic. If the conditional distribution of V , for a given U , does not involve θ , then U is said to be a sufficient statistic for θ .

The search for a sufficient statistic is usually carried out by the guidance of the following theorem.

Fisher-Neymann Theorem 5.2.1. *Let $X_1, X_2, \dots, X_n$ be a random sample with a common pdf $f_X(x|\theta)$. Let $Y_1 = u_1(X_1, \dots, X_n)$ be a statistic with a pdf $g_1(y_1|\theta)$. Y_1 is a sufficient statistic for θ if and only if the joint pdf of the sample can be factored as a product of the pdf of the statistic and a function of the sample, independent of the parameter, such that $\prod_{i=1}^n f_X(x_i|\theta) = g_1(y_1|\theta)H(x_1, \dots, x_n)$.*

Proof. Assume that $\prod_{i=1}^n f_X(x_i|\theta)$ can be factored as stated above.

Now, introduce a change of variables $y_i = u_i(x_1, x_2, \dots, x_n)$ and inverse transformations $x_i = w_i(y_1, \dots, y_n)$, for $i = 1, 2, \dots, n$.

Let $g(y_1, \dots, y_n)$ be the joint pdf of $Y_1, Y_2, \dots, Y_n$.

Thus, $g(y_1, \dots, y_n) = g_1(y_1|\theta)H(w_1(y_1, \dots, y_n), \dots, w_n(y_1, \dots, y_n)) |J|$.

By assumption, $g_1(y_1|\theta)$ is the marginal pdf of Y_1 , thus $H(w_1, \dots, w_n) |J|$ must be the conditional pdf of $Y_2, \dots, Y_n$, for given $Y_1 = y_1$, say $h(y_2, \dots, y_n|y_1, \theta)$.

Since $H(x_1, \dots, x_n)$ does not depend upon θ , and θ does not appear in transformations $\{u_i\}$, then the Jacobian $|J|$ is also free from θ . Hence, $h(y_2, \dots, y_n|y_1, \theta)$ is free from θ . Therefore, Y_1 is a sufficient statistic.

Now, assume that Y_1 is a sufficient statistic, such that

$$g(y_1, \dots, y_n) = g_1(y_1|\theta)h(y_2, \dots, y_n|y_1).$$

The joint pdf of $X_1, \dots, X_n$ would be

$$\begin{aligned} \prod_{i=1}^n f_X(x_i|\theta) &= g(u_1(x_1, \dots, x_n), \dots, u_n(x_1, \dots, x_n)) |J^{-1}| \\ &= g_1(y_1|\theta)h(u_1, \dots, u_n) |J^{-1}|. \end{aligned}$$

Both $h(u_1, \dots, u_n)$ and $|J^{-1}|$ are free from θ , thus

$$H(x_1, \dots, x_n) = h(u_1(x_1, \dots, x_n), \dots, u_n(x_1, \dots, x_n)) |J^{-1}|,$$

is also free from the parameter. Therefore,

$$\prod_{i=1}^n f_X(x_i|\theta) = g_1(y_1|\theta)H(x_1, \dots, x_n). \quad (5.1)$$

□

Example 5.1

Let $X_1, \dots, X_n$ be a random sample coming from a normal distribution with mean μ and with a known variance σ^2 .

The joint pdf of the sample would be

$$\begin{aligned} g(x_1, \dots, x_n | \theta) &= (2\pi\sigma^2)^{-\frac{n}{2}} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2\right) \\ &= (2\pi\sigma^2)^{-\frac{n}{2}} \exp\left(-\frac{1}{2\sigma^2} (\sum_{i=1}^n (x_i - \bar{X}_n)^2 + n(\bar{X}_n - \mu)^2)\right) \\ &= (2\pi\sigma^2)^{-\frac{n}{2}} \exp\left(-\frac{1}{2\sigma^2} n(\bar{X}_n - \mu)^2\right) \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \bar{X}_n)^2\right). \end{aligned}$$

Therefore, $g(x_1, \dots, x_n | \theta)$ is a product of two factors, such that the first one is the marginal pdf of,

$$\bar{X}_n = (2\pi\sigma^2)^{-\frac{n}{2}} \exp\left(-\frac{n}{2\sigma^2} (\bar{X}_n - \mu)^2\right),$$

which is a function that depends upon the unknown parameter, μ , for a fixed σ^2 , and the other factor is a function free from the unknown parameter μ , for a fixed σ^2 ,

$$\exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \bar{X}_n)^2\right).$$

Hence, $\bar{X}_n$ is a sufficient statistic for μ .

There is an extended form of Theorem 5.2.1 that relaxes the need to know the pdf of the sufficient statistic.

Theorem 5.2.2. *Let $X_1, \dots, X_n$ be a random sample with a common pdf $f_X(x|\theta)$. Also, let $Y_1 = u_1(X_1, \dots, X_n)$ be a statistic for θ . Y_1 is a sufficient statistic, if and only if there exist two nonnegative functions, say k_1 and k_2 , such that the joint pdf of the sample $\prod_{i=1}^n f_X(x_i|\theta) = k_1(y_1|\theta)k_2(x_1, \dots, x_n)$, where k_2 is free from θ .*

Proof. Assume that $\prod_{i=1}^n f_X(x_i|\theta) = k_1(y_1|\theta)k_2(x_1, \dots, x_n)$.

As in Theorem 5.2.1, the joint pdf of $Y_1, \dots, Y_n$ is

$$g(y_1, \dots, y_n) = k_1(y_1|\theta)k_2(w_1(y_1, \dots, y_n), \dots, w_n(y_1, \dots, y_n)) |J|,$$

and the marginal pdf of Y_1 would be

$$\begin{aligned} g_1(y_1|\theta) &= \int \dots \int g(y_1, \dots, y_n) dy_2 \dots dy_n \\ &= k_1(y_1|\theta) \int \dots \int |J| k_2(w_1, \dots, w_n) dy_2 \dots dy_n. \end{aligned}$$

By the hypothesis, k_2 is free from θ , the Jacobian J is free from θ , and the integration limits are free from θ . Hence, $g_1(y_1|\theta) = k_1(y_1|\theta)\phi(y_1)$, where $\phi(y_1)$ is a function of y_1 only.

This in turn implies that,

$$k_1(u_1(x_1, \dots, x_n)|\theta) = g_1(u_1(x_1, \dots, x_n)|\theta)/\phi(u_1(x_1, \dots, x_n)).$$

Hence,

$$\begin{aligned} \prod_{i=1}^n f_X(x_i|\theta) &= [g_1(u_1(x_1, \dots, x_n)|\theta)/\phi(u_1(x_1, \dots, x_n))] k_2(x_1, \dots, x_n) \\ &= g_1(u_1(x_1, \dots, x_n)|\theta)[k_2(x_1, \dots, x_n)/\phi(u_1(x_1, \dots, x_n))]. \end{aligned}$$

Therefore, by Theorem 5.2.1, Y_1 is a sufficient statistic.

On the other hand, if Y_1 is a sufficient statistic, then let

$$k_1(x_1, \dots, x_n|\theta) = g_1(u_1(x_1, \dots, x_n)|\theta), \text{ and}$$

$$k_2(x_1, x_2, \dots, x_n) = H(x_1, \dots, x_n), \text{ be nonnegative functions.} \quad \square$$

A statistic can be an estimator, thus an estimator can be a sufficient statistic as well. Yet, in general an estimator is a function of a statistic.

Theorem 5.2.3. *Let $X_1, \dots, X_n$ be a random sample with a common pdf $f_X(x|\theta)$. Also, let $Y = u(X_1, \dots, X_n)$ be a sufficient statistic for θ . If $Z = v(Y)$ is a one-to-one invertible function of Y that does not depend on θ , then Z is also a sufficient statistic for θ .*

Proof. Since Y is a sufficient statistic for θ , we have

$$\prod_{i=1}^n f_X(x_i|\theta) = k_1(y|\theta)k_2(x_1, \dots, x_n), \text{ by Theorem 5.2.2.}$$

Let the inverse transformation $y = v^{-1}(z)$ be single valued, then

$$k_1(y|\theta) = k_1(v^{-1}(z)|\theta) = k_1^*(z|\theta).$$

Consequently, $\prod_{i=1}^n f_X(x_i|\theta) = k_1^*(z|\theta)k_2(x_1, \dots, x_n)$.

Therefore, Z is a sufficient statistic for θ . $\square$

It must be understood by now that our objective is to acquire relevant information from a sample about the unknown characteristics of a population, and we would like to do so by using the smallest possible sample without losing any necessary information.

A sufficient statistic provides that kind of information, when it exists, yet there may be a multitude of sufficient statistics. Thus, we look for the minimal sufficient statistic for the unknown parameter, that is a sufficient statistic that we cannot reduce it any further without losing the relevant information.

5.3 Likelihood Function

Let $\{X_1, X_2, \dots, X_n\}$ be a random sample coming from a probability distribution, that is $\{X_i | i = 1, 2, \dots, n\}$ be a set of i.i.d. random variables with the common cdf $F(x|\theta)$, without loss of generality, with a common pdf $f(x|\theta)$.

The joint pdf of the sample is given as the n -multiple of the pdf at particular realizations, i.e.,

$$f_X(x_1, x_2, \dots, x_n) = \prod_{i=1}^n f(x_i|\theta).$$

Whenever we have a realization of the sample, $(x_1^i, x_2^i, \dots, x_n^i)$, there would be no more randomness left in the joint pdf of the sample, yet the unknown parameter, θ , as a real variable (a number, or a vector) can get any feasible value.

Thus we could consider this joint pdf of the sample, for a fixed realization of the sample, as a function of the unknown parameter and call it the likelihood function of the parameter, denoted by $L(\theta)$, with a reversal of roles, such that the sample plays the role of a known fixed parameter and θ plays the role of the argument of the likelihood function,

$$L(\theta) = L(\theta|x) = \prod_{i=1}^n f(x_i|\theta). \quad (5.2)$$

For more information please see [2].

Here, $f(x_i|\theta) > 0$, for all θ and $i = 1, 2, \dots, n$.

Therefore we could also introduce the natural logarithm (inverse function of the exponential function) of the likelihood function, called the loglikelihood function, denoted by $l(\theta)$,

$$l(\theta) = \log L(\theta) = \sum_{i=1}^n \log f(x_i|\theta). \quad (5.3)$$

Moreover, properties of the logarithm function may help us to estimate the unknown parameter. This construct also generalizes to the multivariate cases, by introducing the set of i.i.d. random vectors for the sample, $\{\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n\}$, with a multivariate joint pdf $\mathbf{f}(\mathbf{X}_i|\theta)$.

5.4 Fisher's Information Index

Fisher introduced the idea of measuring the amount of information which is generated by a random sample [2].

Assume that $f(x|\theta)$ is twice differentiable with respect to θ . Also assume that differentiation and integration can be interchanged for $f(x|\theta)$.

Since $f(x|\theta)$ is a pdf, we have $\int_{-\infty}^{\infty} f(x|\theta)dx = 1$.

By taking the derivative of both sides, with respect to θ , we have

$$\int_{-\infty}^{\infty} \frac{\partial f(x|\theta)}{\partial \theta} dx = 0. \quad (5.4)$$

With a little bit of audacity, we can rewrite this equation as

$$\int_{-\infty}^{\infty} \frac{\partial f(x|\theta)}{\partial \theta} \frac{f(x|\theta)}{f(x|\theta)} dx = 0. \quad (5.5)$$

Moreover, rearranging the integrand terms yields,

$$\int_{-\infty}^{\infty} \frac{\partial f(x|\theta)/\partial \theta}{f(x|\theta)} f(x|\theta) dx = 0, \quad (5.6)$$

that could be simplified as,

$$\int_{-\infty}^{\infty} \frac{\partial \log f(x|\theta)}{\partial \theta} f(x|\theta) dx = 0. \quad (5.7)$$

Since the left hand side is the expectation operator over a function of a random variable X , we have

$$E\left[\frac{\partial \log f(X|\theta)}{\partial \theta}\right] = 0.$$

Again, by taking the derivative of both sides with respect to θ , we have

$$\int_{-\infty}^{\infty} \frac{\partial^2 \log f(x|\theta)}{\partial \theta^2} f(x|\theta) dx + \int_{-\infty}^{\infty} \frac{\partial f(x|\theta)}{\partial \theta} \frac{\partial \log f(x|\theta)}{\partial \theta} f(x|\theta) dx = 0. \quad (5.8)$$

and by repeating the same argument, we derive the equality,

$$\int_{-\infty}^{\infty} \frac{\partial^2 \log f(x|\theta)}{\partial \theta^2} f(x|\theta) dx + \int_{-\infty}^{\infty} \frac{\partial \log f(x|\theta)}{\partial \theta} \frac{\partial \log f(x|\theta)}{\partial \theta} f(x|\theta) dx = 0, \quad (5.9)$$

or,

$$\int_{-\infty}^{\infty} \frac{\partial^2 \log f(x|\theta)}{\partial \theta^2} f(x|\theta) dx + \int_{-\infty}^{\infty} \left(\frac{\partial \log f(x|\theta)}{\partial \theta}\right)^2 f(x|\theta) dx = 0. \quad (5.10)$$

But, these integrals on the left hand side are expectations of functions of the random variable X , that is

$$E\left[\frac{\partial^2 \log f(X|\theta)}{\partial \theta^2}\right] + E\left[\left(\frac{\partial \log f(X|\theta)}{\partial \theta}\right)^2\right] = 0. \quad (5.11)$$

Hence, we have

$$E\left[\left(\frac{\partial \log f(X|\theta)}{\partial \theta}\right)^2\right] = -E\left[\frac{\partial^2 \log f(X|\theta)}{\partial \theta^2}\right]. \quad (5.12)$$

This expectation operator is known as the Fisher's information index, and denoted as

$$I(\theta) = E\left[\left(\frac{\partial \log f(X|\theta)}{\partial \theta}\right)^2\right] = -E\left[\frac{\partial^2 \log f(X|\theta)}{\partial \theta^2}\right]. \quad (5.13)$$

On the other hand, since

$$E\left[\frac{\partial \log f(X|\theta)}{\partial \theta}\right] = 0, \quad (5.14)$$

we also have

$$Var\left(\frac{\partial \log f(X|\theta)}{\partial \theta}\right) = E\left[\left(\frac{\partial \log f(X|\theta)}{\partial \theta}\right)^2\right] - (E\left[\frac{\partial \log f(X|\theta)}{\partial \theta}\right])^2. \quad (5.15)$$

Thus,

$$Var\left(\frac{\partial \log f(X|\theta)}{\partial \theta}\right) = E\left[\left(\frac{\partial \log f(X|\theta)}{\partial \theta}\right)^2\right] = I(\theta). \quad (5.16)$$

The amount of information generated by the sample, for an unknown θ , is measured with this index, subject to the common pdf $f(x|\theta)$.

The higher index values indicate more information is extracted from an observation.

When we have a random sample, that is a set of i.i.d. random variables with a common pdf, then they all have the same mean and the variance, thus the Fisher's information index for a random sample would be

$$E\left[\sum_{i=1}^n \left(\frac{\partial \log f(X|\theta)}{\partial \theta}\right)^2\right] = \sum_{i=1}^n E\left[\left(\frac{\partial \log f(X|\theta)}{\partial \theta}\right)^2\right] = nI(\theta). \quad (5.17)$$

Similarly,

$$Var\left(\sum_{i=1}^n \frac{\partial \log f(X_i|\theta)}{\partial \theta}\right) = \sum_{i=1}^n Var\left(\frac{\partial \log f(X_i|\theta)}{\partial \theta}\right) = nI(\theta), \quad (5.18)$$

i.e. the sample size linearly augments the amount of information acquired by each observation for the unknown θ , subject to the common pdf $f(x|\theta)$.

5.5 Minimum Variance Unbiased Estimator

We prefer to use reliable estimators, thus we seek to minimize estimator variations, as a degree of closeness to the unknown parameter value.

5.5.1 Cramer-Rao Lower Bound

An estimator is expected to estimate the unknown parameter θ .

Let $Y = u(X_1, \dots, X_n)$ be an estimator of θ .

Also, let $v(\theta) = E[u(X_1, \dots, X_n)] = E[Y] < \infty$. Thus we have

$$v(\theta) = \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} u(x_1, \dots, x_n) L(\theta) dx_1 dx_2 \dots dx_n. \quad (5.19)$$

If we differentiate both sides with respect to θ , we would have

$$\frac{\partial v}{\partial \theta} = \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} u(x_1, \dots, x_n) \frac{\partial L(\theta)}{\partial \theta} dx_1 dx_2 \dots dx_n. \quad (5.20)$$

That implies,

$$\frac{\partial v}{\partial \theta} = \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} u(x_1, \dots, x_n) \frac{\partial \log L(\theta)}{\partial \theta} L(\theta) dx_1 dx_2 \dots dx_n. \quad (5.21)$$

Since $L(\theta) = \prod_{i=1}^n f_X(x_i|\theta)$ and $\log L(\theta) = \sum_{i=1}^n \log f_X(x_i|\theta)$, we have

$$\frac{\partial \log L(\theta)}{\partial \theta} = \sum_{i=1}^n \frac{\partial \log f(x_i|\theta)}{\partial \theta}. \quad (5.22)$$

Therefore,

$$\frac{\partial v}{\partial \theta} = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} u(x_1, \dots, x_n) \left(\sum_{i=1}^n \frac{\partial \log f(x_i|\theta)}{\partial \theta} \right) \left(\prod_{i=1}^n f_X(x_i|\theta) \right) dx_1 dx_2 \cdots dx_n. \quad (5.23)$$

Since the sample is a collection of i.i.d. random variables and

$$E\left[\frac{\partial \log f(X_i|\theta)}{\partial \theta}\right] = 0, \text{ clearly, } \sum_{i=1}^n E\left[\frac{\partial \log f(X_i|\theta)}{\partial \theta}\right] = 0.$$

Thus,

$$\begin{aligned} \text{Var}\left(\sum_{i=1}^n \frac{\partial \log f(X_i|\theta)}{\partial \theta}\right) &= \sum_{i=1}^n \text{Var}\left(\frac{\partial \log f(X_i|\theta)}{\partial \theta}\right) = \sum_{i=1}^n E\left[\left(\frac{\partial \log f(X_i|\theta)}{\partial \theta}\right)^2\right] \\ &= nE\left[\left(\frac{\partial \log f(X_i|\theta)}{\partial \theta}\right)^2\right] = nI(\theta). \end{aligned} \quad (5.24)$$

Let $Z = \sum_{i=1}^n \frac{\partial \log f(X_i|\theta)}{\partial \theta}$ be a random variable, such that

$$E[Z] = 0 \text{ and } \sigma_Z^2 = \text{Var}(Z) = nI(\theta).$$

Hence,

$$\frac{\partial v}{\partial \theta} = E[YZ] = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} yz \prod_{i=1}^n f_X(x_i|\theta) dx_1 dx_2 \cdots dx_n. \quad (5.25)$$

$$E[YZ] = E[Y]E[Z] + \rho\sigma_Y\sigma_Z,$$

and $E[Y] = v(\theta)$, we have

$$\frac{\partial v}{\partial \theta} = v(\theta) \cdot 0 + \rho\sigma_Y\sigma_Z,$$

that is, $\rho = \frac{\partial v}{\partial \theta} / (\sigma_Y\sigma_Z)$.

Thus we have,

$$\rho^2 = \left(\frac{\partial v}{\partial \theta}\right)^2 / (\sigma_Y^2 \sigma_Z^2). \quad (5.26)$$

Since the correlation coefficient, ρ , is the cosine of an angle, i.e.

$-1 \leq \rho \leq 1$, we have $0 \leq \rho^2 \leq 1$, and this implies that we have

$$0 \leq \left(\frac{\partial v}{\partial \theta}\right)^2 / (\sigma_Y^2 \sigma_Z^2) \leq 1, \quad (5.27)$$

where the middle term of this inequality is known as the efficiency of the estimator, Y , therefore,

$$\left(\frac{\partial v}{\partial \theta}\right)^2 / \sigma_Z^2 \leq \sigma_Y^2, \quad (5.28)$$

or,

$$\sigma_Y^2 \geq \left(\frac{\partial v}{\partial \theta}\right)^2 / nI(\theta). \quad (5.29)$$

This relationship for the variance of an estimator (here, σ_Y^2) and the sample information is known as the Cramer-Rao lower bound.

When Y is an unbiased estimator, we have $v(\theta) = \theta$ and $\sigma_Y^2 = E[(Y - \theta)^2]$.

Hence, $\frac{\partial v(\theta)}{\partial \theta} = 1$ and

$$\sigma_Y^2 \geq 1/nI(\theta), \quad (5.30)$$

where,

$$I(\theta) = E\left[\left(\frac{\partial \log f(X|\theta)}{\partial \theta}\right)^2\right] = -E\left[\frac{\partial^2 \log f(X|\theta)}{\partial \theta^2}\right] = \text{Var}\left(\frac{\partial \log f(X|\theta)}{\partial \theta}\right).$$

5.6 Efficient Estimator

When Y is an unbiased estimator of θ , Y would be called an efficient estimator of θ if and only if $\sigma_Y^2 = 1/nI(\theta)$.

Cramer-Rao Theorem 5.6.1. *Let $X_1, X_2, \dots, X_n$ be a random sample and $L(\theta)$ be the likelihood function of the sample, which is differentiable with respect to θ . Also, let $\hat{\theta}$ be an unbiased estimator of the unknown parameter θ . Then*

$$\text{Var}(\hat{\theta}) \geq 1/E\left[\left(\frac{\partial \log L(\theta)}{\partial \theta}\right)^2\right], \text{ or}$$

$$\text{Var}(\hat{\theta}) \geq -1/E\left[\frac{\partial^2 \log L(\theta)}{\partial \theta^2}\right], \text{ or, equivalently}$$

$$\text{Var}(\hat{\theta}) \geq 1/\text{Var}\left(\frac{\partial \log L(\theta)}{\partial \theta}\right).$$

The variance of an unbiased estimator can be at the Cramer-Rao lower bound level, if

$$\frac{\partial \log L(\theta)}{\partial \theta} = \beta(\theta)(\hat{\theta} - \theta),$$

with $\beta(\theta)$ as a function of the θ , and independent of the sample.

Therefore, $E[(\frac{\partial \log L(\theta)}{\partial \theta})^2] = \beta^2(\theta)E[(\hat{\theta} - \theta)^2]$.

Hence, the minimum variance unbiased estimator is a sufficient statistic.

However, when the Cramer-Rao lower bound cannot be reached, we still look for minimum variance estimators.

Since $E[X] = E[E[X|Y]]$ and $\text{Var}(X) \geq \text{Var}(E[X|Y])$,

we have the following relationship between the variance of a sufficient statistic and the variance of an unbiased estimator.

Rao-Blackwell Theorem 5.6.2. *Let $X_1, X_2, \dots, X_n$ be a random sample coming from a probability distribution with a common pdf $f_X(x|\theta)$. Also, let $Y_1 = u_1(X_1, \dots, X_n)$ be a sufficient statistic for θ , and $Y_2 = u_2(X_1, \dots, X_n)$ be an unbiased estimator of θ . Then, $\phi(y_1) = E[Y_2|y_1]$ identifies a statistic $\phi(Y_1)$, as a function of the sufficient statistic for θ , which is an unbiased estimator of θ with a smaller variance than $\text{Var}(Y_2)$, called a minimum variance unbiased estimator, or, MVUE.*

Proof. $E[Y_2|y_1] = \int_{-\infty}^{\infty} y_2 f(y_2|y_1) dy_2$

$$= \int_{-\infty}^{\infty} y_2 \frac{f(y_1, y_2)}{f(y_1)} dy_2 = \phi(y_1), \text{ i.e.}$$

$$\int_{-\infty}^{\infty} y_2 f(y_1, y_2) dy_2 = \phi(y_1) f(y_1).$$

Thus,

$$\begin{aligned} E[\phi(Y_1)] &= \int_{-\infty}^{\infty} \phi(y_1) f(y_1) dy_1 \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y_2 f(y_1, y_2) dy_2 dy_1 \\ &= \int_{-\infty}^{\infty} y_2 \int_{-\infty}^{\infty} f(y_1, y_2) dy_1 dy_2 \\ &= \int_{-\infty}^{\infty} y_2 f(y_2) dy_2 = E[Y_2] = \theta. \\ &\Rightarrow E[\phi(Y_1)] = \theta. \end{aligned}$$

Since, $\sigma_{Y_2}^2 = E[(Y_2 - \theta)^2]$,

we have

$$\begin{aligned} \sigma_{Y_2}^2 &= E[(Y_2 + \phi(Y_1) - \phi(Y_1) - \theta)^2] \\ &= E[(Y_2 - \phi(Y_1))^2] + E[(\phi(Y_1) - \theta)^2] + 2E[(Y_2 - \phi(Y_1))(\phi(Y_1) - \theta)]. \end{aligned}$$

Therefore,

$$\sigma_{Y_2}^2 = \sigma_{\phi(Y_1)}^2 + E[(Y_2 - \phi(Y_1))^2] + 2E[(Y_2 - \phi(Y_1))(\phi(Y_1) - \theta)].$$

Since,

$$E[(Y_2 - \phi(Y_1))^2] \geq 0, \text{ and}$$

$$\begin{aligned} E[(Y_2 - \phi(Y_1))(\phi(Y_1) - \theta)] &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (y_2 - \phi(y_1))(\phi(y_1) - \theta) f(y_1, y_2) dy_2 dy_1 \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (y_2 - \phi(y_1))(\phi(y_1) - \theta) f(y_2|y_1) f(y_1) dy_2 dy_1 \\ &= \int_{-\infty}^{\infty} (\phi(y_1) - \theta) \int_{-\infty}^{\infty} (y_2 - \phi(y_1)) f(y_2|y_1) dy_2 f(y_1) dy_1 = 0, \end{aligned}$$

we have $\sigma_{Y_2}^2 = \sigma_{\phi(Y_1)}^2 + E[(Y_2 - \phi(Y_1))^2]$.

Therefore, $\sigma_{Y_2}^2 \geq \sigma_{\phi(Y_1)}^2$. □

5.7 Method of Moments

The equality of polynomials requires that the coefficients of the same power terms of these polynomials with the same order must be equal, that is if two polynomials $P(x)$ and $Q(x)$ are equal, say,

$a_n x^n + a_{n-1} x^{n-1} + \dots + a_0 x^0 = b_n x^n + b_{n-1} x^{n-1} + \dots + b_0 x^0$, then $a_i = b_i$, for $i = 0, 1, 2, \dots, n$.

We employ this observation to match two moment generating functions, such that one of them is the manifestation of reality, that is the sample moments, m_a^k , and the other is the representation of the mathematical model we constructed, that is the theoretical moments of the assumed probability law, μ_a^k , about the point a , for all k , where

$$m_a^k = \frac{1}{n} \sum_{i=1}^n (X_i - a)^k, \quad (5.31)$$

and

$$\mu_a^k = E[(X - a)^k], \quad (5.32)$$

for $k = 1, 2, 3, \dots$. Thus, we force our model to fit to the reality, to a degree of significance provided by the random sample.

When the theoretical moments do not exist, we do not estimate them by using a finite sample. Also, method of moment estimators are easy to construct and utilize, yet they suffer from bias.

Example 5.2

Let $\{X_1, X_2, \dots, X_n\}$ be a random sample, coming from a normal probability distribution with an unknown mean, μ , and an unknown variance, σ^2 .

Since, $m_0^1 = \frac{1}{n} \sum_{i=1}^n X_i = \bar{X}_n$ and $\mu_0^1 = E[X] = \mu$, we have $\hat{\mu} = \bar{X}_n$.

Here, the sample mean, $\bar{X}_n$, is the method of moments estimator of the unknown population mean, μ .

Similarly, $m_{\bar{X}_n}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2$ and $\mu_\mu^2 = E[(X - \mu)^2] = \sigma^2$.

Therefore, $\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2$ is the method of moments estimator of the unknown parameter σ^2 , and it is a biased estimator, as we have seen it before. These two parameters, μ and σ^2 , are sufficient for describing a normal cdf, thus we stop here.

For other cdfs, more moments may be necessary to completely describe a probability distribution, thus we estimate them by using the method of moments.

Example 5.3

Let $\{X_1, X_2, \dots, X_n\}$ be a random sample coming from a Bernoulli probability distribution, with an unknown parameter $\theta = p$, where p is the probability of success, such that $0 \leq p \leq 1$. We can construct a method of moments estimator, for θ , as follows:

$m_0^1 = \frac{1}{n} \sum_{i=1}^n X_i = \bar{X}_n$, and $\mu_0^1 = E[X] = p$, for Bernoulli random variables. Therefore, $\hat{\theta} = \bar{X}_n$, would be the method of moments estimator of the parameter θ , i.e. the probability of success.

Example 5.4

Let $\{X_1, X_2, \dots, X_n\}$ be a random sample coming from a Uniform probability distribution, $U(0, \theta)$, with an unknown parameter θ . We can construct a method of moments estimator for θ , as follows.

Since $f_X(x|\theta) = \frac{1}{\theta}$, for $0 \leq x \leq \theta$, we have

$$\mu_0^1 = E[X] = \int_0^\theta \frac{1}{\theta} x dx = \left(\frac{1}{\theta}\right)\left(\frac{\theta^2}{2}\right) = \frac{1}{2}\theta, \text{ for } X \sim U(0, \theta).$$

Also, $m_0^1 = \frac{1}{n} \sum_{i=1}^n X_i = \bar{X}_n$.

Therefore, $\hat{\theta} = 2\bar{X}_n$ would be the method of moments estimator of the unknown parameter θ .

5.8 Maximum Likelihood Method

Let $\{X_1, X_2, \dots, X_n\}$ be a random sample coming from a probability distribution, that is $\{X_i | i = 1, 2, \dots, n\}$ be a set of i.i.d. random variables with the common cdf $F(x|\theta)$, without loss of generality, with a common pdf $f(x|\theta)$.

Therefore, the joint pdf of the sample would be,

$$f_X(x_1, x_2, \dots, x_n) = \prod_{i=1}^n f(x_i|\theta).$$

Whenever we have a realization of the sample, $(x_1^i, x_2^i, \dots, x_n^i)$, there would be no more randomness left in the joint pdf of the sample, yet the unknown parameter, θ , as a real variable (a number, or a vector) can get any feasible value (a variable) in the parameter space, thus we could consider this joint pdf of the sample for a fixed realization of the sample, as a function of the unknown parameter, and call it the likelihood function of the parameter, for a given sample, denoted by $L(\theta)$.

With an exchange of roles such that the sample plays the role of a known fixed parameter, and θ plays the role of argument of the likelihood function,

$$L(\theta) = L(\theta|x) = \prod_{i=1}^n f(x_i|\theta). \quad (5.33)$$

Here, $f(x_i|\theta) > 0$, for all θ and $i = 1, 2, \dots, n$, therefore we could introduce the natural logarithm (inverse function of the exponential function) of the likelihood function, called the loglikelihood function, denoted by $l(\theta)$,

$$l(\theta) = \log L(\theta) = \sum_{i=1}^n \log f(x_i|\theta). \quad (5.34)$$

Moreover, properties of the logarithm function would help us to estimate the unknown parameter, as best as one could.

This construct also generalizes to the multivariate cases, by introducing the set of i.i.d. random vectors for the sample, $\{\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n\}$, with a multivariate joint pdf $\mathbf{f}(\mathbf{X}_i|\theta)$.

Now, we can search for the best possible value for the unknown parameter, θ , that would maximize the likelihood function.

By doing so, we will find a possible value for the unknown parameter, which maximizes the probability of obtaining the given random sample from the assumed probability distribution.

Hence, the maximum likelihood estimator, $\hat{\theta}$, of the unknown parameter, θ , is the value of θ that maximizes the likelihood function, $L(\theta)$, such that

$$\hat{\theta} = \arg \max \{L(\theta|x)\}, \quad (5.35)$$

for all possible θ .

The logarithm function is a smooth, continuous concave function and the maximizer of the likelihood function is also the maximizer of the loglikelihood function.

Therefore, the maximum likelihood estimator of the unknown parameter would be also the value of θ that maximizes the loglikelihood function, $l(\theta)$, such that

$$\hat{\theta} = \arg \max \{l(\theta|x)\}. \quad (5.36)$$

This implies that for a maximum likelihood estimator (mle), $\hat{\theta}$, we must have

$$\frac{\partial l(\theta|x)}{\partial \theta} = 0, \text{ for all possible } \theta.$$

Also, the second order maximality condition must be met, that is

$$\frac{\partial^2 l(\theta|x)}{\partial \theta^2} < 0, \text{ for all possible } \theta.$$

The maximum likelihood estimator may not exist, or if it exists, it may not be unique.

Example 5.5

Let $\{X_1, X_2, \dots, X_n\}$ be a random sample coming from a normal probability distribution, with an unknown finite mean, μ , and an unknown finite variance, σ^2 .

What are the maximum likelihood estimators for μ and σ^2 ?

Here, the parameter is a vector, $\theta = (\mu, \sigma^2)$, and the likelihood function,

$$L(\mu, \sigma^2|x) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2} \frac{(x_i - \mu)^2}{\sigma^2}\right\}.$$

Thus, the loglikelihood function would be,

$$l(\mu, \sigma^2|x) = \sum_{i=1}^n \left\{ \log \frac{1}{\sqrt{2\pi\sigma^2}} - \frac{1}{2} \frac{(x_i - \mu)^2}{\sigma^2} \right\} = -\frac{1}{2} \sum_{i=1}^n \log 2\pi\sigma^2 - \frac{1}{2} \sum_{i=1}^n \frac{(x_i - \mu)^2}{\sigma^2}.$$

If we take the partial derivative of the loglikelihood function with respect to μ and set it equal to zero, after some simplifications we get

$$\sum_{i=1}^n \frac{x_i - \mu}{\sigma^2} = 0, \text{ which implies that}$$

$$\frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu) = 0.$$

Since μ and σ^2 are finite, we have $\sum_{i=1}^n (x_i - \mu) = 0$.

Now, we can solve for the unknown μ , in terms of the observed sample $\{x_i | i = 1, 2, \dots, n\}$, such that

$$\sum_{i=1}^n x_i - n\mu = 0, \text{ that is } \hat{\mu} = \frac{1}{n} \sum_{i=1}^n x_i.$$

Thus, the sample mean $\bar{X}_n$ is the maximum likelihood estimator of μ .

Similarly, if we take the partial derivative of the loglikelihood function with respect to σ^2 and set it equal to zero, we have

$$\sum_{i=1}^n \left(-\frac{1}{2\sigma^2} + \frac{1}{2} \frac{(x_i - \mu)^2}{\sigma^4} \right) = 0.$$

This implies that,

$$-\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_{i=1}^n (x_i - \mu)^2 = 0.$$

Therefore, when we solve for the unknown σ^2 , after some simplifications we have

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2.$$

Since we do not know μ , we replace it with its mle, $\bar{X}_n$, and we obtain the maximum likelihood estimator of σ^2 ,

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2.$$

A simple analysis indicates that the second order conditions were met for the maximum.

Here $\hat{\theta} = (\hat{\mu}, \hat{\sigma}^2)$.

The maximum likelihood estimation is also a valid construct for discrete probability distributions.

Example 5.6

Let $\{X_1, X_2, \dots, X_n\}$ be a random sample coming from a Bernoulli probability distribution, with an unknown parameter $\theta = p$, where p is the probability of success, $0 \leq p \leq 1$.

What is the maximum likelihood estimator of θ ?

The common pmf of the sample is $f_X(x|\theta) = \theta^x(1 - \theta)^{1-x}$, for $x = 0, 1$.

Thus, the likelihood function would be

$$L(\theta|x) = \prod_{i=1}^n \theta^{x_i} (1 - \theta)^{(1-x_i)} = \theta^{\sum_{i=1}^n x_i} (1 - \theta)^{\sum_{i=1}^n (1-x_i)},$$

and the loglikelihood function would be

$$l(\theta|x) = \log \theta (\sum_{i=1}^n x_i) + \log(1 - \theta) (n - \sum_{i=1}^n x_i).$$

Hence, $\frac{\partial l(\theta|x)}{\partial \theta} = 0$ implies that $\frac{\sum_{i=1}^n x_i}{\theta} - \frac{n - \sum_{i=1}^n x_i}{1 - \theta} = 0$.

When we solve this equation for the unknown θ , we have the maximum likelihood estimator, $\hat{\theta} = \frac{1}{n} \sum_{i=1}^n X_i$.

That is, the maximum likelihood estimator of p is the sample mean, $\bar{X}_n$, with $X_i = 1$ for success and $X_i = 0$ for failure identification yields the success ratio in the sample, which is also the sample mean.

Example 5.7

Let $\{X_1, X_2, \dots, X_n\}$ be a random sample coming from a Uniform probability distribution, $U(0, \theta)$, with an unknown parameter θ .

We can construct a maximum likelihood estimator for θ , as follows.

Since $f_X(x|\theta) = \frac{1}{\theta}$, for $0 \leq x \leq \theta$, we have

$$L(\theta|x) = \prod_{i=1}^n \frac{1}{\theta} = \frac{1}{\theta^n}, \text{ for } X \sim U(0, \theta).$$

Thus, $\frac{\partial \log L(\theta|x)}{\partial \theta} = -\frac{n}{\theta}$, has no solution for an mle.

Therefore, we use order statistic to estimate the unknown θ , as follows.

Order the sample observations from the minimum value to the maximum value, $X_{[1]}, X_{[2]}, \dots, X_{[n]}$, where $X_{[1]}$ is the minimum and $X_{[n]}$ is the maximum value of the sample.

Then $\hat{\theta} = X_{[n]}$ would be the maximum likelihood estimator of the unknown parameter θ .

5.9 Invariance

A one-to-one, monotonic function preserves the maximum likelihood estimator, such that if $\hat{\theta}$ is the mle of a parameter θ , and $u(\theta)$ is a one-to-one monotonic function of θ , then $u(\hat{\theta})$ is the mle of $u(\theta)$.

In general, the invariance property of the maximum likelihood estimators is not valid for other types of parameter estimators.

Example 5.8

The maximum likelihood estimator of the standard deviation of a normal population would be the positive square root (one-to-one, monotonic function) of the maximum likelihood estimator of the variance,

$$\hat{\sigma} = \sqrt{\hat{\sigma}^2} = \sqrt{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2}.$$

The maximum likelihood estimator is a consistent estimator.

Example 5.9

The mle $\hat{\mu}_n = \frac{1}{n} \sum_{i=1}^n X_i$, and the mle $\hat{\sigma}_n^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2$ are consistent estimators for the μ and σ^2 , respectively.

The maximum likelihood estimator is asymptotically normal, with mean $\mu_{\hat{\theta}} = \theta$ and variance $\sigma_{\hat{\theta}}^2 = \frac{1}{nI(\theta)}$.

That implies that the maximum likelihood estimator is asymptotically efficient. Moreover, while maximum likelihood estimators are biased, they are asymptotically unbiased.

5.10 Interval Estimators

The sample, $X = \{X_1, \dots, X_n\}$, yields information about the unknown parameter, θ , through a point estimator, $\hat{\theta}$.

Yet, a sample statistic, $Y = u(X_1, \dots, X_n)$, varies from realization to realization and this variation must be incorporated into the parameter estimation, $\hat{\theta} = \phi(Y)$.

In fact one must ask to find an answer to,

What is $P(\hat{\theta} = \theta|X)$?

For this reason, we use an interval estimator for the unknown parameter, such that an interval estimate of the unknown parameter is an interval constructed around the sample statistic that would capture the true parameter in it, to a desired degree of probability measure, also known as the confidence level, $P_d(\hat{\theta} - \delta_n \leq \theta \leq \hat{\theta} + \delta_n|X)$, for a real $\delta_n > 0$, which is also known as the margin of error, since it is the tolerable degree of imprecision in the estimate, based on the given sample.

5.10.1 Confidence Interval for the Mean

Assume that we have a normal population with a finite mean and a finite variance. Let the population mean, μ , be the unknown parameter, yet the population variance, σ^2 , be known, by a very strange twist of fate.

In addition, let a random sample, $X = \{X_1, \dots, X_n\}$, such that $X_i \sim \mathcal{N}(\mu, \sigma^2)$, be taken from the population and a point estimator, say $\hat{\mu} = \bar{X}_n$, is constructed for μ .

Clearly, $\bar{X}_n \sim \mathcal{N}(\mu, \frac{\sigma^2}{n})$ and its standardization would be,

$$\frac{\bar{X}_n - \mu}{\frac{\sigma}{\sqrt{n}}} \sim \mathcal{N}(0, 1).$$

Hence, we can introduce a desired degree of probability for the closeness of the sample mean to the unknown population mean, such that

$P_d(-z \leq \frac{\bar{X}_n - \mu}{\frac{\sigma}{\sqrt{n}}} \leq z)$, for a standard normal variate $z > 0$, being the critical point, or cut-off point, for the confidence interval, as in Figure 5.1.

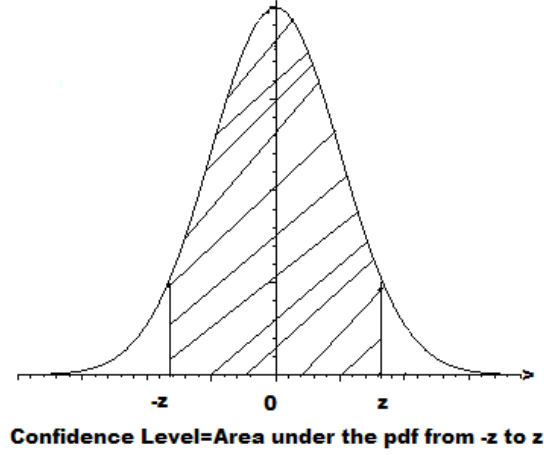


Figure 5.1: Confidence Level on Standard Normal pdf

Since $\frac{\sigma}{\sqrt{n}} > 0$ is a known constant for a given sample, this is equivalent to $P_d(-z \frac{\sigma}{\sqrt{n}} \leq \bar{X}_n - \mu \leq z \frac{\sigma}{\sqrt{n}})$, with the margin of error being $z \frac{\sigma}{\sqrt{n}}$.

Moreover, this is equivalent to $P_d(-\bar{X}_n - z \frac{\sigma}{\sqrt{n}} \leq -\mu \leq -\bar{X}_n + z \frac{\sigma}{\sqrt{n}})$.

Finally, this is equivalent to $P_d(\bar{X}_n - z \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X}_n + z \frac{\sigma}{\sqrt{n}})$.

Before a realization happens, the random interval that is expected to capture the actual unknown parameter in it, would be

$$[\bar{X}_n - z \frac{\sigma}{\sqrt{n}}, \bar{X}_n + z \frac{\sigma}{\sqrt{n}}], \quad (5.37)$$

and it is called the confidence interval estimator for the unknown parameter μ , and the desired degree of probability that the actual parameter would be in this interval is called the confidence level. Here, σ is a known constant, but the end points of the confidence interval are statistics (random variables) since sample mean generates the randomness.

After a sample realization, we would have a known fixed real value for the sample mean, such as $\bar{x}_n$, i.e. no more randomness left for the sample mean, and the confidence interval would be a known segment of the real line, $[\bar{x}_n - z \frac{\sigma}{\sqrt{n}}, \bar{x}_n + z \frac{\sigma}{\sqrt{n}}]$.

Thus, the desired probability measure now refers to the proportion of the confidence intervals, based on the sample realizations with a fixed sample size n , that would capture the actual parameter, when the sampling is repeated infinitely many times. The margin of error in this case would be $\delta_n = z \frac{\sigma}{\sqrt{n}}$, thus for a given margin of error and a desired level of confidence, one can find the corresponding sample size, $n = (\frac{z\sigma}{\delta_n})^2$.

Example 5.10

Let the sample mean be 50 for a random sample of 25 observations coming from a normal distribution with a known variance of 100.

The confidence interval for the unknown parameter μ , at the 95% confidence level would be $P_d(-z \leq \frac{\bar{X}_n - \mu}{\frac{\sigma}{\sqrt{n}}} \leq z) = 0.95$ and since $Z = \frac{\bar{X}_n - \mu}{\frac{\sigma}{\sqrt{n}}}$, we compute

$P_d(-z \leq Z \leq z) = 0.95$, and by using the standard normal distribution, we found that $z = 1.96$.

Therefore, the confidence interval for the unknown μ , based on this realization would be $[50 - 1.96(\frac{10}{5}), 50 + 1.96(\frac{10}{5})] = [46.08, 53.92]$.

When we have large samples, due to the central limit theorem, we can also construct confidence interval estimators similar to what we have done above, for the non-normal populations.

On the other hand, if we are not blessed with the knowledge of the population standard deviation, then we can use the sample standard deviation, S_n , as an estimator for the unknown σ , and the Student's t-distribution would be used for the confidence interval for μ , such that $T_{df} = \frac{\bar{X}_n - \mu}{(\frac{S_n}{\sqrt{n}})}$, with $df = n - 1$ degrees of freedom. The ratio $\frac{S_n}{\sqrt{n}}$ is also known as the standard error.

Example 5.11

Let the sample mean be 50 and the sample standard deviation be 10, for a random sample of 25 observations coming from a normal distribution with an unknown variance.

The confidence interval for the unknown parameter μ , at the 95% confidence level would be $P_d(-t \leq \frac{\bar{X}_n - \mu}{\frac{S_n}{\sqrt{n}}} \leq t) = 0.95$ and since $T_{df} = \frac{\bar{X}_n - \mu}{\frac{S_n}{\sqrt{n}}}$, we have $P_d(-t \leq T_{24} \leq t) = 0.95$, and by using the t-distribution, we found that $t = 2.064$.

Therefore, the confidence interval for the unknown μ , based on this realization would be $[50 - 2.064(\frac{10}{5}), 50 + 2.064(\frac{10}{5})] = [45.872, 54.128]$.

The confidence interval with an unknown population variance is larger than the confidence interval with a known population variance, for the same level of confidence and with the same sample statistic. This was expected since we have less information about the population when we do not know the σ^2 and the utilization of the standard error introduces more uncertainties into our interval estimator. Hence, a larger confidence interval for the unknown μ , when we have less information about the population.

When the underlying population distribution is not normal, we can still use the Student's t-distribution to construct an interval estimator for the unknown population mean. That is $T_{df} = \frac{\bar{X}_n - \mu}{\frac{S_n}{\sqrt{n}}}$ be a statistic to measure the closeness of the estimator, $\bar{X}_n$ to the unknown population mean, μ .

5.10.2 Confidence Interval for the Difference of Means

When we are interested in comparing the unknown means of two independent normal populations, $\mathcal{N}(\mu_1, \sigma_1^2)$ and $\mathcal{N}(\mu_2, \sigma_2^2)$, we can construct a confidence interval estimator for the difference of the means, $\mu_1 - \mu_2$, when the means and the variances are finite.

Let $X = \{X_1, \dots, X_{n_1}\}$ be a random sample, coming from the first population and $Y = \{Y_1, \dots, Y_{n_2}\}$ be a random sample, coming from the second population, independent from each other. Such that $\bar{X}_{n_1} \sim \mathcal{N}(\mu_1, \frac{\sigma_1^2}{n_1})$ and $\bar{Y}_{n_2} \sim \mathcal{N}(\mu_2, \frac{\sigma_2^2}{n_2})$.

Since $E[\bar{X}_{n_1} - \bar{Y}_{n_2}] = E[\bar{X}_{n_1}] - E[\bar{Y}_{n_2}] = \mu_1 - \mu_2$, then $\bar{X}_{n_1} - \bar{Y}_{n_2}$ is an unbiased estimator of $\mu_1 - \mu_2$.

On the other hand,

$$Var(\bar{X}_{n_1} - \bar{Y}_{n_2}) = Var(\bar{X}_{n_1}) + Var(\bar{Y}_{n_2}) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2},$$

due to the independence of random samples.

Thus, $\bar{X}_{n_1} - \bar{Y}_{n_2} \sim \mathcal{N}(\mu_1 - \mu_2, \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2})$,

as a difference of two independent normal random variables (similar to the sum of two independent normal r.v.s).

Therefore, $\frac{(\bar{X}_{n_1} - \bar{Y}_{n_2}) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \xrightarrow{D} \mathcal{N}(0, 1)$.

Hence, at the desired level of confidence,

$$\begin{aligned}
P_d(-z \leq \frac{(\bar{X}_{n_1} - \bar{Y}_{n_2}) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \leq z) = \\
P_d(-z \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} \leq (\bar{X}_{n_1} - \bar{Y}_{n_2}) - (\mu_1 - \mu_2) \leq z \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}) = \\
P_d((\bar{X}_{n_1} - \bar{Y}_{n_2}) - z \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} \leq (\mu_1 - \mu_2) \leq (\bar{X}_{n_1} - \bar{Y}_{n_2}) + z \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}),
\end{aligned}$$

the confidence interval estimator for the difference of two means, would be as follows:

$$[(\bar{X}_{n_1} - \bar{Y}_{n_2}) - z \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}, (\bar{X}_{n_1} - \bar{Y}_{n_2}) + z \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}]. \quad (5.38)$$

When these two populations have the same common variance, σ^2 , then the level of confidence would be,

$$P_d(-z \leq \frac{(\bar{X}_{n_1} - \bar{Y}_{n_2}) - (\mu_1 - \mu_2)}{\sigma \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \leq z),$$

and the confidence interval estimator would be,

$$[(\bar{X}_{n_1} - \bar{Y}_{n_2}) - z\sigma \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}, (\bar{X}_{n_1} - \bar{Y}_{n_2}) + z\sigma \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}]. \quad (5.39)$$

When the common population variance is unknown, we can use an estimator to substitute its value in the confidence interval estimator, such that

$$S_{pooled}^2 = \frac{df_1 S_1^2 + df_2 S_2^2}{df_1 + df_2}, \text{ as a pooled sample variance.}$$

where S_{pooled}^2 is the weighted average of two sample variances with weights as being the degrees of freedoms of the samples, $df_1 = n_1 - 1$ and $df_2 = n_2 - 1$.

Hence, the confidence level would be,

$$P_d(-t \leq \frac{(\bar{X}_{n_1} - \bar{Y}_{n_2}) - (\mu_1 - \mu_2)}{S_{pooled} \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \leq t), \text{ where } \frac{(\bar{X}_{n_1} - \bar{Y}_{n_2}) - (\mu_1 - \mu_2)}{S_{pooled} \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim T_{df_1 + df_2}.$$

Therefore, the confidence interval estimator would be,

$$[(\bar{X}_{n_1} - \bar{Y}_{n_2}) - t S_{pooled} \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}, (\bar{X}_{n_1} - \bar{Y}_{n_2}) + t S_{pooled} \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}]. \quad (5.40)$$

The above construct is a good approximation for a confidence interval, even when the populations are not normal, yet share a common variance.

Example 5.12

When we collect random samples from two independent normal populations, with known variances $\sigma_1^2 = 100$ and $\sigma_2^2 = 625$, such that $\bar{X}_{10} = 50$ and $\bar{Y}_{25} = 70$, with $n_1 = 10$ and $n_2 = 25$, we can construct a 95% confidence interval for the difference of population means as follows.

Since $z = 1.96$, the confidence interval would be

$$\begin{aligned} & [(50 - 70) - 1.96\sqrt{(\frac{100}{10} + \frac{625}{25})}, (50 - 70) + 1.96\sqrt{(\frac{100}{10} + \frac{625}{25})}] \\ & = [-31.5956, -8.4044]. \end{aligned}$$

This indicates that the mean of the second population is larger than the mean of the first population.

Example 5.13

Assume that the above populations have the same variance, σ^2 , but it is unknown to us. With sample variances $S_{10}^2 = 125$ and $S_{25}^2 = 110$. Thus a 95% confidence interval for the difference of the population means would be,

$$\begin{aligned} & [(50 - 70) - (2.0369)(10.6813)\sqrt{\frac{1}{10} + \frac{1}{25}}, (50 - 70) + (2.0369)(10.6813)\sqrt{\frac{1}{10} + \frac{1}{25}}] \\ & = [-28.1406, -11.8594]. \end{aligned}$$

Here, $S_{Pooled}^2 = \frac{9 \times 125 + 24 \times 110}{33} = 114.09$, thus $S_{Pooled} = 10.6813$, and also, $t = 2.0369$. This indicates that the second population mean is bigger than the first population mean.

5.10.3 Confidence Interval for the Proportion of Success

When a population consists of only successes and failures with a fixed, yet unknown probability distribution, we can construct a confidence interval estimator for the unknown probability of success, p .

Let $\{X_1, \dots, X_n\}$ be a random sample coming from the population, with a known fixed n , but with an unknown parameter p , such that $X_i = 1$ for success and $X_i = 0$ for failure.

Also, let $X = \sum_{i=1}^n X_i$, be the number of successes in the sample.

Therefore, $E[X] = np$, $Var(X) = np(1 - p)$, and $\frac{X - np}{\sqrt{np(1-p)}} \xrightarrow{D} \mathcal{N}(0, 1)$.

Hence, with a desired level of confidence,

$$\begin{aligned}
P_d(-z \leq \frac{X - np}{\sqrt{np(1-p)}} \leq z) &= P_d(-z \leq \frac{(X - np)/n}{(\sqrt{np(1-p)})/n} \leq z) \\
&= P_d(-z \leq \frac{\frac{X}{n} - p}{\sqrt{\frac{p(1-p)}{n}}} \leq z).
\end{aligned}$$

We can now build a confidence interval estimator for the unknown parameter as follows:

$$[\frac{X}{n} - z\sqrt{\frac{\frac{X}{n}(1 - \frac{X}{n})}{n}}, \frac{X}{n} + z\sqrt{\frac{\frac{X}{n}(1 - \frac{X}{n})}{n}}]. \quad (5.41)$$

Let $\hat{p} = \frac{X}{n}$, thus the confidence interval for p would be,

$$[\hat{p} - z\sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}, \hat{p} + z\sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}]. \quad (5.42)$$

Here, the unknown parameter, p , also appeared in the variance, thus we assumed that we do not know the variance either, and it is estimated by the sample.

On the other hand, if we do not know p , yet if the variance is known, then that would be absurd, since all we need to do is to solve for the positive square root of the variance.

That is to say that there would be no need for an estimator.

Example 5.14

Assume that a random sample of 1,000 medical doctors indicates that 67% of them are in favor of a global health coverage. We can build a 95% confidence interval for the proportion of the MDs who would be in favor of the global health coverage as follows:

$$[0.67 - 1.96\sqrt{\frac{0.67(1-0.67)}{1000}}, 0.67 + 1.96\sqrt{\frac{0.67(1-0.67)}{1000}}] = [0.6608, 0.6992].$$

5.10.4 Confidence Interval for the Difference of Proportions

When X_1 and X_2 are two independent binomial random variables with unknown parameters p_1 and p_2 , as the probability of success, respectively, then we can construct a confidence interval estimator for the difference of these two probabilities of success, or proportions of success, as follows.

Let there be two populations with successes and failures only. Also, let $\{X_{11}, \dots, X_{1n_1}\}$ be a random sample coming from the first population and $\{X_{21}, \dots, X_{2n_2}\}$ be a random sample coming from the second population, such that

$X_1 = \sum_{i=1}^{n_1} X_{1i}$ and $X_2 = \sum_{i=1}^{n_2} X_{2i}$ be the numbers of successes in these two independent random samples, respectively. Hence,

$\hat{p}_1 = \frac{X_1}{n_1}$ and $\hat{p}_2 = \frac{X_2}{n_2}$ would be the estimators of the proportion of successes in these two independent populations. We can construct a confidence interval estimator for the difference of the proportions of successes, $p_1 - p_2$, when n_1 and n_2 are known, as follows. The confidence interval estimator for $p_1 - p_2$, at the desired level of confidence would be as follows:

$$[(\hat{p}_1 - \hat{p}_2) - ME, (\hat{p}_1 - \hat{p}_2) + ME], \quad (5.43)$$

where the margin of error, ME, is

$$ME = z \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}},$$

and z is the critical point for the level of confidence, based on the standard normal distribution.

Example 5.15

Assume that a random sample of 1,000 medical doctors indicates that 67% of them are in favor of a global health coverage. On the other hand, a random sample of 100 health care management company managers indicates that 25% of them are in favor of a global health coverage.

We can construct a 95% confidence interval for the difference of proportions as follows:

$$ME = 1.96 \sqrt{\frac{0.67(1-0.67)}{1000} + \frac{0.25(1-0.25)}{100}}$$

$$[(0.67 - 0.25) - ME, (0.67 - 0.25) + ME] = [0.3303, 0.5097].$$

Since this confidence interval does not include zero, or a sign change from the lower bound to the upper bound, and includes only positive numbers, there is a significant difference between two population proportions, such that the first population proportion is greater than the second population proportion (proportions themselves are nonnegative numbers between 0% to 100%).

What would you conclude if this confidence interval consists of only negative numbers?

5.10.5 Confidence Interval for the Variance

Let $X_1, \dots, X_n$ be i.i.d. random variables, such that $X_i \sim \mathcal{N}(\mu, \sigma^2)$, with a known μ , but with an unknown σ^2 .

Since $\sum_{i=1}^n Z_i^2 \sim \chi_n^2$, we would have

$$\sum_{i=1}^n \left(\frac{X_i - \mu}{\sigma}\right)^2 = \frac{1}{\sigma^2} \sum_{i=1}^n (X_i - \mu)^2 \sim \chi_n^2, \quad (5.44)$$

for a known μ .

Therefore, we have

$$\frac{n\hat{\sigma}_n^2}{\sigma^2} \sim \chi_n^2. \quad (5.45)$$

Hence, we can build a confidence interval for the unknown σ^2 , based on an estimator of σ^2 , denoted by $\hat{\sigma}_n^2$.

Let $\hat{\sigma}_n^2$ be the S_n^2 , that is the unbiased variance estimator,

$$S_n^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2.$$

Therefore,

$$\frac{(n-1)S_n^2}{\sigma^2} \sim \chi_{(n-1)}^2, \quad (5.46)$$

and at a desired level of confidence, we would have

$$P_d(x_1 \leq \frac{(n-1)S_n^2}{\sigma^2} \leq x_2) = 1 - \alpha,$$

where $x_1 < x_2$ and α is the estimation error probability, such that the confidence interval estimator will not include the unknown parameter.

Thus we have $P_d(\frac{(n-1)S_n^2}{x_2} \leq \sigma^2 \leq \frac{(n-1)S_n^2}{x_1}) = 1 - \alpha$.

Here, we assigned equal likelihood to exclude the actual parameter from the confidence interval (tail probabilities = $\frac{1}{2}\alpha$).

Hence, we have the term symmetric confidence interval, such that

$$P(\frac{(n-1)S_n^2}{\sigma^2} \leq x_1) = P(\frac{(n-1)S_n^2}{\sigma^2} \geq x_2) = \frac{1}{2}\alpha.$$

Therefore, we would have the following symmetric confidence interval estimator for the unknown parameter σ^2 ,

$$\left[\frac{(n-1)S_n^2}{x_2}, \frac{(n-1)S_n^2}{x_1} \right], \quad (5.47)$$

which is based on a non-symmetric Chi-square distribution with $(n - 1)$ degrees of freedom.

We can also build a symmetric confidence interval for the unknown standard deviation, σ , by taking the positive square roots, such that

$$\left[\sqrt{\frac{(n-1)S_n^2}{x_2}}, \sqrt{\frac{(n-1)S_n^2}{x_1}} \right]. \quad (5.48)$$

That would be the symmetric confidence interval estimator of σ , at the $1 - \alpha$ level of confidence. There are infinitely many intervals that generate this probability measure under the $\chi_{(n-1)}^2$ distribution.

Hence, we had to impose some restrictions on our search for x_1 and x_2 , such that they would be the critical points to generate same tail probabilities (areas under the pdf of $\chi_{(n-1)}^2$). Thus, this confidence interval is known as the symmetric confidence interval, represented by Figure 5.2.

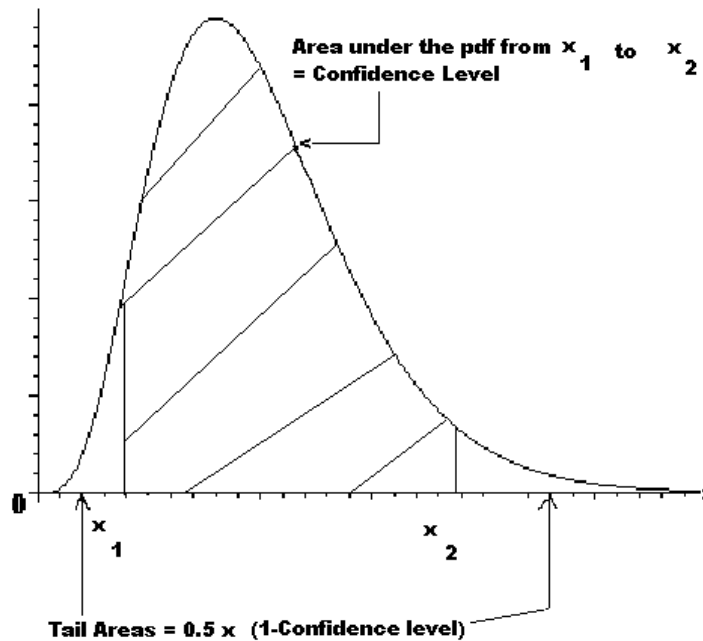


Figure 5.2: Confidence Level on Chi-Square pdf

5.11 Stochastic Simulation

In the eighteenth century, Count de Buffon, a natural philosopher, designed a random experiment in order to estimate the value of π . Hence, this method of estimation of π is known as Buffon's needle problem:

A table is ruled by equidistant lines, and a needle of unit length is randomly tossed on the table. What is the probability that the needle will intersect a line?

Let the needle's position is identified by the distance, D , from the center of the needle, C , to the nearest line and the smallest angle, α , between the needle and that line. If the needle intersects the line at a point, Q , then the orthogonal projection of C on the line, R , together with Q and C form a triangle such that its hypotenuse is equal to $\frac{D}{\sin \alpha}$, by utilizing Thales' theorem, as seen in Figure 5.3.

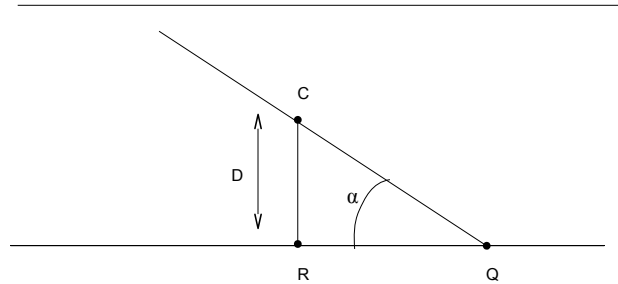


Figure 5.3: Buffon's Needle

On the other hand, if the needle does not intersect the line, then we would extend the needle by a line such that this extension line would intersect the line, and we form a similar triangle.

Hence, the needle would intersect the line if and only if $\frac{D}{\sin \alpha} \leq \frac{1}{2}$, where $D \in [0, \frac{1}{2}]$, and $\alpha \in [0, \frac{\pi}{2}]$.

In this needle toss experiment, we assume that α and D are independent uniform random variables, on their corresponding supports. Therefore, we would have the probability of intersection,

$$\begin{aligned}
P\left(\frac{D}{\sin \alpha} \leq \frac{1}{2}\right) &= \int_0^{\frac{\pi}{2}} P\left(\frac{D}{\sin a} \leq \frac{1}{2}\right) f_{\alpha}(a) da \\
&= \frac{2}{\pi} \int_0^{\frac{\pi}{2}} P\left(D \leq \frac{\sin a}{2}\right) da = \frac{2}{\pi} \int_0^{\frac{\pi}{2}} (\sin a) da = \frac{2}{\pi}. \quad (5.49)
\end{aligned}$$

This experiment, repeated over and over again, provides a computational instrument to compute the value of π . Why? Contemplate on the number of intersections, over a very large number of repetitions of the experiment.

5.11.1 Monte-Carlo Simulation

Assume that we wish to find the finite area of an object, when it is very hard or impossible to do the area computations. In this case replace the object in a unit square, i.e. each side of the square is of unit length (in any unit), and spray this unit square completely with uniformly generated, sufficiently large but finite number of random points.

Then, count and find out the number of points that are falling on the object of interest. This will give us the necessary information to estimate the area of this object, since we know how many uniformly generated points are in the unit square.

We construct the ratio of the number of points in the object, to the total number of points that fell on the unit square, and this ratio will give us the estimated area of the object, since this ratio of number of points in the object to the number of points on the unit square should be preserved by the ratio of area of the object to the area of the unit square.

We can generalize this method in order to estimate the volume of objects that are almost impossible to compute, or integrals that we cannot compute by analytical methods.

The story goes that the name of this stochastic simulation, for computing the value of integrals, is coming from the uniform random number generator that was originally used in the design of this experiment.

In the 1940's, during the Manhattan Project work, some researchers, including John Von Neumann (who contributed to the contemporary computer systems layout, and also did influential work on the game theory), needed to compute integrals by using numerical methods, and had to generate random numbers for this kind of simulation. To do this, they "borrowed" roulette tables from nearby casinos in Las Vegas. Consequently, the name "Monte-Carlo" appeared in this simulation method.

This is the basis of most commonly used stochastic simulation methods, and many versions of it are available as computer softwares. For example the Markov-Chain Monte-Carlo Simulation is one of the most widely used versions of this method.

5.12 Exercises

1. Let $X_1, X_2, \dots, X_n$ be a random sample coming from a normal population, $X_i \sim \mathcal{N}(\mu, \sigma^2)$, with unknown μ and unknown but finite σ^2 . Construct the method of moments estimators for the unknown mean and for the unknown variance.
2. Let $X_1, X_2, \dots, X_n$ be a random sample coming from a normal population, $X_i \sim \mathcal{N}(\mu, \sigma^2)$, with unknown μ and unknown but finite σ^2 . Construct the maximum likelihood estimators for the unknown mean and for the unknown variance.
3. Let $X_1, X_2, \dots, X_{100}$ be a random sample coming from a normal population, $X_i \sim \mathcal{N}(\mu, \sigma^2)$, with unknown μ and unknown but finite σ^2 . The sample average is computed as $\bar{X}_{100} = 75$ and the sample variance is computed as $S_{100}^2 = 100$. Construct a 95% confidence interval for the unknown mean of the population.
4. Let $X_1, X_2, \dots, X_{100}$ be a random sample coming from a normal population, $X_i \sim \mathcal{N}(\mu, \sigma^2)$, with known μ and unknown but finite σ^2 . The sample variance is computed as $S_{100}^2 = 100$. Construct a 95% confidence interval for the unknown variance of the population.
5. Let $X_1, X_2, \dots, X_{100}$ be a random sample of size 100 with a total number of successes is 75; coming from a binomial population with an unknown p . Construct the 90% confidence interval for the unknown probability of success.
6. Let $MSE(\hat{\Theta})$ be $E[\hat{\Theta} - \theta]$, i.e. the mean squared error of the parameter estimator. Show that MSE could be written as $Var(\hat{\Theta}) + (E[\hat{\Theta} - \theta])^2$, where $(E[\hat{\Theta} - \theta])$ is the $Bias(\hat{\Theta})$.
7. Let $S_n^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$. Show that $E[s_n^2] = (\frac{n-1}{n})\sigma^2$, thus S_n^2 is a biased estimator of σ^2 .
8. Let S^2 and S_n^2 be two estimators of σ^2 . What is the ratio of $\frac{MSE(S_n^2)}{MSE(S^2)}$? Elaborate.

9. Show that S_n^2 becomes unbiased as $n \rightarrow \infty$.
10. Show that $\frac{X}{n}$ is an unbiased estimator of p , when X is a binomial random variable with parameter p .

Chapter 6

Hypothesis Testing

“Flamingoes and mustard both bite. And the moral of that is: Birds of a feather flock together,” said the Duchess. “Only mustard is not a bird,” Alice remarked...” - Alice in Wonderland.

6.1 Classical Hypothesis Testing

Inferential statistics, that is, making generalizations based on the evidence, sometimes takes a strange turn. Before we have any evidence, we may construct a hypothesis (a statement yet unproven) for the value of an unknown parameter which begs the question for validity, then we will design an experiment and collect observations to test whether the hypothesized value of the parameter is acceptable, beyond a reasonable doubt, or not acceptable at all, under the weight of the evidence.

In the classical hypothesis testing [12], we construct a dichotomy such that two mutually exclusive and collectively exhaustive statements will be tested under the weight of the evidence and only one of them will be accepted. The status quo (current belief) is represented by the test hypothesis, also known as the null hypothesis, which always refers to the expected population parameter, but never to a sample statistic, and it is denoted by H_0 . As soon as a null hypothesis is constructed, its negation would be the alternative hypothesis, which is denoted by H_a , or, by H_1 . Thus, $H_0 : \theta \in A$ and $H_a : \theta \in A^c$, where A^c is the complement (or, negation) of A , and $A \cup A^c$ is the space of all possible values for θ .

The objective of the test is to examine whether the available evidence supports the rejection of the null hypothesis, as a highly unlikely counterexample to the test hypothesis.

The fundamental difference between parameter estimation and hypothesis testing is that in hypothesis testing we have two competing models, we assume that the null hypothesis model is valid and we utilize evidence to decide whether this model should be rejected. The hypothesis testing is designed so that the rejection probability of the null hypothesis is conditional upon the sample evidence, where the rejection of H_0 indicates that the probability of a counterexample is significantly small, when the null hypothesis were true, thus H_a is more likely to be true.

Beware that the conclusion of an hypothesis test is not a mathematical proof, and if we cannot reject the null hypothesis, it is because we do not have sufficient evidence to reject it yet. That is to say, if there is no evidence for an event, that does not mean that the event did not happen, it only means that we do not know about it.

Let $\mathcal{D}$ be the space of the sample. A test of H_0 vs. H_a is based on a subset of $\mathcal{D}$, which is known as the critical region, $\mathcal{C}$. We construct a decision rule by using the critical region and a function of the sample (statistic), such that we reject H_0 if $f(X_1, \dots, X_n) \in \mathcal{C}$. For a given critical region, we have the following decision table:

DECISION:	H_0 IS TRUE	H_a IS TRUE
Reject H_0 :	Type-I Error	Correct Decision
Accept H_0 :	Correct Decision	Type-II Error

Table 6.1: Decision Table

We wish to choose a critical region that minimizes the probability of these two types of errors simultaneously, but this is not possible.

Hence, we select critical regions so that the probability of Type-I error is under control.

$$\text{Let } \alpha = P(\text{Type} - I \text{ Error}) = P(\text{Reject } H_0 \mid \text{True } H_0),$$

$$\text{and } \beta = P(\text{Type} - II \text{ Error}) = P(\text{Accept } H_0 \mid \text{True } H_a),$$

where the size of a critical region is related to α , which is also known as the significance level of the test.

The power of the hypothesis test is the probability that the test selects the alternative hypothesis when alternative is the true model, $1 - \beta$. Thus, minimizing β is equivalent to maximizing the power of a hypothesis test.

The power function of a critical region is defined as the probability measure that the sample will fall into the critical region, when the alternative

model is true,

$$\gamma(\mathcal{C}) = P((X_1, X_2, \dots, X_n) \in \mathcal{C} \mid \text{True } H_a).$$

Consequently, when we have two critical regions of the same size, α (also known as the significance level of the test associated with the critical region), we may say that the critical region with a dominating (greater) power function is better than the other.

6.1.1 Hypothesis Testing for Mean

Let X be a random variable with a known finite variance. We want to test the hypothesis:

$$\begin{aligned} H_0 : \mu &\leq \mu_0, \text{ versus,} \\ H_a : \mu &> \mu_0, \text{ where } \mu_0 \text{ is identified in } H_0. \end{aligned}$$

Let $X_1, X_2, \dots, X_n$ be a random sample from the distribution of X , and let us also denote the sample mean and sample variance by $\bar{X}_n$ and S_n^2 , respectively.

Since $\bar{X}_n$ converges in probability to μ , intuitively a decision rule could be: reject H_0 if $\bar{X}_n$ is significantly greater than μ_0 .

Hence, by the central limit theorem,

$$\frac{\bar{X}_n - \mu}{\left(\frac{\sigma}{\sqrt{n}}\right)} \xrightarrow{D} \mathcal{N}(0, 1). \quad (6.1)$$

That is, as n goes to infinity, the test statistics $\frac{\bar{X}_n - \mu}{\left(\frac{S_n}{\sqrt{n}}\right)}$ converges in distribution to the standard normal random variable (a.k.a z-variate).

Therefore, practically we have a decision rule for all large samples, when the significance level α is selected:

$$\text{Reject } H_0 \text{ if } \frac{\bar{X}_n - \mu_0}{\left(\frac{S_n}{\sqrt{n}}\right)} \geq z_\alpha, \quad (6.2)$$

with z_α as the boundary point of the critical region with size α and μ_0 as the H_0 assumption of μ .

This hypothesis test can lead to a false rejection of H_0 , only if the test statistic exceeds the critical value, z_α , due to a strange sample with a misleading sample average. Hence this test is called a one tailed-test, where the tail indicates a rejection region for H_0 , thus a possible false rejection can occur only in this region.

Example 6.1

We believe that the average income of male professors is greater than the average income of female professors, which is \$100,000 per year. Hence, we build a hypothesis test to reject the null hypothesis, at the 5% level of significance.

$$H_0 : \mu \leq \$100,000,$$

$$H_1 : \mu > \$100,000.$$

Let the average income of 100 randomly selected male professors would be \$145,000, with a known standard deviation of annual income, \$10,000.

Also, since $\alpha = 5\%$, the critical value would be $z_\alpha = 1.645$.

Thus, the test statistic would be $\frac{145000 - 100000}{\left(\frac{10000}{\sqrt{100}}\right)} = 45$.

According to the decision rule, i.e. if the test statistic exceeds the critical value, then reject the null hypothesis; we conclude that the average annual income of the male professors is greater than the average annual income of female professors, with the given evidence and at the 5% level of significance.

We may construct another version of the hypothesis test for the mean, such that

$$H_0 : \mu = \mu_0 \text{ versus,}$$

$$H_1 : \mu \neq \mu_0.$$

As above, the test statistic is

$$\frac{\bar{X}_n - \mu}{\left(\frac{S_n}{\sqrt{n}}\right)} \xrightarrow{D} \mathcal{N}(0, 1). \quad (6.3)$$

Therefore, the decision rule would be;

$$\text{Reject } H_0 \text{ if } \left| \frac{\bar{X}_n - \mu_0}{\left(\frac{S_n}{\sqrt{n}}\right)} \right| \geq z_{\frac{\alpha}{2}}, \quad (6.4)$$

where $z_{\frac{\alpha}{2}}$ is the boundary point of the critical region with size $\frac{\alpha}{2}$ and μ_0 is the H_0 assumption of μ , for a desired level of significance α . Here, we split the significance level into two equal size regions of rejection for H_0 , since we are testing for the significant difference from the hypothesized value and this could only happen in two possible ways; either the sample average is too small, or the sample average is too large than the hypothesized value of μ_0 . Hence, this test is known as a two-tailed test, since there are two possible ways to wrongly reject H_0 .

Example 6.2

We believe that the average income of male professors is different than the average income of female professors, which is \$ 100,000 per year. Hence, we build a hypothesis test to reject the null hypothesis, at the 5% level of significance.

$$H_0 : \mu = \$100,000,$$

$$H_1 : \mu \neq \$100,000.$$

Let the average income of 100 randomly selected male professors would be \$145,000, with a known standard deviation of annual income, \$10,000. Also, since $\alpha = 5\%$, the critical value would be $z_{\frac{\alpha}{2}} = 1.96$.

Thus, the test statistic would be $\frac{145000-100000}{(\frac{10000}{10})} = 45$.

According to the decision rule, i.e. if the magnitude of the test statistic exceeds the critical value, then we reject the null hypothesis.

We conclude that the average annual income of the male professors is different than the average annual income of female professors, with the given evidence and at the 5% level of significance, or with 95% confidence.

It is clear that in this case of a two-tailed hypothesis test we only questioned equality, and the conclusion does not inform us which average is greater.

6.1.2 Hypothesis Testing for Mean Under Normality

Let the random variable X have a normal probability distribution with a finite variance. Consider the following construct for the one-tailed hypothesis testing, with a desired significance level of α .

$$H_0 : \mu \leq \mu_0, \text{ versus,}$$

$$H_a : \mu > \mu_0, \text{ where } \mu_0 \text{ is specified in } H_0.$$

Let $X_1, X_2, \dots, X_n$ be a random sample, coming from the parent distribution, which is normal (or, gaussian) with an unknown mean μ .

An intuitive decision rule would be to reject the null hypothesis if the sample mean is significantly greater than the hypothesized population mean, μ_0 , as declared in the null hypothesis. Thus, under the H_0 , the test statistic would be

$$T = \frac{\bar{X}_n - \mu_0}{(\frac{S_n}{\sqrt{n}})}, \quad (6.5)$$

which has a Student's t-distribution with $n - 1$ degrees of freedom (df). Hence, the decision rule for rejecting the null hypothesis can be constructed

as follows:

$$\text{Reject } H_0 \text{ if } T > t_{\alpha, df}, \quad (6.6)$$

with $t_{\alpha, df}$ as the upper critical point of a Student's t-distribution with $n - 1$ degrees of freedom.

Example 6.3

We believe that the average income of male professors is greater than the average income of female professors, which is \$100,000 per year. Hence, we build a hypothesis test to reject the null hypothesis, at the 5% level of significance.

$$H_0 : \mu \leq \$100,000,$$

$$H_1 : \mu > \$100,000.$$

Let the annual income of male professors have a normal distribution with a finite yet unknown variance. Also, let the average income of 25 randomly selected male professors would be \$145,000, with a sample standard deviation of annual income, \$10,000. Also, since $\alpha = 5\%$, the critical value would be $t_{\alpha, 24} = 1.711$, from the Student's t-distribution with 24 degrees of freedom.

Thus, the test statistic would be $\frac{145000 - 100000}{\left(\frac{10000}{5}\right)} = 22.5$.

According to the decision rule, i.e. if the test statistic exceeds the critical value, then reject the null hypothesis; we conclude that the average annual income of the male professors is greater than the average annual income of female professors, with the given evidence and at the 5% level of significance.

If we have constructed a two-tailed test, then the critical point would be $t_{\frac{\alpha}{2}, df}$ and the decision rule would be,

$$\text{Reject } H_0 \text{ if } |T| > t_{\frac{\alpha}{2}, df}. \quad (6.7)$$

Example 6.4

We believe that the average income of male professors is different than the average income of female professors, which is \$100,000 per year. Hence, we build a hypothesis test to reject the null hypothesis, at the 5% level of significance.

$$H_0 : \mu = \$100,000,$$

$$H_1 : \mu \neq \$100,000,$$

Let the annual income of male professors has a normal distribution with a finite but unknown variance and the average income of 25 randomly selected male professors would be \$145,000, with a sample standard deviation

of annual income, \$10,000. Also, since $\alpha = 5\%$, the critical value would be $t_{\frac{\alpha}{2}, 24} = 2.064$.

Thus, the test statistic would be $\frac{145000 - 100000}{(\frac{10000}{5})} = 22.5$.

According to the decision rule, i.e. if the magnitude of the test statistic exceeds the critical value, then reject the null hypothesis; we conclude that the average annual income of the male professors is different than the average annual income of female professors, with the given evidence and at the 5% level of significance.

6.1.3 Large Sample Hypothesis Test for Mean

When the sample is large, with usually 25 or more observations are considered large samples, we could apply the asymptotic behavior of random variables to the test statistic, thus $T = \frac{\bar{X}_n - \mu}{(\frac{S_n}{\sqrt{n}})}$ converges to the standard normal in distribution.

Hence, we can substitute the Student's t-distribution with the standard normal distribution, and the critical point $t_{\alpha, df}$ with z_{α} , for large samples.

6.2 p-Value of a Hypothesis Test

The p-value of a test is the observed significance level, defined as the probability of the random variable representing the test statistic, which is subject to a probability law F_X , will exceed the magnitude of the test statistic, given that the H_0 is true, or, it is the counterexample probability that the null hypothesis would be true, when the evidence rejects it,

$$p - value = P(X \geq |\text{test statistic}| \mid H_0 \text{ is true}). \quad (6.8)$$

Therefore, the decision rule for a one-tailed test would be: reject H_0 if the p-value is less than the significance level (α), and for a two-tailed test: reject H_0 if the p-value is less than the half of the significance level ($\frac{\alpha}{2}$).

Example 6.5

We believe that the average income of male professors is greater than the average income of female professors, which is \$100,000 per annum. Hence, we build a hypothesis test to reject the null hypothesis, at the 5% level of significance.

$$H_0 : \mu \leq \$100,000,$$

$$H_1 : \mu > \$100,000.$$

Let the average income of 100 randomly selected male professors be \$145,000, with a known standard deviation of annual income, \$10,000. Also, $\alpha = 5\%$.

Thus, the test statistic would be $\frac{145000-100000}{(\frac{10000}{10})} = 45$, and the p-value would be $P(Z \geq 45) \approx 0$, based on the standard normal distribution.

According to the decision rule for a one-tailed test, we conclude that the average annual income of the male professors is greater than the average annual income of female professors with the given evidence and at the 5% level of significance.

Example 6.6

We believe that the average income of male professors is different than the average income of female professors, which is \$100,000 per annum. Hence, we build a hypothesis test to reject the null hypothesis, at the 5% level of significance.

$$H_0 : \mu = \$100,000,$$

$$H_1 : \mu \neq \$100,000.$$

Let the annual income of male professors have a normal distribution with a finite but unknown variance and the average income of 25 randomly selected male professors would be \$145,000, with a sample standard deviation of \$10,000.

Thus, the test statistic would be $\frac{145000-100000}{(\frac{10000}{5})} = 22.5$, and at 24 degrees of freedom, the p-value would be $P(T \geq 22.5) \approx 0$, based on the Student's t-distribution.

According to the decision rule for a two-tailed test, we conclude that the average annual income of the male professors is different than the average annual income of female professors, with the given evidence and at the 5% level of significance.

6.3 Comparing Means of Two Normal Populations

Sometimes we are interested in comparing two normal population means, in order to determine whether these two populations on the average behave similarly or not.

6.3.1 Independent Populations

Let the population X_1 and the population X_2 be two independent normal populations with unknown means, yet their finite variances are known, such as $\sigma_{X_1}^2$ and $\sigma_{X_2}^2$, respectively.

Since they are independent their covariance is zero, and the variance of their sums, or differences, would be the sum of their individual variances, $\sigma_{X_1}^2 + \sigma_{X_2}^2$.

Their means, say $\bar{X}_1$ and $\bar{X}_2$, are also independent.

Assume that we collect n_1 random observations from the first population and n_2 random observations from the second population.

Thus, $\bar{X}_1 = \frac{1}{n_1} \sum_{i=1}^{n_1} X_{1i}$ and $\bar{X}_2 = \frac{1}{n_2} \sum_{j=1}^{n_2} X_{2j}$.

Since, X_{1i} s are i.i.d. random variables with a common variance of $\sigma_{X_1}^2$

and X_{2j} s are i.i.d. r.v.s with a common variance of $\sigma_{X_2}^2$,

the variance of $\bar{X}_1$ is $\frac{1}{n_1^2} \times n_1 \sigma_{X_1}^2 = \frac{\sigma_{X_1}^2}{n_1}$,

and the variance of $\bar{X}_2$ is $\frac{1}{n_2^2} \times n_2 \sigma_{X_2}^2 = \frac{\sigma_{X_2}^2}{n_2}$.

Therefore, the variance of the difference of these two independent means would be $\frac{\sigma_{X_1}^2}{n_1} + \frac{\sigma_{X_2}^2}{n_2}$.

Now, let us show some interest in the difference of the mean of these two populations, for the same measurable characteristic, such that if the evidence indicates a significant difference among the means, we may say that on the average these two populations do not have the same degree of that characteristic.

Let $X_1 = \{X_{11}, X_{12}, \dots, X_{1n_1}\}$ and $X_2 = \{X_{21}, X_{22}, \dots, X_{2n_2}\}$ be two random samples coming from the two independent normal populations, indicating the measure of independent individuals in their respective parent populations, with known variances σ_1^2 and σ_2^2 , respectively. Let μ_1 and μ_2 be the unknown means of these two populations. Also let $\bar{X}_1$ and $\bar{X}_2$ be the sample averages of X_1 and X_2 . Is there a significant difference in the means of these two populations?

We can construct a hypothesis test to answer this question, as follows:

$$H_0: \mu_1 = \mu_2,$$

$$H_1: \mu_1 \neq \mu_2.$$

That is, we are testing for $\mu_1 - \mu_2 = 0$ and this is a two-tailed test, since inequality is possible when $\mu_1 > \mu_2$, or, $\mu_1 < \mu_2$.

Also,

$$E[X_1] - E[X_2] \sim \mathcal{N}(\mu_1 - \mu_2, \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}). \quad (6.9)$$

At a desired level of significance, say α , we have a critical z-variate, $z_{\frac{\alpha}{2}}$. Thus, the test statistic would be

$$Z_{Test} = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}. \quad (6.10)$$

If the magnitude of the test statistic exceeds the critical z-variate, then we reject the null hypothesis.

Moreover, if the test statistic is positive, that indicates that the first population mean is larger than the mean of the second one; or, if the test statistic is negative, that implies that the first population mean is smaller than the mean of the second one.

Example 6.7

Let the annual income level of professors at a state university follow a normal distribution with unknown mean and a known standard deviation of \$5,000. Also, the annual income level of professors at the next door private college follows a normal distribution with unknown mean and a known standard deviation of \$50,000. A random sample of 100 professors from the state university indicates an average annual income of \$100,000, and a random sample of 50 professors from the private college indicates an average annual income of \$150,000. At the 5% level of significance, is there a difference between the mean annual incomes?

$$H_0: \mu_1 = \mu_2,$$

$$H_1: \mu_1 \neq \mu_2,$$

where μ_1 is the mean annual income for the state university professors and μ_2 is the mean annual income for the private college professors.

The critical value is $z_{\frac{\alpha}{2}} = 1.96$, and the test statistic is

$$Z_{Test} = \frac{100000 - 150000}{\sqrt{\frac{25000000}{100} + \frac{2500000000}{50}}} = -7.053456159.$$

Because the magnitude of the test statistic exceeds the critical value, we reject the null hypothesis, at the 5% significance level. That is we found evidence that the mean annual income levels are not the same, yet there is a 0.05 probability that this decision may be wrong.

Intuitively, the negative value of the test statistic indicates that the mean annual income for the state university professors is less than the mean annual income of their private college colleagues.

But, we also know that the variations among the annual income levels for the private college is larger than the state university income levels.

We can construct one-tailed hypothesis tests as well,

Case A

$$H_0: \mu_1 \geq \mu_2,$$

$$H_1: \mu_1 < \mu_2.$$

Case B

$$H_0: \mu_1 \leq \mu_2,$$

$$H_1: \mu_1 > \mu_2.$$

In these one-tailed hypothesis tests the test statistics will be the same as defined above, yet the critical z-variate would be z_α and the decision rule would be to reject the null hypothesis if the test statistic is less than the negative z-variate value, in the case A, and if the test statistic is greater than the z-variate value, in the case B.

Example 6.8

Same as the above example, yet this time we want to test our hunch that the mean annual income for the private college is higher than the mean average income for the state university.

$$H_0: \mu_1 \geq \mu_2,$$

$$H_1: \mu_1 < \mu_2.$$

The test statistic would be the same value as above, yet the critical value for this one-tailed test would be $z_\alpha = 1.65$. Since the test statistic is less than the negative of the critical value ($-7.05 < -1.65$), we reject the null hypothesis in favor of the alternative hypothesis, at the 5% level of significance.

In the case of unknown variances, we use consistent estimators to substitute the unknown variances, that is we may use the sample variances, S_1^2 and S_2^2 . Thus, for large samples the test statistic would be

$$Z_{Test} = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}. \quad (6.11)$$

In the special case, where the population variances are known and identical, say σ^2 , we have

$$E[X_1] - E[X_2] \sim \mathcal{N}(\mu_1 - \mu_2, \frac{\sigma^2}{n_1} + \frac{\sigma^2}{n_2}), \quad (6.12)$$

and the test statistic would be

$$Z_{Test} = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{\sigma^2}{n_1} + \frac{\sigma^2}{n_2}}} = \frac{\bar{X}_1 - \bar{X}_2}{\sigma \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}. \quad (6.13)$$

In case of unknown common variance, we can use an estimator, pooled sample variance (which is a weighted average of two sample variances where the weights are the degrees of freedoms)

$$S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{(n_1 - 1) + (n_2 - 1)}, \quad (6.14)$$

in order to estimate it. Thus, the test statistic would follow a Student's t-distribution with $n_1 + n_2 - 2$ degrees of freedom,

$$T_{Test} = \frac{\bar{X}_1 - \bar{X}_2}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}, \quad (6.15)$$

and the critical value can be found by using the suitable Student's t-distribution, for a desired level of significance.

Example 6.9

Let a random sample of 13 movie theaters indicates an average of \$1250000 revenue with a sample standard deviation of \$25000 for the opening weekend of an action flick. A random sample of 17 movie theaters also indicates an average of \$990000 revenue with a sample standard deviation of \$40000 for the opening weekend of a comedy movie, but the unknown variance of the weekend revenues are assumed to be the same. Is there a difference between the average revenues generated by these two movies, at their opening weekend for 5% significance level? Consider the hypotheses

$$H_0: \mu_1 = \mu_2,$$

$$H_1: \mu_1 \neq \mu_2,$$

where μ_1 is the average revenue generated by the action movie and μ_2 is the average revenue generated by the comedy movie.

$$\text{Test statistic is } T_{Test} = \frac{1250000 - 990000}{S_p \sqrt{\frac{1}{13} + \frac{1}{17}}} = 20.5246.$$

$$\text{Here, } S_p = \sqrt{\frac{(12 \times 625000000 + 16 \times 1600000000)}{13 + 17 - 2}} = 34382.30442.$$

Also, the critical point for this two-tailed test, at the 28 degrees of freedom, would be $t_{\frac{\alpha}{2}} = 2.0484$.

Since the test statistic exceeds the critical point, we reject the null hypothesis, at 5% level of significance.

6.3.2 Dependent Populations

These tests are of importance when we are looking into the effect of treatments on a population, thus we collect pre-treatment random samples from the population and measure the behavior of it. For example, when we have a pre-treatment sample $X = \{X_1, X_2, \dots, X_n\}$, then treat them and measure the post-treatment behavior of this random sample, say $Y = \{Y_1, Y_2, \dots, Y_n\}$.

In fact we have one random sample, a.k.a. paired sample, before and after the treatment, so that we may be able to deduce the treatment effect on the population.

If the treatment has no significant effect on the population, then before and after average behaviors of the population would be similar. Hence, the mean difference, say μ_D , would be zero. Otherwise, if there is a significant effect of the treatment, then μ_D would be significantly different than zero.

Let the difference between after and before treatment values be a normal random variable $D = Y - X$, such that $D_i = Y_i - X_i$, for $i = 1, 2, \dots, n$.

Is there a significant effect of the treatment?

In order to answer this question, we construct a hypothesis test,

$$\begin{aligned} H_0: \mu_D &= 0, \\ H_1: \mu_D &\neq 0. \end{aligned}$$

The significance level of this two-tailed hypothesis test, say α , is set by the experimenter. The test statistic is the weight of the empirical evidence that manifests itself in the observed average differences,

$$\bar{D} = \frac{1}{n} \sum_{i=1}^n (Y_i - X_i), \quad (6.16)$$

together with its sample variance,

$$S_D^2 = \frac{1}{n-1} \sum_{i=1}^n (D_i - \bar{D})^2, \quad (6.17)$$

where the test statistics follows a Student's t-distribution with $n-1$ degrees of freedom, such that

$$t_{Test} = \frac{\bar{D}}{\left(\frac{S_D}{\sqrt{n}}\right)} = \sqrt{n}(\bar{D}/S_D). \quad (6.18)$$

Hence, one can find the critical value from the Student's t-distribution with $n-1$ degrees of freedom, for a desired level of significance and decide whether

there is a significant effect of the treatment, or not, by using the decision rule for a two-tailed hypothesis test.

Example 6.10

Let there be twenty five randomly selected students who are going to take the Road Driving test for their driver's licenses. They are in a drivers ed class and they have been given driver knowledge points before and after their courses. The average improvement, based on these points, is 75 out of a possible 100 points with a standard deviation of 25 points per person.

At the 5% level of significance is there conclusive evidence that this drivers ed course is effective to teach? Then,

$$H_0: \mu_D = 0,$$

$$H_1: \mu_D \neq 0.$$

Here, $\alpha = 0.05$ and at the 24 degrees of freedom the critical value would be 2.064. Also, $t_{Test} = \frac{75}{(\frac{25}{5})} = 15$. According to the decision rule we reject the null hypothesis, since t_{Test} is greater than the critical value, at the 5% level of significance. Thus, we have evidence that the drivers ed course significantly improves the students' driving skills, with 0.95 probability.

6.4 Goodness of Fit Test

We construct a goodness of fit test to see how well our model fits into the reality and in general there are two different approaches to construct a goodness of fit test.

6.4.1 Chi-Square Test

Karl Pearson constructed the χ^2 test as a goodness of fit test.

Let $X_i \sim \mathcal{N}(\mu_i, \sigma_i^2)$, for $i = 1, 2, \dots, n$, be independent random variables. Therefore,

$$\sum_{i=1}^n \frac{(X_i - \mu_i)^2}{\sigma_i^2} \sim \chi^2(n). \quad (6.19)$$

Consider a sample space that has a partition of $A_1, A_2, \dots, A_k$ with a probability measure induced on each partition, $P(A_j) = p_j$, such that $\sum_{j=1}^k p_j = 1$ for all k classes (a.k.a. categories).

When we repeat the experiment, say n times, we may observe the frequency of each partition, denoted by $X_i = \text{frequency}(A_i)$ with $\sum_{i=1}^k X_i = n$.

Then construct the hypothesis test as follows:

$$\begin{aligned} H_0 : p_1 = p_{1e}, p_2 = p_{2e}, \dots, p_k = p_{ke}, \\ H_1 : H_0 \text{ is not true,} \end{aligned}$$

here p_j is the actual probability and p_{je} is the expected probability from the null hypothesis, for the partition A_j . The test statistic is

$$C = \sum_{j=1}^k \frac{(f_{jo} - f_{je})^2}{f_{je}} \sim \chi^2(k-1), \quad (6.20)$$

with f_{jo} as the observed frequency and f_{je} as the expected frequency from the null hypothesis, of the partition A_j .

Thus, there is a critical point $\chi_\alpha^2(df)$ with $df = k - 1$, such that

$$P(\chi_\alpha^2(df)) = P(\chi^2 \leq \chi_\alpha^2(df)) = 1 - \alpha.$$

Hence, we can construct a decision rule; reject H_0 if the test statistics exceeds the critical point, $C > \chi_\alpha^2(k-1)$.

Example 6.11

Let us have a die with six faces (six categories) and we believe that it is a fair die, i.e. the face distribution is uniform. Assume that we rolled it sixty times and observed the frequencies for each category; one 5 times, two 5 times, three 10 times, four 10 times, five 15 times, and six 15 times. We wish to test how close our belief is to the reality, thus we construct a χ^2 -test with 5% level of significance, under the weight of the empirical evidence, as follows:

$$\begin{aligned} H_0 : p_1 = p_2 = \dots = p_6 = \frac{1}{6}, \\ H_1 : \text{The die is not fair.} \end{aligned}$$

Here, $\alpha = 0.05$ and $k = 6$, therefore $df = 5$ and the critical point would be $\chi_{0.05}^2(5) = 11.0705$, also under the null hypothesis we have $p_{je} = \frac{1}{6}$, thus the expected frequency for each category is $f_{je} = (p_{je})(60) = 10$, for $j = 1, 2, \dots, 6$.

The weight of the evidence is the test statistic:

$$C = \frac{(5-10)^2}{10} + \frac{(5-10)^2}{10} + \frac{(10-10)^2}{10} + \frac{(10-10)^2}{10} + \frac{(15-10)^2}{10} + \frac{(15-10)^2}{10} = 10.$$

By employing the decision rule, we see that the test statistic does not exceed the critical point, thus we cannot reject the null hypothesis at the given significance level. Hence, we don't have sufficient evidence to reject the fairness of the die assumption.

On the other hand, if there are unknown parameters that has to be used for the χ^2 -test, then the degrees of freedom would be reduced by the number

of estimated parameters, denoted by $\#\hat{\theta}$, by using the random sample, that is $df = k - 1 - \#\hat{\theta}$.

Example 6.12

Assume that $Y \sim \mathcal{N}(\mu, \sigma^2)$ be the grades of a class with unknown μ and unknown σ^2 . We want to assign letter grades on the *curve*, i.e. on the normal distribution, such that

LETTER GRADE	Y
A	90 or more
B ⁺	from 80 to 89
B	from 70 to 79
C	from 60 to 69
C ⁻	from 50 to 59
D	from 40 to 49
F	less than 40

Since we have seven grade categories and two unknown parameters to be estimated by the random sample, the degrees of freedom would be four, $df = 7 - 1 - 2$.

Now, also assume that one hundred students have an average grade of 64 out of 100, with a standard deviation of 10.

Thus we can estimate the unknown μ by the sample average of 64 and also estimate the variance by the sample variance of 100.

Since $f_{ej} = n \cdot p_{ej}$, for $j = 1, 2, \dots, k$, we can generate the expected frequencies for the grade categories, under the null hypothesis, by using the normal distribution with a mean of 64 and with a variance of 100.

Thus, for a normal random variable $Y \sim \mathcal{N}(64, 100)$ we may find the probabilities, i.e. proportions, of the letter grades,

$$\begin{aligned}
 P(A) &= P(Y \geq 90) = 0.004661188, \\
 P(B^+) &= P(80 \leq Y \leq 89) = 0.050138104, \\
 P(B) &= P(70 \leq Y \leq 79) = 0.219453826, \\
 P(C) &= P(60 \leq Y \leq 69) = 0.381168624, \\
 P(C^-) &= P(50 \leq Y \leq 59) = 0.263821599, \\
 P(D) &= P(40 \leq Y \leq 49) = 0.072559123, \\
 P(F) &= P(Y < 40) = 0.008198.
 \end{aligned}$$

Hence, the expected frequency under the null hypothesis, for the letter grades, can be computed as,

$$\begin{aligned}
 f_{eA} &= (100)(0.004661188) = 1, \\
 f_{eB^+} &= (100)(0.050138104) = 5,
 \end{aligned}$$

$f_{eB} = (100)(0.219453826) = 22,$
 $f_{eC} = (100)(0.381168624) = 38,$
 $f_{eC^-} = (100)(0.263821599) = 26,$
 $f_{eD} = (100)(0.072559123) = 7,$
 $f_{eF} = (100)(0.008198) = 1.$ Now, assume that the actual observed frequencies for these grade categories would be

$$\begin{aligned}
 f_{oA} &= 10, \\
 f_{oB^+} &= 15, \\
 f_{oB} &= 15, \\
 f_{oC} &= 25, \\
 f_{oC^-} &= 25, \\
 f_{oD} &= 5, \\
 f_{oF} &= 5.
 \end{aligned}$$

Thus, we can construct a χ^2 -test for the normality of the grade distribution, at the 5% significance level.

$$\begin{aligned}
 H_0 &: Y \sim \mathcal{N}(64, 100), \\
 H_1 &: H_0 \text{ is false.}
 \end{aligned}$$

At $\alpha = 0.05$, $k = 7$ and $\#\hat{\theta} = 2$, we have $df = 7 - 1 - 2 = 4$ and the critical point is $\chi_{0.05}^2(4) = 9.4877$.

Thus we construct the test statistic;

$$\frac{(10-1)^2}{1} + \frac{(15-5)^2}{5} + \frac{(15-22)^2}{22} + \frac{(25-38)^2}{38} + \frac{(25-26)^2}{26} + \frac{(5-7)^2}{7} + \frac{(5-1)^2}{1} = 124.2845313.$$

According to the decision rule, since the test statistic exceeds the critical point, we reject the null hypothesis at the 5% significance level, thus the evidence indicates that the grades do not follow the assumed normal distribution.

6.4.2 Contingency Tables

The chi-square test is also suitable to resolve the question of independence of two random variables.

Let there are two factors, say X and Y , such that they collectively generate an event, yet it is not known that these two factors are dependent, or independent, to each other.

Assume that n observations were taken from a sample space, S , which was partitioned by the mutually exclusive categories of X and Y , $\{X_1, X_2, \dots, X_r\}$ and $\{Y_1, Y_2, \dots, Y_c\}$, such that $X_i \cap X_j = \emptyset$ and $Y_i \cap Y_j = \emptyset$, for all $i \neq j$. Here, $\bigcup_{i=1}^r X_i = S$ and $\bigcup_{j=1}^c Y_j = S$.

Also let Q_{ij} be the number of observations, due to X_i and Y_j . We want to find out whether the X_i s and the Y_j s are independent or not.

The following contingency table, Table 6.2, indicates the two factors of the events, X and Y , and the number of observations, where q_{ij} denotes the observed numbers of the Q_{ij} .

Factor	Y_1	$\dots$	Y_c	Row Total
X_1	q_{11}	$\dots$	q_{1c}	R_1
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
X_r	q_{r1}	$\dots$	q_{rc}	R_r
Column Total	C_1	$\dots$	C_c	n

Table 6.2: Contingency Table

If the two random variables, X and Y , are independent we have $P(X_i \cap Y_j) = P(X_i)P(Y_j)$, thus the expected frequency of Q_{ij} is $nP(X_i)P(Y_j)$. Hence, the estimated frequency for Q_{ij} , when the factors are independent, would be

$$n\left(\frac{R_i}{n}\right)\left(\frac{C_j}{n}\right) = \frac{R_i C_j}{n}, \text{ for } i = 1, 2, \dots, r \text{ and } j = 1, 2, \dots, c.$$

Therefore, in each interaction of these factors in the row-column combination of the contingency table, we have an observed and an estimated frequency, due to the independence assumption.

We have a chi-square hypothesis test suitable to test the independence of X and Y , with a test statistic

$$C = \sum_{i=1}^r \sum_{j=1}^c \frac{(Q_{ij} - nP(X_i)P(Y_j))^2}{nP(X_i)P(Y_j)}, \quad (6.21)$$

in which significantly large values of the test statistic would be taken up as evidence against the independence hypothesis.

In this construct, the degrees of freedom would be equal to the number of categories, rc , minus one, minus the number of estimated parameters, $(r-1)+(c-1)$.

When $(n-1)$ of the $P(X_i)$ s are estimated, we can find the remaining estimated value by using $\sum_{i=1}^r P(X_i) = 1$. Similarly for the estimated values of $P(Y_j)$ s.

Therefore, $df = rc - 1 - ((r-1) + (c-1)) = rc - 1 - r + 1 - c + 1 = rc - r - c + 1$, which can be organized as $df = (r-1)(c-1)$. Show this equality.

The hypothesis test for the independence of two factors would be,

H_0 : X and Y are independent random variables,

H_1 : H_0 is false.

The test statistic, based on a realization, is

$$C = \sum_{i=1}^r \sum_{j=1}^c \frac{(q_{ij} - \frac{R_i C_j}{n})^2}{\frac{R_i C_j}{n}}. \quad (6.22)$$

The critical point, for a desired level of significance, is $\chi_\alpha^2(df)$ with $df = (r-1)(c-1)$, such that $F(\chi_\alpha^2(df)) = P(\chi^2 \leq \chi_\alpha^2(df)) = 1 - \alpha$.

Hence, we can construct a decision rule; reject H_0 if the test statistics exceeds the critical point, $C > \chi_\alpha^2(df)$.

Note. We provide robustness to the chi-square test, by combining categories, when a category has less than five observations in it, and making extension of categories such that each has five or more observations in it (re-categorize the sample space).

6.4.3 Kolmogorov-Smirnov Test

There is another approach to construct a hypothesis test for the assumed distribution of a random variable, designed by Kolmogorov [7]. The idea was to design a distance metric between the empirical and the theoretical probability distribution, $\|F_X - F_X^e\|$, where the F_X denotes the assumed theoretical cdf and the F_X^e is the empirical cdf of a random variable X , i.e. the cumulative relative distribution function, such that reality manifests itself in the empirical cumulative distribution functions, when we have n ordered-data points, $X_{[1]}, \dots, X_{[n]}$,

$$F_X^e = \frac{\#\{X \leq X_{[i]}\}}{n}. \quad (6.23)$$

$\#\{X \leq X_{[i]}\}$ denotes the number of points less than or equal to $X_{[i]}$, for $i = 1, \dots, n$, and it is obvious that this is a step function that jumps by an amount of $\frac{1}{n}$ for each point, at the ordered data points.

By the law of large numbers, the empirical cdf converges to the cdf of the random variable, as n gets larger.

Although the Kolmogorov-Smirnov test does not need a special probability distribution for the test statistic, the distribution must be fully declared.

Kolmogorov-Smirnov goodness-of-fit test structure:

H_0 : Data set has the cdf F_X ,

H_1 : H_0 is false.

Test statistic:

$$D = \sup_{i \in [1, n]} \{|F_{X_{[i]}} - F_{X_{[i]}}^e|\}. \quad (6.24)$$

Decision rule: Reject H_0 , if the test statistic (distance between reality and the null hypothesis assumption) exceeds the critical point for a desired level of significance, α , that is $D > \frac{D_\alpha}{\sqrt{n}}$,

with $P(D \leq D_\alpha) = 1 - \alpha$, and

$$F_D(x) = \frac{\sqrt{2\pi}}{x} \sum_{i=1}^{\infty} e^{-[(2i-1)\pi]^2/8x^2}.$$

6.5 Power of a Test

The performance of a test is measured by the probability of Type-I error, α , and the probability of Type-II error, β . In fact, α is selected before the test begins, based on the desired risk level, or, rejection region for H_0 .

In this context, a measure of the test performance would be the probability of rejecting the null hypothesis, known as the power of the hypothesis test, obviously related to α .

Let

$$H_0 : \theta = \theta_0,$$

$$H_a : \theta = \theta_a,$$

be the test hypotheses.

Thus, the power of the test is defined as

$$power = P(\text{Rejecting } H_0 \mid \theta = \theta_a). \quad (6.25)$$

On the other hand,

$$\beta = P(\text{Accepting } H_0 \mid \theta = \theta_a). \quad (6.26)$$

Hence, power of the test is $1 - \beta$.

If a sample is coming from a population with parameter θ , a hypothesis would be called a simple hypothesis, when this statement completely specifies the distribution. Otherwise, an hypothesis would be called a composite hypothesis.

6.5.1 Neyman-Pearson Lemma

If we test the simple null hypothesis, $H_0 : \theta = \theta_0$ versus the simple alternative hypothesis, $H_a : \theta = \theta_a$, based on a sample $X_1, \dots, X_n$, coming from a population with parameter θ , then we can construct a likelihood ratio of the sample, under these two parameter specifications,

$$\frac{L(\theta_0)}{L(\theta_a)},$$

such that a test that maximizes the power, for a fixed α , would have the rejection region

$$\left\{ \frac{L(\theta_0)}{L(\theta_a)} < c \right\},$$

where the value of c is selected for the desired value of α .

This test is the most powerful test at the α level.

Example 6.13

Let X be a single observation, coming from a population subject to a pdf, $f(x|\theta) = \theta x^{\theta-1}$, for $0 < x < 1$, and $f(x|\theta) = 0$, elsewhere.

Consider $H_0 : \theta = 1$, versus $H_a : \theta = 2$, at $\alpha = 0.05$, the most powerful test has the rejection region,

$$\frac{L(\theta_0)}{L(\theta_a)} < c = \frac{f(x|\theta_0)}{f(x|\theta_a)} < c = \frac{1}{2x} < c, \text{ for } 0 < x < 1.$$

This implies that the rejection region for H_0 is $R = \left\{ \frac{1}{2c} < x \mid 0 < x < 1 \right\}$,

$$\text{and } \alpha = P(X \in R \mid \theta = 1) = \int_{\frac{1}{2c}}^1 1 dx = 1 - \frac{1}{2c} = 0.05,$$

$$\text{or, } c = \frac{1}{(2)(0.95)} = 0.526316.$$

This test would have the maximum power,

$$P(X \in R \mid \theta = 2) = \int_{0.95}^1 2x dx = 1 - (0.95)^2 = 0.0975.$$

This degree of “maximal” power may be unacceptably low by many, thus it has to be increased, but how? The only feasible way to do so is to increase the sample size from one to many observations.

If the value assigned to the parameter, under the alternative hypothesis, does not change the test statistic, nor the rejection region, for a fixed α level, then the most powerful test would be the uniformly most powerful test, for the $H_0 : \theta = \theta_0$ versus $H_a : \theta \neq \theta_0$.

6.6 Likelihood Ratio Test

Let the likelihood function $L(\Theta) = L(\theta_1, \dots, \theta_k | x_1, \dots, x_n)$ be a function of the parameter vector, $\Theta = (\theta_1, \dots, \theta_k)$, for a given sample realization $x_1, \dots, x_n$.

If we are interested only in one parameter, without loss of generality, say θ_1 , we can take the others as nuisance parameters, thus the likelihood function is a function of the target parameter and the nuisance parameters. Suppose the null hypothesis declares that $H_0 : \Theta \in \Omega_0$ and the alternative hypothesis declares $H_a : \Theta \in \Omega_a$, such that the parameter space Ω is partitioned into Ω_0 and Ω_a , two mutually exclusive sub-spaces, and $\Omega = \Omega_0 \cup \Omega_a$.

Let $L(\hat{\Omega}_0)$ be the $\sup_{\Theta \in \Omega_0} L(\Theta)$, and $L(\hat{\Omega})$ be the $\sup_{\Theta \in \Omega} L(\Theta)$.

If $L(\hat{\Omega}_0) = L(\hat{\Omega})$, then the best explanation for the observed sample is provided by the H_0 .

On the other hand, if $L(\hat{\Omega}_0) < L(\hat{\Omega})$, then the best explanation for the given sample would be outside of the null hypothesis. Here, the $\sup$ denotes the supremum, or the least upper bound, and it may be equal to the maximum.

Now, we can build a likelihood ratio test, based on the ratio of

$$\lambda = \frac{L(\hat{\Omega}_0)}{L(\hat{\Omega})}, \quad (6.27)$$

and with a rejection region of $\{\lambda < c\}$, for $0 < \lambda < 1$. (Why is this likelihood ratio less than one?)

6.6.1 Asymptotic Likelihood Ratio

Let $X_1, \dots, X_n$ be a random sample with a likelihood function $L(\Theta)$. Also, let r_0 be the number of free parameters declared by H_0 , and r be the number of free parameters declared by H_a . Thus, for a large sample size, we have $-2 \log(\lambda)$ subject to a chi-square distribution with $r_0 - r$ degrees of freedom.

Here, the likelihood ratio test has a rejection region $R = \{\lambda < c\}$, thus this is equivalent to $R = \{-2 \log(\lambda) > c^*\}$. That is to say, if we select an α significance level for our test, then $c^* = \chi^2_{\alpha}$, with $r_0 - r$ degrees of freedom.

6.7 Exercises

1. A random sample of 100 death records during the past year indicates an average lifetime of 69 years. Assume a population standard deviation of 10 years for lifetime, at the 5% level of significance, can we say that the average lifetime is shorter than 70 years?
2. GE manufactures light bulbs and claims a normally distributed lifetime of 1000 hours with a standard deviation of 100 hours. Test this claim when a random sample of 100 bulbs have an average lifetime of 888 hours, at the 5% level of significance.
3. Standard test designers believe that the time required to finish their test is a normal random variable with a mean of 55 minutes. A random sample of 25 test takers yields an average time of 44 minutes with a standard deviation of 5 minutes. At the 5% level of significance, is the average test time shorter than 55 minutes?
4. Assume that a random sample of 500 associate professors from research institutions has an average annual income of \$70,000 with a standard deviation of \$10,000. Also, assume that a random sample of 500 associate professors from non-research institutions has an average annual income of \$60,000 with a standard deviation of \$5,000. At the 5% level of significance can we say that the average annual income of associate professors for the research institutions is \$10,000 higher than the others?
5. A marketer believes that 50% of the customers prefer blue color cars. If 900 out of 2000 cars are blue colored, at the 1% level of significance what can you say about the marketer's belief?
6. At a small liberal arts college, it is believed that 60% of the students ride bicycles to lectures. At the 5% level of significance, if a random sample of 100 students yields 45 of them riding a bicycle, what can you say about this common belief ?
7. It is believed that the time to complete college education is a normal random variable with a standard deviation of two years. Test this hypothesis at the 5% level of significance, if a random sample of 36 college graduates yields a standard deviation of 1.25 years.

8. A random sample of 100 people, categorized according to sex and level of hours of watching TV during a week, is as follows:

Degree:	High	Low	Right Margin
Female	25	30	55
Male	25	20	45
Bottom Margin	50	50	100

Table 6.3: Contingency Table for Exercise 8

At the 5% level of significance, can you say that the degree of watching TV time is independent of sex?

9. The grade distribution for a course were as follows:

LETTER GRADE	FREQUENCY
<i>A</i>	10
<i>B</i> ⁺	11
<i>B</i>	15
<i>C</i>	25
<i>C</i> ⁻	13
<i>D</i>	12
<i>F</i>	14

At the 5% level of significance, test the hypothesis that the grade distribution is uniform.

10. Repeat the Example 6.11, by using the Kolmogorov-Smirnov test.
11. Repeat the Example 6.12, by using the Kolmogorov-Smirnov test.
12. Search the R software for the above mentioned hypothesis tests, and repeat the examples by using the R software, whenever possible.

Chapter 7

Nonparametric Statistics

7.1 Introduction

The common assumption that the data set consists of normal observations is the most convenient approach to assume a statistical model, in order to perform data analysis.

We have to ask ourselves, what if this model is wrong?

The answer depends upon what you want. You may wish to construct an instrument that is robust enough to carry the analysis of the data set, even if the assumptions are not satisfied anymore, or use a test statistic such that its probability law stays the same, regardless of the population behavior.

This second approach is known as distribution free, or nonparametric statistics.

A parametric hypothesis test has at least one nonparametric analogue. Hence, which nonparametric test to choose is a primary consideration of the decision maker.

Let us begin with location tests.

One constructs a point estimator and a confidence interval, but the estimator is not distribution free, therefore the term “rank based” is commonly used for these tests.

Without loss of generality, we will consider one sample, generated by a continuous random variable X . We will look at a parameter as a function of the cdf.

For example, consider the expected value of X , also known as the mean of X , as a parameter

$$\mu_X = g(F_X) = \int_{-\infty}^{\infty} x dF_X(x) = \int_{-\infty}^{\infty} x f_X(x) dx.$$

Or, similarly, consider the median of X , as a parameter, m_X , where

$$m_X = h(F_X) \text{ such that } F_X(m_X) = \frac{1}{2}, \text{ thus } m_X = F_X^{-1}(\frac{1}{2}).$$

Hence, in this context, a parameter is a composite function, or a functional (function of a function), of the random variable X .

Let $X_1, \dots, X_n$ be a random sample. Also, define the empirical cumulative distribution function of the sample,

$$\hat{F}_n(x) = F_X^e(x), \text{ for } -\infty < x < \infty.$$

Let T be a functional. Since $\hat{F}_n(x)$ is a cdf, $T(\hat{F}_n)$ is a well defined parameter, depending upon the sample. Therefore, $T(\hat{F}_n)$ is a statistic.

Let X be a continuous random variable. Also, let $0 < q < 1$ be a proper positive proportion, such that for a fixed q ,

$$F(x_q) = P(X \leq x_q) = q, \quad (7.1)$$

has a unique solution for x_q . Here, x_q is called the q th quantile of this probability distribution (similar to the q th percentile).

7.1.1 Quantile as a Parameter

The sentence, $F(x_{\frac{1}{2}}) = \frac{1}{2}$, implies that the median is the $\frac{1}{2}$ th quantile, or, the 50th percentile.

If we work with ordered data sets and the quantile is the parameter of interest, then the functions of order statistics would be suitable to conduct nonparametric inferences.

In order to construct a confidence interval for x_q , we collect a random sample, $X_1, \dots, X_n$, and generate an order statistic of the sample, $Y_1, \dots, Y_n$.

Let Y_k be the number of successes for the k th ordered observation, in n identical and independent Bernoulli trials with a common success probability for each trial.

Also, let $\{Y_i < x_q < Y_j\}$ be the event such that in these trials, there would be at least Y_i and at most Y_j number of successes, thus

$$P(Y_i < x_q < Y_j) = \sum_{k=i+1}^{j-1} \binom{n}{n-k} q^k (1-q)^{n-k}. \quad (7.2)$$

That means that $P(Y_i < x_q < Y_j) = 1 - \alpha$ is the probability measure that the random interval (Y_i, Y_j) would capture the q th quantile, which would yield a $100(1 - \alpha)\%$ confidence interval of x_q , for a given realization of (y_i, y_j) .

Example 7.1

Let a random sample of size 50, coming from a continuous probability distribution, yield an estimator of the median, $x_{0.5}$ th quantile, i.e., $q = P(X \leq x_{0.5})$, then the twentyfifth order statistics (since $nq = (50)(0.5) = 25$), y_{25} , would be the point estimator for $x_{0.5}$.

Thus, in order to build a confidence interval for $x_{0.5}$, take two order statistics; one less than y_{25} , say y_{10} , and one greater than y_{25} , say y_{40} .

Therefore, (y_{10}, y_{40}) would be a $100(1 - \alpha)\%$ confidence interval of $x_{0.5}$, at a confidence level generated by this sample,

$$P(Y_{10} < x_{0.5} < Y_{40}) = \sum_{k=11}^{39} \binom{50}{50-k} (0.5)^k (0.5)^{50-k} = 1 - \alpha.$$

Conversely, when a desired confidence level is selected, the corresponding order statistics, as limits of a symmetric interval which is centered at the point estimator, can be found. Moreover, for large samples, an approximation to the standard normal distribution would simplify computations.

This indicates that we can compute the probability measure for a random interval which includes a selected percentage of the distribution.

Let X be a continuous random variable and let $Z = F(X)$ be another random variable with the pdf $h(z) = 1$, for $0 < z < 1$ and $h(z) = 0$, otherwise.

Thus, if $0 < p < 1$, then $P(F(X) \leq p) = \int_0^p dz = p$.

Because $P(X = x) = 0$ and $F(x) = P(X \leq x)$, $F(x)$ becomes the fraction of the probability distribution of X , from $-\infty$ to x .

On the other hand, $P(F(X) \leq p) = p$, i.e., p is the proportion of the probability distribution, which is in between $-\infty$ and X , or, the random interval $(-\infty, X)$ which captures $100p\%$ of the distribution has a probability measure of p .

Let $X_1, \dots, X_n$ be a random sample, coming from a positive continuous distribution. Now, $F(X_1), \dots, F(X_n)$ are independent random variables with an identical uniform distribution over $(0, 1)$. Therefore, $F(X_1), \dots, F(X_n)$ is a random sample, coming from $U(0, 1)$.

Let $Z_1, \dots, Z_n$ be the ordered set of these $F(X_i)$ s. If $Y_1, \dots, Y_n$ is the order statistics of the random sample $X_1, \dots, X_n$, then $Z_i = F(Y_i)$.

Hence, the pdf of $Z = (Z_1, \dots, Z_n)$ would be $h(z_1, z_2, \dots, z_n) = n!$, for $0 < z_1 < z_2 < \dots < z_n < 1$, and $h(z) = 0$, otherwise.

In general, we have the following theorem.

Order Statistics Theorem 1. Let $Y_1, \dots, Y_n$ be the order statistics of a random sample of size n , coming from a continuous distribution, with a cdf $F(X)$.

The joint pdf of $Z_i = F(Y_i)$, for $i = 1, 2, \dots, n$, would be

$h(z_1, \dots, z_n) = n!$, for $0 < z_1 < \dots < z_n < 1$,

$h(z_1, \dots, z_n) = 0$, otherwise.

Example 7.2

Find the smallest integer, n , which satisfies the probability:

$P(Y_1 < x_{0.5} < Y_n) \geq 0.99$, for the order statistics, Y_i s, that are coming from a continuous distribution with a sample size of n .

Since $P(Y_1 < x_{0.5} < Y_n) = \sum_{i=2}^{n-1} \binom{n}{i} (0.5)^i (1 - 0.5)^{n-i} = 1 - 2(0.5)^n$,

we have $1 - 0.99 \geq 2(\frac{1}{2})^n$, that implies $2^{n-1} \geq \frac{1}{0.01}$, or, $2^{n-1} \geq 100$.

Hence, we have either $n = 7$, or, $n = 8$.

Now, let us consider the difference $Z_j - Z_i = F(Y_j) - F(Y_i)$, for $i < j$.

Since $P(X = y_j) = 0$, $P(X = y_i) = 0$, and

$F(y_j) = P(X \leq y_j)$, $F(y_i) = P(X \leq y_i)$,

then, the probability that the event X is in between y_i and y_j would be

$F(y_j) - F(y_i)$.

Let p be a proper percentage (i.e. p is different than zero percent), thus $F(y_j) - F(y_i) \geq p$ implies that at least $100p\%$ of the probability distribution is in between y_i and y_j .

In addition, let $\gamma = P(F(Y_j) - F(Y_i) \geq p)$, then the random interval (Y_i, Y_j) has the probability γ , for capturing at least $100p\%$ of the distribution of X .

If y_i and y_j are realizations of Y_i and Y_j , then the interval (y_i, y_j) is called the $100\gamma\%$ tolerance interval, for the $100p\%$ of the probability distribution of X .

To compute this probability, we observe

$\gamma = P(F(Y_j) - F(Y_i) \geq p)$ and

$Z_i = F(Y_i)$, $Z_j = F(Y_j)$, thus

$\gamma = P(Z_j - Z_i \geq p) = \int_0^{1-p} \int_{p+z_j}^1 h_{ij}(z_i, z_j) dz_j dz_i$,

which is a cumbersome computation, to put it mildly.

Thus, we want to simplify our task and we introduce the coverage of a random interval.

Let

$$W_1 = F(Y_1) = Z_1,$$

$$W_2 = F(Y_2) - F(Y_1) = Z_2 - Z_1,$$

$$W_3 = F(Y_3) - F(Y_2) = Z_3 - Z_2, \dots, W_n = F(Y_n) - F(Y_{n-1}) = Z_n - Z_{n-1}.$$

Here, the random variable W_i is called the coverage of the random interval $\{x \mid x \in (Y_{i-1}, Y_i)\}$, for $i = 2, \dots, n$ and W_1 is the coverage of the random interval $\{x \mid -\infty < x < Y_1\}$.

Observe that, we have

$$Z_1 = W_1, Z_2 = W_1 + W_2, Z_3 = W_1 + W_2 + W_3, \dots, \text{ and } Z_n = \sum_{i=1}^n W_i.$$

The joint pdf of $Z_1, Z_2, \dots, Z_n$ would be $n!$,

for $0 < z_1 < \dots < z_n < 1$, and zero, everywhere else.

Therefore, the joint pdf of coverages is,

$$k(w_1, \dots, w_n) = n!,$$

for $0 < w_i, i = 1, 2, \dots, n$, such that $\sum_{i=1}^n w_i < 1$,

and zero, everywhere else.

The joint pdf, $k(w)$, is symmetric in $w_1, \dots, w_n$, thus the distribution of every sum of the coverages, $\sum_{i=1}^r W_i$, for $r < n$, would be the same.

Let $r = j - i$, for $i < j$.

The distribution of

$$Z_j - Z_i = F(Y_j) - F(Y_i) = W_{i+1} + \dots + W_j,$$

is the same as of

$$Z_{j-i} = F(Y_{j-i}) = W_1 + \dots + W_{j-i}.$$

On the other hand, the distribution of Z_{j-i} is the beta distribution with pdf,

$$h_{j-i}(s) = \frac{\Gamma(n+1)}{\Gamma(j-i)\Gamma(n-j+i+1)} s^{j-i-1} (1-s)^{n-j+i}, \quad (7.3)$$

for $0 < s < 1$, and $h_{j-i}(s) = 0$, everywhere else.

Therefore, $F(Y_j) - F(Y_i)$ is also subject to this pdf.

Hence, $P(F(Y_j) - F(Y_i) \geq p) = \int_p^1 h_{j-i}(s) ds$.

Since $W_1 = Z_1 = F(Y_1)$ has a pdf, $k_1(w) = n(1-w)^{n-1}$, for $0 < w < 1$, and $k_1(w) = 0$, everywhere else, each of the coverages, W_i , for $i = 1, \dots, n$ has the same pdf, $k_1(w)$.

Hence,

$$E[W_i] = \int_0^1 wn(1-w)^{n-1}dw = \frac{1}{n+1}, \quad (7.4)$$

for $i = 1, \dots, n$.

This indicates that the order statistics partition the distribution into $n+1$ parts with equal expectations, and $E[F(Y_j) - F(Y_i)] = \frac{j-i}{n+1}$, for $i < j$.

That is why Y_k is called the 100pth percentile of the sample, since

$$E[F(Y_k)] = \frac{k}{n+1} = \frac{(n+1)p}{n+1} = p. \quad (7.5)$$

The testing of a hypothesis also benefits from nonparametric approach, where the tester selects a significance level, α , and analyses the data set, in order to choose a test statistic.

Suppose there are multiple possible test statistics, $T_1, T_2, \dots, T_m$, together with their corresponding rejection regions, $C_1, C_2, \dots, C_m$, such that

$$\alpha = P(T_i \in C_i \mid H_0 \text{ is true}), \text{ for } i = 1, 2, \dots, m.$$

In addition, let a statistic U , based on the same data set, select one of these T_i s, for testing the H_0 , where the selection rule is based on mutually exclusive, yet collectively exhaustive events; $V_1, \dots, V_m$, such that when $U \in V_i$, then we select the test statistics T_i , for $i = 1, 2, \dots, m$.

Moreover, for all i , we pick a U , such that U and T_i are conditionally independent, for the given true H_0 .

Therefore, the rejection probability of H_0 would be as follows:

$$\begin{aligned} \sum_{i=1}^m P(U \in V_i \cap T_i \in C_i) &= \sum_{i=1}^m P(U \in V_i)P(T_i \in C_i) \\ &= \sum_{i=1}^m P(U \in V_i)\alpha = \alpha \sum_{i=1}^m P(U \in V_i) = \alpha. \end{aligned}$$

Hence, this procedure preserves the desired significance level, α .

The common way of finding a proper U , which is independent of every test statistic T_i , is to use complete sufficient statistics, for a given H_0 . That would have distributions free from the parameters.

It is known that the order statistics, $Y_1 < \dots < Y_n$, of a random sample of size n , coming from a continuous distribution, are sufficient statistics.

Moreover, the family of joint probability distributions is complete, when this family of distributions includes all continuous distributions. Therefore, order statistics are complete sufficient statistics for the parameters of F .

In the context of nonparametric methods the term parameter is confusing, which is the main reason why many prefer to use the term distribution-free.

The order statistics, under H_0 , would be the basis for constructing a selector statistic U . The trade-off for the distribution-free tests is in between the preserved significance level and the power of the test. Thus, adapting the test to the data set (selecting a proper test, based on the data) should increase the power of the test.

Moreover, a hypothesis test that preserves the level of significance, for many different underlying distributions, is known as a robust test.

7.2 Sign Test

Let $F(x)$ be the unknown cdf of a random variable X . We want to test the null hypothesis that the quantile of the distribution of X would be p_0 , versus a one-sided alternative hypothesis (rather than two-sided alternatives) that this particular quantile is greater than p_0 , such that

$$H_0: F(x_{p_0}) = p_0,$$

$$H_1: F(x_{p_0}) > p_0.$$

One approach is to construct a hypothesis test, based on the number of *successes*, as a random variable Y which is the number of observations less than or equal to x_{p_0} , in a sample of size n .

Thus, if H_0 is true, then Y is a binomial random variable with the probability of success p_0 , i.e. $Y \sim B(n, p_0)$.

As a decision rule, reject H_0 if and only if the observed value of y exceeds a critical integer value c , such that $P(Y \geq c \mid H_0) = \alpha$.

Similarly, if a one-sided alternative is set to $H_1: F(x_{p_0}) < p_0$, then the critical region would be $\{y \mid y \leq c\}$.

When we have a random sample of size n , say $X_1, X_2, \dots, X_n$, then we construct $X_1 - x_{p_0}, X_2 - x_{p_0}, \dots, X_n - x_{p_0}$, and the test statistic would be the number of nonpositive signs of this set, that is why this construct is commonly known as the sign test.

7.3 Wilcoxon Test

If we have a symmetric continuous distribution, then the median is equal to the mean of this distribution. Hence, we could use this property of symmetric distributions as follows.

Assume that we wish to test a hypothesis about the mean,

$$H_0: \mu = \mu_0,$$

based on a random sample of size n , such as $X_1, \dots, X_n$, coming from a symmetric and continuous distribution.

Construct the absolute deviations from μ_0 , such that

$$|X_1 - \mu_0|, \dots, |X_n - \mu_0|,$$

and then sort them from the smallest value to the largest value.

This generates a relative *rank*, for the sample observations from the hypothesized mean. Denote this rank as R_k for the observation X_k and also keep track of the side of the deviation from μ_0 , by using an indicator function, such that

$$I_k = 1, \text{ if } X_k > \mu_0, \text{ and } I_k = 0, \text{ otherwise.}$$

Therefore, for each observation we have a pair of *rank* and *indicator* to identify which side of the μ_0 it is located.

The test statistic is constructed as a linear combination of the indicator functions, weighted by their ranks, such that

$$W = \sum_{k=1}^n R_k I_k, \quad (7.6)$$

which is indeed the sum of all ranks above the hypothesized mean.

Here, observe that W ranges from 0 to $\frac{n(n+1)}{2}$.

Thus, W takes values from all observations that are below μ_0 to all observations that are above μ_0 .

Since X_k s are independent random variables, R_k s are independent random variables and also,

$$P(R_k I_k = k) = P(R_k I_k = -k) = \frac{1}{2}.$$

The moment generating function of W is

$$M(t) = E[\exp(t \sum_{k=1}^n R_k I_k)] = \prod_{k=1}^n E[e^{t R_k I_k}] = \prod_{k=1}^n \frac{e^{-kt} + e^{kt}}{2}. \quad (7.7)$$

Consequently, if H_0 is true, then the mean of W would be

$$E[W] = \frac{n(n+1)}{4}, \text{ with a variance of } Var(W) = \frac{n(n+1)(2n+1)}{24}.$$

The distribution of W is tedious to work with, thus we seek simplifications.

Since W is a sum of independent random variables and even if they are not identically distributed, we have an asymptotic relationship between the sample size and the test statistic's probability law, such that

$$\lim_{n \rightarrow \infty} \frac{W - E[W]}{\sqrt{Var(W)}} \xrightarrow{D} \Phi(0, 1). \quad (7.8)$$

Now, we can construct the Wilcoxon Signed Rank Test, as described below.

Let $X_1, \dots, X_n$ be a random sample, coming from a symmetric and continuous distribution with a mean μ .

Hypothesis test for the mean:

$$H_0: \mu = \mu_0$$

$$H_1: \mu \neq \mu_0.$$

Test Statistic: W , as given in Equation (7.6).

Decision Rule: If $W \leq c$, or, $W \geq \frac{n(n+1)}{2} - c$, then reject H_0 .

The significance level is given as $2P(W \leq c) = 2 \sum_{i=0}^c P(W = i)$.

We can investigate the equality of two probability distributions, for continuous random variables X and Y .

Let the cdfs for X and Y be $F(x)$ and $G(y)$, respectively, then we wish to investigate $F(z) = G(z)$, for all z . Let X and Y be independent random variables.

In addition, let there be k mutually exclusive sets, A_i s, such that

$$p_{1i} = P(X \in A_i) \text{ and } p_{2i} = P(Y \in A_i), \text{ for } k = 1, 2, \dots, k.$$

Thus, if $F(z) = G(z)$, for all z , then $p_{1i} = p_{2i}$, for all i .

Hence, we can construct a hypothesis test, that would decide whether the two distributions are equal or not, for all i ,

$$H_0: p_{1i} = p_{2i},$$

$$H_1: p_{1i} \neq p_{2i}.$$

Collect a random sample of size m , from the X population, and also independently collect a random sample of size n , from the Y population, then combine them together, in order to make a pooled sample.

Then, order this pooled sample, from the smallest value to the largest value, and look for the runs of values of x s and y s in these ordered values. Let the random variable R be the number of runs in the ordered pooled sample. Assume that when the two distributions are equal, all arrangements of the m values of X and the n values of Y have equal probabilities.

We can choose m positions from $m+n$ positions in $\binom{m+n}{m}$ different ways, for X values, and each arrangement has the same probability, $\frac{1}{\binom{m+n}{m}}$.

Then, we can find how many of these arrangements produce $R = r$, for a positive integer r in the range of R .

Fix $r = 2k + 1$, for $k + 1$ runs of X and k runs of Y values, among the ordered values.

The number of ways of getting $k + 1$ runs out of the m values of X is $\binom{m-1}{k}$ different ways.

Also, the number of ways of getting k runs out of the n values of Y is $\binom{n-1}{k-1}$ different ways.

Therefore, there are $\binom{m-1}{k} \binom{n-1}{k-1}$ ways for this joint arrangement.

Also, we can have k runs of the values of X and $k + 1$ runs of the values of Y .

Similarly, we say that there are $\binom{m-1}{k-1} \binom{n-1}{k}$ different ways for these arrangement of runs.

Therefore,

$$P(R = 2k + 1) = \frac{\binom{m-1}{k} \binom{n-1}{k-1} + \binom{m-1}{k-1} \binom{n-1}{k}}{\binom{m+n}{m}}. \quad (7.9)$$

Now fix $r = 2k$, that is the number of X runs and the number of Y runs should be k each, with $\binom{m-1}{k-1}$ and $\binom{n-1}{k-1}$ possible ways, respectively.

Since an assembly begins with X runs or with Y runs, we have

$$P(R = 2k) = \frac{2 \binom{m-1}{k-1} \binom{n-1}{k-1}}{\binom{m+n}{m}}. \quad (7.10)$$

Hence, we can construct a hypothesis test for these runs, as follows:

Null Hypothesis H_0 : $F(z) = G(z)$, for all z .

Critical Region $R \leq c$, with $\alpha = P(R \leq c \mid H_0)$.

For large n and m , the distribution of R converges to a normal distribution, $N(\mu, \sigma^2)$, such that

$$\mu = 2 \frac{mn}{m+n} + 1 \quad \text{and} \quad \sigma^2 = \frac{(\mu-1)(\mu-2)}{m+n-1}.$$

7.3.1 Mann-Whitney-Wilcoxon Test

Let X and Y be independent continuous random variables, with cdfs $F(x)$ and $G(y)$, respectively.

Collect independent random samples, $X_1, \dots, X_m$, and $Y_1, \dots, Y_n$, coming from these distributions.

The Mann-Whitney-Wilcoxon test hypothesis is,

$$H_0: F(z) = G(z), \text{ for all } z.$$

Let $U_{ij} = 1$, for $X_i < Y_j$ and $U_{ij} = 0$, when $X_i \geq Y_j$.

Furthermore, let

$$V = \sum_{i=1}^m \sum_{j=1}^n U_{ij}, \quad (7.11)$$

that is, V counts the number of values of $X < Y$ in the sample.

Here, the realization values of V ranges from 0 to mn .

Moreover, if V is large, then the values of X are tend to be smaller than the values of Y , that is $F(z) \geq G(z)$, for all z .

Similarly, a small value of V implies that $F(z) \leq G(z)$, for all z .

Hence, if the alternative hypothesis is $H_1: F(z) \geq G(z)$, for all z , then the critical region would be $\{V \geq c\}$.

Otherwise, if the alternative hypothesis is $H_1: F(z) \leq G(z)$, for all z , then the critical region would be $\{V \leq c\}$.

In order to determine the size of the critical region, the distribution of V must be identified, when H_0 is true.

Let $h(v; m, n)$ denote the probability, $P(V = v)$.

We can fill m positions with the values of X by $\binom{m+n}{m}$ possible ways,

and the remaining n positions would be filled with the values of Y .

Each of these arrangements has the probability of $\frac{1}{\binom{m+n}{m}}$.

Here V can have two mutually exclusive and exhaustive structures; the final position in the $m+n$ values can be filled with an X value, with a probability

of $h(v; m-1, n)$, or, the largest value of the arrangement can be a value of Y , with a probability of $h(v-m; m, n-1)$.

Therefore, the probability of $V = v$ is:

$$h(v; m, n) = \binom{m}{m+n} h(v; m-1, n) + \binom{n}{m+n} h(v-m; m, n-1), \quad (7.12)$$

with the following properties:

$$\begin{aligned} h(v; 0, n) &= 1, \text{ for } v = 0, \quad h(v; 0, n) = 0, \text{ for } v > 0 \text{ and } n \geq 1, \\ h(v; m, 0) &= 1, \text{ for } v = 0, \quad h(v; m, 0) = 0, \text{ for } v > 0 \text{ and } m \geq 1, \\ h(v; m, n) &= 0, \text{ for } v < 0 \text{ and } m \geq 0, n \geq 0. \end{aligned}$$

For large values of m and n , the distribution of

$$\frac{V - \frac{mn}{2}}{\left(\frac{mn(m+n+1)}{12}\right)} \text{ asymptotically converges to } N(0, 1).$$

If T denotes the sum of the ranks of Y_i s, among $m+n$ observations,

$$X_1, \dots, X_m, Y_1, \dots, Y_n, \text{ then } V = T - \frac{n(n+1)}{2}.$$

7.4 Kruskal-Wallis Test

Assume that there are k independent samples of sizes $n_1, n_2, \dots, n_k$, coming from continuous distributions, with $F(x_1 - c_1) = \dots = F(x_k - c_k)$, for fixed $c_1, \dots, c_k$.

We want to construct the following hypothesis test:

$$H_0: \mu_1 = \dots = \mu_k,$$

$$H_1: \text{Not all } \mu_i \text{ s are equal.}$$

First, all of the observations, $\sum_{i=1}^k n_i$, have to be ordered, from the smallest to the largest, then the rank sum for each sample, R_i , has to be computed.

The test statistic is given by,

$$B = \frac{12}{n(n+1)} \sum_{i=1}^k \frac{R_i^2}{n_i} - 3(n+1), \quad (7.13)$$

Note that B gets larger as the difference between μ_i s gets larger.

This test statistic follows a χ_{k-1}^2 distribution, thus

Decision rule: reject H_0 , if the realization of B , say b , exceeds $\chi^2_{1-\alpha; k-1}$.

7.5 Friedman Test

Assume k treatments are ranked independently, within b blocks. Let R_i be the rank sum of the i th treatment.

Thus the test hypothesis;

H_0 : Median of all k treatments are the same.

The test statistic is

$$G = \frac{12}{bk(k+1)} \sum_{i=1}^k R_i^2 - 3b(k+1). \quad (7.14)$$

This test statistic follows a χ^2_{k-1} distribution.

Therefore, the decision rule would be: if the realization of the test statistic exceeds a critical point, $\chi^2_{1-\alpha; k-1}$, then reject the null hypothesis.

7.6 Exercises

1. Let Y_n be the n th order statistic of a sample of size n , coming from a continuous distribution.

What is the smallest n for $P(x_{0.95} < Y_n) \geq 0.75$?

2. Find $P(Y_4 < x_{0.5} < Y_6)$, if $Y_1 < \dots < Y_{10}$ are the order statistics of a sample, coming from a continuous distribution.

3. Let $Y_1 < \dots < Y_{100}$ be the order statistics of a sample, coming from a continuous distribution. What are i and j , such that $i < j$ and $P(Y_i < x_{0.5} < Y_j)$ is approximately 0.95 ?

4. In the definition of Wilcoxon's statistic, let W_1 be the sum of ranks of observations of the sample that are positive, and let W_2 be the sum of the ranks of observations of the sample that are negative. Find the mean and variance of W_1 and W_2 .

5. A random sample of waiting times at a dentist office, given in minutes, is as follows:

15, 20, 24, 13, 18, 26, 15, 17, 24, 15, 17, 22, 28, 13, 11, 18, 15, 10, 9, 11, 15, 18, 22, 10, 5

At the 5% level of significance, test the claim that the median waiting time does not exceed 15 minutes, by using the sign test.

6. The body weight of randomly selected individuals, given in kilograms, were measured before and after they stopped smoking cigarettes:

Before	After
69	69
85	84
79	75
55	57
89	85
75	70
50	54
99	90

By using the signed rank test for paired observations, test the hypothesis that stopping smoking cigarette has no effect on a person's body weight, at the 5% level of significance.

7. A new treatment is designed to cure breast cancer and randomly selected patients, who are at the very advanced stage of cancer, undergo a double blind clinical trial, i.e. some of them are randomly selected to receive the treatment, others are not. The survival times, given in years, were recorded as follows:

Treatment	No Treatment
2.2	2
5.5	1
1.1	2.5
4.5	5
0.5	1
7	0.1
5	0.4
2	1
3	1
0.1	0.5

By using the rank-sum test, at the 5% level of significance, can you say that the treatment is effective, i.e. increases the survival time?

8. A sample of voters was asked whether they are college graduates, and a response of Y is affirmative, while a response of N is not. The sample response was $\{N, N, N, N, Y, Y, Y, Y, N, Y, Y\}$. At the 5% level of significance, by using the run test, can you say that this sample is a random sample?
9. In a production facility, products are randomly inspected for being defective. If an item is found defective, then it is denoted by D , or, if an item is found not defective, then it is denoted by N . A sample of inspected items were listed as: $\{D, D, N, N, D, D, N, N, N, D, N, N, N, N, D, D, D, N, D, N, N, N, N, D, N, N, N, D, N, N, D, N, D, N, N, D, N, N\}$. At the 5% level of significance, by using the large sample version of the run-test, can you say that the defective items are produced randomly?
10. What is the sample size where at least 75% of the population will be less than the largest observation in the sample, at the 99% confidence level?

Chapter 8

Bayesian Statistics

“If there be two subsequent events, the probability that of the second b/N and the probability of both together P/N , and it being first discovered that the second event has happened, from hence I guess that the first event has also happened, the probability I am in the right is P/b ...” - Thomas Bayes.

Bayesian statistics begins with a belief, known as a prior probability of an event, followed by collecting relevant evidence for the probability of the event and refining the prior probability under the weight of the evidence, thus generating a posterior probability of the event.

A family of priors that generate posteriors, belong to the same family, is called the conjugate family, or conjugate priors.

In fact, all models are subjective, that is to say a model is a collection of assumptions such that it may, or, it may not support our decisions, followed by our actions according to how close the model can get to reality. Moreover, even if two different models are close to reality, to some acceptable degree, they may produce significantly different decisions, based on the same data set.

8.1 Bayesian Estimation

Let a random random variable X be subject to a probability law which is shaped by a parameter θ , such that the parameter is an element of a parameter space, Ω .

Up to now, we took the parameter as an unknown point in space, and we tried to estimate its value, from a sample.

Now, we assume that the parameter is a random variable itself, Θ , and it has its probability distribution over Ω . Let us denote the pdf of Θ by $h(\theta)$, or $\Theta \sim h(\theta)$, called the prior probability law of Θ which reflects the subjective belief, thus we have a pdf of X , denoted by $f(x|\theta)$, or $X|\theta \sim f(x|\theta)$, as a conditional probability law subject to $h(\theta)$.

Let $X_1, \dots, X_n$ be a random sample coming from this distribution of X , with the likelihood function $L(\theta) = \prod_{i=1}^n f(x_i|\theta)$, then the joint pdf of $X_1, \dots, X_n$ and the Θ would be $g(x_1, \dots, x_n, \theta) = \prod_{i=1}^n f(x_i|\theta)h(\theta)$, i.e. the product of the likelihood function and the prior of Θ .

The marginal joint pdf of $X_1, \dots, X_n$ would be

$u(x_1, \dots, x_n) = \int_{-\infty}^{\infty} g(x_1, \dots, x_n, \theta) d\theta$, for a continuous Θ . When Θ is a discrete random variable, summation will replace the integration.

Hence, the conditional pdf of Θ , for a given sample realization, $x_1, \dots, x_n$, is (remember Bayes' theorem)

$$v(\theta|x_1, \dots, x_n) = \frac{g(x_1, \dots, x_n, \theta)}{u(x_1, \dots, x_n)} = \frac{\prod_{i=1}^n f(x_i|\theta)h(\theta)}{u(x_1, \dots, x_n)} = \frac{L(\theta)h(\theta)}{\int_{-\infty}^{\infty} L(\theta)h(\theta)d\theta}. \quad (8.1)$$

Thus, $v(\theta|x_1, \dots, x_n)$ is proportional to $g(x_1, \dots, x_n, \theta) = \prod_{i=1}^n f(x_i|\theta)h(\theta)$, or product of the likelihood and the prior.

Here, $v(\theta|x_1, \dots, x_n)$ is called the posterior pdf of Θ , since it comes after the observations were made and it combines the prior belief together with the sample information.

The selection of an admissible prior results the subjective probability.

Example 8.1

Let $X_1, \dots, X_n$ be a sample, coming from a Poisson population with a finite mean θ , such that θ is the realization of a random variable, subject to a gamma probability law with known parameters, α and β .

Hence, $g(x_1, \dots, x_n, \theta) = (\frac{\theta^{x_1} e^{-\theta}}{x_1!} \dots \frac{\theta^{x_n} e^{-\theta}}{x_n!}) (\frac{\theta^{\alpha-1} e^{-\frac{\theta}{\beta}}}{\Gamma(\alpha)\beta^\alpha})$, for $i = 1, 2, \dots, n$, and $x_i = 0, 1, 2, \dots$

Therefore, the marginal pdf is

$$u(x_1, \dots, x_n) = \int_0^{\infty} g(x_1, \dots, x_n, \theta) d\theta = \frac{\Gamma(\sum_{i=1}^n x_i + \alpha)}{x_1! \dots x_n! \Gamma(\alpha) \beta^\alpha (n + \frac{1}{\beta})^{\sum_{i=1}^n x_i + \alpha}}. \quad (8.2)$$

Thus, the conditional pdf of Θ , for the given realization, $x_1, \dots, x_n$, would be

$$v(\theta|x_1, \dots, x_n) = \frac{g(x_1, \dots, x_n, \theta)}{u(x_1, \dots, x_n)} = \frac{\theta^{\sum x_i + \alpha - 1} e^{-\frac{\theta}{[\beta/(n\beta + 1)]}}}{\Gamma(\sum_{i=1}^n x_i + \alpha) [\beta/(n\beta + 1)]^{\sum x_i + \alpha}}. \quad (8.3)$$

This is a gamma pdf with parameters, $\alpha_{new} = \sum_{i=1}^n x_i + \alpha$, and $\beta_{new} = \beta/(n\beta + 1)$.

We could also write this as $v(\theta|x_1, \dots, x_n) \propto \theta^{\sum_{i=1}^n x_i + \alpha - 1} e^{-\frac{\theta}{[\beta/(n\beta + 1)]}}$.

8.1.1 Bayesian Decision Function

Assume a sufficient statistics, $Y = T(X_1, \dots, X_n)$, exists for the parameter with a marginal pdf of $g_Y = \int_{-\infty}^{\infty} g(y|\theta)h(\theta)d\theta$. Thus, the Bayesian approach boils down to selecting a decision function, d , such that $d(y)$ predicts an experimental value of Θ , when a realization for Y is available. But, how one could predict an experimental value for a random variable? The usual answer to this question would be the expected value of the random variable, but it is not the only candidate.

The selection of a suitable decision function depends upon using a loss function, $L(\theta, d(y))$, such that the conditional expectation of the loss function, for a given $Y = y$, is minimized for a properly selected decision function, $\min_d \{E[L(\Theta, d(y))|Y = y]\}$.

For example, if the loss function is defined as $L(\theta, d(y)) = (\theta - d(y))^2$, then the Bayes' decision function would be the conditional expected value of Θ , for a given posterior, $d(y) = E[\Theta|y]$. On the other hand, if the loss function is defined as $L(\theta, d(y)) = |\theta - d(y)|$, then the Bayes' decision function would be the median of $(\Theta|Y = y)$. Thus, these Bayesian decision functions provide optimal predictors for an experimental value of Θ .

Here, the conditional expectation of the loss function, for a given sufficient statistic, Y , identifies a function of the random variable Y . Then, without loss of generality, the expected value of this function is

$$\int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} L(\theta, d(y))v(\theta|y)d\theta \right) g_Y(y)dy = \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} L(\theta, d(y))g(y|\theta)dy \right) h(\theta)d\theta. \quad (8.4)$$

This integral is the expected value of Bayes' risk function,

$$R(\theta, d) = \int_{-\infty}^{\infty} L(\theta, d(y))v(\theta|y)d\theta, \quad (8.5)$$

for a given θ .

Thus, Bayes' decision function also minimizes the expected risk, since it is a minimizer for the loss function, for every sufficient statistic Y with $g_Y(y) > 0$.

Example 8.2

Let $X_1, \dots, X_n$ be a sample coming from a Bernoulli distribution with the probability of success being θ . Find a Bayesian decision function, $d(y)$. Here, the sufficient statistic would be $Y = \sum_{i=1}^n X_i$, such that the Y would be subject to a binomial probability law, $B(n, \theta)$.

Thus, the conditional pdf of Y is $g(y|\theta) = \binom{n}{y} \theta^y (1-\theta)^{n-y}$, for $y = 0, 1, 2, \dots, n$. Let the prior pdf of Θ be $h(\theta) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \theta^{\alpha-1} (1-\theta)^{\beta-1}$.

Therefore, the conditional pdf of Θ , for a realization y , is

$$v(\theta|y) = \frac{\Gamma(n+\alpha+\beta)}{\Gamma(\alpha+y)\Gamma(n+\beta-y)} \theta^{\alpha+y-1} (1-\theta)^{\beta+n-y-1}, \text{ for } y = 0, 1, 2, \dots, n.$$

If we take the risk function $L(\theta, d(y)) = [\theta - d(y)]^2$, we derive the expected risk as,

$$\begin{aligned} \int_0^1 \left(\sum_{y=0}^n [\theta - d(y)]^2 \binom{n}{y} \theta^y (1-\theta)^{n-y} \right) h(\theta) d\theta \\ = \sum_{y=0}^n n \left(\int_0^1 [\theta - d(y)]^2 v(\theta|y) d\theta \right) g_Y(y). \end{aligned}$$

Since the Bayes' solution is the mean of the conditional pdf of Θ , for the realization y , we have

$$d(y) = \int_0^1 \theta v(\theta|y) d\theta = \frac{\alpha + y}{\alpha + \beta + n},$$

which minimizes $\int_0^1 [\theta - d(y)]^2 v(\theta|y) d\theta$, for $y = 0, 1, \dots, n$, and minimizes the expected risk, too.

This optimal decision function can also be written as

$$d(y) = \left[\frac{n}{\alpha + \beta + n} \right] \frac{y}{n} + \left[\frac{\alpha + \beta}{\alpha + \beta + n} \right] \frac{\alpha}{\alpha + \beta},$$

i.e., a weighted average of the m.l.e. of the parameter and the mean of the prior pdf of the parameter. That implies that the worth of the prior information, in a sample of size n , is expressed by the sum, $\alpha + \beta = n$.

Also, observe what happens to this Bayes' estimator when $n \rightarrow \infty$.

In Bayesian statistics, the posterior pdf has all the information that the sample provides. Thus, the Bayes' decision function changes with different loss functions.

Example 8.3

What are the Bayes' estimators for the mean and the variance of a Bernoulli distribution, with a beta prior with given parameters α and β ?

The posterior distribution of the probability of success, p , would be beta distribution with new parameters,

$$\alpha_{new} = \sum_{i=1}^n x_i + \alpha \text{ and } \beta_{new} = (n - \sum_{i=1}^n x_i) + \beta, \text{ i.e.}$$

$$v(p|x_1, \dots, x_n) = \frac{\Gamma(\alpha_{new} + \beta_{new})}{\Gamma(\alpha_{new})\Gamma(\beta_{new})} p^{\alpha_{new}-1} (1-p)^{\beta_{new}-1}, \text{ for } 0 < p < 1.$$

Thus the Bayes' estimate for p is the posterior mean of p ,

$$\hat{p}_B = E[\Theta|x_1, \dots, x_n] = \frac{\alpha_{new}}{\alpha_{new} + \beta_{new}} = \frac{\sum_{i=1}^n x_i + \alpha}{n + \alpha + \beta},$$

and the Bayes' estimator for $p(1-p)$ is

$$\begin{aligned} \hat{p}(1-\hat{p})_B &= E[\Theta(1-\Theta)|x_1, \dots, x_n] = \int_0^1 p(1-p)v(p|x_1, \dots, x_n) \\ &= \frac{\alpha_{new}\beta_{new}}{(\alpha_{new} + \beta_{new} + 1)(\alpha_{new} + \beta_{new})} = \frac{(\sum_{i=1}^n x_i + \alpha)(n - \sum_{i=1}^n x_i + \beta)}{(n + \alpha + \beta + 1)(n + \alpha + \beta)}. \end{aligned}$$

Here, we have observed that if $\hat{\theta}_B$ is a Bayes' estimate of Θ , then a function of $\hat{\theta}_B$, say $g(\hat{\theta}_B)$, would be a Bayes' estimator of $g(\Theta)$.

Example 8.4

Laplace's rule of succession: Assume a Bernoulli trial occurs with an unknown probability. Repeat the experiment many times, say n times, and observe that all the outcomes have turned as successes.

Let the prior for the unknown success probability be a uniform distribution, $\theta \sim U(0, 1)$. Find the Bayes' estimate of the success probability, for the next trial.

Let X be the set of random samples $X_1, \dots, X_n$, coming from this experiment. Thus, we have $f(\theta) = 1$, for $\theta \in [0, 1]$, $P(X|\theta) = \theta^n$, and

$$P(X) = \int_0^1 P(X|\theta)f(\theta)d\theta = \frac{1}{n+1}.$$

Hence, the posterior is

$$f(\theta|X) = \frac{P(X|\theta)f(\theta)}{P(X)} = (n+1)\theta^n, \text{ for } \theta \in [0, 1].$$

Consequently, we have the conditional expected value of the unknown success probability, a.k.a. the posterior expectation, as the Bayes' estimator:

$$E[\Theta|X] = \int_0^1 \theta f(\theta|X) d\theta = (n+1) \int_0^1 \theta^{n+1} d\theta = \frac{n+1}{n+2}.$$

According to Laplace [10], this is the probability that the event will happen again, when the event have happened many times in the past. As an example, he selected the event to be the sunrise, i.e. the probability that the sun will rise again, based on the observations such that the sun has risen every day in the past.

Obviously, this example raises the question of *what is the natural phenomenon governing this experiment?*

Since, possibly, the experimental conditions may change (in the future, sun may not be there to rise).

8.1.2 Bayesian Interval Estimator

We can also construct an interval to capture the random parameter, Θ , for a given sufficient statistic, Y , at a desired level of probability measure, $1 - \alpha$, such that

$$P(a(y) \leq \Theta \leq b(y) | Y = y) = \int_{a(y)}^{b(y)} v(\theta|y) d\theta = 1 - \alpha. \quad (8.6)$$

Here, the sample realization, $x_1, \dots, x_n$, provides the empirical value of y and the interval $(a(y), b(y))$ estimates a region for Θ supported by this region, with a desired probability of $1 - \alpha$, and this interval is known as credible interval, which is a proper probability interval for a given sample realization, i.e. the boundaries of this interval are real numbers but Θ is a random variable, captured by this interval at the assigned probability measure.

On the other hand, if you consider the $100(1 - \alpha)\%$ confidence interval of an unknown but fixed parameter θ , for a given sample realization, there would be no randomness in this confidence interval, thus this begs a non-probabilistic interpretation such that if you construct infinitely many confidence intervals, for a fixed sample size and with different sample realizations, $100(1 - \alpha)\%$ of all those intervals would be expected to capture the unknown parameter in them, but for a given sample realization your confidence interval either contains, or does not contain the unknown parameter in it, there is no more probabilistic meaning there.

8.2 Bayesian Hypothesis Testing

Assume that the joint pdf (likelihood function) of i.i.d. random variables $\{x_1, \dots, x_n\}$ depends on the parameter θ and denoted by $f(\theta|x_1, \dots, x_n)$. Let θ_0, θ_1 be two possible values of θ such that we want to test the hypotheses $H_0 : \theta = \theta_0$, versus $H_1 : \theta = \theta_1$. Hence, we need a decision function, $d(Y)$, that selects one hypothesis, based on the given data and a proper statistic, Y , such that the Bayesian decision process is to decide between two values of d , say $d = 0$ and $d = 1$. Let $L(\theta, d)$ be the loss function for this Bayesian decision process, where $\{\theta = \theta_0, d = 0\}$ and $\{\theta = \theta_1, d = 1\}$ are the correct decision pairs, therefore $L(\theta_0, 0) = L(\theta_1, 1) = 0$. On the other hand, if a wrong decision was made, then $L(\theta_0, 1) > 0$ and $L(\theta_1, 0) > 0$.

Therefore, we can select a region of the sample space (rejection region), C , such that if $Y(x_1, \dots, x_n) \in C$, then $d = 1$. Similarly, if $Y(x_1, \dots, x_n) \notin C$, then $d = 0$. That is to say, we can identify a risk function, $R(\theta, d)$, by utilizing the rejection region, C , such that $R(\theta, d) = \int_{\Omega} L(\theta, d) f(\theta|x_1, \dots, x_n) dy$, where $\Omega = C \cup C^c$ and C^c is the complement of C with respect to Ω .

Moreover, since $d = 1$ for $Y \in C$, we can write

$$R(\theta, C) = \int_C L(\theta, 1) f(\theta|x_1, \dots, x_n) dy + \int_{C^c} L(\theta, 0) f(\theta|x_1, \dots, x_n) dy. \quad (8.7)$$

Thus $R(\theta_0, C) = L(\theta_0, 1) \int_C f(\theta_0|x_1, \dots, x_n) dy$,

or, $R(\theta_1, C) = L(\theta_1, 0) \int_{C^c} f(\theta_1|x_1, \dots, x_n) dy$.

The critical region C that minimizes the $\max R(\theta_0, C), R(\theta_1, C)$ would be the minimax solution such that $C = \frac{f(\theta_0|x_1, \dots, x_n)}{f(\theta_1|x_1, \dots, x_n)} \leq r$, for an $r > 0$, therefore $R(\theta_0, C) = R(\theta_1, C)$.

Now, assume that the parameter is a random variable Θ , with a pdf $h(\theta)$, and with two possible values, θ_0 and θ_1 such that $h(\theta_0) + h(\theta_1) = 1$.

The posterior probability of Θ , for a given sample $X_1, \dots, X_n$, is obtained from Bayes' theorem,

$$k(\theta|x_1, \dots, x_n) = \frac{h(\theta)f(\theta; x_1, \dots, x_n)}{\sum_{\Omega} h(\theta)f(\theta; x_1, \dots, x_n)}. \quad (8.8)$$

A Bayesian solution would be such that $E[L(\theta, d(y)|Y = y)]$ is minimum, for a $d(y)$. If $d = 0$, then the conditional expectation of the loss function would be, since $L(\theta_0, 0) = 0$,

$$\sum_{\Omega} L(\theta, 0)k(\theta|x_1, \dots, x_n) = \frac{L(\theta_1, 0)h(\theta_1)f(\theta_1; x_1, \dots, x_n)}{h(\theta_0)f(\theta_0; x_1, \dots, x_n) + h(\theta_1)f(\theta_1; x_1, \dots, x_n)}. \quad (8.9)$$

Similarly, if $d = 1$, since $L(\theta_1, 1) = 0$, then the conditional expectation is equal to

$$\sum_{\Omega} L(\theta, 1)k(\theta|x_1, \dots, x_n) = \frac{L(\theta_0, 1)h(\theta_0)f(\theta_0; x_1, \dots, x_n)}{h(\theta_0)f(\theta_0; x_1, \dots, x_n) + h(\theta_1)f(\theta_1; x_1, \dots, x_n)}. \quad (8.10)$$

Therefore, the decision would be $d = 1$, when $\frac{f(\theta_0; x_1, \dots, x_n)}{f(\theta_1; x_1, \dots, x_n)} < \frac{L(\theta_1, 0)h(\theta_1)}{L(\theta_0, 1)h(\theta_0)}$.

If the inequality is reversed, then $d = 0$.

A simple Bayesian hypothesis test over the partitioned parameter space, Ω , would be constructed as choosing between two statements,

$$H_0 : \theta \in \Omega_0,$$

$$H_a : \theta \in \Omega_a,$$

such that $\Omega_0 \cup \Omega_a = \Omega$ and $\Omega_0 \cap \Omega_a = \emptyset$.

Now, compute the posterior pdf for a given prior and a sample realization, then construct the posterior probability measures for these two hypothesis, $P(H_0|y)$ and $P(H_a|y)$.

The decision rule to reject H_0 would be: If $P(H_0|y) < P(H_a|y)$, then reject H_0 .

Observe that this approach gives us a generalization, for testing a collection of alternative simple hypotheses.

On the other hand, if the posterior pmf is continuous and the null hypothesis states that $H_0 : \Theta = \theta_0$, then we know that the posterior probability measure of H_0 would always be zero, thus we cannot use this decision rule to test the hypothesis; instead we use the correspondence between two-tailed tests and credible intervals, such that if θ_0 falls into the $100(1 - \alpha)\%$ credible interval, then we cannot reject H_0 , at the α level of significance, otherwise we reject the null hypothesis.

8.3 Mixture Priors

A person may have a strong believe in a prior, but it could be wrong. Maybe this prior is based on a historical observation produced by different conditions than the present ones, or, simply a strong bias may be the cause for this misspecified prior.

If we believe that our prior is misspecified, then we have a vague idea about the parameter. Let $h_1(\theta)$ be the strong belief prior and $h_2(\theta)$ be the vague belief prior, possibly uniform distribution, or, a conjugate prior with very heavy tails, in order to accommodate vagueness.

Also, let $v_1(\theta|x_1, \dots, x_n)$ be the posterior, generated by the strong prior h_1 and the sample realization, $v_2(\theta|x_1, \dots, x_n)$ be the posterior, generated by the vague prior h_2 and the sample realization.

Now, let I be an indicator random variable such that if $I = 1$, then strong prior is the distribution of the parameter. Otherwise, if $I = 2$, then the vague prior is the distribution of the parameter.

Hence, the conditional prior, for a given indicator value, would be

$$h(\theta|I = i) = h_1(\theta), \text{ if } I = 1, \text{ and } h(\theta|I = i) = h_2(\theta), \text{ if } I = 2.$$

Let the probability distribution of indicator variable be

$$P(I = 1) = p_1 \text{ and } P(I = 2) = p_2, \text{ such that } p_1 + p_2 = 1.$$

The joint prior distribution of Θ and I is

$$h(\theta, i) = p_i h_i(\theta), \text{ for } i = 1, 2.$$

Therefore, the mixture prior of Θ is evaluated as

$$h(\theta) = \sum_{i=1}^2 p_i h_i(\theta). \quad (8.11)$$

That leads us to construct a mixture posterior of Θ , as described below.

The joint posterior of Θ and I , for a given realization would be

$$v(\theta, i|x_1, \dots, x_n) = h(\theta, i)f(\theta, i|x_1, \dots, x_n), \text{ for } i = 1, 2.$$

But, the likelihood function does not depend on the indicator variable, thus we have,

$$v(\theta, i|x_1, \dots, x_n) = p_i h_i(\theta)f(\theta|x_1, \dots, x_n), \text{ for } i = 1, 2.$$

Now, we can find the posterior probability of the indicator variable,

$$\begin{aligned}
P(I = i|x_1, \dots, x_n) &= \int v(\theta, i|x_1, \dots, x_n) d\theta \\
&= \frac{p_i \int h_i(\theta) f(\theta|x_1, \dots, x_n) d\theta}{\sum_{i=1}^2 p_i \int h_i(\theta) f(\theta|x_1, \dots, x_n) d\theta}.
\end{aligned} \tag{8.12}$$

The posterior distribution of Θ is found by summing the joint posterior over all possible values of the indicator variable, that is to say,

$$v(\theta|x_1, \dots, x_n) = \sum_{i=1}^2 v(\theta, i|x_1, \dots, x_n). \tag{8.13}$$

This posterior is robust against the wrong prior. Also, here the index variable is a nuisance parameter and marginalized out of the posterior distribution of Θ . This is a common characteristic of Bayesian statistics, to take the unknown parameters as nuisance parameters (random variables) and marginalize them out, in contrast with the frequentist statistics, where an unknown nuisance parameter brings the statistical procedure into a halt.

8.4 Sequential Bayesian Statistics

If desired, the process of filtering the prior under the evidence, can be repeated by using the previous posterior as the prior for the next cycle, and a new sample would help to generate the new posterior. This adaptive process will be repeated under new samples and updated priors, until the prior and the posterior converges, i.e. no new knowledge can be acquired from new samples.

In Bayesian statistics, many computations for generating the posterior distributions, finding the mean of those posterior distributions, and computing posterior probabilities of events, require using a computer and employing some form of the Monte-Carlo simulation, or, special computer softwares. This feature of Bayesian statistics is a consequence of operating on functions, in contrast to operating on real numbers in the case of frequentist statistics.

8.5 Exercises

1. Let $X_1, \dots, X_n$ be a sample coming from a normal distribution, $N(\theta, \sigma^2)$, with a finite and known variance σ^2 . Also, let $Y = \bar{X}$ be the sample

mean. Set the loss function be $L(\theta, d(y)) = |\theta - d(y)|$, and if θ is an observed value of $\Theta \sim N(\mu, \tau^2)$ with known μ and τ . Find the Bayes' solution $d(y)$, for an estimate of θ .

2. Let $X_1, \dots, X_n$ be a sample coming from a Poisson distribution with a finite mean θ . Also, let $Y = \sum_{i=1}^n X_i$, with a loss function $L(\theta, d(y)) = [\theta - d(y)]^2$. If θ is an observed value of Θ , such that the pdf of Θ is $h(\theta) = \frac{\theta^{n-1} e^{-\theta/\beta}}{\Gamma(\alpha)\beta^\alpha}$, where $\alpha > 0$ and $\beta > 0$ are known numbers, then find the Bayes' solution $d(y)$, for an estimate of θ .
3. Let $n = 50$, $\alpha = 20$, and $\beta = 5$ in the above Example 2, such that $d(y) = (20 + y)/75$ would be the Bayes' estimate of θ . If Y is subject to a binomial distribution with $n = 50$ and the probability of success is θ , then find the Bayesian risk, $E[(\theta - d(Y))^2]$.
4. A call center receives telephone calls, according to a Poisson process with λ calls per minute. A random sample of 35 one-minute call periods yields an average of 1.8 calls per minute. If the assumed prior for λ was an exponential distribution with parameter $\lambda = 2$, find the posterior distribution for λ .
5. A study indicates that the percentage of the cigarette smokers, p , who have the lung cancer is subject to a beta distribution with an average of 75% and a standard deviation of 15%. A new study data yield that 90 out of 200 smokers have lung cancer. Find the posterior distribution of the proportions of smokers who have lung cancer.
6. A liberal arts college's administrators believe that the average score for the SAT test varies from from one incoming freshman class to another, and it is also believed that this variation is subject to a normal distribution with a mean of 1200 and a standard deviation of 100. If a random sample of 100 incoming students were observed, with a result of an average score of 1100, and with a standard deviation of 81, construct a 95% Bayesian interval for the average score.
7. A random variable follows an exponential distribution with mean $\frac{1}{\theta}$. Also, assume that the prior distribution of θ is another exponential distribution with mean 2. Find the Bayes estimate of θ .
8. Assume that the time to a failure, T , of a component is exponentially distributed with pdf $f(t) = \theta e^{-\theta t}$, for $t > 0$. It is also believed that θ is an exponential random variable with pdf $g(\theta) = 2e^{-2\theta}$, for $\theta > 0$. When we collect a random sample of n observations, prove that the

posterior distribution of Θ is a gamma distribution with parameters $\alpha = n + 1$ and $\beta = \frac{1}{2 + \sum_{i=1}^n t_i}$, where t_i s are the observed failure times.

9. Assume that an arrival process of number of busses at a bus stop during one hour, a Poisson distribution with mean λ , yield a random sample of $\{6, 5, 6, 5, 4, 7, 9, 10, 5, 6, 5, 3, 4, 5, 6, 7, 8, 9, 10, 4, 5, 6, 6, 7, 5\}$.

Also, assume that λ is subject to a gamma distribution with parameters 6 and 4. Find the Bayes estimate of λ .

10. The daily attendance of a course follows a normal distribution with unknown mean μ and variance σ^2 . Since the attendance will change from day to day, it is believed that the prior $\mu_0 = 30$ and $\sigma_0 = 5$. During the first 30 days an average of 24 students/day attended with a standard deviation of 3 students/day. Find the Bayes estimate of the average daily attendance and construct a 95% Bayes interval for μ .

Chapter 9

Information Theory

“Some erroneously consider information theory as synonymous with communication theory. Information theory is a branch of the mathematical theory of probability and mathematical statistics...” - Solomon Kullback.

9.1 Meaning of Information

Information theory is considered as a subject of mathematical statistics and probability theory. Thus, information theory has applications in communication theory, decision theory, computer science, and some other fields that models relationships as stochastic systems. Also, it has connections with physical concepts, such as entropy in thermodynamics. Information is the critical factor for decision making, thus its impact is imbedded in all aspects of statistics. Whenever we conduct descriptive statistics, inferential statistics, or design of experiments, we collect data to generate information, as best as we could. Briefly, information is a means of transforming raw data into sound decisions.

The statistical measure of the amount of information, generated by data about an estimated parameter, was defined by Fisher [2] in the early 1920s and well known by statisticians. After Fisher's introduction of information measure, communication engineers showed interest and started to define the quality of signals, as well as the amount of information acquired by a transmitted message [5], [13], [17].

9.2 Probabilistic Construction of Information

Let the probability space be identified as a triplet (Ω, F, μ_i) , where Ω is the sample space, F is the smallest σ -field (Borel sigma field) of all subsets of Ω , and μ_i , for $i \in \mathbb{N}$, is a relatively absolutely continuous probability measure over Ω , denoted by $\mu_1 \equiv \mu_2 \equiv \dots$, that is to say if $E \subset \Omega$ and $\mu_1(E) = 0$, then also $\mu_j(E) = 0$, for all $j \neq 1$.

Thus with this probability measure we have a measurable space (Ω, F) . Here, F may be a class of Borel sets in $\mathbb{R}^n$, when we have a sample of n observations and μ_i may be the probability measure of the i th sample, subject to its own parameters.

Let κ be a probability measure such that $\kappa \equiv \mu_i$, then there exists a nonnegative and finite function, $f_i(x)$, which is called a generalized probability measure [4], measurable in κ , such that for any $E \subset \Omega$, we have $\mu_i(E) = \int_E f_i(x) d\kappa(x)$, for all i .

Here, f_i is also known as the Radon-Nikodym derivative such that we have $d\mu_i(x) = f_i(x) d\kappa$, thus $f_i(x) = \frac{d\mu_i}{d\kappa}$.

As an application, let us consider the classical hypothesis test with two mutually exclusive hypotheses such that H_i be the hypothesis that the sample X comes from a population with a probability measure μ_i , for $i = 1, 2$.

Hence,

$$P(H_i|x) = \frac{P(H_i)f_i(x)}{P(H_1)f_1(x) + P(H_2)f_2(x)}, \quad (9.1)$$

for $i = 1, 2$.

Consequently, by taking the natural logarithm of both sides of this equality, we have

$$\log \frac{f_1(x)}{f_2(x)} = \log \frac{P(H_1|x)}{P(H_2|x)} - \log \frac{P(H_1)}{P(H_2)}, \quad (9.2)$$

where $P(H_i)$ is the prior probability and $P(H_i|x)$ is the posterior probability of H_i , for a realization of X .

The right hand side of this equation is the difference between the logarithm of the odds ratios in favor of H_1 , after and before a sample realization, $x \in E$, which provides information for discriminating in favor of H_1 versus H_2 .

The conditional expected value of information based on the event E , for discriminating in favor of H_1 versus H_2 , would be given as

$$\begin{aligned}
E[I(H_1 : H_2)|E] &= \int_E \frac{1}{\mu_1(E)} \log \frac{f_1(x)}{f_2(x)} d\mu_1(x) \\
&= \frac{1}{\mu_1(E)} \int_E f_1(x) \log \frac{f_1(x)}{f_2(x)} d\kappa(x), \tag{9.3}
\end{aligned}$$

with $d\mu_1(x) = f_1(x)d\kappa(x)$.

The expected value of the information over the entire sample space, for discriminating in favor of H_1 versus H_2 , would be

$$\begin{aligned}
E[I(H_1 : H_2)] &= \int_{\Omega} \log \frac{f_1(x)}{f_2(x)} d\mu_1(x) = \int f_1(x) \log \frac{f_1(x)}{f_2(x)} d\kappa(x) \\
&= \int \log \frac{P(H_1|x)}{P(H_2|x)} d\mu_1(x) - \log \frac{P(H_1)}{P(H_2)}. \tag{9.4}
\end{aligned}$$

This would also be considered as the average information of μ_1 with respect to μ_2 .

Similarly,

$$E[I(H_2 : H_1)] = \int f_2(x) \log \frac{f_2(x)}{f_1(x)} d\kappa(x), \tag{9.5}$$

or,

$$-E[I(H_2 : H_1)] = \int f_2(x) \log \frac{f_1(x)}{f_2(x)} d\kappa(x), \tag{9.6}$$

as the average information for discriminating in favor of H_2 versus H_1 . The degree of average discrimination of the hypothesis, based on a sample, would be

$$\begin{aligned}
E[I(H_1 : H_2)] + E[I(H_2 : H_1)] &= \int [f_1(x) - f_2(x)] \log \frac{f_1(x)}{f_2(x)} d\kappa(x) \\
&= \int \log \frac{P(H_1|x)}{P(H_2|x)} d\mu_1(x) - \int \log \frac{P(H_1|x)}{P(H_2|x)} d\mu_2(x), \tag{9.7}
\end{aligned}$$

which is the distance between two hypotheses, for a given sample realization, and it is also known as the Kullback's divergence, or, Kullback-Leibler divergence [9]. Since it does not satisfy the triangle inequality [8], it is not a proper topological metric, therefore it may be categorized as a semi-metric.

If the realization x supports H_1 , that is to say $P(H_1|x) = 1$, then the information in x for H_1 would be $-\log P(H_1)$ and that would be the additional amount of information generated by the sample, that is if the hypothesis was true to begin with, we gain no additional information by the sample, otherwise this measure indicates a degree of information for the probability of true hypothesis.

Furthermore, assume that we have a sequence of hypotheses, $H_1, \dots, H_n$, for which we can select one of them to be true, when a realization is observed. Where, the average information generated by an observation about the hypotheses would be $-\sum_{i=1}^n P(H_i) \log P(H_i)$.

This average information generated by the sample is known as the entropy of the hypotheses. Since we can transform logarithms from one base to another, we could use base two, or, binary base, for the logarithms. Then the unit of information is called a bit, a shorthand writing for binary digit, and one bit of information is capable of deciding between two equiprobable hypotheses, that is to say in a dichotomy with 50% probability of being true or false, $-\frac{1}{2} \log_2 \frac{1}{2} - \frac{1}{2} \log_2 \frac{1}{2} = \log_2 2 = 1$ bit [13].

In the case that when all the hypotheses are equally likely, we have $-\sum_{i=1}^n P(H_i) \log P(H_i) = \log n$. When $P(H_i) > 0$ and $\sum_{i=1}^n P(H_i) = 1$, then $\log n \geq -\sum P(H_i) \log P(H_i)$, and equality holds only if $P(H_i) = \frac{1}{n}$, thus the maximum uncertainty happens when all alternatives have the same likelihood.

We may denote the information generated by an event, A , in order to understand another event, B , as $I(A : B)$.

At this point, some special cases may be interesting to investigate.

Let the sample space be $\mathbb{R}^2$, and let H_1 declare that X and Y are dependent random variables with a joint density function $f(x, y)$, and H_2 declares that X and Y are independent random variables with their own density functions $g(x)$ and $h(y)$, respectively.

Therefore,

$$E[I(H_1 : H_2)] = \int \int f(x, y) \log \frac{f(x, y)}{g(x)h(y)} dx dy, \quad (9.8)$$

which is the average information generated by x about y , or vice versa. Particularly, if H_1 declares a bivariate normal joint density,

$$f(x, y) = \frac{1}{2\pi\sigma_x\sigma_y(1-\rho^2)^{\frac{1}{2}}} \exp\left[-\frac{1}{2(1-\rho^2)}\left(\frac{x^2}{\sigma_x^2} - 2\rho\frac{xy}{\sigma_x\sigma_y} + \frac{y^2}{\sigma_y^2}\right)\right], \quad (9.9)$$

and the marginal distributions for the H_2 would be

$$g(x) = \frac{1}{\sigma_x \sqrt{2\pi}} \exp\left[-\frac{x^2}{2\sigma_x^2}\right], \quad (9.10)$$

and

$$h(y) = \frac{1}{\sigma_y \sqrt{2\pi}} \exp\left[-\frac{y^2}{2\sigma_y^2}\right]. \quad (9.11)$$

Then $E[I(H_1 : H_2)] = -\frac{1}{2} \log(1 - \rho^2)$, that is the average information is a function of the correlation only.

Moreover, the distance between these two hypotheses would be

$$\int \int (f(x, y) - g(x)h(y)) \log \frac{f(x, y)}{g(x)h(y)} dx dy = \frac{\rho^2}{(1 - \rho^2)}, \quad (9.12)$$

which is also a function of the correlation only.

Example 9.1

Let x be the transmitted signal's voltage and y be the received signal's voltage. Assume that $y = x + \eta$, where η is a random variable, known as the noise's voltage. When the noise and the transmitted signal are taken as independent we have $f(x, y) = g(x)h(y|x) = g(x)h(y - x)$.

Moreover, when $f(x, y)$ is assumed as a bivariate normal distribution, we have $\rho_{\sigma_y}^{\sigma_x} = 1$ and $\rho^2 = \frac{\sigma_x^2}{\sigma_y^2} = \frac{E[x^2]}{E[x^2] + E[\eta^2]}$.

Where $E[x^2]$ is the mean transmitted signal power and $E[\eta^2]$ is the average power of the noise. The ratio of $\frac{E[x^2]}{E[\eta^2]}$ is known as the signal to noise ratio and used as a measure of the quality of the signal.

The sampling theorem of Shannon [13] says that in order to identify a function of period T and bandwidth W , we need to have twice the number of period times bandwidth, $2TW$, of samples coming from this function.

Hence, $E[I(H_1 : H_2); W, T] = 2WTE[I(H_1 : H_2)] = WT \log\left(1 + \frac{E[x^2]}{E[\eta^2]}\right)$, which is also known as the channel capacity.

Similarly, we can find the information generated by an experiment. Let θ be the parameter value over a parameter space Θ . Thus, $f(x, \theta)$ be the joint pdf of the sample and the parameter, $h(\theta)$ be the prior pdf of the parameter, $u(x|\theta)$ be the conditional pdf of x , for a fixed θ , and $v(x) = \int_{\Theta} u(x|\theta)h(\theta)d\theta$ be the marginal pdf of x .

Hence, the information provided by an experiment, by using the prior knowledge $h(\theta)$, would be $\int_{\Theta} \int_{\Omega} f(x, \theta) \log \frac{f(x, \theta)}{v(x)h(\theta)} dx d\theta$.

9.3 Properties

Let $I(1 : 2; E)$ be the information generated for the event 1 by the event 2, based on the experiment E .

Also, let $J(1, 2; E) = I(1 : 2; E) + I(2 : 1; E)$ be the divergence of these two events.

a) Information is additive for independent events, i.e. when X and Y are independent events, we have $I(1 : 2; X, Y) = I(1 : 2; X) + I(1 : 2; Y)$.

As an example, consider the random sample of n i.i.d. random variables coming from the same population, this sample provides n times more information than a single observation (remember the Fisher's information index).

For dependent events, we use the conditional information,

$$I(1 : 2; X, Y) = I(1 : 2; X) + I(1 : 2; Y|X) = I(1 : 2; Y) + I(1 : 2; X|Y).$$

Example 9.2

Let us have a bivariate normal distribution with constant variances, σ_X and σ_Y .

The joint pdf would be

$$f_i(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho_i^2}} \exp\left(-\frac{1}{2(1-\rho_i^2)}\left[\frac{(x-\mu_{i;X})^2}{\sigma_X^2} - 2\rho_i\frac{(x-\mu_{i;X})(y-\mu_{i;Y})}{\sigma_X\sigma_Y} + \frac{(y-\mu_{i;Y})^2}{\sigma_Y^2}\right]\right),$$

with marginal pdfs

$$g_i(x) = \frac{1}{\sigma_X\sqrt{2\pi}} \exp\left(-\frac{(x-\mu_{i;X})^2}{2\sigma_X^2}\right),$$

and

$$h_i(y|x) = \frac{1}{\sigma_Y\sqrt{2\pi(1-\rho_i^2)}^{\frac{1}{2}}} \exp\left(-\frac{[y-\mu_{i;Y}-\beta_i(x-\mu_{i;X})]^2}{2\sigma_Y^2(1-\rho_i^2)}\right),$$

where $\beta_i = \rho_i \frac{\sigma_Y}{\sigma_X}$, for $i = 1, 2$.

That implies, $I(1 : 2; X) = \frac{(\mu_{2;X}-\mu_{1;X})^2}{2\sigma_X^2}$, and

$$I(1 : 2; Y|X) = \frac{1}{2} \log \frac{1-\rho_2^2}{1-\rho_1^2} - \frac{\rho_1^2-\rho_2^2}{2(1-\rho_2^2)} + \frac{[(\mu_{2;Y}-\mu_{1;Y})-\beta_2(\mu_{2;X}-\mu_{1;X})]^2}{2\sigma_Y^2(1-\rho_2^2)} + \frac{(\rho_2-\rho_1)^2}{2(1-\rho_2^2)}.$$

Hence, we have

$$I(1 : 2; X, Y) = I(1 : 2; X) + I(1 : 2; Y|X).$$

For the independence of X and Y , we set $\rho_i = 0$.

b) Divergence is additive for independent events, i.e.

$$J(1, 2; X, Y) = J(1, 2; X) + J(1, 2; Y),$$

when X and Y are independent.

For dependent events, we have

$$J(1, 2; X, Y) = J(1, 2; X) + J(1, 2; Y|X) = J(1, 2; Y) + J(1, 2; X|Y).$$

9.4 Exercises

1. Let $N = \frac{n!}{n_1!n_2!\dots n_k!}$, where $n = \sum_{i=1}^k n_i$. Show that when n_i is large $\log N \approx -n \sum_{i=1}^k p_i \log p_i$, where $p_i = \frac{n_i}{n}$.
2. Let $P(n_1, \dots, n_k) = \binom{n}{n_1, \dots, n_k} q_1^{n_1} \dots q_k^{n_k}$,
where $n = \sum_{i=1}^k n_i$, $\sum_{i=1}^k q_i = 1$, and $q_i > 0$ for all i .
Show that $-\log P(n_1, \dots, n_k) \approx n \sum_{i=1}^k p_i \log \frac{p_i}{q_i}$, where $p_i = \frac{n_i}{n}$.
Note: In the above problem; when $q_i = \frac{1}{k}$, for all i , then $-\log P(n_1, \dots, n_k)$ is the information value of the observed k sequence.
3. If there are N identical looking gold coins, yet one is a counterfeit, i.e. lighter than the others, how many measurements on a balance, without using weights, are required to identify the counterfeit coin?
4. Prove that the sample average is a sufficient statistic for normal distributions with a common variance.
5. Let $f(x)$ be a probability density with mean μ and a finite variance σ^2 , such that $f(x) \log f(x)$ is σ -finite.
Prove that $\int_{-\infty}^{\infty} f(x) \log f(x) dx \geq \log \frac{1}{\sigma\sqrt{2\pi e}}$. Equality holds if and only if $f(x)$ is a normal pdf.
6. How would you generalize the above exercise 6 to multivariate probabilities?
7. Let H be a nonempty set of hypotheses, such that one of them is true and H_1 is a member of H . Hence, $P(H) = 1$ and $P(H|x) = 1$. Thus, $\log \frac{P(H_1|x)}{P(H|x)} - \log \frac{P(H_1)}{P(H)}$ yields information of x in favor of H_1 .
When is the value of $\log P(H_1|x) - \log P(H_1)$ is zero?
8. How many “bits” of information on the average are there in a binary selection of probabilities $p = 0.9999$ and $q = 1 - p$?
9. Consider sequences of k different symbols. Prove that the observation of a sequence of n symbols contains $n_1, n_2, \dots, n_k$ of the k symbols is approximately an information value of $n \sum_{i=1}^k \hat{p}_i \log \hat{p}_i + n \log k$, where $n = \sum_{i=1}^k n_i$ and $\hat{p}_i = \frac{n_i}{n}$.

Chapter 10

Linear Regression Model

10.1 Introduction

In many statistical inferences, the target random variable Y may depend upon other variables X (perhaps random). This dependence can be modeled by a functional structure and the model parameters may be estimated by the sample observations.

Once upon a time, Sir Francis Galton, who was a biologist and also a cousin of Charles Darwin, was conducting research on human genetics and collected data about the height of offspring of generations of people, then he observed that the height of an offspring converges to the average height of the population, as generations go by. In other word, the height of an individual regresses to the mean, thus he called this phenomenon the reversion to the mean (actually, he called it as the reversion to mediocrity).

Among the conditional moments of Y , for a given X , there are two that are of special interest; $E[Y|X]$ which is known as the regression to the mean, and $Var(Y|X)$ which is known as the scedasticity (spread) of the regression.

If $E[Y|X]$ is a linear function, then we have a linear regression model.

Similarly, if $Var(Y|X)$ is constant for all X values, then the regression model is called homoscedastic. Otherwise, if $Var(Y|X)$ is not constant for all X values, then it is called heteroscedastic regression.

A regression model represents causality, a relationship between a cause and its effect, or, a stimulus and its response, relationship.

10.2 Linear Regression Model

Here, we are interested in predicting a value for Y , for a given value of X , therefore we can build and use a probabilistic model in order to support these predictions. The simplest model to represent the relationships between X and Y is a linear model,

$$Y = \beta_0 + \beta_1 X + \epsilon, \quad (10.1)$$

with, β_0 and β_1 as unknown parameters.

If X is a random variable, then X and ϵ are assumed to be independent random variables. If X is not random but it is designed to produce a response Y , then X is called the design variable (input), and ϵ is assumed as an additive randomness effect, independent of the X values (this linear model is also known as the general linear model).

Our model supports both cases similarly, thus we do not specify the difference and construct the model by using a design variable X (nonrandom).

In this simple model we shall assume that $\epsilon \sim \mathcal{N}(0, \sigma^2)$, i.e. a zero mean and a constant variance normal random variable. Our model represents the average response level to a desired level of input (a.k.a. stimulus, or, predictor level)

$$E[Y|x] = E[\beta_0 + \beta_1 x + \epsilon|x] = \beta_0 + \beta_1 x + E[\epsilon|x]. \quad (10.2)$$

Observe that the last term is zero, since ϵ is assumed to be an independent normal random variable with zero mean.

Therefore, the average response to a fixed input, x , is given by

$$E[Y|x] = \beta_0 + \beta_1 x, \quad (10.3)$$

with unknown parameters β_0 and β_1 .

Also, observe that if X was a random variable, then $E[X|x] = x$, thus there is no structural difference here, and this leads us to conjure up a statistical method for estimating the unknown parameters, by simultaneously observing the stimulus and response variables, then fitting a line to this data set, $\{(X_i, Y_i) | i = 1, 2, \dots, n\}$, as best as we could. The best fitted line to this data set is known as the regression line.

We can represent the multi-dimensional model as,

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}. \quad (10.4)$$

Here, $\mathbf{y}$ denotes an n dimensional column vector of response observations, $\mathbf{X}$ denotes an $n \times 2$ dimensional design matrix, where the first column consists

of only 1s, the second column consists of the observation values of input x , and β is the 2 dimensional unknown column vector, $(\beta_0, \beta_1)^T$.

Similar to Eq. 10.1, ϵ denotes the n dimensional column vector of independent, zero mean and constant variance, normal random variables. If we desire to fit the line through the origin, then $\mathbf{X}$ would be the n dimensional design vector of observation values of the input x and β would be a singleton vector, of the only unknown parameter β_1 .

Here, our linear model indicates that the linear relationship is due to the unknown parameter vector β .

10.2.1 Best Fitted Line

The method of fitting a line to a set of points was standardized by the work of Gauss [3]. Once we collect a set of data, consisting of pairs of observations of stimulus and response, then we can estimate the unknown parameters of our linear regression model, $E[Y|x] = \beta_0 + \beta_1 x$, by using the least square estimation method of Gauss.

Let the estimated parameters be $\hat{\beta}_0$ and $\hat{\beta}_1$, thus the best fitted line, in the sense of the least square error, is given as $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 x$.

Here, we define the estimation error, also known as the residual, for the best fitted line as the difference between the actual response observation value versus the estimated response value for a given stimulus level, that is $e_i = y_i - \hat{y}_i$, for $i = 1, \dots, n$.

Therefore, we have the total squared error, $SSE = \sum_{i=1}^n e_i^2$, and then we search for the optimal values of the parameters that would minimize this total sum of the squared estimation errors, such that,

$$\{\hat{\beta}_0^*, \hat{\beta}_1^*\} = \arg \min_{\{\hat{\beta}_0, \hat{\beta}_1\}} \{SSE\} = \arg \min_{\{\hat{\beta}_0, \hat{\beta}_1\}} \left\{ \sum_{i=1}^n [y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i)]^2 \right\}. \quad (10.5)$$

Since SSE is a convex function of the estimated parameters, it has a minimum when $\frac{\partial SSE}{\partial \hat{\beta}_0} = 0$ and $\frac{\partial SSE}{\partial \hat{\beta}_1} = 0$, such that the simultaneous solution to these two equations with two unknowns provides the desired values of the least square estimated (l.s.e.) parameter values, $\{\hat{\beta}_0^*, \hat{\beta}_1^*\}$.

Based on the data set, we have computational formulas for these optimal parameters;

$$\hat{\beta}_1^* = \frac{S_{XY}}{S_{XX}}, \quad (10.6)$$

and

$$\hat{\beta}_0^* = \bar{y} - \hat{\beta}_1^* \bar{x}, \quad (10.7)$$

with the sample covariance, $S_{XY} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$,

and the sample variances $S_{XX} = \sum_{i=1}^n (x_i - \bar{x})^2$, $S_{YY} = \sum_{i=1}^n (y_i - \bar{y})^2$,

and sample means $\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$, and $\bar{y} = \frac{\sum_{i=1}^n y_i}{n}$.

of X and Y , respectively.

At last we can build our best fitted line, in the sense of the least squared error, in order to predict an expected value of the response for a given value of the stimulus;

$$\hat{Y} = \hat{\beta}_0^* + \hat{\beta}_1^* x, \quad (10.8)$$

a.k.a. regression line, or regression equation.

This regression line always includes $(\bar{x}, \bar{y})$, since

$$\hat{Y} = \hat{\beta}_0^* + \hat{\beta}_1^* \bar{x}, \quad (10.9)$$

and

$$\hat{\beta}_0^* = \bar{y} - \hat{\beta}_1^* \bar{x}. \quad (10.10)$$

We can now inspect the properties of the regression line in the linear statistical model,

$$Y = \beta_0 + \beta_1 x + \epsilon.$$

Rewriting Eq. 10.6, we have

$$\hat{\beta}_1^* = \frac{S_{XY}}{S_{XX}} = \frac{\sum_{i=1}^n (x_i - \bar{x})(Y_i - \bar{Y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{\sum_{i=1}^n (x_i - \bar{x})Y_i - \bar{Y} \sum_{i=1}^n (x_i - \bar{x})}{S_{XX}} = \frac{\sum_{i=1}^n (x_i - \bar{x})Y_i}{S_{XX}}.$$

Therefore,

$$\begin{aligned} E[\hat{\beta}_1^*] &= \frac{\sum_{i=1}^n (x_i - \bar{x})E[Y_i]}{S_{XX}} = \frac{\sum_{i=1}^n (x_i - \bar{x})(\beta_0 + \beta_1 x_i)}{S_{XX}} \\ &= \beta_0 \frac{\sum_{i=1}^n (x_i - \bar{x})}{S_{XX}} + \beta_1 \frac{\sum_{i=1}^n (x_i - \bar{x})x_i}{S_{XX}}. \end{aligned} \quad (10.11)$$

On the other hand, we know that $\sum_{i=1}^n (x_i - \bar{x}) = 0$ and

$S_{XX} = \sum_{i=1}^n (x_i - \bar{x})^2 = \sum_{i=1}^n (x_i - \bar{x})x_i$, thus we have

$$E[\hat{\beta}_1^*] = 0 + \beta_1 \frac{S_{XX}}{S_{XX}} = \beta_1.$$

That makes $\hat{\beta}_1^*$ as an unbiased estimator.

If we look at the variance,

$$\begin{aligned} Var(\hat{\beta}_1^*) &= Var\left(\frac{\sum_{i=1}^n (x_i - \bar{x})Y_i}{S_{XX}}\right) = \left(\frac{1}{S_{XX}}\right)^2 \sum_{i=1}^n Var((x_i - \bar{x})Y_i) \\ &= \left(\frac{1}{S_{XX}}\right)^2 \sum_{i=1}^n (x_i - \bar{x})^2 Var(Y_i). \end{aligned} \quad (10.12)$$

Since $Var(Y_i) = \sigma^2$, it immediately follows that $Var(\hat{\beta}_1^*) = \frac{\sigma^2}{S_{XX}}$.

Similarly,

$$E[\hat{\beta}_0^*] = E[\bar{Y} - \hat{\beta}_1^* \bar{x}] = \beta_0 + \beta_1 \bar{x} - \beta_1 \bar{x} = \beta_0, \quad (10.13)$$

which makes $\hat{\beta}_0^*$ as an unbiased estimator, with

$$Var(\hat{\beta}_0^*) = Var(\bar{Y}) + (\bar{x})^2 Var(\hat{\beta}_1^*) - 2Cov(\bar{Y}, \hat{\beta}_1^*), \quad (10.14)$$

where $Cov(\bar{Y}, \hat{\beta}_1^*) = \frac{\sigma^2}{n} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{S_{XX}}\right) = 0$.

Thus, we have

$$Var(\hat{\beta}_0^*) = \frac{\sigma^2}{n} + (\bar{x})^2 \left(\frac{\sigma^2}{S_{XX}}\right) = \sigma^2 \left(\frac{1}{n} + \frac{(\bar{x})^2}{S_{XX}}\right). \quad (10.15)$$

In general, we do not know the value of σ^2 , and we want to estimate it. Here, the estimator of σ^2 is

$$S^2 = \frac{1}{n-2} SSE = \frac{1}{n-2} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2. \quad (10.16)$$

We have $n-2$ as the degrees of freedom, since the sample would be used to estimate two unknown parameters. By Eq. 10.16, $E[S^2] = \frac{1}{n-2} E[SSE]$, where

$$\begin{aligned} E[SSE] &= E\left[\sum_{i=1}^n (Y_i - \hat{\beta}_0^* - \hat{\beta}_1^* x_i)^2\right] = E\left[\sum_{i=1}^n (Y_i - \bar{Y} + \hat{\beta}_1^* \bar{x} - \hat{\beta}_1^* x_i)^2\right] \\ &= E\left[\sum_{i=1}^n ((Y_i - \bar{Y}) - \hat{\beta}_1^* (x_i - \bar{x}))^2\right] \\ &= E\left[\sum_{i=1}^n (Y_i - \bar{Y})^2 + (\hat{\beta}_1^*)^2 \sum_{i=1}^n (x_i - \bar{x})^2 - 2\hat{\beta}_1^* \sum_{i=1}^n (x_i - \bar{x})(Y_i - \bar{Y})\right]. \end{aligned} \quad (10.17)$$

Since,

$$\sum (x_i - \bar{x})(Y_i - \bar{Y}) = \sum (x_i - \bar{x})^2 \hat{\beta}_1^*,$$

and

$$\sum (Y_i - \bar{Y})^2 = \sum Y_i^2 - n\bar{Y}^2,$$

we have;

$$\begin{aligned} E[SSE] &= \sum E[Y_i^2] - nE[\bar{Y}^2] - S_{XX}E[(\hat{\beta}_1^*)^2] \\ &= \sum (Var(Y_i) + E[Y_i]^2) - n(Var(\bar{Y}) + E[\bar{Y}]^2) - S_{XX}(Var(\hat{\beta}_1^*) + E[\hat{\beta}_1^*]^2) \\ &= n\sigma^2 + \sum \hat{y}_i^2 - n(\frac{\sigma^2}{n} + \bar{y}^2) - S_{XX}(\frac{\sigma^2}{S_{XX}} + \beta_1^2) \\ &= (n-2)\sigma^2. \end{aligned}$$

Hence, $S^2 = \frac{1}{n-2}SSE$ is an unbiased estimator of σ^2 .

A useful computational formula is, $SSE = S_{YY} - \hat{\beta}_1^* S_{XY}$.

From now on, without loss of generality, we will denote the optimal estimators as $\hat{\beta}_i$, for all i .

Example 10.1

Let the bivariate data be $\{(x, y)\} = \{(1, 0), (2, 0), (3, 1), (4, 1), (5, 3)\}$, coming from a designed experiment such that the stimulus values are set in equal distances around the origin and they are ordered from the smallest value to the largest value.

Let us find the l.s.e. parameters for the best fitted line,

$$\hat{\beta}_1 = \frac{S_{XY}}{S_{XX}} \text{ and } \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

Then we could construct the following Table 10.1 in preparation for the computations.

x	y	x- $\bar{x}$	y- $\bar{y}$	(x- $\bar{x}$)(y- $\bar{y}$)
1	0	-2	-1	2
2	0	-1	-1	1
3	1	0	0	0
4	1	1	0	0
5	3	2	2	4

Table 10.1: Bivariate Data Preparation

Thus, $S_{XX} = 10$, $S_{YY} = 6$, $S_{XY} = 7$, $\bar{x} = 3$, $\bar{y} = 1$, $\hat{\beta}_1 = \frac{7}{10} = 0.8$, $\hat{\beta}_0 = 1 - (0.7)(3) = -1.1$, and the best fitted line is $\hat{y} = -1.1 + 0.7x$.

Regression Line

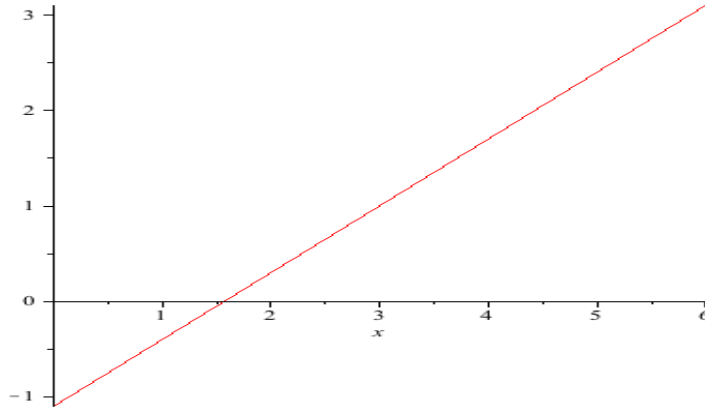


Figure 10.1: Best Fitted Line

The graphical picture of this regression line is given in Figure 10.1. Here, we can estimate the unknown variance σ^2 with the estimator

$$S^2 = \left(\frac{1}{n-2}\right)SSE, \text{ where } SSE = S_{YY} - \hat{\beta}_1 S_{XY} = 6 - (0.7)(7) = 1.1,$$

thus giving $S^2 = \frac{1.1}{3} = 0.3667$.

If we use this linear regression equation to estimate a response level for a fixed stimulus, say $x = 1.5$, then $\hat{y} = -1.1 + (0.7)(1.5) = -0.05$, which would be a reliable estimation.

On the other hand, if we estimate an average response to another level of stimulus, say $x = 10$, then $\hat{y} = -1.1 + (0.7)(10) = 5.9$ would be an unreliable estimation, since the stimulus level is outside of the range of the data observations and that extrapolation assumes the linear regression model stays the same for all regions, which may be a false assumption.

The plot of the raw data and a visual inspection of this scatter plot usually provides a quick decision about the linearity of the relationship between x and y , as given in the following Figure 10.2.

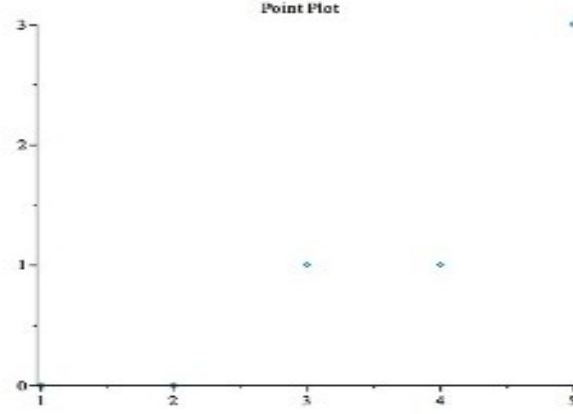


Figure 10.2: Plot of the Raw Data

10.3 Inferences about Linear Regression Model

In the above linear regression model, we have assumed that $\epsilon \sim \mathcal{N}(0, \sigma^2)$, and $\hat{\beta}_i \sim \mathcal{N}(\beta_i, c_{ii}\sigma^2)$, for $i = 0, 1$, where $c_{00} = \frac{\sum_{i=1}^n x_i^2}{nS_{XX}}$, and $c_{11} = \frac{1}{S_{XX}}$.

Hence, both of the unbiased estimators have variances as function of σ^2 .

This information serves us to build hypothesis tests about the model parameters as,

$$H_0 : \beta_i = \beta_{i0}, \text{ for } i = 0, 1.$$

Here, the test statistic is

$$Z_{Test} = \frac{\hat{\beta}_i - \beta_{i0}}{\sigma \sqrt{c_{ii}}},$$

and the decision rule would be, for a given sample realization and a desired level of significance α , reject H_0 if $|z_{Test}| > z_{\frac{\alpha}{2}}$.

In the case of unknown σ^2 , this variance has to be estimated again by using $S^2 = \frac{1}{n-2}SSE$.

Thus the test statistics would follow a Student's t-distribution with $n - 2$ degrees of freedom, where $T_{Test} = \frac{\hat{\beta}_i - \beta_{i0}}{S \sqrt{c_{ii}}}$, and the rejection region would be $\{|t_{Test}| > t_{\frac{\alpha}{2}}\}$.

Moreover, we can build a $100(1 - \alpha)\%$ confidence interval for the model parameters, such that,

$$(\hat{\beta}_i - t_{\frac{\alpha}{2}} S \sqrt{c_{ii}}, \hat{\beta}_i + t_{\frac{\alpha}{2}} S \sqrt{c_{ii}}).$$

The linear regression model represents the relationship between X and Y , as a cause-and-effect relationship, as long as the slope of the best fitted line is not zero.

Thus, β_1 is the critical parameter for the validity of the model.

We can construct a $100(1 - \alpha)\%$ confidence interval for β_1 and if this interval does not contains zero in it, i.e. does not change sign, then at this level of confidence, we may accept the validity of our model. Otherwise, our model would not support a linear relationship between X and Y .

Example 10.2

Consider the data set given in the Example 10.1

If we construct a 95% confidence interval for β_1 , then we would have $(\hat{\beta}_1 - t_{\frac{\alpha}{2}} S \sqrt{c_{11}}, \hat{\beta}_1 + t_{\frac{\alpha}{2}} S \sqrt{c_{11}})$,

where $c_{11} = \frac{1}{S_{XX}} = \frac{1}{10} = 0.1$,

$$\sqrt{c_{11}} = 0.3163, S = \sqrt{0.3667} = 0.6056,$$

and $t_{\frac{\alpha}{2}}$ would be subject to a Student's t-distribution with $5 - 2$, or,

3 degrees of freedom, with a desired tail area of $\frac{\alpha}{2} = 0.025$, thus

$$t_{\frac{\alpha}{2}} = 3.1824.$$

Therefore, our confidence interval would be $(0.09, 1.31)$. Since this interval does not have a sign change, that is zero is not included in this interval, we may say that β_1 is different from zero, at the 95% level of confidence. That is to say, this linear regression model may explain the behavior of the response, with 95% probability, or 5% of the response level variations cannot be explained by the regression model.

10.3.1 Predicting a Response

The main objective of building a linear regression model is to predict an expected response level Y , for a fixed stimulus level $X = x$, and we can build a $100(1 - \alpha)\%$ confidence interval for $E[Y|x]$ as follows:

$$(\hat{\beta}_0 + \hat{\beta}_1 x - t_{\frac{\alpha}{2}} S \sqrt{\frac{1}{n} + \frac{(x-\bar{x})^2}{S_{XX}}}, \hat{\beta}_0 + \hat{\beta}_1 x + t_{\frac{\alpha}{2}} S \sqrt{\frac{1}{n} + \frac{(x-\bar{x})^2}{S_{XX}}}),$$

where x is the fixed value of stimulus and $t_{\frac{\alpha}{2}}$ is subject to a Student's t-distribution with $n - 2$ degrees of freedom.

The variance of $\hat{Y}$ is obtained as,

$$\begin{aligned} Var(\hat{Y}) &= Var(\hat{\beta}_0 + \hat{\beta}_1 x) = Var(\bar{Y} - \hat{\beta}_1 \bar{x} + \hat{\beta}_1 x) \\ &= Var(\bar{Y}) + (x - \bar{x})^2 Var(\hat{\beta}_1) = \frac{1}{n} \sigma^2 + \frac{(x - \bar{x})^2}{S_{XX}} \sigma^2 \\ &= \left(\frac{1}{n} + \frac{(x - \bar{x})^2}{S_{XX}} \right) \sigma^2, \end{aligned}$$

where $\frac{1}{n} + \frac{(x - \bar{x})^2}{S_{XX}}$ is known as the variance inflation factor for the input level x .

A through investigation indicates that when the selected stimulus (predictor) level is $x = \bar{x}$, this confidence interval would be the sharpest (tightest) interval, for the desired level of confidence.

If we interpolate, that is if we estimate the average response within the range of data values, then we have a reliable confidence interval, where its sharpness depends upon the difference between the predictor level and the average predictor levels. But, if we extrapolate and estimate an average response outside of the range of data values, then we have no guarantee that our best fitted line is still valid, thus there is a great risk of making predictive errors.

In addition, as a secondary objective, we may be interested in predicting an individual value of the response Y , for a fixed level of stimulus $X = x$. This is a particularly interesting problem since Y is a random variable which was not involved with computing the best fitted line, thus it is not a parameter. Here, $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 x$ is the predicted value of Y , when $X = x$ is fixed, and Y_x is the actual response to $X = x$. In this case, the prediction error would be $\eta = Y_x - \hat{Y}$.

Obviously, $E[\eta] = 0$, but the variance of the prediction error would be,

$$Var(\eta) = Var(Y_x) + Var(\hat{Y}) - 2Cov(Y_x, \hat{Y}).$$

Here, we are predicting a random variable, new $Y|X = x$ value, which is independent from $\hat{\beta}_i$, for $i = 0, 1$, thus the correlation between Y_x and $\hat{Y}$ would be zero.

That is to say,

$$\begin{aligned} Var(\eta) &= Var(Y_x) + Var(\hat{Y}) \\ &= \sigma^2 + Var(\hat{\beta}_0 + \hat{\beta}_1 x) \\ &= \sigma^2 + \left(\frac{1}{n} + \frac{(x - \bar{x})^2}{S_{XX}} \right) \sigma^2 \\ &= \left(1 + \frac{1}{n} + \frac{(x - \bar{x})^2}{S_{XX}} \right) \sigma^2. \end{aligned}$$

This implies that $\eta \sim \mathcal{N}(0, Var(\eta))$, and when σ^2 is not known, it has to

be estimated by $S^2 = \frac{1}{n-2}SSE$, thus we can build a $100(1 - \alpha)\%$ confidence interval for an individual predicted value of response, for a fixed level of stimulus.

That would be subject to a Student's t-distribution with $n - 2$ degrees of freedom, as follows:

$$(\hat{\beta}_0 + \hat{\beta}_1 x - t_{\frac{\alpha}{2}} S \sqrt{1 + \frac{1}{n} + \frac{(x-\bar{x})^2}{S_{XX}}}, \hat{\beta}_0 + \hat{\beta}_1 x + t_{\frac{\alpha}{2}} S \sqrt{1 + \frac{1}{n} + \frac{(x-\bar{x})^2}{S_{XX}}}).$$

10.3.2 Slope and Correlation

The linear regression model assumes that the bivariate random variable (X, Y) is subject to a normal probability law, with $E[X] = \mu_X$, $E[Y] = \mu_Y$, $Var(X) = \sigma_X^2$, $Var(Y) = \sigma_Y^2$, and with a correlation coefficient of ρ , where $\rho = \frac{\sigma_X}{\sigma_Y} \beta_1$.

Thus, making inferences about ρ is not different than making inferences about β_1 .

The sample correlation coefficient

$$r = \frac{S_{XY}}{\sqrt{S_{XX}S_{YY}}} = \frac{S_{XY}}{\sqrt{S_{XX}S_{YY}}} \left(\sqrt{\frac{S_{XX}}{S_{XX}}} \right) = \hat{\beta}_1 \sqrt{\frac{S_{XX}}{S_{YY}}}, \quad (10.18)$$

is the m.l.e. of ρ and it has the same sign as $\hat{\beta}_1$.

The squared value of the sample correlation coefficient, r , is called the coefficient of determination, and it is denoted by r^2 , a.k.a. "R-SQUARE,"

$$\begin{aligned} r^2 &= \left(\frac{S_{XY}}{\sqrt{S_{XX}S_{YY}}} \right)^2 = \left(\frac{S_{XY}}{S_{XX}} \right) \left(\frac{S_{XY}}{S_{YY}} \right) \\ &= \hat{\beta}_1 \left(\frac{S_{XY}}{S_{YY}} \right) = \frac{S_{YY} - SSE}{S_{YY}} = 1 - \frac{SSE}{S_{YY}}. \end{aligned} \quad (10.19)$$

Hence, r^2 indicates the degree of total variations in Y , explained by X , for the given linear regression model.

Example 10.3

Consider the data set given in the example 10.1

The correlation coefficient is estimated to be,

$$r = \frac{S_{XY}}{\sqrt{S_{XX}S_{YY}}} = \frac{7}{\sqrt{(10)(6)}} = 0.9037.$$

Thus, the coefficient of determination would be $r^2 = 0.8167$, which indicates that the linear regression model does explain about 81% of the response level

variations, or, predicting the response level would be about 80% reliable, based on this model.

10.4 Multilinear Regression Model

A straightforward generalization of our linear regression model is to estimate an average response for a linear combination of stimuli.

Thus our linear model would be $Y = \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k + \epsilon$, known as multilinear model.

When we collect n observations, $\{(x_{i1}, x_{i2}, \dots, x_{ik}, y_i) | i = 1, 2, \dots, n\}$, and we fit our model to this data set:

$$\{y_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_k x_{ik} + \epsilon_i | (x_{i1}, x_{i2}, \dots, x_{ik}, y_i), i = 1, 2, \dots, n\}.$$

This data fitted model can be written as,

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}, \quad (10.20)$$

where

$$\begin{aligned} \mathbf{y} &= (y_1, \dots, y_n)^T, \\ \boldsymbol{\beta} &= (\beta_0, \beta_1, \dots, \beta_k)^T, \\ \boldsymbol{\epsilon} &= (\epsilon_1, \dots, \epsilon_n)^T, \end{aligned}$$

and

$$\mathbf{X} = \begin{pmatrix} x_{10} & x_{11} & \dots & x_{1k} \\ x_{20} & x_{21} & \dots & x_{2k} \\ \vdots & \vdots & \dots & \vdots \\ x_{n0} & x_{n1} & \dots & x_{nk} \end{pmatrix}.$$

Here, we set $x_{i0} = 1$, for $i = 1, \dots, n$.

On the other hand, if for some reason it is desired that the regression model fit through the origin, then we would set $\beta_0 = 0$ and $x_{i0} = 0$, for $i = 1, \dots, n$.

The least square estimation for the unknown parameter vector, $\boldsymbol{\beta}$ is,

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}, \quad (10.21)$$

where $(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T$ is known as the Penrose's generalized left inverse of the data matrix $\mathbf{X}$, as a generator of the orthogonal projection.

Differentiation, or, orthogonal projection of $\mathbf{y}$ on the space of $\mathbf{X}$, yields that $\hat{\boldsymbol{\beta}}$ minimizes the residual sum of squares, $(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^T (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$, thus $\hat{\boldsymbol{\beta}}$ is also the m.l.e. of $\boldsymbol{\beta}$.

Hence, we can construct a multilinear regression model, in order to estimate the expected value of a response, for a fixed level of stimuli, $\mathbf{x}$,

$$E[Y|\mathbf{x}] = \mathbf{x}^T \hat{\boldsymbol{\beta}}, \quad (10.22)$$

where $\mathbf{x}^T = (1, x_1, \dots, x_n)$.

We can compute the sum of the squared errors as $SSE = \mathbf{y}^T \mathbf{y} - \hat{\boldsymbol{\beta}}^T \mathbf{X}^T \mathbf{y}$, in terms of the l.s.e. of parameters and the data.

Therefore we can estimate the unknown σ^2 , by $S^2 = \frac{1}{n-(k+1)} SSE$.

10.4.1 Confidence Interval for the Average Response

The expected (average) response level for a fixed level of a set of stimuli; $x_1, \dots, x_n$, would be captured by $100(1 - \alpha)\%$ confidence interval,

$$(\mathbf{a}^T \hat{\boldsymbol{\beta}} - t_{\frac{\alpha}{2}} S \sqrt{\mathbf{a}^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{a}}, \mathbf{a}^T \hat{\boldsymbol{\beta}} + t_{\frac{\alpha}{2}} S \sqrt{\mathbf{a}^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{a}}). \quad (10.23)$$

where $\mathbf{a} = (1, x_1, \dots, x_n)^T$ is the augmented fixed stimuli vector, and the test statistic t is subject to a Student's t-distribution with $n - (k + 1)$ degrees of freedom.

10.4.2 Prediction Interval for an Individual Response

The prediction interval for an individual response level to a fixed level of a set of stimuli; $x_1, x_2, \dots, x_n$, can be constructed as described below.

Let $\mathbf{a} = (1, x_1, \dots, x_n)^T$ be the augmented fixed stimuli vector, then

$$(\mathbf{a}^T \hat{\boldsymbol{\beta}} - t_{\frac{\alpha}{2}} S \sqrt{1 + \mathbf{a}^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{a}}, \mathbf{a}^T \hat{\boldsymbol{\beta}} + t_{\frac{\alpha}{2}} S \sqrt{1 + \mathbf{a}^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{a}}). \quad (10.24)$$

would be a $100(1 - \alpha)\%$ prediction interval, where t is subject to a Student's t-distribution with $n - (k + 1)$ degrees of freedom.

10.4.3 Validity Test for the Reduced Multilinear Model

We construct a linear regression model in order to predict the response for a linear combination of input variables.

Thus, we want each one of these input variables to explain some part of the behavior of the response, but we do not want that one input variable to provides the same information as provided by another input variable.

For this reason we look at the absolute correlation between input variables, and if we find collinear input variables, then we can remove at least one of them from our linear regression model, since only one of them would be sufficient to provide the necessary information that we want.

A high degree of absolute correlation, say $|r_{X_i X_j}| \geq 0.98$, for $i, j = 1, 2, \dots, k$ and $i \neq j$, may be considered as an evidence of collinearity between the X_i and X_j input variables, thus we may eliminate one of them from our model.

Similarly, if the partial slope, β_i , for an input variable, X_i , is zero, or, very close to zero, we may say that this source of information, i.e. observing the variable X_i , would be a waste of resources, since it would not contribute to explaining the response's behavior. Thus, we can eliminate this input variable from our model, too.

The sum of the squares of residuals, SSE , provides an instrument to make decisions about the linear model parameters, β_i s. We may construct a complete linear model, $Y = \beta_0 + \beta_1 x_1 + \dots + \beta_s x_s + \beta_{s+1} x_{s+1} + \dots + \beta_k x_k + \epsilon$, and a reduced (shortened) linear model, $Y = \beta_0 + \beta_1 x_1 + \dots + \beta_s x_s + \epsilon$.

Here, let us denote the sum of the squares for the complete model by SSE_C , and for the reduced model by SSE_S . Also, observe that our complete model includes the reduced model and some other extra terms, that is $k > s$.

Now, ask yourself, if these extra terms would contribute a substantial degree of information for the predictive power of our complete linear model that was not available in the reduced linear model, then how could we utilize the sum of the squares, for both of our linear models, to detect this phenomenon?

Hint: If our complete model has more explanatory power than our reduced model, then we expect that the predictions generated by the complete model would have smaller errors than the predictions generated by our reduced model.

The degree of difference between the sum of squares, $SSE_S - SSE_C$, would indicate the strength of evidence that these extra terms in our complete model are necessary to explain the behavior of the response.

Hence, we can construct a hypothesis test to decide whether we can utilize a reduced model, or, we have to use a complete model, as follows:

$$H_0 : \beta_{s+1} = \dots = \beta_k = 0,$$

We can partition the SSE_S such that $SSE_S = SSE_C + SSE_D$, where $SSE_D = (SSE_S - SSE_C)$, by Pythagorean theorem, and if H_0 is a correct statement, then

$$\chi^2_S = \frac{SSE_S}{\sigma^2},$$

$$\chi^2_C = \frac{SSE_C}{\sigma^2},$$

$$\chi^2_D = \frac{SSE_D}{\sigma^2},$$

subject to Chi-square distributions with

$n - (s + 1)$, $n - (k + 1)$, and $k - s$ degrees of freedoms, respectively.

Hence, the ratio of $F_{Test} = \frac{\chi^2_D/(k-s)}{\chi^2_C/[n-(k+1)]} = \frac{SSE_D/(k-s)}{SSE_C/[n-(k+1)]}$,

would be subject to an F distribution with $k-s$ numerator degrees of freedom and $n - (k + 1)$ denominator degrees of freedom.

Here, we have observed that large values of F_{Test} would be the evidence for rejecting our null hypothesis. Thus, we select a significance level α and construct the decision rule: if $F_{Test} > F_\alpha$, then reject H_0 .

10.4.4 Testing for the Multilinear Model Validity

A special case of this above mentioned test is known as the $F - Test$:

$$H_0 : \beta_1 = \dots = \beta_k = 0,$$

that is to say our multilinear model does not support any prediction of the response for observing stimuli, and it would be a random number $Y = \beta_0 + \epsilon$.

In this case, the test statistic,

$$F_{Test} = \frac{(S_{YY} - SSE_C)/k}{SSE_C/[n - (k + 1)]}, \quad (10.25)$$

would be subject to an F distribution with k numerator degrees of freedom and $n - (k + 1)$ denominator degrees of freedom.

We have a performance measure of the multilinear model, known as the multiple coefficient of determination,

$$R^2 = \frac{S_{YY} - SSE_C}{S_{YY}} = 1 - \frac{SSE_C}{S_{YY}} \in [0, 1], \quad (10.26)$$

which indicates the degree of variations in the response variable, explained by the multilinear model. This is the squared value of the correlation coefficient of the response variable and the set of input variables, i.e. the cosine of the angle between the $\mathbf{y}$ and the space of $\mathbf{X}$.

Thus, higher the R^2 value, higher the explanatory strength of the model.

Also, the multiple coefficient of determination and testing the validity of the complete model are related through the test statistic,

$$F_{Test} = \frac{n - (k + 1)}{k} \left(\frac{R^2}{1 - R^2} \right). \quad (10.27)$$

Hence, higher the R^2 value is, higher the probability of rejecting the null hypothesis would be.

After finding the evidence for the validity of the model, we have a second wave of questions to answer: do we need every one of the input variables in the model?

The answer to this legitimate follow-up question is provided by the individual t - Tests, conducted for each of the input variables.

10.4.5 Testing for the Necessity of an Input Variable

An hypothesis test for the coefficient of each input variable is different than zero is known as the t - Test, for $i = 1, 2, 3, \dots, k$:

$$H_0 : \beta_i = 0.$$

The test statistic is subject to a Student's t-distribution with $n - k - 1$ degrees of freedom:

$$t_{Test} = \frac{b_i - \beta_i}{S_{b_i}},$$

where b_i is the estimated coefficient of X_i , β_i is the value of the null hypothesis assumption for the coefficient of X_i , and S_{b_i} is the standard error of b_i .

At a desired level of significance α , the critical level to compare the test statistic would be subject to a Student's t-distribution, $t_{critical} = t_{\frac{\alpha}{2}; df}$. Where $df = n - k - 1$.

If $|t_{Test}| > t_{critical}$, then reject H_0 .

If an input variable fails the t-Test, that is its coefficient is not significantly different than zero, then this input variable may be removed from the model.

10.5 Linearized Regression Models

It is not expected that all predictor-response relationships have linear structures, but some of these nonlinear relations may be translated into a linear structure, and these linearized form of the data sets may be described by a linear regression model. We will provide a short list of these transformed linear regression models.

10.5.1 Exponential Regression Model

Assume that the relationship between the predictor, x , and the response, y , has an exponential structure, i.e. $y = ce^{ax}$, where $c > 0$.

Hence, a transformation by the natural logarithm, $\log y = \log c + ax$, would provide a linearized relationship between x and $\log y$. After fitting a linear regression model to the linearized data set, an inverse transformation of the $\log \hat{y}$ would yield the estimated level of response to the fixed level of predictor, such as $e^{\log \hat{y}}$.

10.5.2 Logarithmic Regression Model

Let the relationship between the predictor and the response be $y = cx^a$, for $c > 0$. Hence, by using the logarithm as the translation, we would have

$$\log y = \log c + a \log x.$$

we have a linear relationship between the $\log y$ and the $\log x$, that would be fitted by a regression line.

Again, the expected response can be obtained by the antilogarithm of the $\log \hat{y}$.

10.5.3 Logistic Regression Model

The evolution, or growth, phenomenon may be expressed by the logistic model, a.k.a. S-shaped curve, $y = \frac{c}{1+e^{a+bx}}$, where $c > 0$.

Therefore, $\frac{c}{y} = 1 + e^{a+bx}$, or, $\frac{c-y}{y} = e^{a+bx}$.

We can use the natural logarithm transformation, to yield a linear relationship, $\log \frac{c-y}{y} = a + bx$.

Hence, $\log \frac{c-\hat{y}}{\hat{y}}$ is linearly related to x .

10.5.4 Estimating the Success Probability for a Bernoulli Distribution

The response variable may be a categorical variable that depends on a dichotomous event, a success or a failure. The logistic regression model would help us to predict the probability of the response, for a set of fixed values of the predictor variables.

The odds ratio, $odds(A) = \frac{P(A)}{1-P(A)}$, for a given event A and its probability, $P(A)$, would yield the prediction of the success probability of a Bernoulli distribution.

That is, the natural logarithm of the odds ratio would yield a linear logistic regression equation,

$\log odds(A) = \beta_0 + \beta_1 X_1 + \dots + \beta_k X_k + \epsilon$, where $\epsilon \sim \mathcal{N}(0, \sigma^2)$ is the independent random error, and there are k number of independent predictor variables.

The logarithm of the estimated odds ratio for the event A is obtained from this logistic regression equation, $\log \widehat{odds}(A)$.

Therefore, the estimated odds ratio for the given set of predictor variables would be $\widehat{odds}(A) = e^{\log \widehat{odds}(A)}$.

When we have obtained the estimated odds ratio, then we can find the estimated probability of the event A , that is, $\widehat{P}(A) = \frac{\widehat{odds}(A)}{1 + \widehat{odds}(A)}$.

10.6 Exercises

1. Consider the following bivariate data set.

$$\{(X, Y)\} = \{(-2, 0), (-1, 0), (0, 1), (1, 1), (2, 3)\}.$$

Find the l.s.e. parameters, $\hat{\beta}_1$ and $\hat{\beta}_0$, construct the linear regression equation, estimate the average response $\hat{y}$ when stimulus is fixed at $x = 1.5$ and $x = 10$, test for $\beta_1 = 0$ at the 5% level of significance, find the r^2 and explain your results.

2. Consider the following multivariate data.

$$\mathbf{y} = (0, 0, 1, 1, 3)^T, \mathbf{X} = \begin{pmatrix} 1 & -2 & 1 \\ 1 & -1 & 2 \\ 1 & 0 & 3 \\ 1 & 1 & 4 \\ 1 & 2 & 5 \end{pmatrix}.$$

Find the l.s.e. parameter vector, $\hat{\beta}$, construct the multilinear regression equation, estimate the average response, $\hat{y}$, for the fixed stimulus vector, $\mathbf{x} = (1.5, 15)$, test for the validity of this model at the 5% level of significance, find the R^2 and explain your results.

3. Find the reduced model, for the data set given in the above problem, such that the reduced model's R^2 would be no less than 85% of the full model's R^2 .
4. A designed experiment to study the amount of wear, y , related to the oil viscosity, x_1 , and the load, x_2 , was conducted and yield the following result, as given in Table 10.2.

Estimate the unknown parameters of the multilinear regression model,

$E[Y|x_1, x_2] = \beta_0 + \beta_1 x_1 + \beta_2 x_2$, and predict the amount of wear when oil viscosity is 25 and the load is 1500.

Also, construct the 95% confidence interval for the mean wear, and a 95% prediction interval for the wear when $x_1 = 25$, and $x_2 = 1500$.

5. Using the data given in the above Exercise 4, test for the validity of the linear regression model, at the 5% level of significance.
6. Use the data given in Exercise 4, estimate σ^2 , by using the multiple regression model of y on x_1 and x_2 .
7. Compute the coefficient of multiple determination, for Exercise 4, and interpret it.
8. If we use a linear model with k independent variables, and test the null hypothesis, $H_0 : \beta_1 = \dots = \beta_k = 0$, then show that the proper F distributed test statistic would be $F_{Test} = \frac{n-(k+1)}{k}(\frac{R^2}{1-R^2})$.
9. Consider the general linear model, $Y = \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k + \epsilon$, where $E[\epsilon]=0$ and $Var(\epsilon) = \sigma^2$. Let $\hat{\beta}_i = \mathbf{a}^T \hat{\beta}$, where $a_j = 1$, if $j = i$, zero otherwise.
Show that $E[\hat{\beta}_i] = \beta_i$, and $Var(\hat{\beta}_i) = c_{ii}\sigma^2$, where c_{ii} is the element of row- i and column- i of $(\mathbf{X}^T \mathbf{X})^{-1}$.
10. Show that the least squares prediction equation,
 $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \dots + \hat{\beta}_k x_k$ passes through the point $(\bar{x}_1, \bar{x}_2, \dots, \bar{x}_k, \bar{y})$.

y	x_1	x_2
193	1.6	851
172	22	1058
113	33	1357
230	15.5	816
91	43	1201
125	40	1115
100	25	1001
200	3	900

Table 10.2: Data for Exercise 4

Chapter 11

Analysis of Variance

11.1 Introduction

An experimenter designs an experiment, in order to investigate the effect of stimuli on a response. Independent variables that the experimenter keeps it under control are known as factors. Any data analysis needs the identification of the factors, as well as other categorical variables that are known as blocks. Our experimenter may wish to increase the observational accuracy, by controlling the sources of variations in the response variable, thus perhaps randomizes the blocking effect in the designed experiment.

11.2 Analysis of Variance Method

The analysis of variance, a.k.a. ANOVA, method was constructed by analyzing the set of variations of response, and finding the extent of the effect of each independent variable on the response variations.

In this method, the total sum of squares, $SSTOTAL = \sum_{i=1}^n (y_i - \bar{y})^2$, for a sample of n observations, identifies the response variations. In this method, we partition the total sum of squares into partial sum of squares that are generated by each of the independent variable, and a remainder part of the sum of squares that is generated by an independent random error, known as the SSE . If we consider the case that the independent variables are not related to the response, then each one of the partial sum of the squares, generated by the corresponding independent variable, yields an unbiased and independent estimator for the response's variance σ^2 , similar to the sum of squares of error estimating σ^2 .

On the other hand, if a variable is highly related to the response, then this variable's partial sum of squares will be much higher, i.e. inflated, than the other variables' partial sum of squares.

Here, make an important observation that variance is a function of the mean. Thus, we can search for this inflation by comparing the partial sum of squares with the sum of squares of error, SSE , where the total sum of squares can be partitioned as the collection of partial sum of squares and the sum of squares of error.

The test statistic would be subject to an F distribution, as a ratio of the partial sum of squares to the sum of squares of error, such that

H_0 : *There is no effect due to independent variable*,
shall be rejected for significantly large F values.

11.3 Completely Random Experiment Design Model

If we randomly collect independent samples coming from k independent populations, then we would design a completely randomized experiment.

Assume that $\{Y_{ij}\}$ are independent random variables, coming from normal populations, such that $E[Y_{ij}] = \mu_i$, and $Var(Y_{ij}) = \sigma^2$, for $i = 1, 2, \dots, k$ and $j = 1, 2, \dots, n_i$.

Let

$$Y_{ij} = \mu_i + \epsilon_{ij}, \quad (11.1)$$

where the independent random variables, $\{\epsilon_{ij}\}$, i.e. the random error terms that represent the difference between the observations and the population mean in each sample, would be subject to a stationary normal probability law, $\epsilon_{ij} \sim \mathcal{N}(0, \sigma^2)$. Thus, Y_{ij} be an independent normal random variable, with mean μ_i and variance σ^2 , for all i, j .

Assume that

$$\mu_i = \mu + \tau_i, \quad (11.2)$$

for $i = 1, \dots, k$, such that $\sum_{i=1}^k \tau_i = 0$.

Since, $\sum_{i=1}^k \mu_i = k\mu + \sum_{i=1}^k \tau_i = k\mu$,

we have $\mu = \frac{\sum_{i=1}^k \mu_i}{k}$, i.e. the average of k -populations' means, known as the overall mean.

Here, $\tau_i = \mu_i - \mu$ is the difference between the mean of population- i and the overall mean, thus it is known as the treatment effect, or, factor effect, of the population- i , for $i = 1, \dots, k$.

Hence, there is one variable of interest, i.e. the treatment effect, therefore the corresponding analysis of variance method, based on the completely random experiment design, is called the one-way ANOVA.

For this case, a completely randomized experiment design model is :

$$Y_{ij} = \mu + \tau_i + \epsilon_{ij}, \quad (11.3)$$

for $i = 1, \dots, k$ and $j = 1, \dots, n_i$.

Based on this model, we can also construct equivalent hypothesis tests, such that:

$H_0 : \mu_1 = \dots = \mu_k$, would be equivalent to

$H_0 : \tau_1 = \dots = \tau_k = 0$, with the same test statistic, F_{Test} .

11.3.1 Comparing Means

If we assume that these k independent populations are normally distributed with a commonly shared variance, σ^2 , and with means, $\mu_1, \dots, \mu_k$, respectively, then we can collect random samples coming from each population with sample sizes, n_i , for $i = 1, \dots, k$, such that the total number of observations would be $n = \sum_{i=1}^k n_i$.

Let Y_{ij} be the response of the j th experimental unit in the i th sample, $Y_{i.}$ and $\bar{Y}_{i.}$ be the total and the mean of the n_i responses in the i th sample, where the dot in the second subscript indicates that this quantity is computed by summing over all values in the j variable.

Thus,

$$Y_{i.} = \sum_{j=1}^{n_i} Y_{ij}, \text{ and } \bar{Y}_{i.} = \left(\frac{1}{n_i}\right) \sum_{j=1}^{n_i} Y_{ij} = \left(\frac{1}{n_i}\right) Y_{i.}, \text{ for } i = 1, 2, \dots, k.$$

Once we have this form, we can construct an ANOVA test for the means, as described below.

The total sum of squares is the combination of sum of the squares of treatments (factors) and the sum of the squares of error,

$$SSTOTAL = SST + SSE,$$

where,

$$\begin{aligned}
SSTOTAL &= \sum_{i=1}^k \sum_{j=1}^{n_i} (Y_{ij} - \bar{Y})^2 = \sum_{i=1}^k \sum_{j=1}^{n_i} Y_{ij}^2 - \frac{1}{n} \left(\sum_{i=1}^k \sum_{j=1}^{n_i} Y_{ij} \right)^2 \\
&= \sum_{i=1}^k \sum_{j=1}^{n_i} Y_{ij}^2 - n\bar{Y}^2,
\end{aligned} \tag{11.4}$$

and

$$SST = \sum_{i=1}^k n_i (\bar{Y}_{i.} - \bar{Y})^2 = \sum_{i=1}^k \frac{Y_{i.}^2}{n_i} - n\bar{Y}^2,$$

and

$$SSE = SSTOTAL - SST = \sum_{i=1}^k \sum_{j=1}^{n_i} (Y_{ij} - \bar{Y}_{i.})^2 = \sum_{i=1}^k (n_i - 1) S_i^2,$$

and

$$S_i^2 = \frac{1}{n_i - 1} \sum_{j=1}^{n_i} (Y_{ij} - \bar{Y}_{i.})^2.$$

Here, the SSE is a function of the sample variances, S_i^2 , for $i = 1, 2, \dots, k$, and each one of the sample variances provides an unbiased estimator for σ^2 with $n_i - 1$ degrees of freedom.

Therefore, an unbiased estimator for σ^2 , constructed by combining all the independent samples, based on $\sum_{i=1}^k (n_i - 1) = n - k$ degrees of freedom, would be called the mean square error, MSE , and is denoted by S^2 ,

$$MSE = \frac{SSE}{n - k}. \tag{11.5}$$

The overall sample mean is given as,

$$\bar{Y} = \frac{1}{n} \sum_{i=1}^k \sum_{j=1}^{n_i} Y_{ij} = \frac{1}{n} \sum_{i=1}^k n_i \bar{Y}_{i.}. \tag{11.6}$$

This implies that the SST is a function of the sample averages, $\bar{Y}_{i.}$, for $i = 1, 2, \dots, k$, and the mean square treatment, MST , for all the samples would be,

$$MST = \frac{SST}{k - 1}. \tag{11.7}$$

Therefore, we test the null hypothesis which states that all these independent populations have the same mean, versus the alternative hypothesis which states that at least one of these populations has a different mean than the others.

$$H_0 : \mu_1 = \dots = \mu_k.$$

Based on the test statistic,

$$F_{Test} = \frac{MST}{MSE}, \quad (11.8)$$

with $k - 1$ numerator degrees of freedom and $n - k$ denominator degrees of freedom.

If the test statistic is significantly large, then we would have evidence, beyond a reasonable doubt, that the null hypothesis is a false statement. Hence, the decision rule would be: If $F_{Test} > F_\alpha$, then reject the null hypothesis, at the α level of significance.

The information summary for a completely random experiment design can be represented by a tabular form, known as the ANOVA table, such as the following Table 11.1, that would be used for the test statistics computations.

Source	df	SS	MS	F
Treatments	k-1	$SST = \sum_{i=1}^k n_i (\bar{Y}_i - \bar{Y})^2$	$MST = \frac{SST}{k-1}$	$F_{Test} = \frac{MST}{MSE}$
Error	n-k	$SSE = \sum_{i=1}^k \sum_{j=1}^{n_i} (Y_{ij} - \bar{Y}_i)^2$	$MSE = \frac{SSE}{n-k}$	
TOTAL	n-1	$SSTOTAL = \sum_{i=1}^k \sum_{j=1}^{n_i} (y_{ij} - \bar{y})^2$		

Table 11.1: ANOVA Table for Completely Randomized Experiments

This tabular form is almost standard, and when a person utilizes a computer software for ANOVA, this table (or, a very similar table) will be printed out by the software, usually with additional information about the p -value.

Example 11.1

A quality controller wishes to compare four high-end quality headphones, brands A , B , C , and D , for sound reproduction quality.

A random sample of 5 brand A , 4 brand B , 7 brand C , and 6 brand D headphones are selected for the test.

The measured sound quality, based on the 10×percentage of total harmonic distortion, is as follows:

Observations

$A : \{10, 15, 8, 12, 15\}$

$B : \{14, 18, 21, 15\}$

$C : \{17, 16, 14, 15, 17, 15, 18\}$

$D : \{12, 15, 17, 15, 16, 15\}$

A computer software generated the following ANOVA Table 11.2, also computing a p -value of 0.018, that would be used for comparing the average sound quality of these brands, assuming they all have the same variance for the sound quality.

Source	df	SS	MS	F
Factor	3	68	22.67	4.34
Error	18	94	5.22	
TOTAL	21	162		

Table 11.2: ANOVA Table for Example 11.1

Is there a difference among the average sound quality of these four brands, at the 5% level of significance?

11.3.2 Estimating Treatment Mean

We can construct confidence intervals, for estimating a single treatment effect, as well as comparing pairs of treatments effects, under the completely random experiment model.

The $100(1-\alpha)\%$ confidence interval for the treatment mean of population- i would be

$$(\bar{Y}_i - t_{\frac{\alpha}{2}} S(\frac{1}{\sqrt{n_i}}), \bar{Y}_i + t_{\frac{\alpha}{2}} S(\frac{1}{\sqrt{n_i}})). \quad (11.9)$$

The $100(1-\alpha)\%$ confidence interval for the difference of the treatment means of population- i and population- j is

$$([\bar{Y}_i - \bar{Y}_j] - t_{\frac{\alpha}{2}} S(\sqrt{\frac{1}{n_i + n_j}}), [\bar{Y}_i - \bar{Y}_j] + t_{\frac{\alpha}{2}} S(\sqrt{\frac{1}{n_i + n_j}})), \quad (11.10)$$

where $S = \sqrt{MSE} = \sqrt{\frac{SSE}{n_1 + \dots + n_k - k}}$

and $t_{\frac{\alpha}{2}}$ is subject to Student's t-distribution with $n - k$ degrees of freedom.

11.4 Randomized Block Experiment Design Model

The randomized block design of experiment is for comparing the average effects of k number of treatments (factors), by using b number of blocks

(categories).

Here, the experimental units within each block share the same common characteristic, and the treatments are randomly assigned to blocks (thus, the name randomized block design), such that each treatment would be applied only once in each block.

The total number of observations in this experiment would be $n = (k)(b)$, by using two independent variables, namely the treatment effect and the block effect, thus this analysis of variance method is also called as the two-way ANOVA.

11.4.1 ANOVA for Randomized Block Design

Let the observation of the treatment- i , in the block- j , be Y_{ij} , for $i = 1, \dots, k$ and $j = 1, \dots, b$, and defined as

$$Y_{ij} = \mu + \tau_i + \beta_j + \epsilon_{ij}, \quad (11.11)$$

where μ is the overall mean, τ_i is the effect of treatment- i , such that $\sum_{i=1}^k \tau_i = 0$, β_j is the effect of block- j , such that $\sum_{j=1}^b \beta_j = 0$, and ϵ_{ij} is the independent random error which is normally distributed, with zero mean and a constant variance of σ^2 .

In this model, we assumed that the block effects are not random, but fixed constants, such as the treatment effects being unknown constants. For this reason this model is called fixed block effect model.

Here, we have $E[Y_{ij}] = \mu + \tau_i + \beta_j$, and $Var(Y_{ij}) = \sigma^2$, for $i = 1, \dots, k$ and $j = 1, \dots, b$. Based on this model, we can construct two hypothesis tests, one for the average treatment(factor) effect, and the other for the average blocking effect:

$H_{T0} : \tau_1 = \dots = \tau_k = 0$. The test statistic, F_{TTest} , has $k - 1$ numerator degrees of freedom, and $(k - 1)(b - 1)$ denominator degrees of freedom.

$H_{B0} : \beta_1 = \dots = \beta_b = 0$, The test statistic, F_{BTest} , has $b - 1$ numerator degrees of freedom, and $(k - 1)(b - 1)$ denominator degrees of freedom.

In this model, when two observations received the same treatment, the difference of the average responses would be $E[Y_{iu}] - E[Y_{iv}] = \beta_u - \beta_v$, for $u, v = 1, \dots, b$.

Similarly, if two observations are selected from the same block, the difference of the average responses would be, $E[Y_{yj}] - E[Y_{zj}] = \tau_y - \tau_z$, for $y, z = 1, \dots, k$.

On the other hand, if observations are taken as a combination of different treatments and different blocks, then the difference of the average responses would be, $E[Y_{ij}] - E[Y_{lm}] = (\tau_i - \tau_l) + (\beta_j - \beta_m)$, for $i, l = 1, \dots, k$ and $j, m = 1, \dots, b$.

11.4.2 Comparing Means of Treatments and Blocks

The analysis of variance for the randomized block design is similar to completely randomized design, with additional average effect of the blocks. Thus, ANOVA computations are for comparing the average effect of treatments, as well as comparing the average effect of blocking, based on the $n = (b)(k)$ observations, would be:

$$SSTOTAL = \sum_{i=1}^k \sum_{j=1}^b (Y_{ij} - \bar{Y})^2 - CM = SST + SSB + SSE, \text{ where}$$

$$SST = b \sum_{i=1}^k (\bar{Y}_{i.} - \bar{Y})^2 = \sum_{i=1}^k \frac{Y_{i.}^2}{b} - CM,$$

$$SSB = k \sum_{j=1}^b (\bar{Y}_{.j} - \bar{Y})^2 = \sum_{j=1}^b \frac{Y_{.j}^2}{k} - CM,$$

$$CM = \frac{1}{bk} (\sum_{i=1}^k \sum_{j=1}^b Y_{ij})^2, \text{ and}$$

$$SSE = SSTOTAL - SST - SSB.$$

Those computations are represented by an ANOVA table, such as the following Table 11.3, that would be used for the test statistics computations.

Source	df	SS	MS	F
Treatments	k-1	SST	$MST = \frac{SST}{k-1}$	$F_{TTest} = \frac{MST}{MSE}$
Blocks	b-1	SSB	$MSB = \frac{SSB}{b-1}$	$F_{BTest} = \frac{MSB}{MSE}$
Error	n-k-b+1	SSE	$MSE = \frac{SSE}{n-k-b+1}$	
TOTAL	n-1	SSTOTAL		

Table 11.3: ANOVA Table for Randomized Block Experiments

This tabular form is almost standard, and when a person utilizes a computer software for ANOVA, this table (or, a very similar table) will be printed out by the software, usually with additional information about the corresponding p -values, for testing differences among the average effect of treatments, as well as the average effect of blocks.

The hypothesis test for our two-way ANOVA model would have the following decision rules: If $F_{TTest} > F_{\alpha}$, then reject H_{T0} . Also, if $F_{BTest} > F_{\alpha}$ then reject H_{B0} .

Example 11.2

Consider a randomly blocked experiment data resulted the following ANOVA Table 11.4, as a computer software output.

Source	df	SS	MS	F
Factor	2	71.17	35.58	9.35
Block	3	90.92	30.31	7.96
Error	6	22.83	3.81	
TOTAL	11	184.92		

Table 11.4: ANOVA Table for Example 11.2

The computer printed the p -value of treatments as 0.014, and the p -value of blocks as 0.016. These will be used for comparing the average treatment effects, as well as the average block effects.

Is there a difference among the average treatment effects, as well as the average block effects, at the 5% level of significance?

11.4.3 Estimating Means of Treatments and Blocks

We can construct a $100(1 - \alpha)\%$ confidence interval for the difference of the average effect of treatments, $\mu_i - \mu_l$, such that

$$((\bar{Y}_i - \bar{Y}_l) - t_{\frac{\alpha}{2}} S \sqrt{\frac{2}{b}}, (\bar{Y}_i - \bar{Y}_l) + t_{\frac{\alpha}{2}} S \sqrt{\frac{2}{b}}), \quad (11.12)$$

for any $i, l = 1, \dots, k$, where the number of observations for each treatment is, $n_i = n_l = b$, and $S = \sqrt{MSE}$.

Here, $S^2 = MSE$ is an estimator of σ^2 , and computed in the ANOVA Table 11.3. Also, $t_{\frac{\alpha}{2}}$ is subject to a Student's t-distribution with $n - b - k + 1$ degrees of freedom.

If the confidence interval has a sign change, i.e. it includes zero in it, we say there is no significant difference between the average effects of treatment- i and treatment- l .

Otherwise, we may say there is a difference between the average treatment effects and analyze further, at the given confidence level.

Similarly, we can construct a $100(1 - \alpha)\%$ confidence interval for the difference of the average effect of blocks, $\tau_j - \tau_q$, such that

$$((\bar{Y}_{.j} - \bar{Y}_{.q}) - t_{\frac{\alpha}{2}} S \sqrt{\frac{2}{k}}, (\bar{Y}_{.j} - \bar{Y}_{.q}) + t_{\frac{\alpha}{2}} S \sqrt{\frac{2}{k}}), \quad (11.13)$$

for any $j, q = 1, \dots, b$, where the number of observations for each block is, $n_j = n_q = k$, and $S = \sqrt{MSE}$. Analysis for the average effect of blocking follows, at α significance level.

11.4.4 Two-Way ANOVA with Replications

One may also wish to investigate the average effect of interactions between the treatments and the blocks, and extend the ANOVA to this new case. In this experiment design, we will repeat the each *treatment* $\times$ *block* trial several times, say m replications, for all possible treatment-block combinations. Therefore, the new linear model would be,

$$Y_{ij} = \mu + \tau_i + \beta_j + \gamma_{ij} + \epsilon_{ij}, \quad (11.14)$$

where μ is the overall mean, τ_i is the effect of treatment- i , such that $\sum_{i=1}^k \tau_i = 0$, β_j is the effect of block- j , such that $\sum_{j=1}^b \beta_j = 0$, γ_{ij} is the interaction effect between factor- i and block- j , and ϵ_{ij} is the independent random error which is normally distributed, with zero mean and with a constant variance of σ^2 .

The two-way ANOVA with replications is similar to two-way ANOVA, with additional average effect of the interactions between factors and blocks. Thus, ANOVA computations are for comparing the average effect of treatments, blocks, and interactions, based on the $n = (m)(b)(k)$ observations, would be:

$$SSTOTAL = \sum_{i=1}^k \sum_{j=1}^b \sum_{l=1}^m (Y_{ijl} - \bar{Y})^2 = SST + SSB + SSI + SSE,$$

where

$$SST = bm \sum_{i=1}^k (\bar{Y}_{i..} - \bar{Y})^2,$$

$$SSB = km \sum_{j=1}^b (\bar{Y}_{.j.} - \bar{Y})^2,$$

$$SSI = m \sum_{i=1}^k \sum_{j=1}^b (\bar{Y}_{ij.} - \bar{Y}_{i..} - \bar{Y}_{.j.} + \bar{Y})^2,$$

and

$$SSE = \sum_{i=1}^k \sum_{j=1}^b \sum_{l=1}^m (Y_{ijl} - \bar{Y}_{ij.})^2.$$

Source	df	SS	MS	F
Treatments	k-1	SST	$MST = \frac{SST}{k-1}$	$F_{TTest} = \frac{MST}{MSE}$
Blocks	b-1	SSB	$MSB = \frac{SSB}{b-1}$	$F_{BTest} = \frac{MSB}{MSE}$
Interaction	(k-1)(b-1)	SSI	$MSI = \frac{SSI}{(k-1)(b-1)}$	$F_{ITest} = \frac{MSI}{MSE}$
Error	kb(m-1)	SSE	$MSE = \frac{SSE}{kb(m-1)}$	
TOTAL	n-1	SSTOTAL		

Table 11.5: Two-Way ANOVA Table with Replications

Those computations are represented by an ANOVA table, such as the following Table 11.5, that would be used for the test statistics computations.

The computer will also print the p -values for testing the null hypotheses;

$H_{T0} : \tau_1 = \dots = \tau_k = 0$, with $k-1$ numerator and $kb(m-1)$ denominator degrees of freedom.

$H_{B0} : \beta_1 = \dots = \beta_b = 0$, with $b-1$ numerator and $kb(m-1)$ denominator degrees of freedom.

$H_{I0} : \gamma_1 = \dots = \gamma_{bk} = 0$, with $(k-1)(b-1)$ numerator and $kb(m-1)$ denominator degrees of freedom.

We start analyzing the interaction effects, such that if the computed p -value for the interaction effects is smaller than the desired significance level, α , then we reject the H_{I0} , or, similarly if the computed $F_{I_{Test}}$ is greater than the critical point F_α with $(k-1)(b-1)$ numerator and $kb(m-1)$ denominator degrees of freedom, then we would reject H_{I0} , at the given significance level. In the case of rejecting H_{I0} , we have strong evidence of interaction effects between the treatments and the blocks, thus we stop the analysis.

On the other hand, if we cannot reject H_{I0} , then we start analyzing the data for average treatment effects and average blocking effects, since there is no significant evidence indicating blocks changes treatment effects, and we repeat the individual hypothesis tests for H_{T0} and H_{B0} as before.

11.5 Simultaneous Confidence Intervals

In the case that we want to construct many confidence intervals at once, for the pair of average treatment effects, such that each interval has a $100(1-\alpha)\%$ confidence level, we can set a $100(1-\alpha)\%$ simultaneous confidence interval.

Assume that we wish to find a set of intervals, $\{I_j | j = 1, \dots, m\}$, for a set of corresponding unknown, but fixed parameters, $\{\theta_j | j = 1, \dots, m\}$, such that $P(\theta_j \in I_j) \geq 1 - \alpha$, for $j = 1, \dots, m$.

Here, we can utilize the Bonferroni inequality, $P(\cap_{j=1}^m A_j) \geq 1 - \sum_{j=1}^m P(A_j^c)$, where A_j^c is the complement of A_j .

Hence, we have $P(\theta_1 \in I_1, \theta_2 \in I_2, \dots, \theta_m \in I_m) \geq 1 - \sum_{j=1}^m P(\theta_j \notin I_j) = 1 - \sum_{j=1}^m \alpha_j$, where we can choose all α_j s to be equal to $\frac{\alpha}{m}$, such that $\sum_{j=1}^m \alpha_j = \alpha$.

Simultaneously comparing the pairs of average treatment effects, $\mu_i - \mu_l$, would be based on the following $100(1-\alpha)\%$ confidence intervals:

$$((\bar{Y}_i. - \bar{Y}_l.) - t_{\frac{\alpha}{2m}} S \sqrt{\frac{1}{n_i} + \frac{1}{n_l}}, (\bar{Y}_i. - \bar{Y}_l.) + t_{\frac{\alpha}{2m}} S \sqrt{\frac{1}{n_i} + \frac{1}{n_l}}), \quad (11.15)$$

for $i, l = 1, \dots, k$.

After the set of confidence intervals is constructed we can simultaneously compare pairs of average treatment effects by using this set of confidence intervals, and if any interval does not have a sign change, then this would indicate that the average treatment effects are different, for this interval's pair. Similarly, we can construct simultaneous confidence intervals for pairwise comparing the average effect of blocks.

11.5.1 Tukey's Method

The question is *to be, or not to be*, for a set of follow-up tests of hypothesis which involves pairs of treatment effects, after the original hypothesis test of $H_0 : \mu_1 = \dots = \mu_k$ is completed. There is an affirmative answer to this very important question, but finding a reasonable approach is not that easy.

In Tukey's approach [15], these sub-hypothesis tests, $H_0 : \mu_i = \mu_j$, are carried-out by using confidence intervals of $\mu_i - \mu_j$, for all $i, j = 1, \dots, k$ and $i \neq j$. Here, construction is based on the ratio of $\frac{R}{S}$, where R is the range and S is the unbiased estimator of σ , for a one-way ANOVA random sample, subject to a normal distribution.

Let $\{Y_i\}$ be a set of k independents and normally distributed random variables with a mean, μ , and a variance, σ^2 . Also, let R be the range, such that $R = \max_i\{Y_i\} - \min_i\{Y_i\}$.

Assume that $E[S^2] = \sigma^2$, where S^2 is a random variable, conditionally independent of the Y_i s, and subject to a χ^2 distribution with ν degrees of freedom.

Therefore, the Studentized range, $Q_{k,\nu}$, would be the ratio of $\frac{R}{S}$. The values of $Q_{k,\nu;\alpha}$ represent the $100(1 - \alpha)$ th percentiles (quantiles) of $Q_{k,\nu}$.

Tukey's Theorem 11.5.1. *Let $\{\bar{Y}_j\}$ be the k sample means in a completely randomized (one-way) design of experiment. Also, let n_j be r , common sample size, and let μ_j be the actual means, for $j = 1, \dots, k$.*

The probability would be $1 - \alpha$ such that all $\binom{k}{2}$ differences of $\mu_i - \mu_j$ will simultaneously satisfy the conditions:

$$(\mu_i - \mu_j) \in ([\bar{Y}_{.i} - \bar{Y}_{.j}] - D\sqrt{MSE}, [\bar{Y}_{.i} - \bar{Y}_{.j}] + D\sqrt{MSE}),$$

where

$$D = \frac{Q_{k,k(r-1);\alpha}}{\sqrt{r}}. \quad (11.16)$$

If this interval does not have a sign change from negative to positive, then the null hypothesis of $H_0 : \mu_i = \mu_j$, for all i, j , can be rejected at the α level of significance.

Proof. Let $\{X_t\} = \{\bar{Y}_t - \mu_t\}$ in the above theorem 11.5.1.

Thus, X_t is normally distributed with mean zero and variance $\frac{\sigma^2}{r}$.

Let X_{max} and X_{min} be the maximum and the minimum values of X_t , for all t .

Also, let $\frac{MSE}{r}$ be an unbiased estimator of $\frac{\sigma^2}{r}$.

Here, the Studentized (similar to “standardized”) range, $R_{student}$, would be $\frac{X_{max} - X_{min}}{\sqrt{\frac{MSE}{r}}}$,

and it would be subject to the $Q_{k,k(r-1)}$ probability law.

Some tables for given tail probabilities are provided in **Appendix B**.

Hence, $P(R_{student} < Q_{k,k(r-1);\alpha}) = 1 - \alpha$, or, equivalently,

$$P(X_{max} - X_{min} < D\sqrt{MSE}) = 1 - \alpha, \text{ where } D = \frac{Q_{k,k(r-1);\alpha}}{\sqrt{r}}.$$

Therefore, we have $P(|X_i - X_j| < D\sqrt{MSE})$, and thus

$$P(-D\sqrt{MSE} < X_i - X_j < D\sqrt{MSE}) = 1 - \alpha, \text{ for all } i \text{ and } j.$$

This implies that, since we have $X_t = \bar{Y}_t - \mu_t$, we would have

$$P([\bar{Y}_i - \bar{Y}_j] - D\sqrt{MSE} < \mu_i - \mu_j < [\bar{Y}_i - \bar{Y}_j] + D\sqrt{MSE}) = 1 - \alpha,$$

for all i and j . □

In Tukey’s theorem, we have seen that if you look at all the confidence intervals, for the difference of treatment means, then you will be pleasantly informed that the same common level of confidence works for all of those intervals.

Example 11.3

Assume that there are five competing antibiotic treatments for infection, P, T, S, E, C , and the quality of treatments are measured by the percentage of binding in the serum. Based on the following data, Table 11.6, decide which antibiotics are similar, and which ones are different, at the 5% level of significance.

We shall construct pairwise comparisons such that all the $\binom{5}{2} = 10$ differences of $\mu_i - \mu_j$ will simultaneously be tested by the condition of being less than the critical range.

If any interval is larger than the critical range, then this would indicate a significant difference between the treatment means constituting that interval.

Treatment	P	T	S	E	C
y_{1j}	29.6	27.3	5.8	21.6	29.2
y_{2j}	24.3	32.6	6.2	17.4	32.8
y_{3j}	28.5	30.8	11	18.3	25
y_{4j}	32	34.8	8.3	19	24.2
$\bar{y}_{.j}$	28.6	31.4	7.8	19.1	27.8

Table 11.6: Data Table for Example 11.3

Here $SSE = \sum_{i=1}^4 \sum_{j=1}^5 (y_{ij} - \bar{y}_{.j})^2 = 135.83$, and

$$MSE = \frac{SSE}{k(r-1)} = \frac{135.83}{5(4-1)} = 9.06.$$

Since $\alpha = 0.05$, the corresponding critical point would be,

$$Q_{k,k(r-1);\alpha} = Q_{5,15;0.05},$$

and we could find the value, from **Appendix B**, to be 4.367. Hence, $D = \frac{Q_{5,15;0.05}}{\sqrt{r}} = \frac{4.367}{2} = 2.1835$, and we have

$D\sqrt{MSE} = 6.5723$, as our critical range, for all pairwise differences of the treatment means, $\bar{y}_{.i} - \bar{y}_{.j}$.

Therefore, if any absolute difference of $|\bar{y}_{.i} - \bar{y}_{.j}|$ is greater than our critical range, then this would be strong evidence for the difference of these average treatment effects.

The following Table 11.7 indicates pairwise differences among the sample means, under different treatments.

Pairwise Difference	Absolute Magnitude	Conclusion
$\bar{y}_{.1} - \bar{y}_{.2}$	2.8	No Difference
$\bar{y}_{.1} - \bar{y}_{.3}$	20.8	Different
$\bar{y}_{.1} - \bar{y}_{.4}$	9.5	Different
$\bar{y}_{.1} - \bar{y}_{.5}$	0.8	No Difference
$\bar{y}_{.2} - \bar{y}_{.3}$	23.6	Different
$\bar{y}_{.2} - \bar{y}_{.4}$	12.3	Different
$\bar{y}_{.2} - \bar{y}_{.5}$	3.6	No Difference
$\bar{y}_{.3} - \bar{y}_{.4}$	11.3	Different
$\bar{y}_{.3} - \bar{y}_{.5}$	20	Different
$\bar{y}_{.4} - \bar{y}_{.5}$	8.7	Different

Table 11.7: Pairwise Sample Mean Differences for Example 11.3

Hence, we conclude that there is no difference between average treatment effects for P and T , P and C , T and C . Other treatment means are different, based on the 95% simultaneous Tukey's confidence interval.

The sign of these pairwise differences would yield information about which treatment mean is greater, for the difference of pairs that exceeds the computed critical range. For example, $\bar{y}_{.1} - \bar{y}_{.4} = 9.5$ and $\bar{y}_{.3} - \bar{y}_{.4} = -11.3$, thus the average effect of treatment-1 is greater than the average effect of treatment-4. On the other hand, the average effect of treatment-3 is less than the average effect of treatment-4, at the 95% level of confidence.

In this chapter we took a look at a special case of the design of experiment, and saw how we can investigate the average effect of treatments (factors), as well as the average effect of categories (blocks).

11.6 Exercises

1. A computer software generated the following One-way ANOVA table 1, and computed a p -value of 0.0413656, that would be used for comparing the average treatment effects, assuming they all have the same variance. What would you say about the average treatment effects, at the 5% level of significance?

Source	df	SS	MS	F
Factor	3	63.2855	21.09517	3.461629
Error	16	97.504	6.094	
TOTAL	19	160.7895		

2. A computer software printed out the following Two-way ANOVA table 2, that would be used for the test statistics computations. Construct hypothesis tests for the average treatment effects and average blocking effects, at the 5% level of significance.

Source	df	SS	MS	F
Factor	2	5.48	2.74	36.53
Block	3	3.48	1.16	15.47
Error	6	0.45	0.075	
TOTAL	11	9.41		

What could you say about the average treatment effects and average blocking effects for this two-way ANOVA experiment?

3. Prove that the mean square error for one-way ANOVA, $s^2 = \frac{SSE}{k(n-1)}$, is an unbiased estimator of σ^2 .
4. A computer software generated the following One-way ANOVA table 4, that would be used for comparing the average treatment effects, assuming they all have the same variance. Construct the confidence intervals for all pairs of $\mu_i - \mu_j$, for $i, j = 1, \dots, 4$, with a simultaneous confidence coefficient no less than 95%.

Here, $\{Y_{i.}\} = \{454, 549, 425, 351\}$, $\{n_i\} = \{6, 7, 6, 4\}$,
and $\{\bar{Y}_{i.}\} = \{75.67, 78.43, 70.83, 87.75\}$.

Source	df	SS	MS	F
Factor	3	712.6	237.5	3.77
Error	19	1196.6	63	
TOTAL	22	1909.2		

5. Repeat Example 11.3, with $\alpha = 1\%$ and $\alpha = 10\%$. What are the decision differences among these significance levels?
6. The sample data of five pain-killer drugs provided the relief, measured in hours, are as follows:

D1	D2	D3	D4	D5
5	9	3	2	6
4	7	5	3	7
8	8	2	4	9
6	7	3	3	4
4	8	6	5	8
9	5	7	8	5

Is the average number of hours of relief same, for all five brands of pain-killers, at the 5% level of significance?

7. Show that the randomized block design with k treatments and b blocks, $E[SSB] = (b-1)\sigma^2 + k \sum_{j=1}^b \beta_j^2$.

8. A weight loss program uses two types of treatments, namely self-induced and therapist-controlled, and a control group. A random sample of ten subjects assigned randomly to all the different treatments yields the following body weight changes, given in kilograms:

Control Group	Self-Induced	Therapist-Controlled
1	-2	-10
3	-5	-15
0	-2	0
-0.5	-1.5	-5
-2	-4	-7
-1	-1	1
-1	-7	-10
3	-1	-3
1	0.5	-1
-1	-2	-5

At the 5% level of significance, which treatment is better than others?

Chapter 12

Experiment Design

12.1 Introduction

A researcher pursues a systematic search for unknown facts, by using sound scientific methods. By the nature of this act, these facts that are sought after cannot be found by doing a literature search, though this would help to acquire the available information so that the researcher does not have to replicate the same knowledge.

Most research activities are experimental, but some are not; such that only random observations are available to analyze and conduct a research.

In this context, we may say that an experiment is a study in which some independent variables are controlled, their effects on dependent variables are determined and the levels of the independent variables are randomly assigned to the experimental units of the study, as in the ANOVA or regression experiments. This common experiment structure implies that randomization is essential in experiment design.

Therefore, a researcher must ask the following question:

What courses of action must be taken for a meaningful study?

The answer to this question has three parts:

First: Planning, Second: Design, Third: Analysis.

All three phases must be carried out properly in a coherent manner, in order to conduct an experiment successfully.

The experiment begins with a statement of the problem, in terms that are well understood and that point the course through which the research must be conducted.

Also, this statement includes references to a success criterion, in order to

assess the result of the experiment.

It is necessary to identify the independent variables, or factors, that affect the dependent, or response, variables.

Are the factors to be varied quantitatively, or, qualitatively? How are the different factor levels to be combined?

Sometimes, experimental units come from different sources and the researcher randomly selects units from each source. There may also be multiple critical success factors (decision criteria), such that some of them may create conflict, yet some others may collaborate.

All of these issues should be incorporated in the planning phase.

12.1.1 Optimal Design

How many observations must be collected? How large a difference would be detected? How much variations are present? What magnitude of risks are to be tolerated?

These are all legitimate questions, in order to decide the sample size for the experiment. Otherwise, the researcher would collect an arbitrarily large number of sample observations.

Once a decision has been made about the levels of control variables, the researcher knows that in the experiment there are other variables that cannot be controlled.

Randomization, to some extent, would help to reduce the effect of those uncontrollable variables. Also, randomization would let the researcher assume that the errors of measurements are independent.

Thus, the question would be:

What is the randomization procedure and how are the experimental units arranged for treatments?

After an agreement is reached about the randomization procedure, a mathematical model will be constructed to support and predict the experiment's outcomes.

In this model, relationships between the response and the factors are expressed, and other necessary conditions declared.

The objective of the research is to illuminate the stated problem, thus the researcher can construct testable hypotheses, based on an experiment.

Here, a research hypothesis would capture what the researcher expects to find in the data set. Hence a null hypothesis is stated, based on the experimental model, as the testable statement.

In general, the researcher wishes to maximize the effect of factors, minimize the variance of the model prediction error and the effect of uncon-

trollable variables, subject to the constraints, such as time and monetary resources allocated to the experiment.

Unfortunately, an optimal design of experiment that provides the most information is not available for all types of parameter estimation, indeed, there are very few possibly best experiment designs for specific cases.

Example 12.1

Let an experiment be conducted, in order to estimate the difference between the means of a pair of independent populations. If the researcher has limited amount of resources, for only collecting n observations, then how many observations must be selected from each populations, such that $n_1 + n_2 = n$, so that the relevant information about $\mu_1 - \mu_2$ is maximized?

When the random samples are independent, the unbiased estimator of $\mu_1 - \mu_2$ would be $\bar{X}_1 - \bar{X}_2$, with a standard error of $\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$, where $\bar{X}_1$ and $\bar{X}_2$ are sample averages.

When the standard error gets smaller, the relevant information about $\mu_1 - \mu_2$ gets larger.

If $\sigma_1 = \sigma_2 = \sigma$, then the standard error would be $\sigma\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$, and the standard error will be minimum when $n_1 = n_2$. (Why?) Thus, the researcher allocates sampling resources equally to those two independent populations, as an optimal experiment design.

Example 12.2

When we are comparing the average effect of two treatments, we can design a matched-pairs experiment. In this design we randomly collect a sample of n observations, and we use the sample observations to be the subject of two alternative treatments, and then observe the average difference of responses, for these treatments on the same experimental subject.

By this experiment design, we can control the random sources of variability and reduce the standard error for the average difference estimator,

$$\bar{D} = \bar{Y}_1 - \bar{Y}_2, \text{ such that}$$

$$E[\bar{D}] = \mu_D = \mu_1 - \mu_2, \text{ and}$$

$$Var(\bar{D}) = \frac{\sigma_D^2}{n} = \frac{1}{n}[\sigma_1^2 + \sigma_2^2 - 2Cov(Y_1, Y_2)].$$

Instead, if we had used two independent populations for the treatment effects, with equal sample sizes, $n_1 = n_2 = n$, then the average difference of the sample treatment effects, $E[\bar{Y}_1 - \bar{Y}_2]$, would be our estimator with a standard error of $\frac{1}{n}[\sigma_1^2 + \sigma_2^2]$.

Observe that the matched-pairs experiment has a smaller standard error.

12.2 Properties of Experiment Design

In designing experiments, we randomly assign treatments to experimental subjects. In some experiments, we also block the subjects, according to some common qualitative characteristic, and randomly allocate all the treatments into blocks. This probabilistic design of experiment reduces error and provides a probabilistic inference model.

The experimental units are the objects where the measurements are taken. Factors are variables under control, and the degree of a factor is known its level. In a single factor experiment design, each level denotes a treatment. In general, a treatment is a specified combination of factor levels.

In a completely randomized experiment design to compare k treatments, a group of n experimental units are randomly divided into k subsets of sizes $n_1, \dots, n_k$, such that $n = \sum_{i=1}^k n_i$. All units in each subset receive the same treatment, with each treatment applied to exactly one subset. The observations collected from a completely randomized experiment design are considered to be independent samples coming from the parent population, corresponding to each one of the treatments.

12.2.1 Blocked Experiment Design

A one-way layout to compare k populations is an arrangement where independent samples are collected from each of the populations. Thus this is the independent samples case. In the matched-pair design, the objective was to reduce variations, thus improve precision of the experiment. What can we do if we want to compare more than two treatments? Instead of constructing several matched-pairs experiments, we would form several groups, such that each one of them contains the number of matched treatments objects.

Within each group, we would randomly select an object to receive the treatment-1, another object to receive the treatment-2, $\dots$, and another object to receive the treatment- k . Hence, similar to matched-pairs design, we reduce the variations. This design of experiment is known as the randomized block design. A randomized block design contains b blocks and k treatments, i.e. this experiment contains b blocks of k experimental subjects each. The

treatments are randomly assigned to a unit in each block, such that each treatment appears only once in every block.

The randomized block design is one of the many possible types of blocked experiment design. Blocking in two directions can be obtained by using a *Latin Square* design, where k treatments require a $k \times k$ array of all possible treatment permutations (circular), as represented in Table 12.1.

	1	2	3	4	...	k
1	T_1	T_2	T_3	T_4	...	T_k
2	T_2	T_3	T_4	T_5	...	T_1
3	T_3	T_4	T_5	T_6	...	T_2
4	T_4	T_5	T_6	T_7	...	T_3
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
k	T_k	T_1	T_2	T_3	...	T_{k-1}

Table 12.1: Latin Square Design

Here, the treatment- i is denoted by T_i . Other Latin squares must be constructed, for more observations.

12.3 Analysis of Covariance

In the course of experimental designs, when a study is being carried for the effects of one, or more factors on a response, there may be other variables that are varying together with the response and it may not possible to control these other variables at a constant level, during the experiment.

These variables are called concomitant variables, as they run along with the response, and they may be measurable variables.

Thus, we wish to remove the effects of concomitant variables, in order to be able to make assessments about the treatment effects on the response.

The method of removing the effects of concomitant variables on the response and conducting an analysis of the residuals, for the effects of treatments on the response is known as the covariance analysis.

Example 12.3

There were three treatments that produced the following response data, as given in Table 12.2

Treatment	A	B	C
Random	40	25	27
Observations	38	32	24
of the	30	13	20
Experiment	47	35	13

Table 12.2: Data Table for Example 12.3

Assume a completely randomized design of experiment was conducted for the single factor effect, and represented by the ANOVA model,

$$Y_{ij} = \mu + \tau_j + \epsilon_{ij}, \quad (12.1)$$

where μ is the common average treatment effect, τ_j is the average treatment- j effect, and ϵ_{ij} is the independent random error, for $i = 1, \dots, 4$ and $j = 1, \dots, 3$.

This data produced the following ANOVA Table 12.3,

Source	df	SS	MS	F
Factor	2	665.2	332.6	5.50663
Error	9	543.5	60.4	
TOTAL	11	1208.7		

Table 12.3: ANOVA Table for Example 12.3

At the 5% level of significance, we may say that there is a difference in the average treatment effects, since $F_{critical} = 4.26$, at 2 numerator degrees of freedom and 9 denominator degrees of freedom, and further analysis may be carried out to find the most beneficial treatment.

Here, the main interest is to identify another variable, say X , may be lurking among the treatments, thus induces correlation with the effect of treatments, Y .

If X is present during this experiment, then a covariance analysis is conducted to remove the the effect of X on Y .

A linear regression model expresses this relationship of X and Y :

$$Y_{ij} = \mu + \beta(X_{ij} - \bar{X}) + \epsilon_{ij}, \quad (12.2)$$

here β is the regression line's slope between X and Y , over all the data.

How one can adjust the ANOVA in order to incorporate the removal of the effect of the X covariate?

We can consider the linear regression model: $y = bx + e$, where x and y are deviations from the means, i.e. $x = X_{ij} - \bar{X}$ and $y = Y_{ij} - \bar{Y}$, the slope is $\frac{S_{xy}}{S_{xx}}$, where S_{xx} , S_{yy} , and S_{xy} are computed over the data set.

A hypothesis test is conducted for $H_0 : b = 0$, and if this null hypothesis is rejected, then the effect of the covariate is removed from the response, by using this linear regression model.

Here, an important feature of experiment design is observed, such that the linear regression model and the ANOVA model are similar in terms of analyzing the effect of treatments on a response, when relations are linear.

12.4 Response Surface Design

If the response variable and all the controlled variables of an experiment are measurable, then the response surface would be $Y = f(X_1, \dots, X_k)$.

When there are only two independent variables, X_1 and X_2 , the surface Y can be plotted such that the set of coordinates, $\{(X_1, X_2)\}$, that yields the same Y value, generate contours (equal-yield lines), such as the level curves found on topographical maps, or, air pressure maps.

The use of response surface design is to determine the direction of the optimal point on the underlying response surface, based on the conducted experiment, and then construct a response surface at, or, near to this optimal point. A design of experiment for searching the optimal point of the response surface may be one factor at a time search. For example, if X_1 is fixed and X_2 is varied, one may find the X_2 value to optimize the response, Y . After finding this optimal value of X_2 , experiment can be run for the fixed value of X_2 at this optimal level, and the value for X_1 can be found for the optimal response, Y .

The shortcoming of this approach is such that the search for the optimal point may fail. An alternative method for searching the optimal point is to employ the steepest descent approach. This approach begins with running a simple experiment over a small area of the response surface, and determining the direction of the search for the optimum point, based on this experiment. The next experiment, in this direction, will provide the fastest approach to the optimum point of the response surface, and as a sequence of experiments in the steepest descent path, the optimal is reached.

The response surface for more than two factors (controlled variables) design would be more complex, but many approaches are constructed for estimating the parameters of these surfaces.

12.5 Exercises

1. If you are conducting a study of cause-and-effect relationship between two variables, and decided to utilize a linear regression model to support your study, then how would you design your experiment?
2. Why do we design blocked experiments?
3. What is the purpose of covariance analysis?
4. What is the purpose of response surface design of experiments?
5. What are the reasons for using randomization in any experiment?
6. Find an example, or, a counter-example such that a variable is a factor in one experiment and a nuisance parameter in another.
7. What factors changes the degree of information in an experiment, and what design characteristics control these factors?
8. Assume that you will use the model $Y = \beta_0 + \beta_1 x + \epsilon$, where $E[\epsilon] = 0$ and $Var(\epsilon) = \sigma^2$, to fit a set of n data points. When the experimental design uses $x \in \{-1, 0, 1\}$, what proportions should be assigned to each of x values, in order to minimize the variances?
9. If X is a binomial random variable based on n trials and a success probability of p , such that p has the conjugate beta prior distribution with parameters $\alpha = 1$ and $\beta = 2$, for what value of p is the mean squared error (MSE) of the Bayes estimator for p less than that of the unbiased estimator, $\hat{p} = \frac{X}{n}$?
10. Let $X_1, \dots, X_n$ be a random sample, coming from a normal population with known mean μ and an unknown variance σ^2 . Set $U = \sum_{i=1}^n (X_i - \mu)^2$ is a sufficient statistic for the unknown variance, and $V = \frac{U}{\sigma^2}$ has a χ^2 distribution with n degrees of freedom. If the designer selects the conjugate gamma prior with parameters (α, β) , then the Bayes estimator for σ^2 would be $\frac{\beta U + 2}{\beta(2\alpha + n - 2)}$, and the MLE for σ^2 would be $\frac{U}{n}$. Prove that the Bayes estimator is the weighted average of the MLE and the prior mean of σ^2 , for this design.

Appendix A

Preliminaries

“ “Take some more tea,” the March Hare said to Alice, very earnestly. “I’ve had nothing yet,” Alice replied in an offended tone, “so I can’t take more...”
” - Alice in Wonderland.

A.1 Introduction

Probability and Statistics are a study of relationships, subject to chance, and mathematics is the language to express these relationships. In this introductory chapter a very brief review of basic mathematical concepts such as numbers, sets, and functions will be presented. A learned reader may skip this chapter, yet it is suggested to get acquainted with the symbolism used in this book.

Also, there are many excellent books available on these subjects and must be read for a comprehensive understanding.

A.2 Numbers

Numbers primarily are the instruments to count and the number system that is utilized for counting is the Natural numbers, denoted by $\mathbf{N}$. Natural numbers associate countable entities with a number that identifies the quantity, such as one, two, three, etc. and this answers the question of “How Many”. With the invention of the number “zero”, which answers as “No count”, the natural number system is extended to new horizons such as we can even count negative number of quantities (may be representing how many items we have to give to another party from our pocket). This new and larger than natural numbers system for counting of numbers is called integers, denoted by $\mathbf{Z}$, and

identifies positive, or, negative quantities such as ..., negative two, negative one, zero, one, two,..., etc. If we want to measure uncountable entities, such as length, area, volume, time, etc. then we have to use richer number systems than integers, such as rational numbers, denoted by $\mathbf{Q}$, (constructed as a ratio of two integer numbers) such as $-2/3$, $0/5$, $11/7$, etc. Yet this rational number system is not sufficient to measure non-countable but measurable quantities. If we pose the question; What is the measure for the ratio of the circumference of a circle to its diameter, even though the answer is a constant number for all circles, also known as π , we cannot represent this measure by using rational numbers. Hence, we introduce irrational numbers to identify the measure of entities that require number systems more dense than $\mathbf{Q}$. All of those above mentioned numbers can be embedded in a real number line, identified by the real numbers system, and denoted by $\mathbf{R}$, intuitively this gives us the geometric relationship between order, quantity, and distance. A real number line is in a one to one correspondence with the real numbers, that is each and every point of the real line is the geometric representation of a distinct real number. The most common way of constructing a real number line is to draw a horizontal line and fix (mark) an origin, call it real number “zero,” on it. Then, choose a unit distance between integer points such as negative one, zero, one (also known as unit point), etc. and mark those point relative to the origin. Once the origin and the unit is selected, then we can locate all the integers on the real line, then conceptually, we can locate all the rational, irrational, and real numbers on the same real line.

Real line imposes an order to the numbers, such that we can say points located on the real line are representing numbers, which are greater than the numbers represented by points that are located on their left-hand side. Moreover this ranking of locations provide us a measure of magnitude, called distance, denoted by $d(.,.)$, between two points. For example, the magnitude between point two and point negative two is four units, hence we say the distance between number two and negative number two is number four, or number two is four unit greater than number negative two, denoted by $2 > -2$ and $d(2, (-2)) = 4$. This distance is a valid measurement for any numbers on the real line. The orientation of real line provides an order relationship, represented by $x \leq y$, if $y - x \geq 0$. The geometric order relationship of the real line implies the continuum of real numbers, that is if x, y are two distinct real numbers such that $x < y$, then there are infinitely many real numbers on the line segment identified by the end points, x and y , no matter how big or how small the distance between points x and y , (the length of the line segment). This concept implies that any interval of real numbers, identified by a line segment, has the same number (infinitely many) of real numbers in it.

Thus a point moving from one end of a real interval to another end will always go through real numbers. On the other hand, if we consider integer numbers on the real line and an interval of integer numbers, a point moving in this interval must jump from an integer point to another integer point since there are gaps between two adjacent integer points, relative to the real line. We can call the system of integer numbers as discrete numbers and the system of real numbers as continuous numbers, for this reason.

The geometric construct of a real line also provides us the concept of absolute value, denoted by $|x|$, for x being a real number. The unsigned distance between the number, x , and the origin, number zero, is defined as the absolute value of x . For example, $|2|=2$, $|-3|=3$, etc.

A.3 Sets

The set identifies a collection of entities. Thus, a set is specified by its elements, i.e., members, and usually denoted by capital letters.

For example, $A = \{0, 1, 2\}$ is a collection of numbers 0, 1, 2 and set A denotes this collection.

The set elements are enclosed in brackets to indicate their membership. Sometimes there is a common characteristic for the set members, which is indicated by a vertical strike, “|”, and a statement explaining that common characteristic, in the set notation. For example $A = \{x | -1.5 < x < 2.9, x \in \mathbf{R}\}$ is a set of real numbers with the property that all these numbers take values greater than -1.5 and less than 2.9 on the real line, an open ended interval of real numbers. The symbol $x \in A$ indicates that x is a member of set A . We refer to the set which contains all the possible entities of interest as the universe.

A.3.1 Counting Methods

Multiplication rule:

Let us have two countable finite sets with n and m elements. If we want to select an element from the first set, followed by an element selected from the second set, then we have nm possible pairs of selection.

This sequential selection process of one element from each set can be generalized to multiple sets, i.e. if we want to select an element from each set of a sequence of k sets, then we would have $n_1 n_2 \dots n_k$ possible ways of choosing, where n_i is the cardinality (number of elements) of the set i in this sequence of k sets, for $i = 1, 2, \dots k$.

If we select an element, record it and replace it back, repeatedly, say k times, of a countable set with cardinality n , then by the multiplication rule, we have n^k many possible k -tuples.

On the other hand, if we discard (not replace it back) a previously selected element from the sequential selection process, and if $k \leq n$, then we have n possible alternatives for the first selection, $n - 1$ alternatives for the second selection, $n - 2$ alternatives for the third selection, etc.

Therefore, we can choose k -tuples with distinct elements (no repetitions) in $n \times (n - 1) \times \dots \times (n - (k - 1))$ possible ways, for this sequential selection process, by the multiplication rule.

Any one of these k -tuples is called a permutation, and denoted by σ_k^n , which counts the different ways of choosing k -tuples without repetitions, in the same order of the selection process.

Computationally, $\sigma_k^n = n \times (n - 1) \times \dots \times (n - (k - 1)) = \frac{n!}{(n-k)!}$, where $n! = n \times (n - 1) \times (n - 2) \times \dots \times (n - (n - 1))$ for nonnegative integers and $0! = 1$.

Here, $m!$ is called the factorial of a positive integer number, m , and it has a recursive property, i.e. $m! = m(m - 1)!$, which is computationally efficient.

Whenever m gets larger, it may be cumbersome to compute $m!$. In this case, we can use an approximation, known as the Stirling's formula, to compute $m! \approx \sqrt{2\pi m} m^{m+1/2} e^{-m}$.

If we do not care about the order of selections, but only care about the possible number of k -tuples, coming from the set of n elements, by using the multiplication rule, we may say that there are $k \times k - 1 \times \dots \times 1$, i.e. $k!$, different ways of ordering k -tuples, then if we don't care about the order among elements of these k -tuples, all these different ordered sets would be the same one.

Hence, we say that there are $\sigma_k^n/k!$ different ways one can select k distinct elements from a set of n elements.

This number of selecting k distinct elements from a set of n elements, without paying attention to the order of elements, is known as the combination of k objects taken from a set of n objects, and denoted by C_k^n , where $C_k^n = \sigma_k^n/k!$, i.e. $C_k^n = \frac{n!}{k!(n-k)!}$.

Sometimes C_k^n is denoted as $\binom{n}{k}$, which is also known as the binomial coefficient. A binomial (two names) expansion indicates that we translate a product form of a binomial (sum of two variables) into a summation form of a binomial (product of two variables), as follows: $(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}$.

Here, $\binom{n}{k}$ indicates that we can select x in $\binom{n}{k}$ different ways.

Similarly, we can define the multinomial (many names) coefficient as explained below.

Let there be n objects, distributed into k boxes, such that each box has n_i objects and $\sum_{i=1}^k n_i = n$.

If we pick n_i objects from each box i , for $i = 1, 2, \dots, k$, then there are

$\binom{n}{n_1, n_2, \dots, n_k}$ different ways to select n_i objects, for $i = 1, 2, \dots, k$, from these k boxes.

Computationally, $\binom{n}{n_1, n_2, \dots, n_k} = \frac{n!}{n_1! n_2! \dots n_k!}$.

The multinomial coefficient is used to express a summation form multinomial in multiplication form multinomial given as follows:

$$(x_1 + x_2 + \dots + x_k)^n = \sum_{\{n_1 + \dots + n_k = n\}} \binom{n}{n_1, n_2, \dots, n_k} x_1^{n_1} x_2^{n_2} \dots x_k^{n_k}.$$

A.3.2 Set Operations

The set operations are well defined, as described below.

The complement of a set A is the set of elements which are left in the universe, after we extracted the elements of A from the universe, and the complement of A is denoted by A^c .

The union of two sets A, B is the set that has all the elements of set A and set B , denoted as $A \cup B = \{x | x \in A \text{ OR } x \in B\}$.

The intersection of two sets A, B is the set that has the common elements of set A and set B only, denoted as $A \cap B = \{x | x \in A \text{ AND } x \in B\}$.

An empty set is a set that has no elements, denoted by $\emptyset = \{ \}$.

If every member of the set A is also a member of the set B , we say that set A is a subset of B , denoted by $A \subseteq B$. Similarly if $B \subseteq A$ too, then we say $A = B$.

We can subtract a set, B , from a set, A , defined as follows:

$A \setminus B = A \cap \{A \cap B\}^c$, that is we extract the common elements of the sets A and B from the set A .

We sometimes need the addition of sets, defined as follows.

Let A and B be two sets, the addition of A and B ,

$$A + B = \{a + b | a \in A, b \in B\}.$$

A sequence of sets $\{A_i\}$ is called a monotone sequence, if $A_i \subset A_{i+1}$ (nondecreasing), or, $A_i \supset A_{i+1}$ (nonincreasing), for all i .

We can define open and closed intervals, as follows.

An open interval $(a, b) = \{x | a < x < b\}$, if $b > a$. Else this set will be an empty set.

A closed interval $[a, b] = \{x | a \leq x \leq b\}$, if $b \geq a$. Else this set will be an empty set.

It is obvious that an open interval does not contain its end points. Similarly we can define semi-open or semi-closed intervals, by including and excluding end points at one side.

The convention is such that, although ∞ is not a real number, we include it as a point at infinity, and its symmetric sibling $-\infty$ as a point at minus infinity, on the real line, thus the real numbers is the set $\mathbf{R} = \{x | -\infty < x < \infty\}$.

Since real numbers are associated with the real line, we can add a second dimension to construct a real surface, by pairing every point in the real line with the points on the other real line, known as cartesian product of two real lines.

The orthogonal intersection of the real lines, at their origins, serves as a reference frame for the two dimensional real surface, where each point has two coordinates (distances from the origin of the corresponding real lines). These real lines are called coordinate axes, where usually the horizontal line is called the X axis and the vertical line is called the Y axis. Let P be any point on this real plane, where lines parallel to these axes are drawn from P to X and Y axes, the intersection of these projection lines from P into X and Y are called the points x and y respectively. The pair, (x, y) , is called the coordinates of the point P in this cartesian coordinate system (reference frame) and identifies the position of P , completely with respect to the origin of this two dimensional frame, $\mathbf{R}^2 = \{(x, y) | x \in \mathbf{R}, y \in \mathbf{R}\}$.

We have defined the distance on the real line, as $|x - y|$ between points x and y . Similarly we can define distance between two points on $\mathbf{R}^2$ as given below.

Let P be a point with coordinates (x_1, y_1) and Q be a point with coordinates (x_2, y_2) on $\mathbf{R}^2$. The distance between P and Q is defined as $d(P, Q) = [(x_2 - x_1)^2 + (y_2 - y_1)^2]^{1/2}$.

A property of distance is known as “triangle inequality” and it refers to a generic triangle that the sum of two sides’ lengths is greater than the third side’s length, denoted by $d(P, Q) \leq d(P, R) + d(R, Q)$, for any three corners P, Q, R of a triangle.

A.4 Complex Numbers

There is a problem to solve an algebraic equation of the kind $aX^2 + b = 0$, for a real value of the variable x , when a and b were positive real numbers, since this solution does not exist.

The square root of minus one is defined as the imaginary number unit, i , such that $i = \sqrt{-1}$.

Thus we are able to solve (find a value for the variable x that satisfies) the above mentioned algebraic equation, in terms of complex numbers.

We define a complex number as a two-tuple (ordered pair) of real numbers, with a real part and an imaginary part, say z be a complex number, then we can represent $z = x + iy$, where x and y are real numbers. Here, x is called the real part and y is called the imaginary part of the complex number z , and this representation only indicates combination of two real numbers, as an ordered tuple, but not actually adding these two quantities, since they are in different quality categories. Sometimes, we denote complex numbers as an ordered pair, $z = (x, y)$, with the same meanings for x, y, z , as above.

We can add two complex numbers, by adding their real parts together and adding their imaginary parts together. If $z_1 = x_1 + iy_1$, $z_2 = x_2 + iy_2$, then $z_1 + z_2 = (x_1 + x_2) + i(y_1 + y_2)$.

Example 1

Let $z_1 = -1 + i2$ and $z_2 = 3 - i5$ be two complex numbers, then $z_1 + z_2 = (-1 + 3) + i(2 - 5) = 2 - i3$.

Similarly, we can multiply two complex numbers as follows: $z_1 \times z_2 = (x_1 \times x_2 - y_1 \times y_2) + i(x_1 \times y_2 + x_2 \times y_1)$.

Example 2

Let $z_1 = -i1$ and $z_2 = 1 + i2$, then $z_1 \times z_2 = 2 - i1$.

We define a new operator for the complex numbers, called conjugate operator, and it generates the symmetric complex number with respect to the real axis, denoted by z^* , when z is a complex number, such as $z = x + iy$, then $z^* = x - iy$. That is conjugate complex numbers have the same real parts yet their imaginary components are negative of one another.

We think of complex plane consists of two perpendicular real lines intersect at their common origin, point zero, and they generate two dimensional euclidean space by a real axis (horizontal) and an imaginary axis (vertical). That is, we identify a complex number $z = x + iy$ on this plane as a point z with coordinates (x, y) . Complex plane is denoted by $\mathcal{C}$ and it is in fact the cartesian product of two real lines, $\mathcal{C} = \mathbf{R} \times \mathbf{R}$, a cartesian product of two sets is an ordered set of pairs, where first element of the pair is a member of the first set and the second element of the pair is a member of the second set, for example let $S_1 = \{a, b\}$ and $S_2 = \{c, d\}$, then $S_1 \times S_2 = \{(a, c), (a, d), (b, c), (b, d)\}$. The complex numbers lead to solve any algebraic equation (polynomial equation).

A.4.1 Fundamental Theorem of Algebra

Any algebraic equation (polynomial equation) of the order n has no more than n roots (n values that satisfy the equation, may be repetitions of the same value up to the order n), over the complex numbers.

This theorem indicates the existence of solutions for any polynomial (multiple names) equation.

Example 3

Let $P(x) = x^2 + 2x + 2$ be a second order polynomial, the roots (values of x that satisfy $P(x) = 0$) are $x_1 = -0.5 + i$ and $x_2 = -0.5 - i$. That is there are two complex conjugate roots for this polynomial equation.

Algebraic properties of complex numbers, such as commutativity, $z_1 + z_2 = z_2 + z_1$, $z_1 \cdot z_2 = z_2 \cdot z_1$ and associativity $(z_1 + z_2) + z_3 = z_1 + (z_2 + z_3)$, $(z_1 \cdot z_2) \cdot z_3 = z_1 \cdot (z_2 \cdot z_3)$ are obvious, due to properties of real numbers. Additive identity, $(0, 0)$, and multiplicative identity, $(1, 0)$, exist for complex numbers. Thus, additive inverse, u , of a complex number, z is identified by the subtraction operation, $z + u = (0, 0)$, $u = -z$. Multiplicative inverse, v , of a complex number, z , is less obvious. $z \cdot v = (1, 0)$, $v = z^{-1}$, and for $z = (x, y)$, $z^{-1} = (\frac{x}{x^2+y^2}, \frac{-y}{x^2+y^2})$.

Thus, we can divide complex numbers, $z_1/z_2 = z_1 \cdot z_2^{-1}$, for nonzero z_2 , by using the multiplicative inverse.

The absolute value (magnitude), or modulus, of a complex number $z = (x, y)$ is denoted by $|z|$ and defined as the nonnegative real number, $\sqrt{x^2 + y^2}$, that indicates the distance to the origin. Therefore, $|z|^2 = x^2 + y^2$, and $|z| \geq |x| \geq x$, similarly, $|z| \geq |y| \geq y$. Also, $|z| = |z^*|$.

The distance between two complex numbers, z_1 and z_2 , is measured by,
 $|z_1 - z_2| = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$.

Also, $|z_1 \cdot z_2| = |z_1| \cdot |z_2|$, and $|z_1/z_2| = |z_1|/|z_2|$.

When $z^* = (x, -y)$, i.e. complex conjugate of z , we have $z \cdot z^* = |z|^2$.

The triangle inequality, $|z_1 + z_2| \leq |z_1| + |z_2|$, is intuitive, and is argued as follows. By definition,

$|z_1 + z_2|^2 = (z_1 + z_2) \cdot (z_1 + z_2)^*$, by carrying out the algebraic operations, we have, $|z_1 + z_2|^2 = z_1 z_1^* + (z_1 z_2^* + z_1^* z_2) + z_2 z_2^*$.

On the other hand, $(z_1 z_2^* + z_1^* z_2) \leq |z_1 z_2^*| + |z_1^* z_2| = 2|z_1||z_2|$.

Therefore, $|z_1 + z_2|^2 \leq |z_1|^2 + 2|z_1||z_2| + |z_2|^2$.

That implies $|z_1 + z_2|^2 \leq (|z_1| + |z_2|)^2$, giving the required inequality.

The triangle inequality is also valid for many (may be even infinitely many) complex numbers,

$$|z_1 + z_2 + \dots + z_n| \leq |z_1| + |z_2| + \dots + |z_n|.$$

A complex number can also be represented by using polar coordinates, instead of using cartesian coordinates.

Let $z = (x, y) = x + iy$, thus $x = r \cos \theta$, $y = r \sin \theta$ and we would have $z = r(\cos \theta + i \sin \theta)$.

Here, $r = |z|$ (magnitude) and $\theta = \arg z$ is the argument (angle that z makes with the positive real axis), such that $\tan \theta = y/x$, of a complex number.

The unique value of the argument such that $-\pi < \arg z \leq \pi$ is called the principle value of argument and denoted by $\text{Arg } z$.

There is an important property, $\arg(z_1 z_2) = \arg z_1 + \arg z_2$, but not valid for principle value of the arguments.

Let $z_1 = r_1(\cos \theta_1 + i \sin \theta_1)$ and $z_2 = r_2(\cos \theta_2 + i \sin \theta_2)$, then

$$z_1 z_2 = r_1 r_2 [(\cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2) + i(\sin \theta_1 \cos \theta_2 + \cos \theta_1 \sin \theta_2)].$$

Hence, $z_1 z_2 = r_1 r_2 [\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)]$.

Similarly, for a nonzero complex number $z = r(\cos \theta + i \sin \theta)$, we have its multiplicative inverse, $z^{-1} = \frac{1}{r}(\cos(-\theta) + i \sin(-\theta))$.

A.4.2 Euler Formula

It is much more convenient to use the Euler formula, $e^{i\theta} = \cos \theta + i \sin \theta$, to represent complex numbers and to carry out algebraic operations on complex numbers, since $e^{i\theta_1} e^{i\theta_2} = e^{i(\theta_1 + \theta_2)}$.

Thus, a complex number can be represented by using the Euler formula, $z = r e^{i\theta}$, with its multiplicative inverse, $z^{-1} = \frac{1}{r} e^{-i\theta}$, and $z_1 z_2 = r_1 r_2 e^{i(\theta_1 + \theta_2)}$.

The Euler representation is useful to find the powers of complex numbers, such as, $z^n = r^n e^{in\theta}$, for an integer n . When $n = 0$, we have $z^0 = 1$.

Here, $(e^{i\theta})^n = e^{in\theta}$, or $(\cos \theta + i \sin \theta)^n = (\cos n\theta + i \sin n\theta)$.

Example 4

Solve for $z^n = 1$, for $z \neq 0$ and $n > 0$.

Here, $(r e^{i\theta})^n = 1$ implies $r^n e^{in\theta} = 1^n e^{i0}$.

This equality holds when the moduli are the same for two complex numbers, and the arguments can differ only by an integer multiple of 2π . Thus, $r^n = 1$ and $n\theta = 0 + 2k\pi$. Therefore, $r = 1$ and $\theta = 2k\pi/n$.

The solution, $z = e^{i(2k\pi/n)}$, for $k = 0, 1, 2, \dots, n-1$, are the n roots.

We can generalize this example to the roots of any complex number, $z = r e^{i\theta} = r(\cos \theta + i \sin \theta)$.

The n roots of z are: $r^{1/n} e^{i(\frac{\theta+2k\pi}{n})} = r^{1/n} (\cos \frac{\theta+2k\pi}{n} + i \sin \frac{\theta+2k\pi}{n})$, for

$k = 0, 1, \dots, n-1$.

The complex number is an extension of real numbers, as representation of points in two dimensional Euclidean space. Thus, they carry the real number property of order, that is one real number is said to be greater than another one if it is located more to the righthand side on the real line, than the other. Similarly, we can impose an order among complex numbers by measuring their distance from the origin (magnitude) and also by identifying their locations on the two dimensional real surface by the angle they make with the real axis, counterclockwise sense.

The generalization of real line to real plane may continue to higher dimensional real spaces (even to the infinite dimensional real space). For example, if we represent points in the three dimensional real space, we use three dimensional numbers (triplets of ordered numbers). Also, you may think of complex numbers as two dimensional numbers (twins of ordered numbers). As an extension of complex numbers, these triplets can be denoted as (x, y, z) , or $x + iy + jz$, where i, j are the imaginary unit, $\sqrt{-1}$. We can attempt to do addition and multiplication of these numbers, as we did in for the complex numbers.

Let $u = (x_1, y_1, z_1)$ and $v = (x_2, y_2, z_2)$.

Addition:

If we add $u + v$, we will get component-wise addition of these two triplets, $(x_1 + x_2, y_1 + y_2, z_1 + z_2)$. Thus addition operation does not introduce any difficulty.

Multiplication: If we multiply two complex numbers, we multiply their magnitudes and add their arguments, to find the result. In three dimensional real space we need two arguments to identify the location of a point and a third argument to identify the rotation, together with the magnitude. Hence, we need four dimensions to identify a point, but here we have only three.

A.5 Quarternion

In the 19th century, Hamilton constructed a number system to remedy this by adding another dimension, and represented the original triplet (x, y, z) with a quadruplet (x, y, z, s) , by introducing another imaginary unit $k = \sqrt{-1}$.

Hence, (x, y, z, s) can be denoted as $x + iy + jz + ks$, and this leads to the name quarternion.

Therefore, we may think a quarternion as a higher dimensional complex number, such that it has a real part, x , and an imaginary part, (y, z, s) .

Here, the orthogonality of euclidean space is preserved, that is, i, j, k are noncommutative orthogonal unit vectors, with following properties.

$$ij = k, jk = i, ki = j, ji = -k, kj = -i, ik = -j, \text{ and } ijk = -1.$$

Since quaternions are extended complex numbers, we have a conjugate operator for quaternions.

Let $u = (x_1, y_1, z_1, s_1)$ and $v = (x_2, y_2, z_2, s_2)$, or $u = x_1 + iy_1 + jz_1 + ks_1$ and $v = x_2 + iy_2 + jz_2 + ks_2$.

Example 5

$$u^* = (\text{Real}\{u\}, -\text{Imaginary}\{u\}) = (x_1, -y_1, -z_1, -s_1).$$

We now reattempt to represent addition and multiplication of quaternions.

Addition:

Componentwise addition as before, $u + v = (x_1 + x_2, y_1 + y_2, z_1 + z_2, s_1 + s_2)$,

or

$$u + v = (x_1 + x_2) + i(y_1 + y_2) + j(z_1 + z_2) + k(s_1 + s_2).$$

Subtraction can be seen as addition of two quaternions, but the second one as a negative number, $u - v = u + (-v) = (x_1 - x_2, y_1 - y_2, z_1 - z_2, s_1 - s_2)$.

Multiplication:

If we multiply a quaternion by a real number, this will have only a scaling effect, as in the case of complex numbers.

Example 6

Let c be a real number, then $c.u = (cx_1, cy_1, cz_1, cs_1)$.

The multiplication of two quaternions have special properties, as presented below.

$$\begin{aligned} uv &= (x_1, y_1, z_1, s_1)(x_2, y_2, z_2, s_2) \\ &= (x_1 + iy_1 + jz_1 + ks_1)(x_2 + iy_2 + jz_2 + ks_2) \\ &= x_1v + iy_1v + jz_1v + ks_1v \\ &= x_1x_2 + ix_1y_2 + jx_1z_2 + kx_1s_2 + iy_1x_2 - y_1y_2 + ky_1z_2 - jy_1s_2 + jz_1x_2 - \\ &\quad kz_1y_2 - z_1z_2 + iz_1s_2 + ks_1x_2 + js_1y_2 - is_1z_2 - s_1s_2 \\ &= (x_1x_2 - y_1y_2 - z_1z_2 - s_1s_2, x_1y_2 + y_1x_2 + z_1s_2 - s_1z_2, x_1z_2 - y_1s_2 + z_1x_2 + \\ &\quad s_1y_2, x_1s_2 + y_1z_2 - z_1y_2 + s_1x_2). \end{aligned}$$

Division:

We can define the division as a multiplication by an inverse.

Inverse of a quaternion, q , is defined carefully, as described below.

Since $q^*q = |q|^2 = qq^*$, we have $1 = \frac{qq^*}{|q|^2} = \frac{q^*q}{|q|^2}$.

Therefore, the inverse, $q^{-1} = \frac{q^*}{|q|^2}$.

Henceforth, we may define division in two different ways,

$$u/v = uv^{-1}, \text{ or } u/v = v^{-1}u.$$

Absolute Value:

The absolute value (magnitude) of a quaternion is the square root of the number, multiplied by its conjugate.

Example 7

$$|u| = \sqrt{uu^*} = \sqrt{x_1^2 + y_1^2 + z_1^2 + s_1^2}.$$

Thus, real and complex numbers are closed under addition and multiplication, but quaternions are not.

A.6 Functions

One of our objectives is to establish a rule to explain relationships between a desired quantity that depends upon another quantity. Therefore it is necessary to understand the concept of a function.

A function represents a relationship of an independent variable x , defined on the domain $\mathcal{D}$ of the function $x \in \mathcal{D}$, and a dependent variable y , defined on the range $\mathcal{R}$ of the function $y \in \mathcal{R}$. The set of the pair $f = \{(x, y) | x \in \mathcal{D}, y \in \mathcal{R}\}$ is the function representing the relationship between x and y , denoted by f . If x is an element of the domain of f , then the corresponding element of the range of f is denoted by $f(x)$, which is the value of f at x . The domain and the range of functions may be a set of any kind of objects, even set of sets, but we will use set of real numbers mostly.

Thus, a point set that represents a function in $\mathbf{R}^2$ is called as the graph of the function f , denoted by $\text{graph}(f) = \{(x, y) | y = f(x)\}$. A picture of a graph of a function explains the behavior of f .

A linear function, defined by the equation $y = ax + b$, where a and b are constant real numbers, known as the parameters, and x and y are real variables related with the above equation, is known as a linear structure. The graph of a linear function is a line with a slope of a and with an intercept (Y axis crossing point) of b . A line parallel to the X axis is the graph of the constant function, $y = b$, and a line parallel to the Y axis is a graph of equation, $x = c$, where c is a constant real number indicating the distance between the Y axis and the line.

A special type of a linear function is called homogeneous linear function, in the form of $f = ax$, which passes through the origin. They have following properties.

$$f(x_1 + x_2) = f(x_1) + f(x_2), \text{ and } f(cx) = cf(x).$$

Where c is a real constant, x_1, x_2 are two distinct variables in the domain of f .

$$\text{Since } f(cx) = a(cx) = c(ax) = cf(x) \text{ and } f(x_1 + x_2) = a(x_1 + x_2) = ax_1 + ax_2 = f(x_1) + f(x_2).$$

These properties are only valid for homogeneous linear functions (not all functions have these properties). As an exercise, find different functions and check for these properties.

In general, two lines are parallel if and only if their slopes are equal, and two lines are orthogonal if and only if the product of their slopes is equal to negative one.

A.7 Limits

When we draw a picture of a graph of a function f , we plot the pair of points $(x, f(x))$ when a point c is fixed on the domain, we look for the Y coordinate value of that graph which should intersect with the line $x = c$. This value is the limit of f at c . A limit may or may not exist.

If a function is continuous at the point c , then the limit of f at c exists, denoted by $\lim_{x \rightarrow c} f(x) = f(c)$.

For continuous functions f, g we can do operations with limits as follows.

$$\lim_{x \rightarrow c} [f(x) + g(x)] = \lim_{x \rightarrow c} f(x) + \lim_{x \rightarrow c} g(x).$$

$$\lim_{x \rightarrow c} g(f(x)) = g(\lim_{x \rightarrow c} f(x)).$$

$$\lim_{x \rightarrow c} f(x)g(x) = \lim_{x \rightarrow c} f(x) \lim_{x \rightarrow c} g(x).$$

$$\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow c} f(x)}{\lim_{x \rightarrow c} g(x)}, \text{ if } \lim_{x \rightarrow c} g(x) \neq 0.$$

For example, if $A_i \subset A_{i+1}$ (nondecreasing sequence), for $i = 1, 2, \dots$, then $\lim_{i \rightarrow \infty} A_i$ would be $A_1 \cup A_2 \cup \dots$. Similarly, if $A_i \supset A_{i+1}$ (nonincreasing sequence), for $i = 1, 2, \dots$, then $\lim_{i \rightarrow \infty} A_i$ would be $A_1 \cap A_2 \cap \dots$.

A.8 Derivative

Now since we know the slope of a linear function, let us try to define the slope of a function. The slope of a line at any one point is the slope of the line, which is the amount of change occurs at the Y axis corresponding to the unit amount of change in the X axis.

The slope of a continuous function is also the rate of change in the tangent line to the graph of the function $f(x)$, at a given point x . This slope of the tangent line is obviously derived from the function f for all the points of the domain of f , and it has different values according to the contour (shape) of f . Thus the slope of the tangent line is a function of $x \in \text{Domain}(f)$ and it is called the derivative of f , usually denoted by f' . Hence $f'(x)$ is the slope of the graph of f , at the point x .

If the following limit exists, $\lim_{h \rightarrow 0} \frac{f(x+h)-f(x)}{h}$, then this limit is equal to $f'(x)$, at a point $x \in \text{Domain}(f)$. This derivative is the rate of change of the function f at the point x .

A.8.1 Mean Value Theorem for Derivative

If $f(x)$ is differentiable in the interval (a, b) (f has derivative at every point in (a, b)), then there is a point m in (a, b) such that $f'(m) = \frac{f(b)-f(a)}{b-a}$.

A.8.2 L'Hospital Rule

If f and g are differentiable in the neighborhood of a point a , $\lim_{x \rightarrow a} f(x) = 0$, $\lim_{x \rightarrow a} g(x) = 0$, and $g'(x) \neq 0$ in the neighborhood, then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}, \text{ when the right hand side exists.}$$

This theorem has many practical implications, such as Newton's method of finding the root of a continuous function, or Taylor series approximation to a function at a given point.

A.8.3 Some Properties of the Differentiation Operator

Let D_x denotes the differentiation operator with respect to x , that is $D_x f(x) = f'(x)$.

$$D_x x^r = r x^{r-1}, \text{ where } r \text{ is a real number.}$$

$D_x c f(x) = c D_x f(x)$, where c is a real constant and $f(x)$ is a differentiable function.

$D_x [f(x) + g(x)] = D_x f(x) + D_x g(x)$, where f and g are two differentiable functions.

$D_x f(x) g(x) = f(x) D_x g(x) + g(x) D_x f(x)$, where f, g are two differentiable functions.

$D_x \frac{f(x)}{g(x)} = \frac{g(x) D_x f(x) - f(x) D_x g(x)}{g^2(x)}$, where f, g are two differentiable functions and $g(x) \neq 0$.

A.8.4 Chain Rule

If f and g are two differentiable functions, let $u = g(x)$, thus $D_x f(u) = D_u f(u) D_x u$.

A.9 Integral

A dual problem arises if we are interested with finding the area of a region bounded by a graph of a function, over an interval.

Thus we introduce the integral of a function over an interval $[a, b]$, where the integral is equivalent to the area of the region, bounded by the graph of f , the X axis and the interval $[a, b]$ on the X axis.

We could measure the area by tracing the closed region on a graph paper and counting the squares in this projected figure on the graph paper. Number of squares times the area of a square, on this figure, will provide us the desired area of the region. Hence, the value of the integral of the function over this interval.

But, in general we may not have an integer number of complete squares that cover this traced area of the region, thus we have to estimate the area by utilizing infinitesimal squares that lie in the region. It is obvious that if we utilize smaller and smaller area elements to fill the region area, the estimated integral value will be more and more reliable, although the number of area elements that we have to count will increase.

Declare an area (or, volume) element such as $dA = \Delta x f(x)$, where Δx represents a very small amount of perturbation from point x in the domain of function f , thus dA is a very small amount of an “area” located in the closed region, at the point $x \in (a, b)$. If we sum all these area elements in the region, we will estimate the total area with some precision as illustrated below. Let S be the closed surface that we are trying to estimate its area,

$Area(S) = \lim_{dA \rightarrow 0} \sum_S dA = \lim_{\Delta x \rightarrow 0} \sum_{a \leq x \leq b} f(x) \Delta x = \int_a^b f(x) dx$, if the limit exists.

Where dx is a measure of change for $x \in Domain(f)$, $\sum$ is the symbol for summation operator, $\int$ is the symbol for integral operator (similar to summation operator, in the infinitesimal scale, or the first letter of “Summa”).

A.9.1 Some Properties of the Integral Operator

$$\int_a^a f(x) dx = 0,$$

$$\int_a^b f(x) dx = - \int_b^a f(x) dx.$$

$$\int_a^b c f(x) dx = c \int_a^b f(x) dx,$$

$$\int_a^b [f(x) + g(x)] dx = \int_a^b f(x) dx + \int_a^b g(x) dx.$$

If $[a, b]$ can be split into two disjoint closed intervals such as $[a, c]$ and $[c, b]$, then $\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$

with a, b, c as real constants, f and g as functions of x ,

A.9.2 Mean Value Theorem for Integral

If f is a continuous function of x on the closed interval $[a, b]$, then there is a point, m , in this interval such that $f(m) = \frac{1}{b-a} \int_a^b f(x) dx$,

where $f(m)$ is called the mean (average) value of f on the interval $[a, b]$.

There is a relationship between the derivative and the integral operators, through a differentiable function f , given as follows:

$$\int_a^b f'(x)dx = f(b) - f(a), \text{ where } x \in [a, b].$$

That is, $\int_a^b f'(x)dx = f(x)|_a^b$, where f is evaluated at the integration limits.

For example, if $f'(x) = x^k$, then $\int_a^b f'(x)dx = f(b) - f(a)$, where $f(x) = \frac{x^{k+1}}{k+1}$, and $k \neq -1$ is a rational number.

If $k = -1$ then $f'(x) = \frac{1}{x}$ and we define, $\int_a^b f'(x)dx = \log(b) - \log(a)$, where $f(x) = \log(x)$ is called the logarithm (in fact, natural logarithm) of $x \in \mathbf{R}^+$.

That is, as a real valued function, $\log(x)$ is defined for positive real numbers.

Basic properties of the $\log(x)$ are as follows:

$$\log(xz) = \log(x) + \log(z),$$

$$\log(x^r) = r\log(x),$$

$$\log\left(\frac{x}{z}\right) = \log(x) - \log(z),$$

x, z are positive real numbers, r is a real number.

A.10 Inverse Function

The inverse function, denoted by f^{-1} , is a function that pulls the image of a function f back from the $Range(f)$ to the $Domain(f)$.

That is if $y = f(x)$ where $x \in Domain(f)$, and $y \in Range(f)$, then $f^{-1}(y) = x$, or $f^{-1}(f(x)) = x$.

For example, the inverse function of the logarithm function is called the exponential function, denoted by $f(x) = \exp(x)$, and $y = \log(x)$ if and only if $x = \exp(y)$. Thus $\exp(\log(x)) = x$ and $\log(\exp(y)) = y$.

Properties of the exponential function.

$$\exp(x + z) = \exp(x)\exp(z)$$

$$\exp(px) = (\exp(x))^p.$$

$$f'(x) = f(x) \text{ for } f(x) = \exp(x).$$

$$\int_a^b \exp(x)dx = \exp(x)|_a^b = \exp(b) - \exp(a).$$

A particularly useful irrational number (a kind of number such as π) is the $e = \exp(1)$, which has a value between 2.718 and 2.719, and called base e for the natural logarithm that we defined before.

Since for real positive x, y , we have $\log_b x = y$ when b is a positive base, different than 1, we have the following relationship.

$\log_e x = y$ if and only if $x = e^y$, where $\log_e x = \log(x)$ and $e^y = \exp(y)$, in our notation and both set of notations can be substituted for each other.

In the sense of inverse relationships, integral is considered to be the inverse operator of differential, that is an integral reconstructs the original function f , after f being differentiated, such as $f = \int df = \int f'(x)dx$.

An important case emerges when the limits of the integral and the differentiation operation involve with the same variable, known as the Leibniz integral rule:

$$\frac{\partial}{\partial s} \int_{a(s)}^{b(s)} f(x, s)dx = \int_{a(s)}^{b(s)} \frac{\partial f}{\partial s} dx + f(b(s), s) \frac{\partial b}{\partial s} - f(a(s), s) \frac{\partial a}{\partial s}.$$

A.11 Multivariate Functions

When the domain of a function consists of more than one variable (many independent variables) the function is called a multivariable function. The function maps collective values of these arguments into an image in its range. Thus a multivariate function is denoted as $f(x_1, x_2, \dots, x_n)$, where n represents the number of arguments that this function has.

A.11.1 Partial Derivatives

If a multivariate function is differentiable at its k^{th} argument, for $k = 1, 2, \dots, n$, when all the other arguments are fixed, then the derivative at this argument is called the partial derivative of the function and is denoted by $\frac{\partial f}{\partial x_k}$.

This indicates the rate of change in the function value with respect to the k^{th} argument, when all other arguments are fixed.

We impose an order of differentiation with respect to individual arguments, and denote this order by $\frac{\partial f}{\partial x_1 \partial x_2 \dots \partial x_n}$, indicating the first derivative is due to the argument x_n and differentiation operations proceed until the last one, due to x_1 , takes place.

Also, we have multiple integrals of a multivariate function, with respect to its arguments, denoted by $\int \int \dots \int f(x_1, x_2, \dots, x_n) dx_1 dx_2 \dots dx_n$, as an ordered sequence of integration operations, beginning with respect to the argument x_1 and continue until the last one, due to x_n , takes place. These operations follow the same rules of differentiation and integration for a function of one argument.

A.12 Complex Functions

A function that maps complex numbers into complex numbers is called a complex function, such as $f : z \rightarrow u$, or, $u = f(z)$, where z, u are complex numbers. Hence, these functions map complex numbers in their domains to complex numbers in their ranges.

Example 8

$u = f(z)$, where f is a linear form, i.e. $f(z) = az + b$, and a, b are real numbers.

Thus, $u = a(x + iy) + b = ax + b + iay = (ax + b, y)$. Hence, this function maps the complex point (x, y) into the complex point $(ax + b, y)$.

The function $f(z) = a_n z^n + a_{n-1} z^{n-1} + \dots + a_1 z + a_0$ is called a polynomial of degree n , and sometimes denoted by $P(z)$, where a_i s are complex numbers, such that $a_n \neq 0$ and n is an integer number.

Ratios of polynomials, $P(z)/Q(z)$, are generally called as rational functions, for $Q(z) \neq 0$.

Properties of complex functions cannot be illustrated by a graphical representation, contrary to our experience with the case of real functions.

Yet, we can treat the limit of a complex function as an extension of the limits of real functions, such that $f(z) = u(x, y) + iv(x, y)$, when we have $z = x + iy$, $\lim_{z \rightarrow z_0} f(z) = u_0 + iv_0$ if and only if $\lim_{(x,y) \rightarrow (x_0,y_0)} u(x, y) = u_0$ and $\lim_{(x,y) \rightarrow (x_0,y_0)} v(x, y) = v_0$.

Also, we have $\lim_{z \rightarrow z_0} [f(z) + g(z)] = \lim_{z \rightarrow z_0} f(z) + \lim_{z \rightarrow z_0} g(z)$,
 $\lim_{z \rightarrow z_0} [f(z)g(z)] = \lim_{z \rightarrow z_0} f(z) \lim_{z \rightarrow z_0} g(z)$, and
 $\lim_{z \rightarrow z_0} [f(z)/g(z)] = \lim_{z \rightarrow z_0} f(z) / \lim_{z \rightarrow z_0} g(z)$, when $g(z) \neq 0$.

When $\lim_{z \rightarrow z_0} f(z)$ and $f(z_0)$ exists, then we say the function $f(z)$ is continuous at z_0 , if $\lim_{z \rightarrow z_0} f(z) = f(z_0)$.

Thus, derivative of a complex function $f(z)$, at point z_0 , is defined as $\frac{d}{dz} f(z) = \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0}$.

If this limit exists, then the function is called differentiable at point z_0 . In this construct, we can see that the differentiation of complex functions follow the same rules of real functions.

Beware of the fact that a complex function may have its real and imaginary parts having continuous partial derivatives at a point, yet the function may not be differentiable at that point.

Also, note that the continuity of a function at a point does not imply the differentiability of the function at that point, but differentiability of a function at a point implies continuity of the function at that point.

Let a function $f(z) = u(x, y) + iv(x, y)$ exists on a neighborhood of a point z_0 .

In addition, let $\frac{d}{dz}f(z_0) = \lim_{\Delta z \rightarrow 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z}$ exists.

Here, $z_0 = x_0 + iy_0$, $\Delta z = \Delta x + i\Delta y$, and $\Delta x = x - x_0$, $\Delta y = y - y_0$.

Thus, $Re\{\frac{d}{dz}f(z_0)\} = \lim_{(\Delta x, \Delta y) \rightarrow (0,0)} Re\{\frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z}\},$

$$Im\{\frac{d}{dz}f(z_0)\} = \lim_{(\Delta x, \Delta y) \rightarrow (0,0)} Im\{\frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z}\}.$$

If $\Delta z = \Delta x$, then

$$Re\{\frac{d}{dz}f(z_0)\} = \lim_{\Delta x \rightarrow 0} \frac{u(x_0 + \Delta x, y_0) - u(x_0, y_0)}{\Delta x},$$

$$Im\{\frac{d}{dz}f(z_0)\} = \lim_{\Delta x \rightarrow 0} \frac{v(x_0 + \Delta x, y_0) - v(x_0, y_0)}{\Delta x}.$$

Hence, $f'(z_0) = u_x(x_0, y_0) + iv_x(x_0, y_0)$, where $u_x(x_0, y_0), v_x(x_0, y_0)$ are partial derivatives of u, v with respect to x , at the point (x_0, y_0) .

Similarly, if $\Delta z = \Delta y$, then $f'(z_0) = v_y(x_0, y_0) - iu_y(x_0, y_0)$, and $v_y(x_0, y_0), u_y(x_0, y_0)$ are partial derivatives of v, u with respect to y , at the point (x_0, y_0) .

Therefore, after setting real parts equal and imaginary parts equal, we obtain that $u_x(x_0, y_0) = v_y(x_0, y_0)$ and $u_y(x_0, y_0) = -v_x(x_0, y_0)$.

These are called Cauchy-Reimann equations for the existence of $f'(z_0)$.

In the polar coordinate form, $x = r\cos\theta$, $y = r\sin\theta$ and $z = re^{i\theta}$.

$f(z) = u + iv$ is differentiable at $z_0 = r_0 e^{i\theta_0}$, if $u_r(r_0, \theta_0) = \frac{1}{r_0}v_\theta(r_0, \theta_0)$ and $\frac{1}{r_0}u_\theta(r_0, \theta_0) = -v_r(r_0, \theta_0)$.

Here, u_r, v_r are partial derivatives of u, v with respect to r and u_θ, v_θ are partial derivatives of u, v with respect to θ .

A.13 Linear Algebra

An algebra, which is a set of elements and their operations, was constructed to express linear relations among its members. Let A^n be an n -dimensional space (set of points with n -components). Then any point in A (an n -tuple) is a “vector” in A , with respect to a reference frame (a fixed point in A with a coordinate system). A vector is the position of a point in A , with a distance (magnitude) and with a direction (orientation), relative to the reference frame. For example, $a \in A$ is an n -tuple point in A , denoted $a = (a_1, a_2, \dots, a_n)$ and in this form it is called a “row vector.” Which is the convenient way of writing it. If we write the components of a , stacked as a column, then it is called a “column vector.” The reference frame is

identified by a basis, which is a collection of vectors in A , such that any point (vector) in A can be expressed as the linear combination of these basis vectors. Let $B = (b_1, b_2, \dots, b_n)$ be a basis of A , then any $a \in A$ can be identified by using B , such that $a = (c_1 b_1, c_2 b_2, \dots, c_n b_n)$, where $c_1, \dots, c_n$ are real numbers (coordinates of a , in terms of B).

The magnitude of a real vector, a , is called the norm of a and denoted by $\|a\|$, with the following properties.

- n1. $\|a\| > 0$, and $\|a\| = 0$ if and only is $a = 0$.
- n2. $\|c.a\| = |c|. \|a\|$, for a real constant c .
- n3. $\|a + b\| \leq \|a\| + \|b\|$, for any two real vectors a and b (Triangle inequality).

Note; if $\|a\| = 0$ does not imply that $a = 0$ with n2 and n3, then the norm is called a semi-norm.

A commonly used norm for real vectors is the euclidean norm, such that $\|a\| = \sqrt{(a_1^2 + a_2^2 + \dots + a_n^2)}$.

We can impose an algebra, on the space A , by introducing some operations and particular points in A , relative to these operations.

(1) Addition:

There is a point in A , such that it is located at the fixed point (origin) of A and it is called the “zero vector,” denoted by 0 .

If a is a point in A , then $a + 0 = a$.

If a and b are two points in A , then $a + b$ is a point in A .

If a is a point in A , then there exists another point c (additive inverse) in A , such that $a + c = 0$.

(2) Scalar Multiplication:

If λ is a real number (scalar) and a is a point in A , then $\lambda.a$ is a point in A , where it is the component-wise multiplication of a with λ , that is, $(\lambda a_1, \lambda a_2, \dots, \lambda a_n)$.

(3) Transpose:

If $a \in A$ is a row (column) vector, then its transpose $a^T \in A$ is a column (row) vector.

(4) Inner Product:

If $a \in A$ and $b \in A$, then $\langle a, b \rangle$ is the componentwise multiplication of a and b , then summing up all the multiplications, that is

$$\langle a, b \rangle = a_1 b_1 + a_2 b_2 + a_3 b_3 + \dots + a_n b_n.$$

(5) Outer Product:

If a and b are two row vectors in A , then $a \times b$ is the ordered collection of all scalar products of b with the components of a , taken as scalars, that is

$$a \times b = \begin{pmatrix} a_1b_1 & a_1b_2 & a_1b_3 & \dots & a_1b_n \\ a_2b_1 & a_2b_2 & a_2b_3 & \dots & a_2b_n \\ \vdots & \vdots & \vdots & \dots & \vdots \\ a_nb_1 & a_nb_2 & a_nb_3 & \dots & a_nb_n \end{pmatrix}.$$

Outer product generates a set of specific points in A , which can be enclosed as a single entity, called a “matrix.”

In fact, a matrix is an ordered collection of vectors, $M = (a_1, a_2, a_3, \dots, a_m)$, where each a_i is an n -dimensional column vector, and M is also known as an $n \times m$ -dimensional matrix, denoted by $M = [a_{ij}]$ that identifies the elements of M , located at the i – th row and j – th column of M .

Matrices are also elements of the linear algebra, together with their operations.

(6) Matrix Transpose:

M^T denotes the transpose of the matrix M , which is the collection of the transpose of a_i s (row vectors). Therefore, M^T is an $m \times n$ -dimensional matrix, when M is $n \times m$ -dimensional.

(7) Scalar Matrix Product:

If λ is a scalar (real number), then λM is the multiplication of all elements of M by λ .

(8) Matrix Addition:

If M_1 and M_2 are two $n \times m$ -dimensional matrices, then $M_1 + M_2$ is another $n \times m$ -dimensional matrix, with component-wise addition of M_1 and M_2 .

(9) Matrix Product:

We multiply two matrices, M_1 and M_2 , if the number of columns of M_1 is the same as the number of rows of M_2 , as the inner product of rows of M_1 with columns of M_2 .

Zero Matrix:

A zero matrix, O , is the collection of zero vectors. Square Matrix:

If $n = m$, then the $n \times m$ -dimensional matrix is called a square matrix.

Identity Matrix:

If all diagonal elements of a $n \times n$ -dimensional square matrix are one, $[a_{ii}] = 1$, for $i = 1, 2, \dots, n$ and all other elements are zero, then this is an n -dimensional identity matrix, denoted by $I_{n \times n}$.

Determinant of a Square Matrix:

Determinant of a square matrix, M , is the volume generated by its constituent vectors with a positive, or, a negative sign assigned, denoted by $\det(M)$, or $|M|$.

Determinant of M is also known as the sum of all possible signed permutation of the elements of M .

Computation of determinant:

$$|M| = \sum_{j=1, \dots, n} (-1)^{i+j} \text{minor}(M_{ij}), \text{ for any } i = 1, 2, \dots, n,$$

where, $\text{minor}(M_{ij}) = |M \text{ remove column } a_i \text{ and row } a_j|$.

(10) Direct Sum:

The direct sum of matrices, $M_1, M_2, M_3, \dots$, would be a new matrix, such that $[a_{ij}] = M_1 \oplus M_2 \oplus M_3 \dots$, where the major diagonal elements, a_{ii} for $i = 1, \dots$, are the constituent matrices, $M_1, M_2, \dots$, and other elements of $[a_{ij}]$ are zero.

The direct sum of vectors, $a = (a_1, \dots, a_n)$, $b = (b_1, \dots, b_m)$, is a new vector $a \oplus b = (a_1, \dots, a_n, b_1, \dots, b_m)$.

(11) Tensor Product:

The set of numbers $T = [T_{i_1, \dots, i_m}]$ with m indices spread over n coordinates is known as a tensor of degree m with dimension n .

Scalars, vectors and matrices are tensors of degrees 0, 1 and 2, respectively.

The tensor product of tensors A of degree p and B of degree q is a new tensor C of degree $p + q$, denoted by $C = A \otimes B$.

If λ is a scalar and a, b are vectors, then $\lambda \otimes a = \lambda a$ and $a \otimes b = ab^T$. The matrix of $A \otimes B$ is also known as the Kronecker product of matrices A and B , in order to construct the Kronecker product of A and B , multiply each element of the matrix A by the matrix B , such that

$$A \otimes B = \begin{pmatrix} a_{11}B & \dots & a_{1n}B \\ \vdots & & \vdots \\ a_{m1}B & \dots & a_{mn}B \end{pmatrix}.$$

A.14 Exercises

1. Find $\lim_{x \rightarrow 0} \frac{\sin x}{x}$
2. Find $f'(x)$ where $f(x) = c$, c is a constant real number.
3. Find $f'(x)$ where $f(x) = ax + b$, a, b are constant real numbers.
4. Find $f'(x)$ where $f(x) = ax^2 + bx + c$, a, b, c are real constants.
5. Find $\int_0^1 f(x)dx$ where $f(x) = c$, c is a real constant.
6. Find $\int_0^1 f(x)dx$ where $f(x) = ax + b$, a, b are real constants.
7. Find $\int_0^1 f(x)dx$ where $f(x) = ax^2 + bx + c$, a, b, c are real constants.
8. Find $\int_0^1 (1/x)dx$.
9. Find $\int_0^1 e^x dx$.
10. Add $a_1 = (1, 1, 1)$ and $a_2 = (2, 1, 0)$.
11. Multiply $c = 1.5$ and $a = (1, 2, 3, 4)$.
12. Find the transpose of $a = (1, 2, 3)$.
13. Find the inner and outer products of $a_1 = (1, 2, 3)$ and $a_2 = (0, -1, -2)$.
14. Find the determinant of $M = [a_{ij}]$,
where $a_{11} = 1, a_{12} = 1, a_{21} = 0, a_{22} = 1$.
15. Find the direct sum and Kronecker product of two Identity Matrices,
 $I_{n \times n}$ and $I_{m \times m}$.
16. Find the direct sum of two vectors, $a = (1, 2, 3)$ and $b = (0, 1)$.

Appendix B

Tables for Commonly Used cdfs

All these tables were generated by either using the Excel software, which is commonly available on many personal computers, or the “R” software, which can be downloaded from the internet, and we strongly suggest using computers to get the numerical values for probabilities, inverse probabilities, cdfs, and the like, for many different distributions. If you cannot have access to a computer, then the following tables may serve you to some degree.

B.1 Standard Normal Distribution.

The cdf for the standard normal random variable, z , is given in the following table, where a particular z value is equal to the left-margin value (boldface), plus top-margin value (boldface) and corresponding $P(Z \leq z)$ is given at the intersection of the left-margin value and the top-margin value, rounded to four digits after the decimal point. Since standard normal pdf is symmetric around zero and $P(Z \leq 0) = 0.5$, it is sufficient to tabulate for $z \geq 0$. For $z \leq 0$, use $P(Z \leq -z) = 1 - P(Z \leq z)$.

For example, $P(Z \leq 1.95) = 0.9744$ and $P(Z \leq -1.95) = 1 - 0.9744 = 0.0256$, from the table.

When you know the probability, but not the value of z , you can use the table to find the inverse probability as well. For example, $P(Z \leq z) = 0.9744$ implies that $z = 1.95$. Similarly, if $P(Z \leq z) = 0.0256$, then $z = -1.95$.

In case of missing values, you can use interpolation (choosing a value between two known values) to approximate it, since this pdf is smoothly continuous.

z	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998
3.5	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9999

B.2 Student's t-Distribution.

In this table, for a fixed degrees of freedom value (left-margin, boldface) and a given cdf, $P(T \leq t)$, value (top-margin, boldface), we can find the generating t value, rounded to four digits after the decimal point, as the intersection of the left-margin value and the top-margin value.

For example, $P(T \leq t) = 0.975$, for $df = 20$, we found that $t = 2.086$

The last row of the table is for infinite degrees of freedom, and for that value of the degrees of freedom, the Student's t-distribution behaves like a standard normal distribution.

For the missing values, you can use interpolation to approximate them. Since this pdf is also smoothly continuous.

When necessary, we can use $t_{df,\alpha} = -t_{df,1-\alpha}$ where $P(T \geq t_{df,\alpha}) = \alpha$ is the tail area, generated by the point $t_{df,\alpha}$.

<i>df</i>	0.9	0.95	0.975	0.99	0.995	0.999
1	3.0777	6.3138	12.7062	31.8205	63.6567	318.3088
2	1.8856	2.9200	4.3027	6.9646	9.9248	22.3271
3	1.6377	2.3534	3.1824	4.5407	5.8409	10.2145
4	1.5332	2.1318	2.7764	3.7469	4.6041	7.1732
5	1.4759	2.0150	2.5706	3.3649	4.0321	5.8934
6	1.4398	1.9432	2.4469	3.1427	3.7074	5.2076
7	1.4149	1.8946	2.3646	2.9980	3.4995	4.7853
8	1.3968	1.8595	2.3060	2.8965	3.3554	4.5008
9	1.3830	1.8331	2.2622	2.8214	3.2498	4.2968
10	1.3722	1.8125	2.2281	2.7638	3.1693	4.1437
11	1.3634	1.7959	2.2010	2.7181	3.1058	4.0247
12	1.3562	1.7823	2.1788	2.6810	3.0545	3.9296
13	1.3502	1.7709	2.1604	2.6503	3.0123	3.8520
14	1.3450	1.7613	2.1448	2.6245	2.9768	3.7874
15	1.3406	1.7531	2.1314	2.6025	2.9467	3.7328
16	1.3368	1.7459	2.1199	2.5835	2.9208	3.6862
17	1.3334	1.7396	2.1098	2.5669	2.8982	3.6458
18	1.3304	1.7341	2.1009	2.5524	2.8784	3.6105
19	1.3277	1.7291	2.0930	2.5395	2.8609	3.5794
20	1.3253	1.7247	2.0860	2.5280	2.8453	3.5518
21	1.3232	1.7207	2.0796	2.5176	2.8314	3.5272
22	1.3212	1.7171	2.0739	2.5083	2.8188	3.5050
23	1.3195	1.7139	2.0687	2.4999	2.8073	3.4850
24	1.3178	1.7109	2.0639	2.4922	2.7969	3.4668
25	1.3163	1.7081	2.0595	2.4851	2.7874	3.4502
26	1.3150	1.7056	2.0555	2.4786	2.7787	3.4350
27	1.3137	1.7033	2.0518	2.4727	2.7707	3.4210
28	1.3125	1.7011	2.0484	2.4671	2.7633	3.4082
29	1.3114	1.6991	2.0452	2.4620	2.7564	3.3962
30	1.3104	1.6973	2.0423	2.4573	2.7500	3.3852
31	1.3095	1.6955	2.0395	2.4528	2.7440	3.3749
32	1.3086	1.6939	2.0369	2.4487	2.7385	3.3653
33	1.3077	1.6924	2.0345	2.4448	2.7333	3.3563
34	1.3070	1.6909	2.0322	2.4411	2.7284	3.3479
35	1.3062	1.6896	2.0301	2.4377	2.7238	3.3400
∞	1.2816	1.6449	1.9600	2.3264	2.5758	3.0902

B.3 χ^2 Distribution.

In this table, for a fixed degrees of freedom value (left-margin, boldface) and a given cdf, $P(X \leq x)$, value (top-margin, boldface), we can find the generating chi-square (x) value, rounded with four digits after the decimal point, at the intersection of the top-margin value and the left-margin value.

For example, when the degrees of freedom is 10, and the cdf value is 0.95, the chi-square value would be 18.3070.

<i>df</i>	0.01	0.025	0.05	0.1	0.9	0.95	0.975	0.99
1	0.0002	0.0010	0.0039	0.0158	2.7055	3.8415	5.0239	6.6349
2	0.0201	0.0506	0.1026	0.2107	4.6052	5.9915	7.3778	9.2103
3	0.1148	0.2158	0.3518	0.5844	6.2514	7.8147	9.3484	11.3449
4	0.2971	0.4844	0.7107	1.0636	7.7794	9.4877	11.1433	13.2767
5	0.5543	0.8312	1.1455	1.6103	9.2364	11.0705	12.8325	15.0863
6	0.8721	1.2373	1.6354	2.2041	10.6446	12.5916	14.4494	16.8119
7	1.2390	1.6899	2.1673	2.8331	12.0170	14.0671	16.0128	18.4753
8	1.6465	2.1797	2.7326	3.4895	13.3616	15.5073	17.5345	20.0902
9	2.0879	2.7004	3.3251	4.1682	14.6837	16.9190	19.0228	21.6660
10	2.5582	3.2470	3.9403	4.8652	15.9872	18.3070	20.4832	23.2093
11	3.0535	3.8157	4.5748	5.5778	17.2750	19.6751	21.9200	24.7250
12	3.5706	4.4038	5.2260	6.3038	18.5493	21.0261	23.3367	26.2170
13	4.1069	5.0088	5.8919	7.0415	19.8119	22.3620	24.7356	27.6882
14	4.6604	5.6287	6.5706	7.7895	21.0641	23.6848	26.1189	29.1412
15	5.2293	6.2621	7.2609	8.5468	22.3071	24.9958	27.4884	30.5779
16	5.8122	6.9077	7.9616	9.3122	23.5418	26.2962	28.8454	31.9999
17	6.4078	7.5642	8.6718	10.0852	24.7690	27.5871	30.1910	33.4087
18	6.4078	7.5642	8.6718	10.0852	24.7690	27.5871	30.1910	33.4087
19	7.6327	8.9065	10.1170	11.6509	27.2036	30.1435	32.8523	36.1909
20	8.2604	9.5908	10.8508	12.4426	28.4120	31.4104	34.1696	37.5662
21	8.8972	10.2829	11.5913	13.2396	29.6151	32.6706	35.4789	38.9322
22	9.5425	10.9823	12.3380	14.0415	30.8133	33.9244	36.7807	40.2894
23	10.1957	11.6886	13.0905	14.8480	32.0069	35.1725	38.0756	41.6384
24	10.8564	12.4012	13.8484	15.6587	33.1962	36.4150	39.3641	42.9798
25	11.5240	13.1197	14.6114	16.4734	34.3816	37.6525	40.6465	44.3141
26	12.1981	13.8439	15.3792	17.2919	35.5632	38.8851	41.9232	45.6417
27	12.8785	14.5734	16.1514	18.1139	36.7412	40.1133	43.1945	46.9629
28	13.5647	15.3079	16.9279	18.9392	37.9159	41.3371	44.4608	48.2782
29	14.2565	16.0471	17.7084	19.7677	39.0875	42.5570	45.7223	49.5879
30	14.9535	16.7908	18.4927	20.5992	40.2560	43.7730	46.9792	50.8922
31	15.6555	17.5387	19.2806	21.4336	41.4217	44.9853	48.2319	52.1914
32	16.3622	18.2908	20.0719	22.2706	42.5847	46.1943	49.4804	53.4858
33	17.0735	19.0467	20.8665	23.1102	43.7452	47.3999	50.7251	54.7755
34	17.7891	19.8063	21.6643	23.9523	44.9032	48.6024	51.9660	56.0609
35	18.5089	20.5694	22.4650	24.7967	46.0588	49.8018	53.2033	57.3421
36	19.2327	21.3359	23.2686	25.6433	47.2122	50.9985	54.4373	58.6192
37	19.9602	22.1056	24.0749	26.4921	48.3634	52.1923	55.6680	59.8925
38	20.6914	22.8785	24.8839	27.3430	49.5126	53.3835	56.8955	61.1621
39	21.4262	23.6543	25.6954	28.1958	50.6598	54.5722	58.1201	62.4281
40	22.1643	24.4330	26.5093	29.0505	51.8051	55.7585	59.3417	63.6907

B.4 F-Distribution.

These tables are for the cdf of F-distribution, $P(F \leq f) = cdf(f)$, where the top-margin values (boldface) of the table refer to the numerator degrees of freedom, and the left-margin values (boldface) of the table refer to the denominator degrees of freedom. The cdf value (boldface) is printed on the top left corner of a table. The numbers on the tables are the f values to generate the given cdf, for the intersection of numerator and denominator degrees of freedom, rounded up to two digits after the decimal point.

cdf=0.9	1	2	3	4	5	10	20	30	60	120	∞
1	39.86	49.50	53.59	55.83	57.24	60.19	61.74	62.26	62.79	63.06	63.33
2	8.53	9.00	9.16	9.24	9.29	9.39	9.44	9.46	9.47	9.48	9.49
3	5.54	5.46	5.39	5.34	5.31	5.23	5.18	5.17	5.15	5.14	5.13
4	4.54	4.32	4.19	4.11	4.05	3.92	3.84	3.82	3.79	3.78	3.76
5	4.06	3.78	3.62	3.52	3.45	3.30	3.21	3.17	3.14	3.12	3.10
10	3.29	2.92	2.73	2.61	2.52	2.32	2.20	2.16	2.11	2.08	2.06
20	2.97	2.59	2.38	2.25	2.16	1.94	1.79	1.74	1.68	1.64	1.61
30	2.88	2.49	2.28	2.14	2.05	1.82	1.67	1.61	1.54	1.50	1.46
60	2.79	2.39	2.18	2.04	1.95	1.71	1.54	1.48	1.40	1.35	1.29
120	2.75	2.35	2.13	1.99	1.90	1.65	1.48	1.41	1.32	1.26	1.19
∞	2.71	2.30	2.08	1.94	1.85	1.60	1.42	1.34	1.24	1.17	1.00

cdf=0.95	1	2	3	4	5	10	20	30	60	120	∞
1	161.45	199.50	215.71	224.58	230.16	241.88	248.01	250.10	252.20	253.25	254.31
2	18.51	19.00	19.16	19.25	19.30	19.40	19.45	19.46	19.48	19.49	19.50
3	10.13	9.55	9.28	9.12	9.01	8.79	8.66	8.62	8.57	8.55	8.53
4	7.71	6.94	6.59	6.39	6.26	5.96	5.80	5.75	5.69	5.66	5.63
5	6.61	5.79	5.41	5.19	5.05	4.74	4.56	4.50	4.43	4.40	4.36
10	4.96	4.10	3.71	3.48	3.33	2.98	2.77	2.70	2.62	2.58	2.54
20	4.35	3.49	3.10	2.87	2.71	2.35	2.12	2.04	1.95	1.90	1.84
30	4.17	3.32	2.92	2.69	2.53	2.16	1.93	1.84	1.74	1.68	1.62
60	4.00	3.15	2.76	2.53	2.37	1.99	1.75	1.65	1.53	1.47	1.39
120	3.92	3.07	2.68	2.45	2.29	1.91	1.66	1.55	1.43	1.35	1.25
∞	3.84	3.00	2.60	2.37	2.21	1.83	1.57	1.46	1.32	1.22	1.00

cdf=0.975	1	2	3	4	5	10	20	30	60	120	∞
1	647.79	799.50	864.16	899.58	921.85	968.63	993.10	1001.41	1009.80	1014.02	1018.23
2	38.51	39.00	39.17	39.25	39.30	39.40	39.45	39.46	39.48	39.49	39.50
3	17.44	16.04	15.44	15.10	14.88	14.42	14.17	14.08	13.99	13.95	13.90
4	12.22	10.65	9.98	9.60	9.36	8.84	8.56	8.46	8.36	8.31	8.26
5	10.01	8.43	7.76	7.39	7.15	6.62	6.33	6.23	6.12	6.07	6.02
10	6.94	5.46	4.83	4.47	4.24	3.72	3.42	3.31	3.20	3.14	3.08
20	5.87	4.46	3.86	3.51	3.29	2.77	2.46	2.35	2.22	2.16	2.09
30	5.57	4.18	3.59	3.25	3.03	2.51	2.20	2.07	1.94	1.87	1.79
60	5.29	3.93	3.34	3.01	2.79	2.27	1.94	1.82	1.67	1.58	1.48
120	5.15	3.80	3.23	2.89	2.67	2.16	1.82	1.69	1.53	1.43	1.31
∞	5.02	3.69	3.12	2.79	2.57	2.05	1.71	1.57	1.39	1.27	1.00

cdf=0.99	1	2	3	4	5	10	20	30	60	120	∞
1	4052.18	4999.50	5403.35	5624.58	5763.65	6055.85	6208.73	6260.65	6313.03	6339.39	6365.68
2	98.50	99.00	99.17	99.25	99.30	99.40	99.45	99.47	99.48	99.49	99.50
3	34.12	30.82	29.46	28.71	28.24	27.23	26.69	26.50	26.32	26.22	26.13
4	21.20	18.00	16.69	15.98	15.52	14.55	14.02	13.84	13.65	13.56	13.46
5	16.26	13.27	12.06	11.39	10.97	10.05	9.55	9.38	9.20	9.11	9.02
10	10.04	7.56	6.55	5.99	5.64	4.85	4.41	4.25	4.08	4.00	3.91
20	8.10	5.85	4.94	4.43	4.10	3.37	2.94	2.78	2.61	2.52	2.42
30	7.56	5.39	4.51	4.02	3.70	2.98	2.55	2.39	2.21	2.11	2.01
60	7.08	4.98	4.13	3.65	3.34	2.63	2.20	2.03	1.84	1.73	1.60
120	6.85	4.79	3.95	3.48	3.17	2.47	2.03	1.86	1.66	1.53	1.38
∞	6.63	4.61	3.78	3.32	3.02	2.32	1.88	1.70	1.47	1.32	1.00

B.5 Standardized Range Distribution

These tables are for the Tukey's method for simultaneous pairwise comparison of average treatment effects. There are only three tables provided for three different levels of significances; $\alpha = 0.10$, 0.05 , and 0.01 . If more information is needed, there is an "R" software package available for the Tukey's method.

The Studentized range upper quantiles $Q(k, df; 0.10)$

df	k→ 2	3	4	5	6	7	8	9	10
1	8.929	13.437	16.358	18.488	20.150	21.504	22.642	23.621	24.477
2	4.129	5.733	6.772	7.538	8.139	8.633	9.049	9.409	9.725
3	3.328	4.467	5.199	5.738	6.162	6.511	6.806	7.062	7.287
4	3.015	3.976	4.586	5.035	5.388	5.679	5.926	6.139	6.327
5	2.850	3.717	4.264	4.664	4.979	5.238	5.458	5.648	5.816
6	2.748	3.558	4.065	4.435	4.726	4.966	5.168	5.344	5.499
7	2.679	3.451	3.931	4.280	4.555	4.780	4.971	5.137	5.283
8	2.630	3.374	3.834	4.169	4.431	4.646	4.829	4.987	5.126
9	2.592	3.316	3.761	4.084	4.337	4.545	4.721	4.873	5.007
10	2.563	3.270	3.704	4.018	4.264	4.465	4.636	4.783	4.913
11	2.540	3.234	3.658	3.965	4.205	4.401	4.567	4.711	4.838
12	2.521	3.204	3.621	3.921	4.156	4.349	4.511	4.652	4.776
13	2.504	3.179	3.589	3.885	4.116	4.304	4.464	4.602	4.724
14	2.491	3.158	3.563	3.854	4.081	4.267	4.424	4.560	4.679
15	2.479	3.140	3.540	3.828	4.052	4.235	4.390	4.524	4.641
16	2.469	3.124	3.520	3.804	4.026	4.207	4.360	4.492	4.608
17	2.460	3.110	3.503	3.784	4.003	4.182	4.334	4.464	4.579
18	2.452	3.098	3.487	3.766	3.984	4.161	4.310	4.440	4.553
19	2.445	3.087	3.474	3.751	3.966	4.142	4.290	4.418	4.530
20	2.439	3.077	3.462	3.736	3.950	4.124	4.271	4.398	4.510
21	2.433	3.069	3.451	3.724	3.936	4.109	4.255	4.380	4.491
22	2.428	3.061	3.441	3.712	3.923	4.095	4.239	4.364	4.474
23	2.424	3.054	3.432	3.701	3.911	4.082	4.226	4.350	4.459
24	2.420	3.047	3.423	3.692	3.900	4.070	4.213	4.336	4.445
25	2.416	3.041	3.416	3.683	3.890	4.059	4.201	4.324	4.432
26	2.412	3.036	3.409	3.675	3.881	4.049	4.191	4.313	4.420
27	2.409	3.030	3.402	3.667	3.873	4.040	4.181	4.302	4.409
28	2.406	3.026	3.396	3.660	3.865	4.032	4.172	4.293	4.399
29	2.403	3.021	3.391	3.654	3.858	4.024	4.163	4.284	4.389
30	2.400	3.017	3.386	3.648	3.851	4.016	4.155	4.275	4.381
31	2.398	3.013	3.381	3.642	3.845	4.009	4.148	4.268	4.372
32	2.396	3.010	3.376	3.637	3.839	4.003	4.141	4.260	4.365
33	2.393	3.006	3.372	3.632	3.833	3.997	4.135	4.253	4.357
34	2.391	3.003	3.368	3.627	3.828	3.991	4.129	4.247	4.351
35	2.389	3.000	3.364	3.623	3.823	3.986	4.123	4.241	4.344
36	2.388	2.998	3.361	3.619	3.819	3.981	4.117	4.235	4.338
37	2.386	2.995	3.357	3.615	3.814	3.976	4.112	4.230	4.332
38	2.384	2.992	3.354	3.611	3.810	3.972	4.107	4.224	4.327
39	2.383	2.990	3.351	3.608	3.806	3.967	4.103	4.220	4.322
40	2.381	2.988	3.348	3.605	3.802	3.963	4.099	4.215	4.317
48	2.372	2.973	3.330	3.583	3.778	3.937	4.070	4.185	4.285
60	2.363	2.959	3.312	3.562	3.755	3.911	4.042	4.155	4.254
80	2.353	2.945	3.294	3.541	3.731	3.885	4.014	4.125	4.223
120	2.344	2.930	3.276	3.520	3.707	3.859	3.986	4.096	4.191
240	2.335	2.916	3.258	3.499	3.684	3.834	3.959	4.066	4.160
∞	2.326	2.902	3.240	3.478	3.661	3.808	3.931	4.037	4.129

The Studentized range upper quantiles $Q(k, df; 0.05)$

df	k-> 2	3	4	5	6	7	8	9	10
1	17.969	26.976	32.819	37.082	40.408	43.119	45.397	47.357	49.071
2	6.085	8.331	9.798	10.881	11.734	12.435	13.027	13.539	13.988
3	4.501	5.910	6.825	7.502	8.037	8.478	8.852	9.177	9.462
4	3.926	5.040	5.757	6.287	6.706	7.053	7.347	7.602	7.826
5	3.635	4.602	5.218	5.673	6.033	6.330	6.582	6.801	6.995
6	3.460	4.339	4.896	5.305	5.628	5.895	6.122	6.319	6.493
7	3.344	4.165	4.681	5.060	5.359	5.606	5.815	5.997	6.158
8	3.261	4.041	4.529	4.886	5.167	5.399	5.596	5.767	5.918
9	3.199	3.948	4.415	4.755	5.024	5.244	5.432	5.595	5.738
10	3.151	3.877	4.327	4.654	4.912	5.124	5.304	5.460	5.598
11	3.113	3.820	4.256	4.574	4.823	5.028	5.202	5.353	5.486
12	3.081	3.773	4.199	4.508	4.750	4.950	5.119	5.265	5.395
13	3.055	3.734	4.151	4.453	4.690	4.884	5.049	5.192	5.318
14	3.033	3.701	4.111	4.407	4.639	4.829	4.990	5.130	5.253
15	3.014	3.673	4.076	4.367	4.595	4.782	4.940	5.077	5.198
16	2.998	3.649	4.046	4.333	4.557	4.741	4.896	5.031	5.150
17	2.984	3.628	4.020	4.303	4.524	4.705	4.858	4.991	5.108
18	2.971	3.609	3.997	4.276	4.494	4.673	4.824	4.955	5.071
19	2.960	3.593	3.977	4.253	4.468	4.645	4.794	4.924	5.037
20	2.950	3.578	3.958	4.232	4.445	4.620	4.768	4.895	5.008
21	2.941	3.565	3.942	4.213	4.424	4.597	4.743	4.870	4.981
22	2.933	3.553	3.927	4.196	4.405	4.577	4.722	4.847	4.957
23	2.926	3.542	3.914	4.180	4.388	4.558	4.702	4.826	4.935
24	2.919	3.532	3.901	4.166	4.373	4.541	4.684	4.807	4.915
25	2.913	3.523	3.890	4.153	4.358	4.526	4.667	4.789	4.897
26	2.907	3.514	3.880	4.141	4.345	4.511	4.652	4.773	4.880
27	2.902	3.506	3.870	4.130	4.333	4.498	4.638	4.758	4.864
28	2.897	3.499	3.861	4.120	4.322	4.486	4.625	4.745	4.850
29	2.892	3.493	3.853	4.111	4.311	4.475	4.613	4.732	4.837
30	2.888	3.486	3.845	4.102	4.301	4.464	4.601	4.720	4.824
31	2.884	3.481	3.838	4.094	4.292	4.454	4.591	4.709	4.812
32	2.881	3.475	3.832	4.086	4.284	4.445	4.581	4.698	4.802
33	2.877	3.470	3.825	4.079	4.276	4.436	4.572	4.689	4.791
34	2.874	3.465	3.820	4.072	4.268	4.428	4.563	4.680	4.782
35	2.871	3.461	3.814	4.066	4.261	4.421	4.555	4.671	4.773
36	2.868	3.457	3.809	4.060	4.255	4.414	4.547	4.663	4.764
37	2.865	3.453	3.804	4.054	4.249	4.407	4.540	4.655	4.756
38	2.863	3.449	3.799	4.049	4.243	4.400	4.533	4.648	4.749
39	2.861	3.445	3.795	4.044	4.237	4.394	4.527	4.641	4.741
40	2.858	3.442	3.791	4.039	4.232	4.388	4.521	4.634	4.735
48	2.843	3.420	3.764	4.008	4.197	4.351	4.481	4.592	4.690
60	2.829	3.399	3.737	3.977	4.163	4.314	4.441	4.550	4.646
80	2.814	3.377	3.711	3.947	4.129	4.277	4.402	4.509	4.603
120	2.800	3.356	3.685	3.917	4.096	4.241	4.363	4.468	4.560
240	2.786	3.335	3.659	3.887	4.063	4.205	4.324	4.427	4.517
∞	2.772	3.314	3.633	3.858	4.030	4.170	4.286	4.387	4.474

The Studentized range upper quantiles $Q(k, df; 0.01)$

df	k-> 2	3	4	5	6	7	8	9	10
1	90.024	135.041	164.258	185.575	202.210	215.769	227.166	236.966	245.542
2	14.036	19.019	22.294	24.717	26.629	28.201	29.530	30.679	31.689
3	8.260	10.619	12.170	13.324	14.241	14.998	15.641	16.199	16.691
4	6.511	8.120	9.173	9.958	10.583	11.101	11.542	11.925	12.264
5	5.702	6.976	7.804	8.421	8.913	9.321	9.669	9.971	10.239
6	5.243	6.331	7.033	7.556	7.972	8.318	8.612	8.869	9.097
7	4.949	5.919	6.542	7.005	7.373	7.678	7.939	8.166	8.367
8	4.745	5.635	6.204	6.625	6.959	7.237	7.474	7.680	7.863
9	4.596	5.428	5.957	6.347	6.657	6.915	7.134	7.325	7.494
10	4.482	5.270	5.769	6.136	6.428	6.669	6.875	7.054	7.213
11	4.392	5.146	5.621	5.970	6.247	6.476	6.671	6.841	6.992
12	4.320	5.046	5.502	5.836	6.101	6.320	6.507	6.670	6.814
13	4.260	4.964	5.404	5.726	5.981	6.192	6.372	6.528	6.666
14	4.210	4.895	5.322	5.634	5.881	6.085	6.258	6.409	6.543
15	4.167	4.836	5.252	5.556	5.796	5.994	6.162	6.309	6.438
16	4.131	4.786	5.192	5.489	5.722	5.915	6.079	6.222	6.348
17	4.099	4.742	5.140	5.430	5.659	5.847	6.007	6.147	6.270
18	4.071	4.703	5.094	5.379	5.603	5.787	5.944	6.081	6.201
19	4.046	4.669	5.054	5.334	5.553	5.735	5.889	6.022	6.141
20	4.024	4.639	5.018	5.293	5.510	5.688	5.839	5.970	6.086
21	4.004	4.612	4.986	5.257	5.470	5.646	5.794	5.924	6.038
22	3.986	4.588	4.957	5.225	5.435	5.608	5.754	5.882	5.994
23	3.970	4.566	4.931	5.195	5.403	5.573	5.718	5.844	5.955
24	3.955	4.546	4.907	5.168	5.373	5.542	5.685	5.809	5.919
25	3.942	4.527	4.885	5.144	5.347	5.513	5.655	5.778	5.886
26	3.930	4.510	4.865	5.121	5.322	5.487	5.627	5.749	5.856
27	3.918	4.495	4.847	5.101	5.300	5.463	5.602	5.722	5.828
28	3.908	4.481	4.830	5.082	5.279	5.441	5.578	5.697	5.802
29	3.898	4.467	4.814	5.064	5.260	5.420	5.556	5.674	5.778
30	3.889	4.455	4.799	5.048	5.242	5.401	5.536	5.653	5.756
31	3.881	4.443	4.786	5.032	5.225	5.383	5.517	5.633	5.736
32	3.873	4.433	4.773	5.018	5.210	5.367	5.500	5.615	5.716
33	3.865	4.423	4.761	5.005	5.195	5.351	5.483	5.598	5.698
34	3.859	4.413	4.750	4.992	5.181	5.336	5.468	5.581	5.682
35	3.852	4.404	4.739	4.980	5.169	5.323	5.453	5.566	5.666
36	3.846	4.396	4.729	4.969	5.156	5.310	5.439	5.552	5.651
37	3.840	4.388	4.720	4.959	5.145	5.298	5.427	5.538	5.637
38	3.835	4.381	4.711	4.949	5.134	5.286	5.414	5.526	5.623
39	3.830	4.374	4.703	4.940	5.124	5.275	5.403	5.513	5.611
40	3.825	4.367	4.695	4.931	5.114	5.265	5.392	5.502	5.599
48	3.793	4.324	4.644	4.874	5.052	5.198	5.322	5.428	5.522
60	3.762	4.282	4.594	4.818	4.991	5.133	5.253	5.356	5.447
80	3.732	4.241	4.545	4.763	4.931	5.069	5.185	5.284	5.372
120	3.702	4.200	4.497	4.709	4.872	5.005	5.118	5.214	5.299
240	3.672	4.160	4.450	4.655	4.814	4.943	5.052	5.145	5.227
∞	3.643	4.120	4.403	4.603	4.757	4.882	4.987	5.078	5.157

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